

Problems in quantum mechanics

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A collection of quantum mechanics problems from various sources, loosely arranged by topic.
Some solutions are much older than others; any errors are my own!

Contents

1	Review of linear algebra	2
2	Review of classical mechanics	23
3	Matrix representations; diagonalization	26
4	Wave mechanics	35
5	Time evolution; Heisenberg and Schrödinger pictures	68
6	Harmonic oscillator	74
7	Path integrals	101
8	Uncertainty relations	103
9	Systems with N degrees of freedom	110
10	Symmetries	126
11	Rotations and spin	129
12	Scattering theory	162

1 Review of linear algebra

1-1 (Shankar, Exercise 1.4.1)

In a space $\mathbb{V}^n$, prove that the set of all vectors $\{|V_\perp^1\rangle, |V_\perp^2\rangle, \dots\}$, orthogonal to any $|V\rangle \neq 0$, form a subspace $\mathbb{V}^{n-1}$.

Proof. It's clear that $|0\rangle$ is in this set, since $\langle 0|V\rangle = 0$. This set is also closed under addition and scalar multiplication, since

$$\begin{aligned}\langle V|(\alpha|V_\perp^i\rangle + |V_\perp^j\rangle) &= \alpha\langle V|V_\perp^i\rangle + \langle V|V_\perp^j\rangle \\ &= \alpha \cdot 0 + 0 \\ &= 0.\end{aligned}$$

Thus $\{|V_\perp^1\rangle, |V_\perp^2\rangle, \dots\}$ is a vector subspace of $\mathbb{V}^n$. To show that this subspace is $n-1$ dimensional, we will exhibit a basis. Take any set of n linearly independent vectors in $\mathbb{V}^n$ containing $|V\rangle$, and apply the Gram-Schmidt procedure with $|V\rangle$ as the first vector to get $\{|\tilde{V}\rangle, |\tilde{V}_\perp^1\rangle, \dots, |\tilde{V}_\perp^{n-1}\rangle\}$. Then if $|U\rangle$ is any vector orthogonal to $|V\rangle$, we have

$$|U\rangle = \alpha_1|\tilde{V}_\perp^1\rangle + \dots + \alpha_{n-1}|\tilde{V}_\perp^{n-1}\rangle,$$

because these vectors are a basis for $\mathbb{V}^n$, and a nonzero component along $|\tilde{V}\rangle$ would imply that $\langle U|V\rangle \neq 0$, a contradiction. Thus the vectors $\{|\tilde{V}_\perp^1\rangle, \dots, |\tilde{V}_\perp^{n-1}\rangle\}$ span the subspace, and since they are linearly independent by construction, they form a basis. Thus the subspace is $n-1$ dimensional. $\square$

1-2 (Shankar, Exercise 1.4.2)

Suppose $\mathbb{V}_1^{n_1}$ and $\mathbb{V}_2^{n_2}$ are two subspaces such that any element of $\mathbb{V}_1$ is orthogonal to any element of $\mathbb{V}_2$. Show that the dimensionality of $\mathbb{V}_1 \oplus \mathbb{V}_2$ is $n_1 + n_2$.

Proof. By Theorem 4, the subspaces $\mathbb{V}_1^{n_1}$ and $\mathbb{V}_2^{n_2}$ admit a basis of n_1 and n_2 mutually orthogonal vectors, respectively, so by the definition of the subspaces, all $n_1 + n_2$ of these vectors are mutually orthogonal – in particular, they're linearly independent. And these clearly span $\mathbb{V}_1 \oplus \mathbb{V}_2$, so $\dim(\mathbb{V}_1 \oplus \mathbb{V}_2) = n_1 + n_2$. $\square$

1-3 (Shankar, Exercise 1.6.2)

Given Ω and Λ are Hermitian, what can you say about (1) $\Omega\Lambda$; (2) $\Omega\Lambda + \Lambda\Omega$; (3) $[\Omega, \Lambda]$; and (4) $i[\Omega, \Lambda]$?

Solution. (1) The operator $\Omega\Lambda$ is not generally Hermitian or anti-Hermitian:

$$\begin{aligned}(\Omega\Lambda)^\dagger &= \Lambda^\dagger\Omega^\dagger \\ &= \Lambda\Omega.\end{aligned}$$

(2) The operator $\Omega\Lambda + \Lambda\Omega$ is Hermitian:

$$\begin{aligned}(\Omega\Lambda + \Lambda\Omega)^\dagger &= \Lambda^\dagger\Omega^\dagger + \Omega^\dagger\Lambda^\dagger \\ &= \Lambda\Omega + \Omega\Lambda \\ &= \Omega\Lambda + \Lambda\Omega.\end{aligned}$$

(3) The operator $[\Omega, \Lambda]$ is anti-Hermitian:

$$\begin{aligned}(\Omega\Lambda - \Lambda\Omega)^\dagger &= \Lambda^\dagger\Omega^\dagger - \Omega^\dagger\Lambda^\dagger \\ &= \Lambda\Omega - \Omega\Lambda \\ &= -(\Omega\Lambda - \Lambda\Omega).\end{aligned}$$

(4) But tacking on an i turns the operator in (3) into one that's Hermitian:

$$\begin{aligned}\left(i(\Omega\Lambda - \Lambda\Omega)\right)^\dagger &= -i(\Omega\Lambda - \Lambda\Omega)^\dagger \\ &= i(\Omega\Lambda - \Lambda\Omega).\end{aligned}$$

In general, you can turn any Hermitian operator into an anti-Hermitian operator, or vice versa, by multiplying an i . □

1-4 (Shankar, Exercise 1.7.1)

The *trace* of a matrix is defined to be the sum of its diagonal matrix elements

$$\text{Tr } \Omega = \sum_i \Omega_{ii}.$$

Show that

- (1) $\text{Tr}(\Omega\Lambda) = \text{Tr}(\Lambda\Omega)$
 - (2) $\text{Tr}(\Omega\Lambda\theta) = \text{Tr}(\Lambda\theta\Omega) = \text{Tr}(\theta\Omega\Lambda)$ (The permutations are *cyclic*).
 - (3) The trace of an operator is unaffected by a unitary change of basis $|i\rangle \rightarrow U|i\rangle$. [Equivalently, show $\text{Tr } \Omega = \text{Tr}(U^\dagger \Omega U)$].
-

Solution. (1) Summing on repeated indices, we have

$$\begin{aligned}\text{Tr}(\Omega\Lambda) &= \Omega_{ik}\Lambda_{ki} \\ &= \Lambda_{ki}\Omega_{ik} \\ &= \Lambda_{ik}\Omega_{ki} \\ &= \text{Tr}(\Lambda\Omega).\end{aligned}$$

There's not much content here – we simply relabel bound indices in the third line.

(2) Since

$$\begin{aligned}(\Omega\Lambda\theta)_{ij} &= \Omega_{ik}(\Lambda\theta)_{kj} \\ &= \Omega_{ik}\Lambda_{kl}\theta_{lj} \\ &= \Omega_{ik}\Lambda_{kl}\theta_{lj},\end{aligned}$$

we have (cycling indices backward, every other line)

$$\begin{aligned}\text{Tr}(\Omega\Lambda\theta) &= \Omega_{ik}\Lambda_{kl}\theta_{li} \\ &= \Lambda_{kl}\theta_{li}\Omega_{ik} \\ &\stackrel{1}{=} \Lambda_{ik}\theta_{kl}\Omega_{li} \\ &= \theta_{kl}\Omega_{li}\Lambda_{ik} \\ &\stackrel{2}{=} \theta_{ik}\Omega_{kl}\Lambda_{li}.\end{aligned}$$

The expressions in lines 1 and 2 are $\text{Tr}(\Lambda\theta\Omega)$ and $\text{Tr}(\theta\Omega\Lambda)$, respectively.

(3) Finally, if U is unitary, then

$$\begin{aligned}\mathrm{Tr}(U^\dagger \Omega U) &= \mathrm{Tr}(\Omega U U^\dagger) \\ &= \mathrm{Tr}(\Omega I) \\ &= \mathrm{Tr} \Omega,\end{aligned}$$

where we've used the result of (2). □

1-5 (Shankar, Exercise 1.7.2)

Show that the determinant of a matrix is unaffected by a unitary change of basis.

Solution. Use the fact that $\det(AB) = \det(A) \det(B)$, as well as $\det I = 1$:

$$\begin{aligned}\det(U^\dagger \Omega U) &= \det(U^\dagger) \det(\Omega) \det(U) \\ &= \det(U^\dagger) \det(U) \det(\Omega) \\ &= \det(U^\dagger U) \det(\Omega) \\ &= \det(I) \det(\Omega) \\ &= \det(\Omega).\end{aligned}$$

□

1-6 (Shankar, Exercise 1.7.3)

- (1) We have seen that the determinant of a matrix is unchanged by a unitary change of basis. Argue now that

$$\det \Omega = \text{product of eigenvalues of } \Omega = \prod_{i=1}^n \omega_i$$

for a Hermitian or unitary Ω .

- (2) Using the invariance of the trace under the same transformation, show that

$$\text{Tr } \Omega = \sum_{i=1}^n \omega_i.$$

Solution. (1) The determinant of a diagonal matrix is the product of its diagonals; this is easily proven from the definitions. Furthermore, we know that any Hermitian or unitary matrix Ω can be diagonalized by a suitable unitary change of basis. So since we can write

$$\det \Omega = \det(U^\dagger \Omega U),$$

where the matrix on the right is diagonal, with the eigenvalues of Ω as the diagonals, we conclude

$$\det \Omega = \text{product of eigenvalues of } \Omega = \prod_{i=1}^n \omega_i.$$

- (2) By the same reasoning

$$\begin{aligned} \text{Tr } \Omega &= \text{Tr}(U^\dagger \Omega U) \\ &= \sum_{i=1}^n \omega_i. \end{aligned}$$

And actually these identities are true in general, but the proofs can be made especially short when Ω is Hermitian or unitary, as we did here. □

1-7 (Shankar, Exercise 1.8.8)

Consider Hermitian matrices M^1, M^2, M^3, M^4 that obey

$$M^i M^j + M^j M^i = 2\delta^{ij} I, \quad i, j = 1, \dots, 4$$

- (1) Show that the eigenvalues of M^i are ± 1 .
- (2) By considering the relation

$$M^i M^j = -M^j M^i \quad \text{for } i \neq j$$

show that M^i are traceless.

- (3) Show that they cannot be odd-dimensional matrices.

Solution. (1) Let $|\omega\rangle$ be an eigenvector of M^i , so that

$$M^i |\omega\rangle = \omega |\omega\rangle.$$

Then

$$\begin{aligned} M^i (M^i |\omega\rangle) &= M^i (\omega |\omega\rangle) \\ &= \omega M^i |\omega\rangle \\ &= \omega^2 |\omega\rangle. \end{aligned}$$

But by the equation defining M^i , we have $(M^i)^2 = I$, thus $M^i (M^i |\omega\rangle) = |\omega\rangle$. Comparing our two expressions, we find $\omega^2 = 1$, or $\omega = \pm 1$.

- (2) Since $M^i M^j = -M^j M^i$ for $i \neq j$, we have (using Exercise 1.7.1)

$$\begin{aligned} \text{Tr } M^i &= \text{Tr}(M^i M^j M^j) \\ &= \text{Tr}(M^j M^i M^j) \\ &= -\text{Tr}(M^j M^j M^i) \\ &= -\text{Tr } M^i, \end{aligned}$$

and the result follows.

(3) Using $\det(AB) = \det(A)\det(B)$ to rearrange the order of matrix multiplication,

$$\begin{aligned}\det M^i &= \det(M^i M^j M^j) \\ &= \det(M^j M^i M^j) \\ &= \det(-M^j M^j M^i) \\ &= \det(-M^i) \\ &= (-1)^n \det M^i,\end{aligned}$$

where n is the size of the matrix. If n were odd, then this would imply $\det M^i = 0$, contradicting the fact that $(\det(M^i))^2 = \det(M^i M^i) = \det I = 1$. [Nothing here seems to require that \$M^i\$ be a Hermitian matrix.](#) $\square$

Here's another argument I learned in Physics 180. Recall that the matrices M^i have eigenvalues $\omega = \pm 1$, and that they are traceless. But the trace of a matrix is given by

$$\text{Tr } M^i = \sum_{i=1}^n \omega_i,$$

so if n is odd, there's no way for these to sum to zero; each $+1$ needs to pair against a -1 .

1-8 (Shankar, Exercise 1.8.9)

A collection of masses m_α , located at $\mathbf{r}_\alpha$ and rotating with angular velocity $\boldsymbol{\omega}$ around a common axis has an angular momentum

$$\mathbf{l} = \sum_{\alpha} m_{\alpha} (\mathbf{r}_{\alpha} \times \mathbf{v}_{\alpha})$$

where $\mathbf{v}_{\alpha} = \boldsymbol{\omega} \times \mathbf{r}_{\alpha}$ is the velocity of m_{α} . By using the identity

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

show that each Cartesian component l_i of $\mathbf{l}$ is given by

$$l_i = \sum_j M_{ij} \omega_j$$

where

$$M_{ij} = \sum_{\alpha} m_{\alpha} [r_{\alpha}^2 \delta_{ij} - (\mathbf{r}_{\alpha})_i (\mathbf{r}_{\alpha})_j]$$

or in Dirac notation

$$|l\rangle = M|\omega\rangle.$$

- (1) Will the angular momentum and angular velocity always be parallel?
- (2) Show that the moment of inertia matrix M_{ij} is Hermitian.
- (3) Argue now that there exist three directions for $\boldsymbol{\omega}$ such that $\mathbf{l}$ and $\boldsymbol{\omega}$ will be parallel. How are these directions to be found?
- (4) Consider the moment of inertia matrix of a sphere. Due to the complete symmetry of the sphere, it is clear that every direction is its eigendirection for rotation. What does this say about the three eigenvalues of the matrix M ?

Solution. From the definitions,

$$\begin{aligned} \mathbf{l} &= \sum_{\alpha} m_{\alpha} (\mathbf{r}_{\alpha} \times \mathbf{v}_{\alpha}) \\ &= \sum_{\alpha} m_{\alpha} (\mathbf{r}_{\alpha} \times (\boldsymbol{\omega} \times \mathbf{r}_{\alpha})) \\ &= \sum_{\alpha} m_{\alpha} [\boldsymbol{\omega} r_{\alpha}^2 - \mathbf{r}_{\alpha} (\mathbf{r}_{\alpha} \cdot \boldsymbol{\omega})] \\ &= \sum_{\alpha} \sum_{i,j} m_{\alpha} [r_{\alpha}^2 \omega_i - (\mathbf{r}_{\alpha})_i (\mathbf{r}_{\alpha})_j \omega_j] \mathbf{e}_i \\ &= \sum_{\alpha} \sum_{i,j} m_{\alpha} [r_{\alpha}^2 \delta_{ij} - (\mathbf{r}_{\alpha})_i (\mathbf{r}_{\alpha})_j] \omega_j \mathbf{e}_i, \end{aligned}$$

which implies $l_i = \sum_j M_{ij} \omega_j$, with M_{ij} as defined above.

- (1) No. This is only true when $|\omega\rangle$ is an eigenvector of M , and most vectors aren't.
- (2) This follows because

$$\begin{aligned}
M_{ij}^\dagger &= \left(\sum_{\alpha} m_{\alpha} [r_{\alpha}^2 \delta_{ij} - (\mathbf{r}_{\alpha})_i (\mathbf{r}_{\alpha})_j] \right)^\dagger \\
&= \sum_{\alpha} m_{\alpha} [r_{\alpha}^2 \delta_{ji} - (\mathbf{r}_{\alpha})_j (\mathbf{r}_{\alpha})_i] \\
&= \sum_{\alpha} m_{\alpha} [r_{\alpha}^2 \delta_{ij} - (\mathbf{r}_{\alpha})_i (\mathbf{r}_{\alpha})_j] \\
&= M_{ij}.
\end{aligned}$$

- (3) Since M is Hermitian, it has three linearly independent eigenvectors $|\omega\rangle$. And for these $|\omega\rangle$,

$$|l\rangle = M|\omega\rangle = \omega|\omega\rangle.$$

In other words, there are three directions for ω such that $\mathbf{l}$ and ω are parallel. To find these, one can solve for the eigenvectors of the operator in the usual way.

- (4) The three eigenvalues are equal. To see this, choose an arbitrary direction $|\omega\rangle$. Since every direction is an eigendirection, we have

$$M|\omega\rangle = \omega|\omega\rangle.$$

Suppose that $|\omega_1\rangle$, $|\omega_2\rangle$, and $|\omega_3\rangle$ are three orthogonal directions. Since these comprise a basis,

$$|\omega\rangle = c_1|\omega_1\rangle + c_2|\omega_2\rangle + c_3|\omega_3\rangle$$

for some constants c_1 , c_2 , and c_3 . Since M is linear and $|\omega_1\rangle$, $|\omega_2\rangle$, $|\omega_3\rangle$ are also eigendirections, we have

$$M|\omega\rangle = c_1\omega_1|\omega_1\rangle + c_2\omega_2|\omega_2\rangle + c_3\omega_3|\omega_3\rangle.$$

By the first equation, we also have

$$M|\omega\rangle = \omega(c_1|\omega_1\rangle + c_2|\omega_2\rangle + c_3|\omega_3\rangle),$$

so that we obtain

$$0 = c_1(\omega_1 - \omega)|\omega_1\rangle + c_2(\omega_2 - \omega)|\omega_2\rangle + c_3(\omega_3 - \omega)|\omega_3\rangle.$$

Since this is a combination of linearly independent vectors, it follows that each of the coefficients is zero. Now, we can always choose our three orthogonal directions such that none of the c_i are zero (i.e., $|\omega\rangle$ is not parallel to any of the directions), which then implies that $\omega_1 = \omega_2 = \omega_3 = \omega$, as we wanted to show. This argument generalizes to any number of dimensions. $\square$

1-9 (Shankar, Exercise 1.9.1)

We know that the series

$$f(x) = \sum_{n=0}^{\infty} x^n$$

may be equated to the function $f(x) = (1-x)^{-1}$ if $|x| < 1$. By going to the eigenbasis, examine when the q number power series

$$f(\Omega) = \sum_{n=0}^{\infty} \Omega^n$$

of a Hermitian operator Ω may be identified with $(1-\Omega)^{-1}$.

Solution. Take any eigenvector $|\omega_i\rangle$ of Ω . Then

$$\left(\sum_{n=0}^{\infty} \Omega^n\right)(1-\Omega)|\omega_i\rangle = (1-\Omega)\left(\sum_{n=0}^{\infty} \omega_i^n\right)|\omega_i\rangle,$$

and the sum on the right converges when $|\omega_i| < 1$, in which case

$$\begin{aligned} &= (1-\omega_i)(1-\omega_i)^{-1}|\omega_i\rangle \\ &= |\omega_i\rangle. \end{aligned}$$

Applying the same argument to an eigenbasis of Ω , we conclude that $f(\Omega) = (1-\Omega)^{-1}$ when all the eigenvalues of Ω have magnitude less than 1. $\square$

1-10 (Shankar, Exercise 1.9.2)

If H is a Hermitian operator, show that $U = e^{iH}$ is unitary.

Solution. By definition,

$$U = \sum_{n=0}^{\infty} \frac{(iH)^n}{n!},$$

which implies

$$U^\dagger = \left(\sum_{n=0}^{\infty} \frac{(iH)^n}{n!}\right)^\dagger = \sum_{n=0}^{\infty} \frac{(-iH)^n}{n!} = e^{-iH},$$

where we've used the Hermiticity of H . Then since $-iH$ and iH commute,

$$U^\dagger U = e^{-iH} e^{iH} = e^{-iH+iH} = e^0 = I,$$

thus U is unitary. $\square$

1-11 (Shankar, Exercise 1.9.3)

For the case above, show that $\det U = e^{i \text{Tr } H}$.

Solution. If $|\omega_k\rangle$ is an eigenvector of H with corresponding eigenvalue ω_k , then it's also an eigenvector of $U = e^{iH}$ with corresponding eigenvalue $e^{i\omega_k}$; this is easy to show by working from the definitions. So by the results from Exercise 1.7.2,

$$\begin{aligned}\det U &= \prod_{k=1}^n e^{i\omega_k} \\ &= e^{i \sum_{k=1}^n \omega_k} \\ &= e^{i \text{Tr } H}.\end{aligned}$$

□

1-12 (Nielsen and Chuang, Exercise 2.13)

If $|w\rangle$ and $|v\rangle$ are any two vectors, show that $(|w\rangle\langle v|)^\dagger \equiv |v\rangle\langle w|$.

Solution. [Feels like a “physicist’s proof”](#). If we treat both $|w\rangle$ and $\langle v|$ formally as linear operators, then we can use the fact that $(AB)^\dagger = B^\dagger A^\dagger$ to get

$$(|w\rangle\langle v|)^\dagger = \langle v|^\dagger |w\rangle^\dagger = |v\rangle\langle w|.$$

Since we defined $|v\rangle^\dagger = \langle v|$, this makes implicit use of $(A^\dagger)^\dagger = A$, which is proven in 2.15. □

1-13 (Nielsen and Chuang, Exercise 2.14)

Show that the adjoint operator is anti-linear,

$$\left(\sum_i a_i A_i \right)^\dagger = \sum_i a_i^* A_i^\dagger.$$

Solution. By definition of the adjoint,

$$\left(|v\rangle, \left(\sum_i a_i A_i \right) |w\rangle \right) = \left(\left(\sum_i a_i A_i \right)^\dagger |v\rangle, |w\rangle \right).$$

But

$$\begin{aligned}\left(|v\rangle, \left(\sum_i a_i A_i\right)|w\rangle\right) &= \sum_i a_i \left(|v\rangle, A_i|w\rangle\right) \\ &= \sum_i a_i \left(A_i^\dagger|v\rangle, |w\rangle\right) \\ &= \left(\left(\sum_i a_i^* A_i^\dagger\right)|v\rangle, |w\rangle\right),\end{aligned}$$

where line 1 uses linearity in the second slot, line 2 uses the definition of the adjoint, and line 3 uses conjugate-linearity in the first slot. Hence

$$\left(\left(\sum_i a_i A_i\right)^\dagger|v\rangle, |w\rangle\right) = \left(\left(\sum_i a_i^* A_i^\dagger\right)|v\rangle, |w\rangle\right),$$

and since $|v\rangle$ and $|w\rangle$ are arbitrary, the desired identity follows. $\square$

1-14 (Nielsen and Chuang, Exercise 2.15)

Show that $(A^\dagger)^\dagger = A$.

Solution. We have

$$\left(|v\rangle, A^\dagger|w\rangle\right) = \left((A^\dagger)^\dagger|v\rangle, |w\rangle\right),$$

and also

$$\begin{aligned}\left(|v\rangle, A^\dagger|w\rangle\right) &= \left(A^\dagger|w\rangle, |v\rangle\right)^* \\ &= (|w\rangle, A|v\rangle)^* \\ &= (A|v\rangle, |w\rangle),\end{aligned}$$

where line 1 uses property (2) of the inner product, line 2 uses the definition of the adjoint, and line 3 uses property (2) again. Since $|v\rangle$ and $|w\rangle$ are arbitrary, this proves the identity. $\square$

1-15 (Nielsen and Chuang, Exercise 2.16)

Show that any projector P satisfies the equation $P^2 = P$.

Solution. We have

$$\begin{aligned}P^2 &= \left(\sum_{i=1}^k |i\rangle\langle i|\right) \left(\sum_{j=1}^k |j\rangle\langle j|\right) \\ &= \sum_{i=1}^k \sum_{j=1}^k \delta_{ij} |i\rangle\langle j| \\ &= \sum_{i=1}^k |i\rangle\langle i| \\ &= P,\end{aligned}$$

where line 2 follows because $\{|i\rangle\}$ is an orthonormal basis. $\square$

1-16 (Nielsen and Chuang, Exercise 2.17)

Show that a normal matrix is Hermitian if and only if it has real eigenvalues.

Solution. For the forward implication, suppose that A is normal and Hermitian. Let $|a\rangle$ be an eigenvector of A with eigenvalue a , so that

$$A|a\rangle = a|a\rangle.$$

Since A is Hermitian, taking the adjoint of this equation gives

$$\langle a|A = a^*\langle a|.$$

Acting $\langle a|$ on the first equation and $|a\rangle$ on the second, then subtracting, we obtain

$$0 = (a - a^*)\langle a|a\rangle.$$

Thus either $\langle a|a\rangle = 0$ or $a = a^*$. But the former cannot be true, since $|a\rangle$ is an eigenvector and therefore not equal to the zero vector. Thus $a = a^*$, and since $|a\rangle$ was an arbitrary eigenvector of A , it follows that all eigenvalues of A are real.

For the converse, suppose A is normal and that it has real eigenvalues. Since A is normal, it follows from the spectral decomposition that

$$A = U^\dagger D U$$

for some unitary matrix U and diagonal matrix D , where the entries are the eigenvalues of A . Then (using $(AB)^\dagger = B^\dagger A^\dagger$ and $(A^\dagger)^\dagger = A$)

$$A^\dagger = U^\dagger D^\dagger U,$$

but since D is diagonal and has real entries (because the eigenvalues of A are real), we have $D = D^\dagger$. This then implies $A = A^\dagger$; that is, A is Hermitian. $\square$

1-17 (Nielsen and Chuang, Exercise 2.21)

Repeat the proof of the spectral decomposition in Box 2.2 for the case when M is Hermitian, simplifying the proof wherever possible.

Theorem 2.1. (Spectral decomposition) Any normal operator M on a vector space V is diagonal with respect to some orthonormal basis for V . Conversely, any diagonalizable operator is normal.

Proof. The converse of the theorem is not true; not all diagonalizable operators are Hermitian. We showed in Exercise 2.17 that a diagonalizable operator is Hermitian if and only if all its eigenvalues are real. [The proof used the forward implication of the spectral decomposition theorem, so there is no circular reasoning here.](#)

We prove the forward implication by induction on the dimension d of V . As before, the case $d = 1$ is trivial. For the forward implication, let λ be an eigenvalue of M , P the projector onto the λ eigenspace, and Q the projector onto the orthogonal complement. Then

$$\begin{aligned} M &= (P + Q)M(P + Q) \\ &= PMP + QMP + PMQ + QMQ. \end{aligned}$$

Clearly $PMP = \lambda P$. We have $QMP = 0$, since M maps the P subspace into itself. Also, since

P , M , and Q are all Hermitian, we have

$$(QMP)^\dagger = PMQ,$$

which shows that $PMQ = 0$, hence

$$M = \lambda P + QMQ.$$

The matrix λP is diagonal with respect to any orthonormal basis for P . (To see this, note that P acts as the identity on the P subspace.) Furthermore, QMQ is normal, because it is Hermitian. To see this, note that

$$(QMQ)^\dagger = QMQ,$$

because M is Hermitian. Thus, by induction, QMQ is diagonal with respect to some orthonormal basis for Q . Then we can diagonalize M : choose any orthonormal basis for P , and some orthonormal basis for Q that diagonalizes QMQ . Then join these together to form an orthonormal basis for the full vector space V . By construction, M is diagonal with respect to this basis. $\square$

1-18 (Nielsen and Chuang, Exercise 2.24)

Show that a positive operator is necessarily Hermitian.

Solution. This requires a bit of setup; we note the following results.

- (1) Any linear operator A can be written in the form $A = B + iC$ with B, C Hermitian. This follows from the fact that

$$\begin{aligned} A &= \frac{A + A^\dagger}{2} + \frac{A - A^\dagger}{2} \\ &= \frac{A + A^\dagger}{2} + i \left[\frac{i(A^\dagger - A)}{2} \right] \\ &\equiv B + iC. \end{aligned}$$

Note that in general, A fails to be Hermitian due to the operator C .

- (2) A linear operator A on a *complex* inner product space is zero if and only if $(|v\rangle, A|v\rangle) = 0$ for all $|v\rangle$. The proof uses the polarization identity to turn the quadratic form (v, Av) into a bilinear one (w, Av) , then uses the fact that $(w, Av) = 0$ for all v, w if and only if $A = 0$. See the discussion starting on page 138 of Halmos (2nd ed.). Note that this requires a complex space; the result is not true for real inner product spaces. For instance, if our vector space is $\mathbb{R}^2$, the inner product is the ordinary scalar product, and A is a rotation by 90 degrees, then $(x, Ax) = 0$ for all x .
- (3) A linear operator A on a *complex* inner product space is Hermitian if and only if $(|v\rangle, A|v\rangle)$ is real for all $|v\rangle$. The proof is as follows.

Suppose $A = A^\dagger$. Then

$$\begin{aligned}(|v\rangle, A|v\rangle) &= (A^\dagger|v\rangle, |v\rangle) \\ &= (A|v\rangle, |v\rangle) \\ &= (|v\rangle, A|v\rangle)^*,\end{aligned}$$

using the definition of the adjoint and properties of the inner product. For the converse implication, suppose $(|v\rangle, A|v\rangle)$ is real. Then

$$\begin{aligned}(|v\rangle, A|v\rangle) &= (|v\rangle, A|v\rangle)^* \\ &= (A^\dagger|v\rangle, |v\rangle)^* \\ &= (|v\rangle, A^\dagger|v\rangle)\end{aligned}$$

by the same steps as before, which by linearity in the second slot implies

$$(|v\rangle, (A - A^\dagger)|v\rangle) = 0.$$

Since $|v\rangle$ was arbitrary, this implies by result (2) that $A = A^\dagger$.

The original claim is proved as follows. Suppose A is a positive operator, so that $(|v\rangle, A|v\rangle) \geq 0$ for all $|v\rangle$ in V . By result (1), we have

$$(|v\rangle, B|v\rangle) + i(|v\rangle, C|v\rangle) \geq 0.$$

But since B and C are Hermitian, result (3) implies that both inner products are real. This leads to a contradiction unless

$$(|v\rangle, C|v\rangle) = 0.$$

By result (2), we conclude that $C = 0$. Hence $A = B$, which shows that A is Hermitian. □

1-19 (Nielsen and Chuang, Exercise 2.25)

Show that for any operator A , $A^\dagger A$ is positive.

Solution. We have

$$(|v\rangle, A^\dagger A|v\rangle) = ((A^\dagger)^\dagger|v\rangle, A|v\rangle) = (A|v\rangle, A|v\rangle) \geq 0,$$

using the definition of the adjoint, Exercise 2.15, and the properties of the inner product. Notice that if A does not map any vector (aside from $|v\rangle = 0$) to the zero vector, then $A^\dagger A$ is in fact positive definite. □

1-20 (Sakurai, Problem 1.1)

Prove

$$[AB, CD] = -AC\{D, B\} + A\{C, B\}D - C\{D, A\}B + \{C, A\}DB.$$

Solution. Expand the right-hand side and see that the necessary cancellations occur. $\square$

1-21 (Sakurai, Problem 1.2)

Suppose a 2×2 matrix X (not necessarily Hermitian, nor unitary) is written as

$$X = a_0 + \boldsymbol{\sigma} \cdot \mathbf{a},$$

where a_0 and $a_{1,2,3}$ are numbers.

- How are a_0 and a_k ($k = 1, 2, 3$) related to $\text{tr}(X)$ and $\text{tr}(\sigma_k X)$?
- Obtain a_0 and a_k in terms of the matrix elements X_{ij} .

Solution. (a) Provided that these are the Pauli matrices, we have $\text{tr}(\sigma_k) = 0$, $\sigma_i^2 = I$ and $\sigma_i \sigma_j + \sigma_j \sigma_i = 0$ for $j \neq k$. The latter implies (together with the fact $\text{tr}(XY) = \text{tr}(YX)$) that $\text{tr}(\sigma_i \sigma_j) = 0$ for different Pauli matrices σ_i and σ_j . Thus, summing over repeated indices,

$$\text{tr}(X) = a_0 \text{tr}(I) + a_k \cancel{\text{tr}(\sigma_k)} = 2a_0$$

and

$$\text{tr}(\sigma_k X) = a_0 \cancel{\text{tr}(\sigma_k)} + a_j \text{tr}(\sigma_k \sigma_j) = a_k \text{tr}(\sigma_k^2) = 2a_k$$

(no sum on the second-to-last equality). That is, $\boxed{a_0 = \frac{1}{2} \text{tr}(X)}$ and $\boxed{a_k = \frac{1}{2} \text{tr}(\sigma_k X)}$.

(b) A bit of algebra shows that

$$\boxed{\begin{aligned} a_0 &= \frac{1}{2}(X_{11} + X_{22}), \\ a_1 &= \frac{1}{2}(X_{12} + X_{21}), \\ a_2 &= \frac{i}{2}(X_{12} - X_{21}), \\ a_3 &= \frac{1}{2}(X_{11} - X_{22}) \end{aligned}}$$

I missed the point of this problem the first time around; what this does is show that the three Pauli matrices, together with the identity, comprise a basis for the space of 2×2 complex matrices (not necessarily Hermitian, nor unitary). It is shown by explicit construction in part (b) that they span the space; that they are linearly independent follows from the fact that the unique solution to $X_{11} = X_{12} = X_{21} = X_{22} = 0$ is to set all four coefficients a_0, a_1, a_2, a_3 to zero. $\square$

1-22 (Sakurai, Problem 1.3)

Show that the determinant of a 2×2 matrix $\boldsymbol{\sigma} \cdot \mathbf{a}$ is invariant under

$$\boldsymbol{\sigma} \cdot \mathbf{a} \rightarrow \boldsymbol{\sigma} \cdot \mathbf{a}' \equiv \exp\left(\frac{i\boldsymbol{\sigma} \cdot \hat{\mathbf{n}}\phi}{2}\right)(\boldsymbol{\sigma} \cdot \mathbf{a})\exp\left(\frac{-i\boldsymbol{\sigma} \cdot \hat{\mathbf{n}}\phi}{2}\right).$$

Find a'_k in terms of a_k when $\hat{\mathbf{n}}$ is in the positive z -direction and interpret your result.

Solution. Since $\det(ABC) = \det(A)\det(B)\det(C)$, we may rearrange terms to get

$$\begin{aligned}\det(\boldsymbol{\sigma} \cdot \mathbf{a}') &= \det\left[\exp\left(\frac{i\boldsymbol{\sigma} \cdot \hat{\mathbf{n}}\phi}{2}\right)\exp\left(\frac{-i\boldsymbol{\sigma} \cdot \hat{\mathbf{n}}\phi}{2}\right)(\boldsymbol{\sigma} \cdot \mathbf{a})\right] \\ &= \det(\boldsymbol{\sigma} \cdot \mathbf{a}).\end{aligned}$$

The second line uses the fact that if A and B are commuting matrices, then $e^A e^B = e^{A+B}$. Now suppose that $\hat{\mathbf{n}} = \hat{\mathbf{z}}$, so that $\boldsymbol{\sigma} \cdot \hat{\mathbf{n}} = \sigma_z$. Then

$$\begin{aligned}\exp\left(\frac{i\sigma_z\phi}{2}\right) &= \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{i\sigma_z\phi}{2}\right)^n \\ &= \left[\sum_{n=0}^{\infty} \frac{1}{(2n+1)!} \left(\frac{i\phi}{2}\right)^{2n+1}\right]\sigma_z + \left[\sum_{n=0}^{\infty} \frac{1}{(2n)!} \left(\frac{i\phi}{2}\right)^{2n}\right]I \\ &= i\left[\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \left(\frac{\phi}{2}\right)^{2n+1}\right]\sigma_z + \left[\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \left(\frac{\phi}{2}\right)^{2n}\right]I \\ &= i\sin(\phi/2)\sigma_z + \cos(\phi/2)I \\ &= \begin{pmatrix} e^{i\phi/2} & 0 \\ 0 & e^{-i\phi/2} \end{pmatrix},\end{aligned}$$

where we've used $\sigma_z^2 = I$. Now since

$$\boldsymbol{\sigma} \cdot \mathbf{a} = \begin{pmatrix} a_3 & a_1 - ia_2 \\ a_1 + ia_2 & -a_3 \end{pmatrix},$$

it follows that

$$\begin{aligned}\boldsymbol{\sigma} \cdot \mathbf{a}' &= \begin{pmatrix} e^{i\phi/2} & 0 \\ 0 & e^{-i\phi/2} \end{pmatrix} \begin{pmatrix} a_3 & a_1 - ia_2 \\ a_1 + ia_2 & -a_3 \end{pmatrix} \begin{pmatrix} e^{-i\phi/2} & 0 \\ 0 & e^{i\phi/2} \end{pmatrix} \\ &= \begin{pmatrix} a_3 & (a_1 - ia_2)e^{i\phi} \\ (a_1 + ia_2)e^{-i\phi} & -a_3 \end{pmatrix} \equiv \begin{pmatrix} a'_3 & a'_1 - ia'_2 \\ a'_1 + ia'_2 & -a'_3 \end{pmatrix}.\end{aligned}$$

By comparing terms, we obtain $a'_3 = a_3$, and the pair of equations

$$\begin{cases} a'_1 - ia'_2 = (a_1 - ia_2)e^{i\phi}, \\ a'_1 + ia'_2 = (a_1 + ia_2)e^{-i\phi}. \end{cases}$$

Adding these two equations gives

$$\begin{aligned} 2a'_1 &= a_1(e^{i\phi} + e^{-i\phi}) - ia_2(e^{i\phi} - e^{-i\phi}) \\ &= a_1(2\cos\phi) - ia_2(2i\sin\phi) \\ &= 2a_1\cos\phi + 2a_2\sin\phi, \end{aligned}$$

while subtracting the first equation from the second gives

$$2ia'_2 = -2i\sin\phi a_1 + 2i\cos\phi a_2.$$

All together, we have

$$\boxed{a'_1 = a_1\cos\phi + a_2\sin\phi, \quad a'_2 = -a_1\sin\phi + a_2\cos\phi, \quad a'_3 = a_3}.$$

Later, we will see that the operator $\exp(-iS_z\phi/\hbar)$ implements a rotation by ϕ about the z -axis, so the matrix $\boldsymbol{\sigma} \cdot \mathbf{a}'$ represents the action of rotating to a new basis, acting $\boldsymbol{\sigma} \cdot \mathbf{a}$, and then returning to the original basis. We expect the outcomes to be different, because $\boldsymbol{\sigma} \cdot \mathbf{a}$ acts with respect to the new coordinate axes. $\square$

1-23 (Sakurai, Problem 1.6)

Suppose $|i\rangle$ and $|j\rangle$ are eigenkets of some Hermitian operator A . Under what condition can we conclude that $|i\rangle + |j\rangle$ is also an eigenket of A ? Justify your answer.

Solution. This is trivially satisfied if $|i\rangle$ and $|j\rangle$ are linearly dependent, so suppose otherwise. This combination will also be an eigenket if $|i\rangle + |j\rangle$ satisfies

$$A(|i\rangle + |j\rangle) = \lambda(|i\rangle + |j\rangle)$$

for some number λ . But since $|i\rangle$ and $|j\rangle$ are eigenkets, say $A|i\rangle = i|i\rangle$ and $A|j\rangle = j|j\rangle$, this condition amounts to the requirement that

$$(\lambda - i)|i\rangle + (\lambda - j)|j\rangle = 0.$$

But we assumed that $|i\rangle$ and $|j\rangle$ were linearly independent, so we must have $\lambda - i = \lambda - j = 0$, which implies $i = j$. That is, $|i\rangle$ and $|j\rangle$ must correspond to the same eigenvalue. $\square$

1-24 (Sakurai, Problem 1.7)

Consider a ket space spanned by the eigenkets $\{|a'\rangle\}$ of a Hermitian operator A . There is no degeneracy.

a. Prove that

$$\prod_{a'} (A - a')$$

is the null operator.

b. What is the significance of

$$\prod_{a'' \neq a'} \frac{(A - a'')}{(a' - a'')}?$$

c. Illustrate (a) and (b) using A set equal to S_z of a spin $\frac{1}{2}$ system.

Solution. (a) Let $|\alpha\rangle$ be an arbitrary ket. Then we can write

$$|\alpha\rangle = \sum_{a'} |a'\rangle \langle a'|\alpha\rangle.$$

Now consider the projection of $|\alpha\rangle$ onto $|a''\rangle$, namely

$$|a''\rangle \langle a''|\alpha\rangle.$$

Since A commutes with itself and commutes trivially with scalars, we can rearrange the factors in the operator so that it appears as

$$\left(\prod_{a' \neq a''} (A - a') \right) (A - a'').$$

But this clearly kills the projection of $|\alpha\rangle$ onto $|a''\rangle$. Since $|a''\rangle$ was arbitrary, this operator must kill all the projections of $|\alpha\rangle$ onto the $\{|a'\rangle\}$ basis, i.e., it is the null operator. [The lack of degeneracies is important because it allows us to label each basis ket uniquely by its eigenvalue, but otherwise the argument seems to work fine even with degeneracies.](#)

(b) Using the same reasoning as in part (a), this operator projects onto the eigenket $|a'\rangle$. The factors in the denominator “undo” the action of $A - a''$ for each a'' .

(c) For a spin $\frac{1}{2}$ system, the corresponding operator is

$$(S_z - \hbar/2)(S_z + \hbar/2).$$

Then for an arbitrary ket $|\alpha\rangle$ in the space,

$$|\alpha\rangle = |+\rangle \langle +|\alpha\rangle + |-\rangle \langle -|\alpha\rangle$$

which implies

$$\begin{aligned}
(S_z - \hbar/2)(S_z + \hbar/2)|\alpha\rangle &= (S_z - \hbar/2)(S_z + \hbar/2)\left(|+\rangle\langle +|\alpha\rangle + |-\rangle\langle -|\alpha\rangle\right) \\
&= (S_z + \hbar/2)\underbrace{\left(S_z|+\rangle - \hbar/2|+\rangle\right)}_0\langle +|\alpha\rangle \\
&\quad + (S_z - \hbar/2)\underbrace{\left(S_z|-\rangle + \hbar/2|-\rangle\right)}_0\langle -|\alpha\rangle \\
&= 0.
\end{aligned}$$

Say we want to project onto the ket with eigenvalue $-\hbar/2$. Then by the result of (b), we form the operator

$$\frac{S_z - \hbar/2}{-\hbar/2 - \hbar/2} = -\frac{S_z - \hbar/2}{\hbar},$$

which acts on $|\alpha\rangle$ as follows:

$$\begin{aligned}
-\frac{S_z - \hbar/2}{\hbar}|\alpha\rangle &= -\frac{S_z - \hbar/2}{\hbar}\left(|+\rangle\langle +|\alpha\rangle + |-\rangle\langle -|\alpha\rangle\right) \\
&= -\frac{1}{\hbar}\underbrace{\left(S_z|+\rangle - \hbar/2|+\rangle\right)}_0\langle +|\alpha\rangle - \frac{1}{\hbar}\underbrace{\left(S_z|-\rangle - \hbar/2|-\rangle\right)}_{-\hbar|-\rangle}\langle -|\alpha\rangle \\
&= |-\rangle\langle -|\alpha\rangle.
\end{aligned}$$

This is the projection of $|\alpha\rangle$ onto the ket with eigenvalue $-\hbar/2$, as we wanted. $\square$

Here is where degeneracies become relevant. If there is more than one ket corresponding to the eigenvalue, then this operator merely projects onto the subspace spanned by these eigenkets. For instance, if the eigenvalue is triply degenerate, then this operator will return a linear combination of the three eigenkets corresponding to that eigenvalue.

2 Review of classical mechanics

2-1 (Shankar, Exercise 2.8.7)

Consider the harmonic oscillator, for which the general solution is

$$x(t) = A \cos \omega t + B \sin \omega t.$$

Express the energy in terms of A and B and note that it does not depend on time. Now choose A and B such that $x(0) = x_1$ and $x(T) = x_2$. Write down the energy in terms of x_1 , x_2 and T . Show that the action for the trajectory connecting x_1 and x_2 is

$$S_{\text{cl}}(x_1, x_2, T) = \frac{m\omega}{2 \sin \omega T} [(x_1^2 + x_2^2) \cos \omega T - 2x_1 x_2].$$

Verify that $\partial S_{\text{cl}}/\partial T = -E$.

Solution. Note that the energy E is given by

$$\begin{aligned} E &= \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m\omega^2 x^2 \\ &= \frac{1}{2}m[-\omega A \sin \omega t + \omega B \cos \omega t]^2 + \frac{1}{2}m\omega^2 [A \cos \omega t + B \sin \omega t]^2 \\ &= \frac{1}{2}m\omega^2 [A^2 \sin^2 \omega t - \cancel{2AB \sin \omega t \cos \omega t} + B^2 \cos^2 \omega t \\ &\quad + A^2 \cos^2 \omega t + \cancel{2AB \sin \omega t \cos \omega t} + B^2 \sin^2 \omega t] \\ &= \frac{1}{2}m\omega^2 (A^2 + B^2), \end{aligned}$$

which does not depend on time. If we then fix

$$\begin{aligned} x(0) &= x_1 = A, \\ x(T) &= x_2 = A \cos \omega T + B \sin \omega T, \end{aligned}$$

we get

$$\begin{aligned} B \sin \omega T &= x_2 - A \cos \omega T \\ \implies B &= \frac{x_2 - x_1 \cos \omega T}{\sin \omega T}, \end{aligned}$$

and then

$$\begin{aligned} A^2 + B^2 &= x_1^2 + \frac{(x_2 - x_1 \cos \omega T)^2}{\sin^2 \omega T} \\ &= \frac{x_1^2 \sin^2 \omega T + x_2^2 - 2x_1 x_2 \cos \omega T + x_1^2 \cos^2 \omega T}{\sin^2 \omega T} \\ &= \frac{x_2^2 - 2x_1 x_2 \cos \omega T + x_1^2}{\sin^2 \omega T}, \end{aligned}$$

which implies

$$E = \frac{1}{2}m\omega^2 \left(\frac{x_1^2 + x_2^2 - 2x_1x_2 \cos \omega T}{\sin^2 \omega T} \right).$$

Then we compute the action for this trajectory connecting x_1 and x_2 :

$$S_{\text{cl}} = \int_0^T \mathcal{L}(x_{\text{cl}}, \dot{x}_{\text{cl}}, t) dt,$$

where

$$\begin{aligned} \mathcal{L}(x_{\text{cl}}, \dot{x}_{\text{cl}}, t) &= \frac{1}{2}m[-\omega A \sin \omega t + \omega B \cos \omega t]^2 - \frac{1}{2}m\omega^2[A \cos \omega t + B \sin \omega t]^2 \\ &= \frac{1}{2}m\omega^2[A^2 \sin^2 \omega t - 2AB \sin \omega t \cos \omega t + B^2 \cos^2 \omega t \\ &\quad - A^2 \cos^2 \omega t - 2AB \sin \omega t \cos \omega t - B^2 \sin^2 \omega t] \\ &= \frac{1}{2}m\omega^2[(A^2 - B^2) \sin^2 \omega t - (A^2 - B^2) \cos^2 \omega t - 4AB \sin \omega t \cos \omega t] \\ &= \frac{1}{2}m\omega^2[(B^2 - A^2)(\cos^2 \omega t - \sin^2 \omega t) - 2AB(2 \sin \omega t \cos \omega t)] \\ &= \frac{1}{2}m\omega^2[(B^2 - A^2) \cos 2\omega t - 2AB \sin 2\omega t]. \end{aligned}$$

Thus the integral for the action is

$$\begin{aligned} S_{\text{cl}} &= \int_0^T \mathcal{L}(x_{\text{cl}}, \dot{x}_{\text{cl}}, t) dt \\ &= \int_0^T \frac{1}{2}m\omega^2[(B^2 - A^2) \cos 2\omega t - 2AB \sin 2\omega t] dt \\ &= \frac{1}{2}m\omega^2 \left[(B^2 - A^2) \frac{1}{2\omega} \sin 2\omega t + \frac{AB}{\omega} \cos 2\omega t \right] \Big|_0^T \\ &= \frac{1}{2}m\omega^2 \left[(B^2 - A^2) \frac{1}{2\omega} (2 \sin \omega T \cos \omega T) + \frac{AB}{\omega} \cos 2\omega T - \frac{AB}{\omega} \right] \\ &= \frac{1}{2}m\omega \left[(B^2 - A^2) \sin \omega T \cos \omega T + AB(\cos 2\omega T - 1) \right]. \end{aligned}$$

Using our expressions for A and B in terms of x_1 , x_2 , and T , we have

$$\begin{aligned} B^2 - A^2 &= \frac{(x_2 - x_1 \cos \omega T)^2}{\sin^2 \omega T} - x_1^2 \\ &= \frac{x_2^2 - 2x_1x_2 \cos \omega T + x_1^2 \cos^2 \omega T - x_1^2 \sin^2 \omega T}{\sin^2 \omega T} \\ &= \frac{x_2^2 - 2x_1x_2 \cos \omega T + x_1^2 \cos 2\omega T}{\sin^2 \omega T}, \\ AB &= \frac{x_1(x_2 - x_1 \cos \omega T)}{\sin \omega T}, \end{aligned}$$

hence

$$\begin{aligned}
S_{\text{cl}} &= \frac{m\omega}{2\sin\omega T} \left[(x_2^2 - 2x_1x_2\cos\omega T + x_1^2\cos 2\omega T)\cos\omega T \right. \\
&\quad \left. + x_1(x_2 - x_1\cos\omega T)(\cos 2\omega T - 1) \right] \\
&= \frac{m\omega}{2\sin\omega T} \left[x_2^2\cos\omega T - 2x_1x_2\cos^2\omega T + \cancel{x_1^2\cos\omega T\cos 2\omega T} \right. \\
&\quad \left. + x_1x_2\cos 2\omega T - x_1x_2 - \cancel{x_1^2\cos\omega T\cos 2\omega T} + x_1^2\cos\omega T \right].
\end{aligned}$$

Then use the identity $\cos 2\omega T = 2\cos^2\omega T - 1$ to simplify the expression in brackets:

$$\begin{aligned}
\left[\dots \right] &= x_2^2\cos\omega T - \cancel{2x_1x_2\cos^2\omega T} + x_1x_2(2\cos^2\omega T - 1) - x_1x_2 + x_1^2\cos\omega T \\
&= (x_1^2 + x_2^2)\cos\omega T - 2x_1x_2.
\end{aligned}$$

Thus

$$\boxed{S_{\text{cl}} = \frac{m\omega}{2\sin\omega T} [(x_1^2 + x_2^2)\cos\omega T - 2x_1x_2]},$$

as we wanted to show. Then we verify (2.8.18):

$$\begin{aligned}
\frac{\partial S_{\text{cl}}}{\partial T} &= \frac{\partial}{\partial T} \left\{ \frac{m\omega}{2\sin\omega T} [(x_1^2 + x_2^2)\cos\omega T - 2x_1x_2] \right\} \\
&= \frac{1}{2}m\omega \left\{ -\frac{\omega\cos\omega T}{\sin^2\omega T} [(x_1^2 + x_2^2)\cos\omega T - 2x_1x_2] + \frac{1}{\sin\omega T} [(x_1^2 + x_2^2)(-\omega\sin\omega T)] \right\} \\
&= \frac{1}{2}m\omega^2 \left\{ -\frac{\cos\omega T}{\sin^2\omega T} [(x_1^2 + x_2^2)\cos\omega T - 2x_1x_2] - (x_1^2 + x_2^2) \right\} \\
&= \frac{m\omega^2}{2\sin^2\omega T} \{ -(x_1^2 + x_2^2)\cos^2\omega T + 2x_1x_2\cos\omega T - (x_1^2 + x_2^2)\sin^2\omega T \} \\
&= -\frac{1}{2}m\omega^2 \left(\frac{x_1^2 + x_2^2 - 2x_1x_2\cos\omega T}{\sin^2\omega T} \right) \\
&= -E,
\end{aligned}$$

where the last line follows from our first boxed equation. □

3 Matrix representations; diagonalization

3-1 (Sakurai, Problem 1.5)

- a. Consider two kets $|\alpha\rangle$ and $|\beta\rangle$. Suppose $\langle a'|\alpha\rangle, \langle a''|\alpha\rangle, \dots$ and $\langle a'|\beta\rangle, \langle a''|\beta\rangle, \dots$ are all known, where $|a'\rangle, |a''\rangle, \dots$ form a complete set of base kets. Find the matrix representation of the operator $|\alpha\rangle\langle\beta|$ in that basis.
- b. We now consider a spin $\frac{1}{2}$ system and let $|\alpha\rangle$ and $|\beta\rangle$ be $|s_z = \hbar/2\rangle$ and $|s_x = \hbar/2\rangle$, respectively. Write down explicitly the square matrix that corresponds to $|\alpha\rangle\langle\beta|$ in the usual (s_z diagonal) basis.

Solution. (a) The matrix representation is

$$\begin{pmatrix} \langle a'|\alpha\rangle\langle\beta|a'\rangle & \langle a'|\alpha\rangle\langle\beta|a''\rangle & \langle a'|\alpha\rangle\langle\beta|a'''\rangle & \cdots \\ \langle a''|\alpha\rangle\langle\beta|a'\rangle & \langle a''|\alpha\rangle\langle\beta|a''\rangle & \langle a''|\alpha\rangle\langle\beta|a'''\rangle & \cdots \\ \langle a'''|\alpha\rangle\langle\beta|a'\rangle & \langle a'''|\alpha\rangle\langle\beta|a''\rangle & \langle a'''|\alpha\rangle\langle\beta|a'''\rangle & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

If there are n vectors in the basis, then this is an $n \times n$ matrix.

- (b) Recalling that $|S_x; +\rangle = \frac{1}{\sqrt{2}}|+\rangle + \frac{1}{\sqrt{2}}|-\rangle$, we get

$$\begin{pmatrix} \langle +|+\rangle\langle S_x; +|+\rangle & \langle +|+\rangle\langle S_x; +|-\rangle \\ \langle -|+\rangle\langle S_x; +|+\rangle & \langle -|+\rangle\langle S_x; +|-\rangle \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 \end{pmatrix}.$$

□

3-2 (Sakurai, Problem 1.11)

A two-state system is characterized by the Hamiltonian

$$H = H_{11}|1\rangle\langle 1| + H_{22}|2\rangle\langle 2| + H_{12}(|1\rangle\langle 2| + |2\rangle\langle 1|)$$

where H_{11} , H_{22} , and H_{12} are real numbers with the dimension of energy, and $|1\rangle$ and $|2\rangle$ are eigenkets of some observable ($\neq H$). Find the energy eigenkets and corresponding energy eigenvalues. Make sure that your answer makes good sense for $H_{12} = 0$.

Solution. As in the previous problem, using the orthonormality of the base kets, it is straight-

forward to see that the matrix representation of H in the $|1\rangle, |2\rangle$ basis is

$$\begin{pmatrix} H_{11} & H_{12} \\ H_{12} & H_{22} \end{pmatrix}.$$

Then $\det(H - \lambda I) = (\lambda - H_{11})(\lambda - H_{22}) - H_{12}^2 = \lambda^2 - (H_{11} + H_{22})\lambda + H_{11}H_{22} - H_{12}^2$, so that the eigenvalues are

$$\begin{aligned} \lambda_{\pm} &= \frac{H_{11} + H_{22} \pm \sqrt{(H_{11} + H_{22})^2 - 4(H_{11}H_{22} - H_{12}^2)}}{2} \\ &= \frac{H_{11} + H_{22} \pm \sqrt{(H_{11} - H_{22})^2 + 4H_{12}^2}}{2} \\ &\equiv \frac{H_{11} + H_{22} \pm \alpha}{2}, \end{aligned}$$

where we've defined $\alpha \equiv \sqrt{(H_{11} - H_{22})^2 + 4H_{12}^2}$. Now for the eigenvalue λ_+ , we have

$$H - \lambda_+ I \stackrel{*}{=} \begin{pmatrix} \frac{H_{11} - H_{22} - \alpha}{2} & H_{12} \\ H_{12} & \frac{H_{22} - H_{11} - \alpha}{2} \end{pmatrix}.$$

We claim that an eigenvector corresponding to λ_+ is $\begin{pmatrix} 1 & -\frac{2H_{12}}{H_{22} - H_{11} - \alpha} \end{pmatrix}^T$. This clearly yields zero in the second entry when multiplied to $H - \lambda_+ I$, so it suffices to show that

$$\frac{H_{11} - H_{22} - \alpha}{2} - \frac{2H_{12}^2}{H_{22} - H_{11} - \alpha} = 0. \quad (1)$$

After simplifying, we obtain

$$\frac{-(H_{11} - H_{22} + \alpha)(H_{11} - H_{22} - \alpha) - 4H_{12}^2}{2(H_{22} - H_{11} - \alpha)} = 0,$$

and it suffices to show that the numerator vanishes. But the first term in the numerator can be written

$$\begin{aligned} -(H_{11} - H_{22} + \alpha)(H_{11} - H_{22} - \alpha) &= -((H_{11} - H_{22})^2 - \alpha^2) \\ &\equiv -((H_{11} - H_{22})^2 - (H_{11} - H_{22})^2 - 4H_{12}^2) \\ &= 4H_{12}^2, \end{aligned}$$

which shows that Eq. (1) is satisfied, i.e., we have an eigenvector as claimed. Meanwhile, for the eigenvalue λ_- we have

$$H - \lambda_- I \stackrel{*}{=} \begin{pmatrix} \frac{H_{11} - H_{22} + \alpha}{2} & H_{12} \\ H_{12} & \frac{H_{22} - H_{11} + \alpha}{2} \end{pmatrix}.$$

We claim that an eigenvector corresponding to λ_- is $\begin{pmatrix} -\frac{2H_{12}}{H_{11}-H_{22}+\alpha} & 1 \end{pmatrix}^T$. It suffices to show

$$-\frac{2H_{12}^2}{H_{11}-H_{22}+\alpha} + \frac{H_{22}-H_{11}+\alpha}{2} = 0. \quad (2)$$

After simplifying, we get

$$\frac{-4H_{12}^2 + (H_{22}-H_{11}-\alpha)(H_{11}-H_{22}-\alpha)}{2(H_{11}-H_{22}+\alpha)} = 0,$$

and the second term in the numerator can be written

$$-(H_{22}-H_{11}-\alpha)(H_{22}-H_{11}+\alpha) = -((H_{22}-H_{11})^2 - \alpha^2) = 4H_{12}^2.$$

Thus Eq. (2) is satisfied, and we have an eigenvector as claimed. Moving back to bra-ket notation, we conclude that the energy eigenkets of H are

$$\begin{aligned} &|1\rangle + \frac{2H_{12}}{\sqrt{(H_{11}-H_{22})^2 + 4H_{12}^2} + (H_{11}-H_{22})} |2\rangle, \\ &- \frac{2H_{12}}{\sqrt{(H_{11}-H_{22})^2 + 4H_{12}^2} + (H_{11}-H_{22})} |1\rangle + |2\rangle, \end{aligned}$$

corresponding to the eigenvalues

$$\lambda_{\pm} = \frac{H_{11} + H_{22} \pm \sqrt{(H_{11}-H_{22})^2 + 4H_{12}^2}}{2}.$$

As a check, suppose $H_{12} = 0$, so that $|1\rangle$ and $|2\rangle$ are eigenkets of H . Suppose, without loss of generality, that $|1\rangle$ corresponds to the larger eigenvalue, so that $H_{11} - H_{22} > 0$. Then we have

$$\lambda_{\pm} = \frac{H_{11} + H_{22} \pm (H_{11} - H_{22})}{2} \implies \lambda_+ = H_{11} \text{ and } \lambda_- = H_{22}.$$

It's also straightforward to see that the eigenkets we derived reduce to $|1\rangle$ and $|2\rangle$ in this case. $\square$

3-3 (Sakurai, Problem 1.14)

A certain observable in quantum mechanics has a 3×3 matrix representation as follows:

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

- Find the normalized eigenvectors of this observable and the corresponding eigenvalues. Is there any degeneracy?
- Give a physical example where this is all relevant.

Solution. Problem 3.15 in Townsend.

- The eigenvalues of this observable are 0, 1, and -1 , corresponding to the normalized eigenvectors $\frac{1}{\sqrt{2}}(1, 0, -1)$, $\frac{1}{2}(1, \sqrt{2}, 1)$, and $\frac{1}{2}(1, -\sqrt{2}, 1)$, respectively.
- This happens to be the matrix representation of S_x in the S_z basis for a spin-1 particle. $\square$

3-4 (Sakurai, Problem 1.15)

Let A and B be observables. Suppose the simultaneous eigenkets of A and B $\{|a', b'\rangle\}$ form a *complete* orthonormal set of base kets. Can we always conclude that

$$[A, B] = 0?$$

If your answer is yes, prove the assertion. If your answer is no, give a counterexample.

Solution. Yes, because for any pair of simultaneous eigenkets $|a', b'\rangle$ and $|a'', b''\rangle$, we have

$$\langle a', b' | [A, B] | a'', b'' \rangle = a'' b'' \langle a', b' | a'', b'' \rangle - a' b' \langle a', b' | a'', b'' \rangle = 0,$$

and since these eigenkets form a complete orthonormal set, we have

$$[A, B] = \sum_{a', b', a'', b''} |a', b'\rangle \langle a', b' | [A, B] | a'', b'' \rangle \langle a'', b''| = 0,$$

where we've used the completeness relations. $\square$

3-5 (Sakurai, Problem 1.16)

Two Hermitian operators anticommute:

$$\{A, B\} = AB + BA = 0.$$

Is it possible to have a simultaneous eigenket of A and B ? Prove or illustrate your assertion.

Solution. Yes, but this eigenket must correspond to an eigenvalue of zero for A or B .

For suppose that $|a', b'\rangle$ is a simultaneous eigenket of A and B . Then

$$\langle a', b' | \{A, B\} | a', b' \rangle = 2a'b',$$

but since $\{A, B\} = 0$, this implies $2a'b' = 0$. Hence either $a' = 0$ or $b' = 0$. □

3-6 (Sakurai, Problem 1.17)

Two observables A_1 and A_2 , which do not involve time explicitly, are known not to commute,

$$[A_1, A_2] \neq 0,$$

yet we also know that A_1 and A_2 both commute with the Hamiltonian:

$$[A_1, H] = 0, \quad [A_2, H] = 0.$$

Prove that the energy eigenstates are, in general, degenerate. Are there exceptions? As an example, you may think of the central-force problem $H = \mathbf{p}^2/2m + V(r)$, with $A_1 \rightarrow L_z$, $A_2 \rightarrow L_x$.

Solution. Since H and A_1 commute, we can find a simultaneous eigenstate $|n, a'_1\rangle$ such that $H|n, a'_1\rangle = E_n|n, a'_1\rangle$ and $A_1|n, a'_1\rangle = a'_1|n, a'_1\rangle$. Then since H and A_2 commute, we have

$$H(A_2|n, a'_1\rangle) = A_2H|n, a'_1\rangle = E_n(A_2|n, a'_1\rangle);$$

that is, $A_2|n, a'_1\rangle$ is another eigenket of H with eigenvalue E_n . To prove that the energy eigenstates are degenerate, it suffices to show that $A_2|n, a'_1\rangle$ and $|n, a'_1\rangle$ are linearly independent. Since A_1 and A_2 do not commute, it follows from Problem 1.15 that there is no complete set of simultaneous eigenkets for these operators. Thus, there exists some eigenket $|n, a'_1\rangle$ of A_1 which is not an eigenket of A_2 . But this amounts to the statement that $|n, a'_1\rangle$ and $A_2|n, a'_1\rangle$ are linearly independent, so we are done. □

3-7 (Sakurai, Problem 1.23)

Consider a three-dimensional ket space. If a certain set of orthonormal kets—say, $|1\rangle$, $|2\rangle$, and $|3\rangle$ —are used as the base kets, the operators A and B are represented by

$$A \doteq \begin{pmatrix} a & 0 & 0 \\ 0 & -a & 0 \\ 0 & 0 & -a \end{pmatrix}, \quad B \doteq \begin{pmatrix} b & 0 & 0 \\ 0 & 0 & -ib \\ 0 & ib & 0 \end{pmatrix}$$

with a and b both real.

- Obviously A exhibits a degenerate spectrum. Does B also exhibit a degenerate spectrum?
- Show that A and B commute.
- Find a new set of orthonormal kets which are simultaneous eigenkets of both A and B . Specify the eigenvalues of A and B for each of the three eigenkets. Does your specification of eigenvalues completely characterize each eigenket?

Solution. (a) Yes, because

$$\begin{aligned} \det(B - \lambda I) &= (b - \lambda) \begin{vmatrix} -\lambda & -ib \\ ib & -\lambda \end{vmatrix} = (b - \lambda)(\lambda^2 + i^2 b^2) \\ &= -(\lambda - b)(\lambda + b)(\lambda - b) \\ &= -(\lambda - b)^2(\lambda + b), \end{aligned}$$

which shows that the eigenvalues of B are $-b$ and b (with multiplicity 2).

(b) A direct calculation shows that

$$AB = BA = \begin{pmatrix} ab & 0 & 0 \\ 0 & 0 & iba \\ 0 & -iba & 0 \end{pmatrix}.$$

(c) Since A is already diagonal, we will find a combination of the degenerate eigenvectors of A that diagonalizes B . For this it suffices to diagonalize the lower diagonal block of B . Since

$$\begin{vmatrix} -\lambda & -ib \\ ib & -\lambda \end{vmatrix} = \lambda^2 - b^2,$$

it follows that the eigenvalues of the block matrix are b and $-b$. One can see that $\frac{1}{\sqrt{2}}(1, i)^T$ is a normalized eigenvector corresponding to b , and that $\frac{1}{\sqrt{2}}(1, -i)^T$ is a normalized eigenvector corresponding to $-b$. Thus, to get a simultaneous eigenbasis of A and B we should choose

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \frac{i}{\sqrt{2}} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1/\sqrt{2} \\ i/\sqrt{2} \end{pmatrix}$$

and

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} - \frac{i}{\sqrt{2}} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1/\sqrt{2} \\ -i/\sqrt{2} \end{pmatrix}$$

as our last two eigenvectors. (Our problem with $(0, 1, 0)^T$ and $(0, 0, 1)^T$ was that they were not eigenvectors of B . These new vectors are, by construction. Moreover, since they are a linear combination of vectors in the same eigenspace of A , they are also eigenvectors of A .)

Thus, a set of simultaneous eigenkets of A and B are

$$|1\rangle, \quad \frac{1}{\sqrt{2}}(|2\rangle + i|3\rangle), \quad \frac{1}{\sqrt{2}}(|2\rangle - i|3\rangle).$$

One can also check that this works by computing the representation of A and B in this new basis and ensuring that both A and B are both diagonal. In the “collective index” notation of Sakurai, these correspond to

$$|a, b\rangle, \quad |-a, b\rangle, \quad |-a, -b\rangle.$$

This shows that, despite the degeneracy, it is possible to uniquely specify each eigenket by listing its eigenvalues corresponding to A and B . $\square$

3-8 (Sakurai, Problem 1.24)

- Prove that $(1/\sqrt{2})(1 + i\sigma_x)$ acting on a two-component spinor can be regarded as the matrix representation of the rotation operator about the x -axis by angle $-\pi/2$. (The minus sign signifies that the rotation is clockwise.)
- Construct the matrix representation of S_z when the eigenkets of S_y are used as base vectors.

Solution. (a) Let us write

$$\chi = \begin{pmatrix} a \\ b \end{pmatrix}.$$

Then

$$\begin{aligned} \frac{1}{\sqrt{2}}(1 + i\sigma_x)\chi &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} \\ &= \frac{1}{\sqrt{2}} \begin{pmatrix} a + ib \\ ia + b \end{pmatrix}. \end{aligned}$$

For instance, if $a = 1$ and $b = 0$, so that χ represents the ket $|+\rangle$, then the action of this operator is to send $|+\rangle$ to $\frac{1}{\sqrt{2}}(|+\rangle + i|-\rangle) \equiv |S_y; +\rangle$. Meanwhile, if $a = 0$ and $b = 1$, then this operator sends $|+\rangle$ to $\frac{1}{\sqrt{2}}(|+\rangle - i|-\rangle) \equiv |S_y; -\rangle$. Both of these cases are consistent with the interpretation of this matrix as the representation of the rotation operator $R_x(-\pi/2)$.

(b) Since $|S_y; \pm\rangle = \frac{1}{\sqrt{2}}|+\rangle \pm \frac{i}{\sqrt{2}}|-\rangle$, the matrix representation of S_z in the S_y eigenbasis is

$$\frac{\hbar}{2} \begin{pmatrix} 1/\sqrt{2} & -i/\sqrt{2} \\ 1/\sqrt{2} & i/\sqrt{2} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ i/\sqrt{2} & -i/\sqrt{2} \end{pmatrix} = \boxed{\frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}}.$$

This makes sense, if you draw a picture – the z -axis is the same orientation with respect to the y -axis as the x -axis is to the z -axis, so S_z looks like σ_x in the S_y basis. $\square$

3-9 (Sakurai, Problem 1.27)

- a Suppose that $f(A)$ is a function of a Hermitian operator A with the property $A|a'\rangle = a'|a'\rangle$. Evaluate $\langle b''|f(A)|b'\rangle$ when the transformation matrix from the a' basis to the b' basis is known.
- b Using the continuum analogue of the result obtained in (a), evaluate

$$\langle \mathbf{p}''|F(r)|\mathbf{p}'\rangle.$$

Simplify your expression as far as you can. Note that r is $\sqrt{x^2 + y^2 + z^2}$, where x , y , and z are operators.

Solution. (a) We have

$$\begin{aligned} \langle b''|f(A)|b'\rangle &= \sum_{a'} \sum_{a''} \langle b''|a''\rangle \langle a''|f(A)|a'\rangle \langle a'|b'\rangle \\ &= \sum_{a'} \sum_{a''} \langle b''|a''\rangle f(a') \underbrace{\langle a''|a'\rangle}_{\delta_{a''a'}} \langle a'|b'\rangle \\ &= \boxed{\sum_{a'} f(a') \langle b''|a'\rangle \langle a'|b'\rangle}. \end{aligned}$$

(b) Using the continuum analogue of the result above, we get

$$\langle \mathbf{p}''|F(r)|\mathbf{p}'\rangle = \int d^3x' F(r') \langle \mathbf{p}''|\mathbf{x}'\rangle \langle \mathbf{x}'|\mathbf{p}'\rangle,$$

where $r' \equiv |\mathbf{x}'|$. Simplifying further, we have

$$\begin{aligned} \langle \mathbf{p}''|F(r)|\mathbf{p}'\rangle &= \frac{1}{(2\pi\hbar)^3} \int d^3x' F(r') \exp\left(-\frac{i\mathbf{p}'' \cdot \mathbf{x}'}{\hbar}\right) \exp\left(\frac{i\mathbf{p}' \cdot \mathbf{x}'}{\hbar}\right) \\ &= \frac{1}{(2\pi\hbar)^3} \int d^3x' F(r') \exp\left(\frac{i(\mathbf{p}' - \mathbf{p}'') \cdot \mathbf{x}'}{\hbar}\right). \end{aligned}$$

We are free to rotate our axes so that $\mathbf{p}' - \mathbf{p}''$ points along the z' axis, and then by adopting

spherical coordinates, we get

$$(\mathbf{p}' - \mathbf{p}'') \cdot \mathbf{x}' = (p' - p'') r' \cos \theta.$$

This allows us to do the angular integrals, and we get

$$\begin{aligned} \langle \mathbf{p}'' | F(r) | \mathbf{p}' \rangle &= \frac{1}{(2\pi\hbar)^3} \int_0^\infty (r')^2 dr' \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi F(r') \exp\left(\frac{i(p' - p'') r' \cos \theta}{\hbar}\right) \\ &= \frac{2\pi}{(2\pi\hbar)^3} \int_0^\infty (r')^2 dr' \int_0^\pi \sin \theta d\theta F(r') \exp\left(\frac{i(p' - p'') r' \cos \theta}{\hbar}\right) \\ &= \frac{2\pi}{(2\pi\hbar)^3} \int_0^\infty (r')^2 dr' \frac{\hbar}{i(p' - p'') r'} F(r') \int_{-i(p' - p'') r' / \hbar}^{i(p' - p'') r' / \hbar} du e^u \\ &= -\frac{i\hbar}{p' - p''} \frac{4\pi}{(2\pi\hbar)^3} \int_0^\infty dr' r' F(r') \cos\left(\frac{(p' - p'') r'}{\hbar}\right) \\ &= \boxed{-\frac{i}{(p' - p'') 2\pi^2 \hbar^2} \int_0^\infty dr' r' F(r') \cos\left(\frac{(p' - p'') r'}{\hbar}\right)}, \end{aligned}$$

where in the third line we made the substitution $u = i(p' - p'') r' \cos \theta / \hbar$. □

4 Wave mechanics

4-1 (Landau and Lifshitz, §23 Problem 4)

Solution. (a) We would like to sketch the potential

$$U(x) = A(e^{-2\alpha x} - 2e^{-\alpha x}).$$

Let us assume for simplicity that $\alpha > 0$.¹ First, we examine the asymptotic behavior:

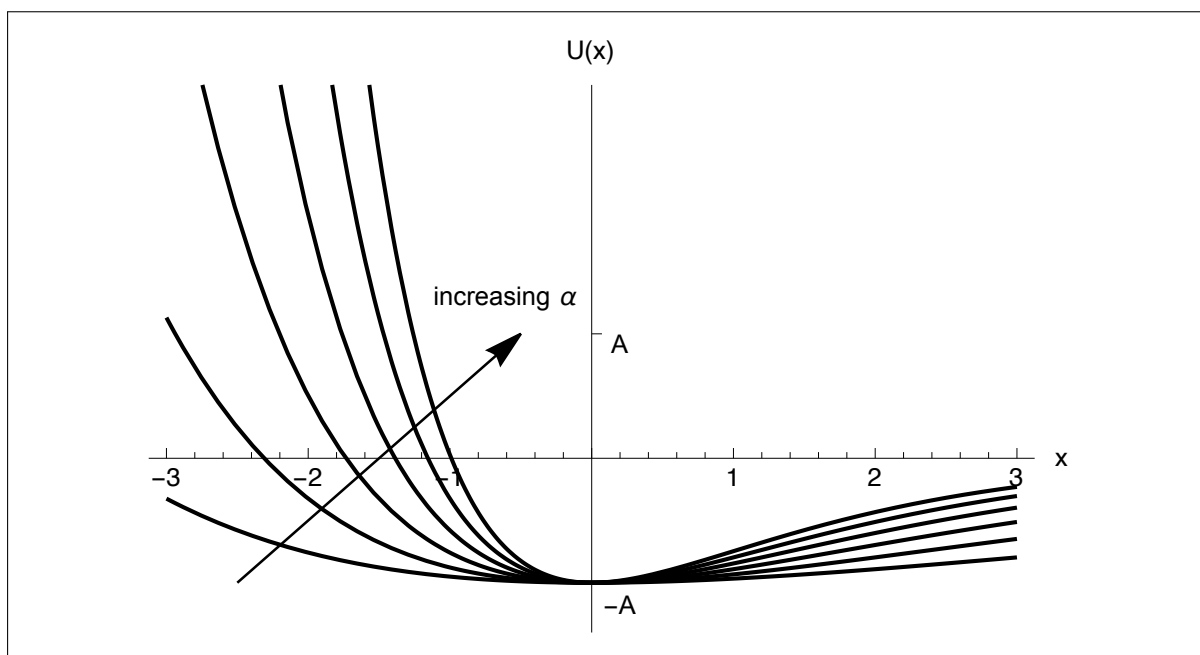
$$U(x) \sim -2Ae^{-\alpha x} \quad \text{as } x \rightarrow +\infty,$$

$$U(x) \sim Ae^{-2\alpha x} \quad \text{as } x \rightarrow -\infty.$$

Then we look at the first derivative:

$$\begin{aligned} U'(x) &= A(-2\alpha e^{-2\alpha x} + 2\alpha e^{-\alpha x}) \\ &= 2\alpha A(-e^{-2\alpha x} + e^{-\alpha x}). \end{aligned}$$

Thus $U(x)$ is decreasing for negative values of x and increasing for positive values. The potential attains its minimum value at $x = 0$, with $U(0) = -A$. See below for a picture:



As α tends to larger values, we get a narrower potential well, still centered at $x = 0$, which is more strongly repulsive for $x < 0$ and dies out more rapidly for $x > 0$. This potential goes to $+\infty$ as $x \rightarrow -\infty$ and 0 as $x \rightarrow +\infty$, so in order to get discrete eigenvalues, we need the energy to be in the range $\boxed{-A < E < 0}$. (The first inequality is merely the requirement that the energy must be larger than the minimum value of the potential.)

¹The treatment should not be essentially different for $\alpha < 0$, since the potential is simply mirrored about the vertical axis when $\alpha \rightarrow -\alpha$.

(b) The Schrödinger equation, when multiplied through by $-2m/\hbar^2$, becomes

$$\frac{d^2\psi}{dx^2} - \frac{2mA}{\hbar^2}(e^{-2\alpha x} - 2e^{-\alpha x})\psi = -\frac{2mE}{\hbar^2}\psi.$$

We may define the dimensionless variable

$$\xi = \frac{2\sqrt{2mA}}{\alpha\hbar} e^{-\alpha x},$$

so that by the chain rule

$$\frac{d}{dx} = \frac{d\xi}{dx} \frac{d}{d\xi} = -\frac{2\sqrt{2mA}}{\hbar} e^{-\alpha x} = -\alpha\xi \frac{d}{d\xi},$$

$$\begin{aligned} \frac{d^2}{dx^2} &= \frac{d\xi}{dx} \frac{d}{d\xi} \left(-\alpha\xi \frac{d}{d\xi} \right) = (-\alpha\xi)(-\alpha) \frac{d}{d\xi} \left(\xi \frac{d}{d\xi} \right) = \alpha^2 \xi \left(\frac{d}{d\xi} + \xi \frac{d^2}{d\xi^2} \right) \\ &= \alpha^2 \xi \frac{d}{d\xi} + \alpha^2 \xi^2 \frac{d^2}{d\xi^2}. \end{aligned}$$

Then, using the definition of ξ we have

$$e^{-2\alpha x} = \frac{\alpha^2 \hbar^2}{8mA} \xi^2, \quad e^{-\alpha x} = \frac{\alpha\hbar}{2\sqrt{2mA}} \xi,$$

and the Schrödinger equation can be rewritten in the form

$$\begin{aligned} \alpha^2 \xi \psi' + \alpha^2 \xi^2 \psi'' - \frac{2mA}{\hbar^2} \left(\frac{\alpha^2 \hbar^2}{8mA} \xi^2 \right) \psi + \frac{2mA}{\hbar^2} \left(2 \frac{\alpha\hbar}{2\sqrt{2mA}} \xi \right) \psi &= -\frac{2mE}{\hbar^2} \psi \\ \iff \alpha^2 \xi \psi' + \alpha^2 \xi^2 \psi'' - \frac{1}{4} \alpha^2 \xi^2 \psi + \frac{\sqrt{2mA}}{\hbar} \alpha \xi \psi &= -\frac{2mE}{\hbar^2} \psi, \end{aligned}$$

where primes denote differentiation with respect to ξ . Since α and ξ are nonzero by definition, we can divide through by $\alpha^2 \xi^2$ to get

$$\psi'' + \frac{1}{\xi} \psi' + \left(-\frac{1}{4} + \frac{\sqrt{2mA}}{\alpha\hbar} \frac{1}{\xi} + \frac{2mE}{\alpha^2 \hbar^2} \frac{1}{\xi^2} \right) \psi = 0.$$

We then define the pair of variables

$$s = \frac{\sqrt{-2mE}}{\alpha\hbar}, \quad n = \frac{\sqrt{2mA}}{\alpha\hbar} - (s + 1/2),$$

so that the above simplifies to

$$\boxed{\psi'' + \frac{1}{\xi} \psi' + \left(-\frac{1}{4} + \frac{n + s + 1/2}{\xi} - \frac{s^2}{\xi^2} \right) \psi = 0} \quad (1)$$

(c) For large ξ , the Schrödinger equation (1) reduces to

$$\psi'' - \frac{1}{4}\psi = 0.$$

Immediately, we can write down the general solution:

$$\psi(\xi) = Ae^{-\xi/2} + Be^{+\xi/2}.$$

Since we want a function belonging to the physical Hilbert space, we discard the second term (since it would imply that the wave function blows up as $x \rightarrow -\infty$), and conclude

$$\boxed{\psi \sim e^{-\xi/2} \quad \text{as} \quad \xi \rightarrow \infty.}$$

For completeness, we briefly discuss the case of small ξ . Since (1) has a regular singular point at $\xi = 0$, we look for a power series solution around 0 using the Frobenius method:

$$\psi(\xi) = \sum_{j=0}^{\infty} a_j \xi^{j+\alpha}$$

Substituting into (1), then examining the lowest-order term, one can show that

$$a_0(\alpha^2 - s^2) = 0,$$

and since a_0 is assumed nonzero, it follows that $\alpha = \pm s$. Thus the solution for small ξ goes like $\xi^{\pm s}$, and here we choose the positive sign, since the negative sign implies that the wave function blows up as $x \rightarrow \infty$. We conclude that

$$\boxed{\psi \sim \xi^s \quad \text{as} \quad \xi \rightarrow 0.}$$

(d) Now we peel off the asymptotic dependence, and work with the ansatz

$$\psi = e^{-\xi/2} \xi^s w(\xi). \tag{2}$$

We substitute this into the Schrödinger equation, and begin by computing derivatives of the

wave function:

$$\begin{aligned}
\psi' &= -\frac{1}{2}e^{-\xi/2}\xi^s w + e^{-\xi/2}s\xi^{s-1}w + e^{-\xi/2}\xi^s w' \\
&= e^{-\xi/2}\left(-\frac{1}{2}\xi^s w + s\xi^{s-1}w + \xi^s w'\right), \\
\psi'' &= -\frac{1}{2}e^{-\xi/2}\left(-\frac{1}{2}\xi^s w + s\xi^{s-1}w + \xi^s w'\right) \\
&\quad + e^{-\xi/2}\left(-\frac{1}{2}s\xi^{s-1}w - \frac{1}{2}\xi^s w' + s(s-1)\xi^{s-2}w + s\xi^{s-1}w' + s\xi^{s-1}w' + \xi^s w''\right) \\
&= e^{-\xi/2}\left(\frac{1}{4}\xi^s w - \frac{1}{2}s\xi^{s-1}w - \frac{1}{2}\xi^s w' \right. \\
&\quad \left. - \frac{1}{2}s\xi^{s-1}w - \frac{1}{2}\xi^s w' + s(s-1)\xi^{s-2}w + s\xi^{s-1}w' + s\xi^{s-1}w' + \xi^s w''\right) \\
&= e^{-\xi/2}\left(\frac{1}{4}\xi^s w - s\xi^{s-1}w - \xi^s w' + s^2\xi^{s-2}w - s\xi^{s-2}w + 2s\xi^{s-1}w' + \xi^s w''\right).
\end{aligned}$$

Then we group like derivatives of w :

$$\begin{aligned}
\psi' &= e^{-\xi/2}\left[\xi^s w' + \left(s\xi^{s-1} - \frac{1}{2}\xi^s\right)w\right], \\
\psi'' &= e^{-\xi/2}\left[\xi^s w'' + (2s\xi^{s-1} - \xi^s)w' + \left(\frac{1}{4}\xi^s - s\xi^{s-1} + s^2\xi^{s-2} - s\xi^{s-2}\right)w\right]. \tag{3}
\end{aligned}$$

Now we can form the terms appearing in the Schrödinger equation:

$$\frac{1}{\xi}\psi' = e^{-\xi/2}\left[\xi^{s-1}w' + \left(s\xi^{s-2} - \frac{1}{2}\xi^{s-1}\right)w\right], \tag{4}$$

$$\begin{aligned}
\left(-\frac{1}{4} + \frac{n+s+1/2}{\xi} - \frac{s^2}{\xi^2}\right)\psi &= e^{-\xi/2}\left(-\frac{1}{4} + \frac{n+s+1/2}{\xi} - \frac{s^2}{\xi^2}\right)\xi^s w \\
&= e^{-\xi/2}\left(-\frac{1}{4}\xi^s + (n+s+1/2)\xi^{s-1} - s^2\xi^{s-2}\right)w. \tag{5}
\end{aligned}$$

Substituting (3)–(5) into (1), we obtain (factoring out the common $e^{-\xi/2}$)

$$\begin{aligned}
&\xi^s w'' + (2s\xi^{s-1} - \xi^s)w' + \left(\frac{1}{4}\xi^s - s\xi^{s-1} + s^2\xi^{s-2} - s\xi^{s-2}\right)w \\
&\quad + \xi^{s-1}w' + \left(s\xi^{s-2} - \frac{1}{2}\xi^{s-1}\right)w \\
&\quad + \left(-\frac{1}{4}\xi^s + (n+s+1/2)\xi^{s-1} - s^2\xi^{s-2}\right)w = 0.
\end{aligned}$$

Dividing through by ξ^{s-1} , this simplifies to

$$\begin{aligned} \xi w'' + (2s - \xi) w' + \left(\frac{1}{4}\xi - s + \frac{s^2}{\xi} - \frac{s}{\xi} \right) w \\ + w' + \left(\frac{s}{\xi} - \frac{1}{2} \right) w \\ + \left(-\frac{1}{4}\xi + (n + s + 1/2) - \frac{s^2}{\xi} \right) w = 0. \end{aligned}$$

Most of the terms appearing here cancel out, leaving us with

$$\boxed{\xi w'' + (2s + 1 - \xi)w' + nw = 0}. \quad (6)$$

This differential equation takes the form

$$zu'' + (\gamma - z)u' - au = 0 \quad (7)$$

with $\boxed{\gamma = 2s + 1}$ and $\boxed{a = -n}$. A solution to (7) is the confluent hypergeometric function

$$F(a, \gamma; z) = 1 + \frac{a}{\gamma} \frac{z}{1!} + \frac{a(a+1)}{\gamma(\gamma+1)} \frac{z^2}{2!} + \dots \quad (8)$$

In general the parameter a is arbitrary, but in order for each term in the series to be well-defined, it is necessary that γ is neither zero nor a negative integer. This is satisfied for the equation (6), since $\gamma = 2s + 1$ is always positive. Then, since ψ represents a physical wave function, we add the requirement that a is either zero or a negative integer. This ensures that the series (8) for $F(a, y; z)$ terminates, which in turn ensures the proper asymptotic behavior for the solution [see the ansatz (2)].

(e) In order for a to be zero or a negative integer, we require that $\boxed{n = 0, 1, 2, 3, \dots}$. Since

$$s = \frac{\sqrt{-2mE}}{\alpha\hbar}, \quad n = \frac{\sqrt{2mA}}{\alpha\hbar} - (s + 1/2),$$

this constrains the allowed energies to take discrete values; in particular,

$$E = -\frac{\alpha^2 \hbar^2 s^2}{2m} \implies E_n = -A \left(1 - \frac{\alpha\hbar}{\sqrt{2mA}} (n + 1/2) \right)^2, \quad n = 0, 1, 2, 3, \dots, n_{\max}.$$

The existence of a maximum n follows from our constraint that $-A < E < 0$. Putting everything together, we conclude that the bound-state solutions are

$$\boxed{\psi_n(\xi) = C e^{-\xi/2} \xi^s F(-n, 2s + 1; \xi)},$$

where C is a normalization constant, and the variable s is given in terms of n by

$$s = 1 - \frac{\alpha\hbar}{\sqrt{2mA}}(n + 1/2) .$$

The wave functions ψ_n correspond to the energies

$$E_n = -A \left(1 - \frac{\alpha\hbar}{\sqrt{2mA}}(n + 1/2) \right)^2, \quad n = 0, 1, 2, 3, \dots, n_{\max} ,$$

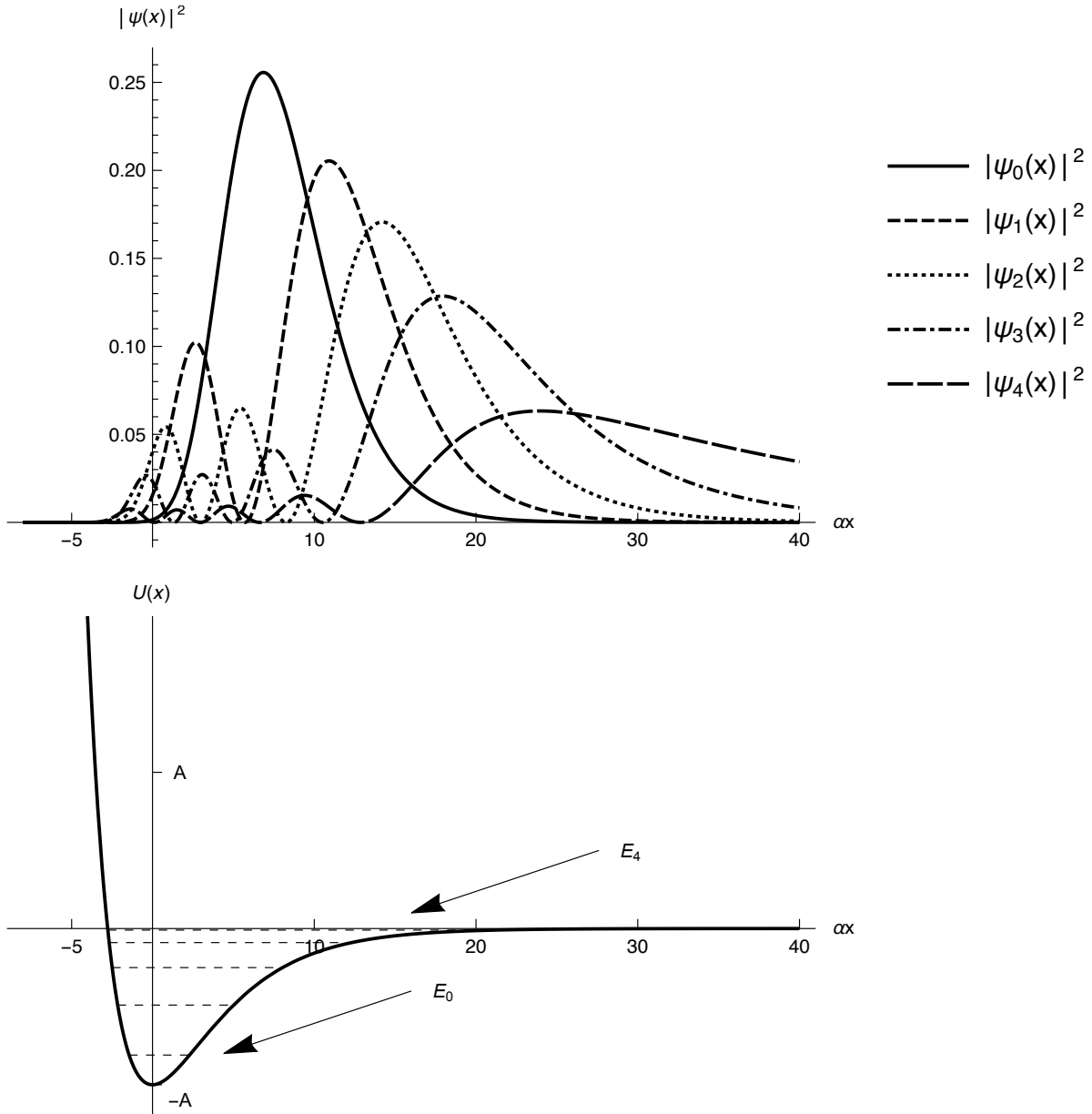
where $n_{\max}$ is determined by the constraint $-A < E < 0$. Some features to note:

1. There are only a finite number of bound states.
2. We have $E_n < E_{n+1}$, since the expression in parentheses decreases with increasing n . The minimum energy is

$$E_0 = -A \left(1 - \frac{\alpha\hbar}{2\sqrt{2mA}} \right)^2 .$$

3. If the dimensionless combination $\alpha\hbar/2\sqrt{2mA}$ is greater than or equal to 1, then the potential supports no bound states, since in this case $s < 0$ for all n , a contradiction. This happens for a potential well that is very narrow (corresponding to large α) or very shallow (corresponding to small A); compare the sketch in part (a).
4. The energy levels are not evenly spaced; as n gets larger, the spacing between adjacent levels decreases.

We can plot the probability densities $|\psi_n(x)|^2$ in terms of the original position variable x , together with the corresponding energies E_n , for a potential with $\alpha\hbar/\sqrt{2mA} = 0.2$ and $\alpha = 0.5$. (The wave functions were normalized using Mathematica.)



Note that the probability density decays rapidly in the region $x < 0$, and that the higher-energy states penetrate further into this region, as we expect. Note also that the expectation value $\langle x \rangle$ moves further out to the right as the energy increases. This is what we expect, at least classically, since the particle spends more of its time in the nearly zero-potential region to the right of the origin. Finally, we see that these wave functions obey Sturm's oscillation theorem; that is, $\psi_n(x)$ has n nodes. $\square$

4-2 (Shankar, Exercise 4.2.1)

Consider the following operators on a Hilbert space $\mathbb{V}^3(C)$:

$$L_x = \frac{1}{2^{1/2}} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad L_y = \frac{1}{2^{1/2}} \begin{bmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{bmatrix}, \quad L_z = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

- (1) What are the possible values one can obtain if L_z is measured?
- (2) Take the state in which $L_z = 1$. In this state what are $\langle L_x \rangle$, $\langle L_x^2 \rangle$, and ΔL_x ?
- (3) Find the normalized eigenstates and the eigenvalues of L_x in the L_z basis.
- (4) If the particle is in the state with $L_z = -1$, and L_x is measured, what are the possible outcomes and their probabilities?
- (5) Consider the state

$$|\psi\rangle = \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2^{1/2} \end{bmatrix}$$

in the L_z basis. If L_z^2 is measured in this state and a result $+1$ is obtained, what is the state after the measurement? How probable was this result? If L_z is measured immediately afterwards, what are the outcomes and respective probabilities?

- (6) A particle is in a state for which the probabilities are $P(L_z = 1) = 1/4$, $P(L_z = 0) = 1/2$, and $P(L_z = -1) = 1/4$. Convince yourself that the most general, normalized state with this property is

$$|\psi\rangle = \frac{e^{i\delta_1}}{2} |L_z = 1\rangle + \frac{e^{i\delta_2}}{2^{1/2}} |L_z = 0\rangle + \frac{e^{i\delta_3}}{2} |L_z = -1\rangle$$

It was stated earlier on that if $|\psi\rangle$ is a normalized state then the state $e^{i\theta}\psi$ is a physically equivalent normalized state. Does this mean that the factors $e^{i\delta_i}$ multiplying the L_z eigenstates are irrelevant? [Calculate for example $P(L_x = 0)$.]

Solution. (1) By inspection of the matrix representation, the possible values that can be obtained from a measurement (i.e. the eigenvalues) of L_z are 1, 0, and -1.

(2) First, we have

$$\begin{aligned}
\langle L_x \rangle &= \langle L_z = 1 | L_x | L_z = 1 \rangle \\
&= \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \\
&= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.
\end{aligned}$$

This shows that $\boxed{\langle L_x \rangle = 0}$. Next, we have

$$\begin{aligned}
\langle L_x^2 \rangle &= \langle L_z = 1 | L_x^2 | L_z = 1 \rangle \\
&= \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} \frac{1}{2} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \\
&= \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix},
\end{aligned}$$

which shows that $\boxed{\langle L_x^2 \rangle = 1/2}$. Then since $\Delta L_x = \sqrt{\langle L_x^2 \rangle - \langle L_x \rangle^2}$, we have $\boxed{\Delta L_x = 1/\sqrt{2}}$.

(3) We did this in Sakurai Problem 1.14 (alternatively, Problem 3.15 in Townsend), and the eigenvalues of L_x are 1, 0, and -1 , corresponding to the normalized eigenvectors

$$\frac{1}{2} \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix}, \quad \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \quad \frac{1}{2} \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix},$$

respectively.

(4) We need to convert $|L_z = -1\rangle$ to the L_x basis using the result of (3):

$$\begin{aligned}
 |L_z = -1\rangle &\stackrel{*}{=} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \xrightarrow{L_x \text{ basis}} \begin{pmatrix} 1/2 & 1/\sqrt{2} & 1/2 \\ \sqrt{2}/2 & 0 & -\sqrt{2}/2 \\ 1/2 & -1/\sqrt{2} & 1/2 \end{pmatrix}^\dagger \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \\
 &= \begin{pmatrix} 1/2 & \sqrt{2}/2 & 1/2 \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 1/2 & -\sqrt{2}/2 & 1/2 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \\
 &= \begin{pmatrix} 1/2 \\ -1/\sqrt{2} \\ 1/2 \end{pmatrix}.
 \end{aligned}$$

The possible outcomes of an L_x measurement are $\boxed{1, 0, \text{ or } -1}$, with corresponding probabilities $\boxed{1/4, 1/2, \text{ and } 1/4}$, respectively. (If we just wanted to ask about the probability of a single outcome, then it would be equivalent – and a bit less work – to compute a dot product with the corresponding eigenvector.)

(5) We have that

$$L_z^2 \stackrel{*}{=} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The eigenvalues of L_z^2 are 0 and 1 (mult. 2). There is a twofold degeneracy. If L_z^2 is measured and a result +1 is obtained, then the state after the measurement is the projection of $|\psi\rangle$ onto the +1 eigenspace,

$$|\psi\rangle \xrightarrow{L_z^2 \text{ measured, } +1 \text{ obtained}} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

The probability of this result is 1/2. (This is the magnitude of the projection of $|\psi\rangle$ onto the +1 eigenspace prior to normalization.) If L_z is measured immediately afterward, the possible outcomes are 1 and -1 , with probabilities of 1/2 each.

(6) We have

$$P(L_z = 1) = |\langle L_z = 1 | \psi \rangle|^2 = \frac{1}{4},$$

so that the only requirement is that the projection of $|\psi\rangle$ onto $|L_z = 1\rangle$ has norm 1/2. Thus,

$$|\psi\rangle = \frac{a + ib}{2} |L_z = 1\rangle + \dots$$

where a, b are arbitrary real numbers satisfying $a^2 + b^2 = 1$. But this last constraint means that $a + ib = \cos \theta + i \sin \theta = e^{i\theta}$, where θ is an arbitrary real number. Applying the same reasoning to the other two projections, we get the claimed result.

However, the factors $e^{i\delta_i}$ are *not* irrelevant. One can generally set one of these phase factors to unity by multiplying $e^{-i\delta_i}$, but the other two terms will pick up a phase difference $e^{i(\delta_j - \delta_i)}$, and these have physical relevance. (Think of the S_x and S_y eigenstates of a spin- $\frac{1}{2}$ particle.) $\square$

4-3 (Shankar, Exercise 4.2.2)

Show that for a real wave function $\psi(x)$, the expectation value of momentum $\langle P \rangle = 0$. (Hint: Show that the probabilities for the momentum $\pm p$ are equal.) Generalize this result to the case $\psi = c\psi_r$, where ψ_r is real and c an arbitrary (real or complex) constant. (Recall that $|\psi\rangle$ and $\alpha|\psi\rangle$ are physically equivalent.)

Solution. We'll use the argument suggested by the hint. Computing this expectation value in the momentum basis, we have

$$\begin{aligned}\langle P \rangle &= \langle \psi | P | \psi \rangle = \int_{-\infty}^{\infty} \langle \psi | p \rangle \langle p | P | \psi \rangle dp \\ &= \int_{-\infty}^{\infty} p |\langle p | \psi \rangle|^2 dp.\end{aligned}$$

To prove that $\langle P \rangle = 0$, it suffices to show that $|\langle p | \psi \rangle|^2$ is an even function of p , or equivalently, that the probabilities for the momenta $\pm p$ are equal. Since we're given information about the wave function in the position basis, we move there:

$$\begin{aligned}\langle p | \psi \rangle &= \int_{-\infty}^{\infty} \langle p | x \rangle \langle x | \psi \rangle dx \\ &= \int_{-\infty}^{\infty} \frac{e^{-ipx/\hbar}}{(2\pi\hbar)^{1/2}} \psi(x) dx.\end{aligned}$$

Then we have

$$\langle \psi | p \rangle = \left(\int_{-\infty}^{\infty} \frac{e^{-ipx/\hbar}}{(2\pi\hbar)^{1/2}} \psi(x) dx \right)^* = \int_{-\infty}^{\infty} \frac{e^{ipx/\hbar}}{(2\pi\hbar)^{1/2}} \psi(x) dx = \langle -p | \psi \rangle.$$

In particular, this implies $\langle p | \psi \rangle = \langle \psi | -p \rangle$. Thus

$$|\langle p | \psi \rangle|^2 = \langle p | \psi \rangle \langle \psi | p \rangle = \langle \psi | -p \rangle \langle -p | \psi \rangle = |\langle -p | \psi \rangle|^2,$$

as we wanted to show. If instead $\psi = c\psi_r$, then the same expansion in the position basis becomes

$$\langle \psi | p \rangle = \left(\int_{-\infty}^{\infty} \frac{e^{-ipx/\hbar}}{(2\pi\hbar)^{1/2}} c\psi_r(x) dx \right)^* = c^* \int_{-\infty}^{\infty} \frac{e^{ipx/\hbar}}{(2\pi\hbar)^{1/2}} \psi_r(x) dx = c^* \langle -p | \psi_r \rangle.$$

This implies that $\langle p|\psi\rangle = c\langle\psi_r|-p\rangle$, and then

$$|\langle p|\psi\rangle|^2 = |c|^2 \langle -p|\psi_r\rangle \langle\psi_r|-p\rangle = |c|^2 |\langle -p|\psi_r\rangle|^2.$$

By the previous result, the last term on the right is an even function of p , which shows that $|\langle p|\psi\rangle|^2$ is also an even function of p . Thus $\langle P\rangle = 0$. $\square$

In other words, you need a nontrivial imaginary component in your position wavefunction (i.e., a wavefunction that cannot be made real by multiplying an overall factor) to get a nonzero expectation value for P .

4-4 (Shankar, Exercise 4.2.3)

Show that if $\psi(x)$ has mean momentum $\langle P\rangle$, $e^{ip_0x/\hbar}\psi(x)$ has mean momentum $\langle P\rangle + p_0$.

Solution. The mean momentum of $e^{ip_0x/\hbar}\psi(x)$ is

$$\begin{aligned} \int_{-\infty}^{\infty} dx e^{-ip_0x/\hbar} \psi(x)^* \left(-i\hbar \frac{d}{dx} \right) e^{ip_0x/\hbar} \psi(x) \\ = -i\hbar \int_{-\infty}^{\infty} dx e^{-ip_0x/\hbar} \psi(x)^* \left(\frac{ip_0}{\hbar} e^{ip_0x/\hbar} \psi(x) + e^{ip_0x/\hbar} \frac{d\psi}{dx} \right) \\ = p_0 \underbrace{\int_{-\infty}^{\infty} dx \psi(x)^* \psi(x)}_{\langle\psi|\psi\rangle} - i\hbar \underbrace{\int_{-\infty}^{\infty} dx \psi(x)^* \frac{d\psi}{dx}}_{\langle P\rangle} \end{aligned}$$

If $|\psi\rangle$ is normalized, then this shows that the mean momentum is $\langle P\rangle + p_0$, as we wanted. $\square$

The function $e^{ip_0x/\hbar}$ is actually the position-space representation of the boost operator.

4-5 (Shankar, Exercise 5.1.1)

Show that Eq. (5.1.9) may be rewritten as an integral over E and a sum over the $\pm$ index as

$$U(t) = \sum_{\alpha=\pm} \int_0^{\infty} \left[\frac{m}{(2mE)^{1/2}} \right] |E, \alpha\rangle \langle E, \alpha| e^{-iEt/\hbar} dE$$

Solution. Equation (5.1.9) reads

$$\begin{aligned} U(t) &= \int_{-\infty}^{\infty} |p\rangle \langle p| e^{-iE(p)t/\hbar} dp \\ &= \int_{-\infty}^{\infty} |p\rangle \langle p| e^{-ip^2t/2m\hbar} dp. \end{aligned}$$

We may divide this into two integrals where $p \geq 0$ and $p \leq 0$, respectively. Then since $p = \pm(2mE)^{1/2}$, where we take the plus sign for the first integral and the minus sign for the

second one, we have that

$$\begin{aligned}
U(t) &= \int_0^\infty |p\rangle\langle p| e^{-ip^2t/2m\hbar} dp + \int_{-\infty}^0 |p\rangle\langle p| e^{-ip^2t/2m\hbar} dp \\
&= \int_0^\infty |p\rangle\langle p| e^{-ip^2t/2m\hbar} dp - \int_0^{-\infty} |p\rangle\langle p| e^{-ip^2t/2m\hbar} dp \\
&= \int_0^\infty \left[\frac{m}{(2mE)^{1/2}} \right] |E, +\rangle\langle E, +| e^{-iEt/\hbar} dE - \left(- \int_0^\infty \left[\frac{m}{(2mE)^{1/2}} \right] |E, -\rangle\langle E, -| e^{-iEt/\hbar} dE \right) \\
&= \sum_{\alpha=\pm} \int_0^\infty \left[\frac{m}{(2mE)^{1/2}} \right] |E, \alpha\rangle\langle E, \alpha| e^{-iEt/\hbar} dE,
\end{aligned}$$

as we wanted to show. $\square$

4-6 (Shankar, Exercise 5.1.2)

By solving the eigenvalue equation (5.1.3) in the X basis, regain Eq. (5.1.8), i.e., show that the general solution of energy E is

$$\psi_E(x) = \beta \frac{\exp[i(2mE)^{1/2}x/\hbar]}{(2\pi\hbar)^{1/2}} + \gamma \frac{\exp[-i(2mE)^{1/2}x/\hbar]}{(2\pi\hbar)^{1/2}}$$

[The factor $(2\pi\hbar)^{-1/2}$ is arbitrary and may be absorbed into β and γ .] Though $\psi_E(x)$ will satisfy the equation even if $E < 0$, are these functions in the Hilbert space?

Solution. In the X basis, the momentum operator P acts like $P|f\rangle \rightarrow -i\hbar(df/dx)$, so that the eigenvalue equation (5.1.3) becomes

$$-\frac{\hbar^2}{2m} \frac{d^2\psi_E(x)}{dx^2} = E \psi_E(x).$$

This has solution

$$\psi_E(x) = \beta e^{i\omega x} + \gamma e^{-i\omega x},$$

where $\omega = (2mE)^{1/2}/\hbar$, which is what's written above. If $E < 0$, neither of these functions is in the Hilbert space because they blow up at one of $x \rightarrow \pm\infty$. $\square$

4-7 (Shankar, Exercise 5.1.3)

(Another Way to Do the Gaussian Problem). We have seen that there exists another formula for $U(t)$, namely, $U(t) = e^{-iHt/\hbar}$. For a free particle this becomes

$$U(t) = \exp \left[\frac{i}{\hbar} \left(\frac{\hbar^2 t}{2m} \frac{d^2}{dx^2} \right) \right] = \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{i\hbar t}{2m} \right)^n \frac{d^{2n}}{dx^{2n}}$$

Consider the initial state in Eq. (5.1.14) with $p_0 = 0$, and set $\Delta = 1$, $t' = 0$:

$$\psi(x, 0) = \frac{e^{-x^2/2}}{(\pi)^{1/4}}$$

Find $\psi(x, t)$ using Eq. (5.1.18) above and compare with Eq. (5.1.15).

Solution. In the X basis, we have

$$\begin{aligned} U(t) &= \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{i\hbar t}{2m} \right)^n \frac{d^{2n}}{dx^{2n}} \\ &= 1 + \left(\frac{i\hbar t}{2m} \right) \frac{d^2}{dx^2} + \left(\frac{i\hbar t}{2m} \right)^2 \frac{d^4}{dx^4} + \cdots \end{aligned}$$

We will compute the action of the first three terms of $U(t)$ on $\psi(x, 0)$ and use this to make a guess for what the remaining terms will be. (It takes a bit more work to make this appear rigorous.) To facilitate this, we rewrite the initial wave function as a power series:

$$\psi(x, 0) = \pi^{-1/4} \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{n! 2^n}.$$

The first term in the propagator does nothing to the wavefunction, and the second term gives

$$\begin{aligned} \left(\frac{i\hbar t}{2m} \right) \frac{d^2}{dx^2} \left(\pi^{-1/4} \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{n! 2^n} \right) &= \frac{i\hbar t}{2m} \pi^{-1/4} \sum_{n=1}^{\infty} \frac{(-1)^n (2n)(2n-1) x^{2n-2}}{n! 2^n} \\ &= \frac{i\hbar t}{2m} \pi^{-1/4} \sum_{n=0}^{\infty} \frac{(-1)^{n+1} (2n+2)(2n+1) x^{2n}}{(n+1)! 2^{n+1}}, \end{aligned}$$

where we've reindexed the sum for easier comparison later. Proceeding similarly for the second

term, we get

$$\begin{aligned}
& \left(\frac{i\hbar t}{2m}\right)^2 \frac{d^4}{dx^4} \left(\pi^{-1/4} \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \frac{x^{2n}}{2^n} \right) \\
&= \left(\frac{i\hbar t}{2m}\right)^2 \pi^{-1/4} \sum_{n=2}^{\infty} \frac{(-1)^n (2n)(2n-1)(2n-2)(2n-3) x^{2n-4}}{n! 2^n} \\
&= \left(\frac{i\hbar t}{2m}\right)^2 \pi^{-1/4} \sum_{n=0}^{\infty} \frac{(-1)^{n+2} (2n+4)(2n+3)(2n+2)(2n+1) x^{2n}}{(n+2)! 2^{n+2}}
\end{aligned}$$

Combining these three expressions (including the first power series) gives

$$\begin{aligned}
\psi(x, t) &= U(t)\psi(x, 0) \\
&= \pi^{-1/4} \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \left(1 - \frac{i\hbar t}{2m} \frac{(2n+2)(2n+1)}{(n+1)2} \right. \\
&\quad \left. + \left(\frac{i\hbar t}{2m}\right)^2 \frac{(2n+4)(2n+3)(2n+2)(2n+1)}{(n+2)(n+1)2^2} + \dots \right) \frac{x^{2n}}{2^n}.
\end{aligned}$$

In the second term, we can factor out two 2's from the numerator, and in the third term, we can factor out four 2's. These serve to cancel out the factors of $n+1$ and $n+2$ in the denominators, as well as the other 2's from the powers of $i\hbar t/2m$. We get

$$\begin{aligned}
\psi(x, t) &= \pi^{-1/4} \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \left(1 - \frac{i\hbar t}{m} (n+1/2) + \left(\frac{i\hbar t}{m}\right)^2 (n+3/2)(n+1/2) + \dots \right) \frac{x^{2n}}{2^n} \\
&= \pi^{-1/4} \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \left(1 + \frac{i\hbar t}{m} \right)^{-n-1/2} \frac{x^{2n}}{2^n} \\
&= \left[\pi^{1/2} \left(1 + \frac{i\hbar t}{m} \right) \right]^{-1/2} \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{-x^2}{2(1+i\hbar t/m)} \right)^n \\
&= \left[\pi^{1/2} \left(1 + \frac{i\hbar t}{m} \right) \right]^{-1/2} \exp \left[\frac{-x^2}{2(1+i\hbar t/m)} \right].
\end{aligned}$$

The second line uses a fairly obscure (to me!) series expansion for $(1 + i\hbar t/m)^{-n-1/2}$. This is precisely Eq. (5.1.15) with $p_0 = 0$ and $\Delta = 1$, as expected. $\square$

4-8 (Shankar, Exercise 5.1.4)

(A Famous Counterexample). Consider the wave function

$$\begin{aligned}\psi(x, 0) &= \sin\left(\frac{\pi x}{L}\right), & |x| \leq L/2 \\ &= 0. & |x| \geq L/2\end{aligned}$$

It is clear that when this function is differentiated any number of times we get another function confined to the interval $|x| \leq L/2$. Consequently the action of

$$U(t) = \exp\left[\frac{i}{\hbar}\left(\frac{\hbar^2 t}{2m} \frac{d^2}{dx^2}\right)\right]$$

on this function is to give a function confined to $|x| \leq L/2$. What about the spreading of the wave packet?

Solution. If you try to compute $\psi(x, t)$ for this initial wave function in the same way as Exercise 5.1.4 (i.e. examine the action of a few terms of $U(t)$), then you get

$$\psi(x, t) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \left[1 - \frac{i\hbar t}{2m} \left(\frac{\pi}{L}\right)^2 + \left(\frac{i\hbar t}{2m} \left(\frac{\pi}{L}\right)^2\right)^2 - \cdots \right] \left(\frac{\pi x}{L}\right)^{2n+1},$$

but the sum in square brackets diverges for nonzero t , so $\psi(x, t)$ is ill-defined. What resolves the “paradox” is noting that the wave function does not actually belong to the Hilbert space; it turns out that this is because the derivatives are singular at the boundary.

The initial claim is supposed to seem counterintuitive, because as we saw for the Gaussian packet, even though the initial expectation value for the momentum is zero (by Exercise 4.2.2), uncertainty in the initial momentum implies that the uncertainty in position grows with time. In other words, it makes no sense for the function to be confined to $|x| \leq L/2$ at all times. $\square$

4-9 (Shankar, Exercise 5.2.1)

A particle is in the ground state of a box of length L . Suddenly the box expands (symmetrically) to twice its size, leaving the wave function undisturbed. Show that the probability of finding the particle in the ground state of the new box is $(8/3\pi)^2$.

Solution. Denote the ground states of the old and new boxes by $|\psi_1\rangle$ and $|\psi'_1\rangle$, respectively, so

that the amplitude we want to compute is

$$\begin{aligned}
\langle \psi'_1 | \psi_1 \rangle &= \int_{-\infty}^{\infty} \langle \psi'_1 | x \rangle \langle x | \psi_1 \rangle dx = \int_{-L/2}^{L/2} \sqrt{\frac{2}{2L}} \cos\left(\frac{\pi x}{2L}\right) \sqrt{\frac{2}{L}} \cos\left(\frac{\pi x}{L}\right) dx \\
&= \frac{2\sqrt{2}}{L} \int_0^{L/2} \cos\left(\frac{\pi x}{2L}\right) \cos\left(\frac{\pi x}{L}\right) dx \\
&= \frac{2\sqrt{2}}{L} \frac{L}{\pi} \int_0^{\pi/4} \cos(u) \cos(2u) du \\
&= \frac{4\sqrt{2}}{\pi} \int_0^{\pi/4} (1 - 2\sin^2(u)) \cos(u) du \\
&= \frac{4\sqrt{2}}{\pi} \int_0^{1/\sqrt{2}} (1 - 2v^2) dv \\
&= \frac{4\sqrt{2}}{\pi} \left(v - \frac{2}{3}v^3 \right) \Big|_0^{1/\sqrt{2}} = \frac{8}{3\pi},
\end{aligned}$$

where in the fourth line we make the substitution $u = \pi x/2L$, and in the sixth line we make the substitution $v = \sin(u)$. This shows that

$$|\langle \psi'_1 | \psi_1 \rangle|^2 = \left(\frac{8}{3\pi} \right)^2,$$

as we wanted. □

4-10 (Shankar, Exercise 5.2.2)

- (a) Show that for any normalized $|\psi\rangle$, $\langle \psi | H | \psi \rangle \geq E_0$, where E_0 is the lowest-energy eigenvalue.
(b) Prove the following theorem: Every attractive potential in one dimension has at least one bound state.

Solution. (a) Since

$$|\psi\rangle = \sum_n |n\rangle \langle n | \psi \rangle,$$

we have

$$\begin{aligned}
\langle \psi | H | \psi \rangle &= \langle \psi | \left(\sum_n E_n |n\rangle \langle n| \right) | \psi \rangle \\
&= \sum_n E_n \langle \psi | n \rangle \langle n | \psi \rangle, \\
&\geq E_0 \sum_n \langle \psi | n \rangle \langle n | \psi \rangle = E_0 \langle \psi | \psi \rangle = E_0,
\end{aligned}$$

where in the last line we use the fact that $E_n \geq E_0$ for all n .

(b) We will exhibit a normalized $|\psi\rangle$ with a negative expectation value for H , which implies by part (a) that there exists some energy eigenstate with a negative eigenvalue, i.e., a bound state. Take as trial wave function

$$\psi_\alpha(x) = \left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\alpha x^2/2},$$

which is normalized. Then note that an attractive potential satisfies $V(x) \leq 0$ everywhere, so $V(x) = -|V(x)|$, and then

$$H = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} - |V(x)|.$$

Since

$$\frac{d^2}{dx^2} e^{-\alpha x^2/2} = \alpha(-1 + \alpha x^2) e^{-\alpha x^2/2},$$

the expectation value of H for the trial wave function is given by

$$\begin{aligned} \langle \psi_\alpha | H | \psi_\alpha \rangle &= \left(\frac{\alpha}{\pi}\right)^{1/2} \int_{-\infty}^{\infty} dx e^{-\alpha x^2/2} \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} - |V(x)| \right) e^{-\alpha x^2/2} \\ &= -\frac{\hbar^2}{2m} \left(\frac{\alpha}{\pi}\right)^{1/2} \int_{-\infty}^{\infty} dx \alpha(-1 + \alpha x^2) e^{-\alpha x^2} - \left(\frac{\alpha}{\pi}\right)^{1/2} \int_{-\infty}^{\infty} dx |V(x)| e^{-\alpha x^2} \\ &= \frac{\hbar^2 \alpha}{4m} - \left(\frac{\alpha}{\pi}\right)^{1/2} \int_{-\infty}^{\infty} dx |V(x)| e^{-\alpha x^2}, \end{aligned}$$

where we used the results of Appendix A.2 in the last line. We want to show that this expectation value can be made negative with an appropriately chosen α . Sometimes the integral can be evaluated explicitly, as in the case of a Dirac delta well, and then the correct choice of α is manifest. For potentials that are finite everywhere, we consider the following argument: take

$$\epsilon = \min(|V(x)|).$$

Then we have

$$\left(\frac{\alpha}{\pi}\right)^{1/2} \int_{-\infty}^{\infty} dx |V(x)| e^{-\alpha x^2} \geq \epsilon \left[\left(\frac{\alpha}{\pi}\right)^{1/2} \int_{-\infty}^{\infty} dx e^{-\alpha x^2} \right] = \epsilon.$$

In particular, this implies that

$$\langle \psi_\alpha | H | \psi_\alpha \rangle \leq \frac{\hbar^2 \alpha}{4m} - \epsilon,$$

so that if we choose α such that

$$\alpha < \frac{4m\epsilon}{\hbar^2},$$

then the expectation value of H for the trial wave function $\psi_\alpha(x)$ must be negative. Together with the result of part (a), we have

$$0 > \langle \psi_\alpha | H | \psi_\alpha \rangle \geq E_0.$$

This implies that $E_0 < 0$, as we wanted to show. □

4-11 (Shankar, Exercise 5.2.3)

Consider $V(x) = -aV_0\delta(x)$. Show that it admits a bound state of energy $E = -ma^2V_0^2/2\hbar^2$. Are there any other bound states?

Solution. We are looking for bound states, so we assume that $E < 0$. In the region $x > 0$, the wave function is given by $\psi_I(x) = Ae^{-\kappa x}$, where $\kappa = (-2mE)^{1/2}/\hbar$, and in the region $x < 0$, we have $\psi_{II}(x) = Be^{\kappa x}$. Continuity of the wave function imposes the condition $\psi_I(0) = \psi_{II}(0)$, which fixes $A = B$. Thus, we have

$$\psi(x) = Ae^{-\kappa|x|}.$$

The value of A is fixed by normalization, whereas κ will be determined from a constraint on $d\psi/dx$ imposed by the Schrödinger equation. Normalizing $\psi(x)$ is straightforward:

$$\langle\psi|\psi\rangle = |A|^2 \int_{-\infty}^{\infty} dx e^{-2\kappa|x|} = 2|A|^2 \int_0^{\infty} dx e^{-2\kappa x} = \frac{|A|^2}{\kappa},$$

so that we may choose $A = \kappa^{1/2}$. Now, we note that $d\psi/dx$ is discontinuous; the wave function has a cusp at the origin. Thus the derivative is undefined at the origin, but we can compute

$$\frac{d\psi}{dx} = \begin{cases} -\kappa^{3/2}e^{-\kappa x} & \text{for } x > 0 \\ \kappa^{3/2}e^{\kappa x} & \text{for } x < 0 \end{cases}$$

Now, integrating the Schrödinger equation over a small interval centered at the origin, we get

$$\int_{-\epsilon}^{\epsilon} dx \frac{d^2\psi}{dx^2} = \int_{-\epsilon}^{\epsilon} dx \frac{2m}{\hbar^2} (V(x) - E)\psi(x) = \int_{-\epsilon}^{\epsilon} dx \frac{2m}{\hbar^2} (-aV_0\delta(x) - E)\psi(x).$$

Applying the Fundamental Theorem of Calculus, then taking the limit $\epsilon \rightarrow 0$, we have

$$\left. \frac{d\psi}{dx} \right|_+ - \left. \frac{d\psi}{dx} \right|_- = -\frac{2maV_0}{\hbar^2} \psi(0) = -\frac{2maV_0}{\hbar^2} \kappa^{1/2}.$$

But from our expression for $d\psi/dx$, the difference on the left is $-2\kappa^{3/2}$, so that

$$2\kappa^{3/2} = \frac{2maV_0}{\hbar^2} \kappa^{1/2} \implies \kappa = \frac{maV_0}{\hbar^2}.$$

Then from the definition $\kappa = (-2mE)^{1/2}/\hbar$, the corresponding energy is

$$E = -\frac{\kappa^2 \hbar^2}{2m} = -\frac{ma^2 V_0^2}{2\hbar^2}.$$

There are no other solutions for E , so that the only bound state for this potential is

$$\boxed{\psi(x) = \frac{\sqrt{maV_0}}{\hbar} e^{-maV_0|x|/\hbar^2}, \text{ where } E = -\frac{ma^2 V_0^2}{2\hbar^2}.}$$

□

4-12 (Shankar, Exercise 5.2.4)

Consider a particle of mass m in the state $|n\rangle$ of a box of length L . Find the force $F = -\partial E/\partial L$ encountered when the walls are slowly pushed in, assuming the particle remains in the n th state of the box as its size changes. Consider a classical particle of energy E_n in this box. Find its velocity, the frequency of collision on a given wall, the momentum transfer per collision, and hence the average force. Compare it to $-\partial E/\partial L$ computed above.

Solution. We have

$$F = -\frac{\partial E}{\partial L} = -\frac{\partial}{\partial L} \left(\frac{\hbar^2 \pi^2 n^2}{2mL^2} \right) = \frac{\hbar^2 \pi^2 n^2}{mL^3}.$$

Now consider a classical particle of energy E_n , so that

$$p = (2mE)^{1/2} = \left(2m \frac{\hbar^2 \pi^2 n^2}{2mL^2} \right)^{1/2} = \frac{\hbar \pi n}{L},$$

with corresponding velocity $v = \hbar \pi n / mL$. This is really small; a particle with mass $m = 0.1$ kg moving in a box of length $L = 1$ m in the ground state $n = 1$ would have a velocity on the order of 10^{-32} meters per second. The same particle with velocity on the order of 1 meter per second would have an energy corresponding to the $n \sim 10^{32}$ bound state of the well. The particle travels a distance $2L$ before returning to the same wall, which takes a time of

$$\frac{2L}{v} = \frac{2L}{\hbar \pi n / mL} = \frac{2mL^2}{\hbar \pi n}.$$

The momentum transfer per collision is $2\hbar \pi n / L$, so the average force is

$$\frac{\Delta p}{\Delta t} = \frac{2\hbar \pi n / L}{2mL^2 / \hbar \pi n} = \frac{\hbar^2 \pi^2 n^2}{mL^3}.$$

This is the same as the force $-\partial E/\partial L$ computed above. □

4-13 (Shankar, Exercise 5.2.5)

If the box extends from $x = 0$ to L (instead of $\pm L/2$), show that $\psi_n(x) = (2/L)^{1/2} \sin(n\pi x/L)$, $n = 1, 2, \dots, \infty$ and $E_n = \hbar^2 \pi^2 n^2 / 2mL^2$.

Solution. The work is nearly identical to what's done in the book; the main difference is that we get $A = -B$ for all n , so that only terms like $\sin(n\pi x/L)$ appear when we combine the exponentials. The energy levels are the same, as expected. □

Alternatively, if $\psi(x)$ is a solution to the time-independent Schrödinger equation with the potential $V(x)$, then $\psi(x - a)$ is a solution with the potential $V(x - a)$.

4-14 (Shankar, Exercise 5.2.6)

Square Well Potential. Consider a particle in a square well potential:

$$V(x) = \begin{cases} 0, & |x| \leq a \\ V_0, & |x| \geq a \end{cases}$$

Since when $V_0 \rightarrow \infty$, we have a box, let us guess what the lowering of the walls does to the states. First of all, all the bound states (which alone we are interested in), will have $E \leq V_0$. Second, the wave functions of the low-lying levels will look like those of the particle in a box, with the obvious difference that ψ will not vanish at the walls but instead will spill out with an exponential tail. The eigenfunctions will still be even, odd, even, etc.

- (1) Show that the even solutions have energies that satisfy the transcendental equation

$$k \tan ka = \kappa \quad (5.2.21)$$

while the odd ones will have energies that satisfy

$$k \cot ka = -\kappa \quad (5.2.24)$$

where k and $i\kappa$ are the real and complex wave numbers inside and outside the well, respectively. Note that k and κ are related by

$$k^2 + \kappa^2 = 2mV_0/\hbar^2 \quad (5.2.25)$$

Verify that as V_0 tends to ∞ , we regain the levels in the box.

- (2) Equations (5.2.23) and (5.2.24) must be solved graphically. In the $(\alpha = ka, \beta = \kappa a)$ plane, imagine a circle that obeys Eq. (5.2.25). The bound states are then given by the intersection of the curve $\alpha \tan \alpha = \beta$ or $\alpha \cot \alpha = -\beta$ with the circle. (Remember α and β are positive.)
- (3) Show that there is always one even solution and that there is no odd solution unless $V_0 \geq \hbar^2\pi^2/8ma^2$. What is E when V_0 just meets this requirement? Note that the general result from Exercise 5.2.2b holds.

Solution. Since the potential $V(x)$ is even, there is no loss of generality in assuming that the wave functions $\psi(x)$ are either odd or even. To see this, note that if $\psi(x)$ is a solution to the time-independent Schrödinger equation, so that

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V(x))\psi(x) = 0,$$

then $\psi(-x)$ is a solution with the same eigenvalue, because $V(x) = V(-x)$. But there are no degeneracies in one-dimensional bound states (Theorem 15), which implies that these solutions are linearly dependent, $\psi(x) = a\psi(-x)$.

Now multiply both sides by $\psi(x)^*$ and integrate to get

$$\int_{-\infty}^{\infty} \psi(x)^* \psi(x) dx = a \int_{-\infty}^{\infty} \psi(x)^* \psi(-x) dx.$$

But the integral on the right is equal to its complex conjugate (make the substitution $z = -x$) and is therefore real. The integral on the left is also real, hence a is real. Finally, normalization fixes $|a|^2 = 1$, so $a = \pm 1$. Taken together, we have

$$\psi(x) = \pm \psi(-x),$$

so that $\psi(x)$ must be either odd or even, as we wanted to show.

(1) We first consider the even solutions. Dividing space into regions I–III as we did for the box potential, we have

$$\psi_I(x) = Ae^{\kappa x},$$

$$\psi_{II}(x) = B \cos(kx),$$

$$\psi_{III}(x) = Ce^{-\kappa x},$$

where $k = \sqrt{2mE}/\hbar$ and $\kappa = \sqrt{2m(V_0 - E)}/\hbar$. Since the wave function is continuous, we have

$$Ae^{-\kappa a} = B \cos(-ka), \tag{1}$$

$$B \cos(ka) = Ce^{-\kappa a}. \tag{2}$$

Since the potential is finite everywhere, $d\psi/dx$ must also be continuous:

$$\kappa Ae^{-\kappa a} = -kB \sin(-ka), \tag{3}$$

$$-kB \sin(ka) = -\kappa Ce^{-\kappa a}. \tag{4}$$

The wave numbers k and $i\kappa$ are determined by solving

$$k^2 + \kappa^2 = \frac{2mV_0}{\hbar^2}, \quad \kappa = k \tan(ka). \tag{5}$$

The first equation follows from the definitions, and the second is obtained by dividing Eq. (3) by Eq. (1). Equations (1) and (2) give $A = C$, and then Eq. (2) implies $B = Ae^{-\kappa a} / \cos(ka)$. Putting everything together, we get

$$\psi_I(x) = Ae^{\kappa x},$$

$$\psi_{II}(x) = \frac{Ae^{-\kappa a}}{\cos(ka)} \cos(kx),$$

$$\psi_{III}(x) = Ae^{-\kappa x},$$

where k and κ are determined by Eq. (5), and A is fixed by normalization. The considerations

are similar for the odd wave functions, and we get

$$Ae^{-\kappa a} = B \sin(-ka), \quad (6)$$

$$B \sin(ka) = Ce^{-\kappa a}, \quad (7)$$

$$\kappa Ae^{-\kappa a} = kB \cos(-ka), \quad (8)$$

$$kB \cos(ka) = -\kappa Ce^{-\kappa a}. \quad (9)$$

Now the wave numbers k and $i\kappa$ are determined by solving

$$k^2 + \kappa^2 = \frac{2mV_0}{\hbar^2}, \quad \kappa = -k \cot(ka). \quad (10)$$

Equations (1) and (2) imply $A = -C$, and then from Eq. (3) we get $B = -Ae^{-\kappa a} / \sin(ka)$. Thus

$$\psi_I(x) = Ae^{\kappa x},$$

$$\psi_{II}(x) = -\frac{Ae^{-\kappa a}}{\sin(ka)} \sin(kx),$$

$$\psi_{III}(x) = -Ae^{-\kappa x}.$$

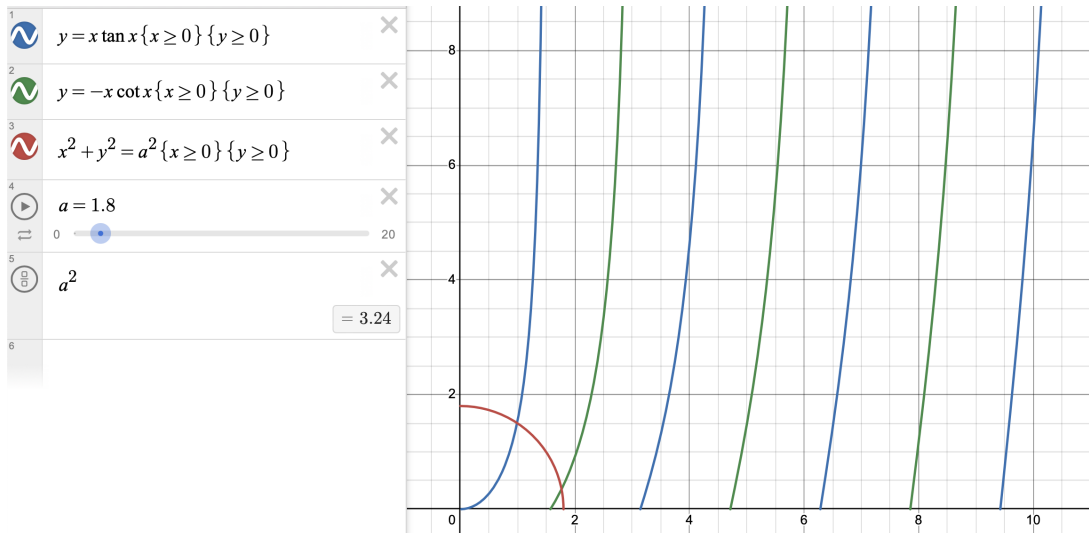
(2) In $\alpha = ka$, $\beta = \kappa a$ coordinates, our equations for k and $i\kappa$ become

$$\alpha^2 + \beta^2 = \frac{2ma^2V_0}{\hbar^2},$$

and

$$\beta = \alpha \tan(\alpha), \quad \beta = -\alpha \cot(\alpha)$$

for the even and odd solutions, respectively. The dimensionless number $2ma^2V_0/\hbar^2$ sets the radius of a circle in the (α, β) plane; a wider, deeper well corresponds to a larger circle, and a narrower, shallower well corresponds to a smaller circle. We plot a graph below, restricting to the physical region $\alpha, \beta \geq 0$:



(3) There is always one even solution because the graph of $\beta = \alpha \tan(\alpha)$ starts at the origin, so there will always be at least one intersection with the circle. For the odd solutions, we note that the green curve starts where $-\alpha \cot(\alpha) = 0$, or when $\alpha = \pi/2$. Thus, the circle's radius must be greater than or equal to $\pi/2$, which implies that the well height must satisfy

$$\frac{2ma^2V_0}{\hbar^2} \geq \frac{\pi^2}{4} \implies V_0 \geq \frac{\hbar^2\pi^2}{8ma^2}.$$

When V_0 just meets this requirement, the corresponding solution is $k = \pi/2a$ and $\kappa = 0$, or

$$\frac{2mE}{\hbar^2} = \frac{\pi^2}{4a^2} \implies E = \frac{\hbar^2\pi^2}{8ma^2}.$$

This makes sense, since $\kappa = 0$ implies (from the definition) that $E = V_0$. We can also see from the graph that as $V_0 \rightarrow \infty$, we recover the energy levels of the box, because the intersections occur at around

$$\alpha = \frac{n\pi}{2}, \quad n = 1, 2, 3, \dots$$

or when

$$\frac{2mE}{\hbar^2} = \frac{n^2\pi^2}{4a^2} \implies E = \frac{\hbar^2\pi^2n^2}{8ma^2}.$$

But for any finite V_0 , there are only a finite number of intersections. To get an estimate of the number of bound states, we have

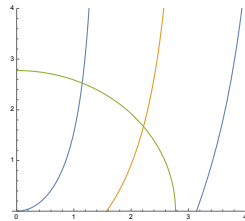
$$\frac{\pi^2n^2}{4} \approx \frac{2ma^2V_0}{\hbar^2} \implies n \approx \sqrt{\frac{8ma^2V_0}{\hbar^2\pi^2}}.$$

□

To apply this to a concrete problem, consider a particle with mass 0.06 times that of the electron in a well of height 240 meV and width 90 Å. We expect to see around 2 bound states, using the estimate given above:

$$\begin{aligned} n &\approx \sqrt{\frac{8(0.06)(9.1 \times 10^{-31} \text{ kg})(45 \times 10^{-10} \text{ m})^2(240 \times 10^{-3} \text{ eV})(1.6 \times 10^{-19} \text{ J})}{(1.05 \times 10^{-34} \text{ kgm}^2/\text{s})^2\pi^2}} \\ &= 1.77. \end{aligned}$$

This estimate can be confirmed by plotting the relevant curves:



There is one even bound state and one odd one. Solving numerically, we find an energy of 41 meV for the ground state, and an energy of 154 meV for the first excited state.

4-15 (Shankar, Exercise 5.3.1)

Consider the case where $V = V_r - iV_i$, where the imaginary part V_i is a constant. Is the Hamiltonian Hermitian? Go through the derivation of the continuity equation and show that the total probability for finding the particle decreases exponentially as $e^{-2V_it/\hbar}$. Such complex potentials are used to describe processes in which particles are absorbed by a sink.

Solution. The Hamiltonian is not Hermitian for nonzero V_i . We mimic the derivation of the continuity equation, writing the time-dependent Schrödinger equation

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi + (V_r - iV_i)\psi$$

and its complex-conjugated version,

$$-i\hbar \frac{\partial \psi^*}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi^* + (V_r + iV_i)\psi^*.$$

Multiplying the first equation by ψ^* and the second by ψ , then taking their difference, we obtain

$$\begin{aligned} i\hbar \frac{\partial(\psi^* \psi)}{\partial t} &= -\frac{\hbar^2}{2m} (\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^*) + V_r \psi^* \psi - iV_i \psi^* \psi - V_r \psi^* \psi - iV_i \psi^* \psi \\ &= -\frac{\hbar^2}{2m} \nabla \cdot (\psi^* \nabla \psi - \psi \nabla \psi^*) - 2iV_i \psi^* \psi. \end{aligned}$$

Rearranging factors and replacing $\psi^* \psi$ with P , we get

$$\frac{\partial P}{\partial t} = -\nabla \cdot \mathbf{j} - \frac{2V_i}{\hbar} P,$$

where we've defined the probability current density as before,

$$\mathbf{j} = \frac{\hbar}{2mi} (\psi^* \nabla \psi - \psi \nabla \psi^*).$$

Integrating both sides of this equation over all space, the integral of $\nabla \cdot \mathbf{j}$ vanishes for the reasons argued in the text, and we get

$$\frac{d}{dt} \int P(\mathbf{r}, t) d^3\mathbf{r} = -\frac{2V_i}{\hbar} \int P(\mathbf{r}, t) d^3\mathbf{r}.$$

Since the integral of $P(\mathbf{r}, t)$ over all space gives the total probability, it follows that the total probability is proportional to $e^{-2V_it/\hbar}$, as we wanted to show. In particular, total probability is not conserved. **On the other hand, we are guaranteed conservation of total probability if H is Hermitian, which is one of the many reasons this property is important.** $\square$

4-16 (Shankar, Exercise 5.3.2)

Convince yourself that if $\psi = c\tilde{\psi}$, where c is constant (real or complex) and $\tilde{\psi}$ is real, the corresponding $\mathbf{j}$ vanishes.

Solution. In this case we get

$$\psi^* \nabla \psi - \psi \nabla \psi^* = (c^* \tilde{\psi}) \nabla (c \tilde{\psi}) - (c \tilde{\psi}) \nabla (c^* \tilde{\psi}) = |c|^2 (\tilde{\psi} \nabla \tilde{\psi} - \tilde{\psi} \nabla \tilde{\psi}) = 0,$$

so $\mathbf{j}$ must also vanish. However, as pointed out in this [Stack Exchange post](#), the wavefunction evolves according to $i\hbar \partial_t \psi = H\psi$, so that a real wavefunction will in general not stay real, and in particular we do not expect $\mathbf{j} = 0$ to hold for all t . A notable exception is when $|\psi\rangle$ represents an eigenstate of the Hamiltonian, so that $|\psi(t)\rangle = e^{-iE_n t/\hbar} |\psi(0)\rangle$; in particular, if the initial wavefunction is real (or can be made real by multiplying an overall factor), then it remains real (up to an overall factor) for all t . This is another reason why we consider eigenstates of the Hamiltonian to be *stationary states*; the probability current is zero everywhere for all t . $\square$

4-17 (Shankar, Exercise 5.3.3)

Consider

$$\psi_{\mathbf{p}} = \left(\frac{1}{2\pi\hbar} \right)^{3/2} e^{i(\mathbf{p} \cdot \mathbf{r})/\hbar}$$

Find $\mathbf{j}$ and P and compare the relation between them to the electromagnetic equation $\mathbf{j} = \rho \mathbf{v}$, $\mathbf{v}$ being the velocity. Since ρ and $\mathbf{j}$ are constant, note that the continuity Eq. (5.3.7) is trivially satisfied.

Solution. For this wave function we have the probability current

$$\mathbf{j} = \frac{\hbar}{2mi} (\psi_{\mathbf{p}}^* \nabla \psi_{\mathbf{p}} - \psi_{\mathbf{p}} \nabla \psi_{\mathbf{p}}^*),$$

and since

$$\begin{aligned} \nabla \psi_{\mathbf{p}} &= \left(\frac{1}{2\pi\hbar} \right)^{3/2} \nabla e^{i(\mathbf{p} \cdot \mathbf{r})/\hbar} \\ &= \left(\frac{1}{2\pi\hbar} \right)^{3/2} \frac{i\mathbf{p}}{\hbar} e^{i(\mathbf{p} \cdot \mathbf{r})/\hbar} \\ &= \frac{i\mathbf{p}}{\hbar} \psi_{\mathbf{p}} \end{aligned}$$

and by similar reasoning $\nabla \psi_{\mathbf{p}}^* = -(i\mathbf{p}/\hbar) \psi_{\mathbf{p}}^*$, we obtain

$$\mathbf{j} = \frac{\hbar}{2mi} \frac{i}{\hbar} (\psi_{\mathbf{p}}^* \psi_{\mathbf{p}} \mathbf{p} + \psi_{\mathbf{p}} \psi_{\mathbf{p}}^* \mathbf{p}) = \frac{1}{m} \psi_{\mathbf{p}}^* \psi_{\mathbf{p}} \mathbf{p},$$

which shows that $\boxed{\mathbf{j} = P(\mathbf{p}/m)}$. To determine P , we take the dot product with $\mathbf{p}$ on both sides

and rearrange terms to obtain $\boxed{P = m(\mathbf{j} \cdot \mathbf{p})/p^2}$. If we make an analogy with electromagnetism, then the quantity P plays the role of the charge density ρ . $\square$

4-18 (Shankar, Exercise 5.3.4)

Consider $\psi = Ae^{ipx/\hbar} + Be^{-ipx/\hbar}$ in one dimension. Show that $j = (|A|^2 - |B|^2)p/m$. The absence of cross terms between the right- and left-moving pieces in ψ allows us to associate the two parts of j with corresponding parts of ψ .

Solution. In one dimension, $\nabla \rightarrow d/dx$. Then to compute the probability current density, we note that

$$\frac{d\psi}{dx} = \frac{ip}{\hbar}Ae^{ipx/\hbar} - \frac{ip}{\hbar}Be^{-ipx/\hbar},$$

which implies that

$$\begin{aligned}\psi^* \frac{d\psi}{dx} &= \frac{ip}{\hbar}(A^*e^{-ipx/\hbar} + B^*e^{ipx/\hbar})(Ae^{ipx/\hbar} - Be^{-ipx/\hbar}) \\ &= \frac{ip}{\hbar}(|A|^2 - A^*Be^{-2ipx/\hbar} + AB^*e^{2ipx/\hbar} - |B|^2)\end{aligned}$$

and (taking the complex conjugate of this result, then multiplying by -1)

$$-\psi \frac{d\psi^*}{dx} = \frac{ip}{\hbar}(|A|^2 - AB^*e^{2ipx/\hbar} + A^*Be^{-2ipx/\hbar} - |B|^2).$$

It follows that

$$\psi^* \frac{d\psi}{dx} - \psi \frac{d\psi^*}{dx} = \frac{2ip}{\hbar}(|A|^2 - |B|^2),$$

which gives a probability current density of

$$j = \frac{\hbar}{2mi} \frac{2ip}{\hbar}(|A|^2 - |B|^2) = (|A|^2 - |B|^2)p/m,$$

as we wanted to show. Notice that these two pieces are precisely what we'd get for the individual probability current densities, by the result of Exercise 5.3.3. $\square$

4-19 (Shankar, Exercise 5.4.3)

Consider a particle subject to a constant force f in one dimension. Solve for the propagator in momentum space and get

$$U(p, t; p', 0) = \delta(p - p' - ft) e^{i(p'^3 - p^3)/6m\hbar f}$$

Transform back to coordinate space and obtain

$$U(x, t; x', 0) = \left(\frac{m}{2\pi\hbar it}\right)^{1/2} \exp\left\{\frac{i}{\hbar}\left[\frac{m(x-x')^2}{2t} + \frac{1}{2}ft(x+x') - \frac{f^2 t^3}{24m}\right]\right\} \quad (5.4.31)$$

[Hint: Normalize $\psi_E(p)$ such that $\langle E|E'\rangle = \delta(E - E')$. Note that E is not restricted to be positive.]

Solution. We sketch the solution. First we obtain the momentum space wave functions for the energy eigenkets $|E\rangle$, and find

$$\langle p|E\rangle = \frac{1}{\sqrt{2\pi\hbar f}} \exp\left[-\frac{ip^3}{6m\hbar f} + \frac{iEp}{\hbar f}\right],$$

where the normalization factor is fixed by demanding that $\langle E|E'\rangle = \delta(E - E')$. Then

$$\begin{aligned} U(p, t; p') &= \langle p|U(t)|p'\rangle \\ &= \int_{-\infty}^{\infty} dE \int_{-\infty}^{\infty} dE' \langle p|E\rangle \underbrace{e^{-iE't/\hbar} \delta(E - E')}_{\langle E|U(t)|E\rangle} \langle E'|p'\rangle \\ &= \delta(p - p' - ft) e^{i(p'^3 - p^3)/6m\hbar f}. \end{aligned}$$

The integration over E is trivial due to the factor $\delta(E - E')$, and the integration over E' involves the same integral for the Dirac delta used in normalizing $\langle p|E\rangle$. From here we compute

$$\begin{aligned} U(x, t; x', 0) &= \langle x|U(t)|x'\rangle \\ &= \int_{-\infty}^{\infty} dp \int_{-\infty}^{\infty} dp' \langle x|p\rangle \langle p|U(t)|p'\rangle \langle p'|x'\rangle \\ &= \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} dp \int_{-\infty}^{\infty} dp' \exp\left(\frac{ipx}{\hbar}\right) \delta(p - p' - ft) \exp\left[\frac{i(p'^3 - p^3)}{6m\hbar f}\right] \exp\left(-\frac{ip'x'}{\hbar}\right). \end{aligned}$$

The algebra is a bit more elaborate, but the idea is straightforward. After integrating over p , which is made trivial by the factor $\delta(p - p' - ft)$, the p'^3 terms cancel out and we have a standard Gaussian integral in p' [see, e.g., Eq. (A.2.5)]. A bit more rearranging then yields

$$U(x, t; x', 0) = \left(\frac{m}{2\pi\hbar it}\right)^{1/2} \exp\left\{\frac{i}{\hbar}\left[\frac{m(x-x')^2}{2t} + \frac{1}{2}ft(x+x') - \frac{f^2 t^3}{24m}\right]\right\},$$

as we wanted. □

4-20 (Sakurai, Problem 2.5)

Consider a particle in one dimension whose Hamiltonian is given by

$$H = \frac{p^2}{2m} + V(x).$$

By calculating $[[H, x], x]$ prove

$$\sum_{a'} |\langle a'' | x | a' \rangle|^2 (E_{a'} - E_{a''}) = \frac{\hbar^2}{2m},$$

where $|a'\rangle$ is an energy eigenket with eigenvalue $E_{a'}$.

Solution. We have

$$[H, x] = \frac{1}{2m} [p^2, x] = -\frac{1}{2m} [x, p^2] = -\frac{1}{2m} (2i\hbar p) = \frac{\hbar}{mi} p,$$

and then

$$[[H, x], x] = \frac{\hbar}{mi} [p, x] = \frac{\hbar}{mi} (-i\hbar) = -\frac{\hbar^2}{m}.$$

Now observe that

$$\begin{aligned} \langle a' | [H, x] | a'' \rangle &= \langle a' | Hx - xH | a'' \rangle \\ &= E_{a'} \langle a' | x | a'' \rangle - E_{a''} \langle a' | x | a'' \rangle \\ &= (E_{a'} - E_{a''}) \langle a' | x | a'' \rangle. \end{aligned}$$

Multiplying both sides by $\langle a'' | x | a' \rangle$ and summing over a' , we get

$$\langle a'' | x [H, x] | a'' \rangle = \sum_{a'} |\langle a' | x | a'' \rangle|^2 (E_{a'} - E_{a''}).$$

Now since $[H, x]$ is anti-Hermitian (see Exercise 1.6.2 of Shankar), taking the Hermitian conjugate of both sides and then adding gives

$$\langle a'' | [x, [H, x]] | a'' \rangle = 2 \sum_{a'} |\langle a' | x | a'' \rangle|^2 (E_{a'} - E_{a''}).$$

Then using the commutator we just evaluated and the fact that $\langle a'' | a'' \rangle = 1$, we have

$$\frac{\hbar^2}{2m} = \sum_{a'} |\langle a' | x | a'' \rangle|^2 (E_{a'} - E_{a''}),$$

as we wanted. □

4-21 (Sakurai, Problem 2.6)

Consider a particle in three dimensions whose Hamiltonian is given by

$$H = \frac{\mathbf{p}^2}{2m} + V(\mathbf{x}).$$

By calculating $[\mathbf{x} \cdot \mathbf{p}, H]$ obtain

$$\frac{d}{dt}\langle \mathbf{x} \cdot \mathbf{p} \rangle = \left\langle \frac{\mathbf{p}^2}{m} \right\rangle - \langle \mathbf{x} \cdot \nabla V \rangle.$$

To identify the preceding relation with the quantum-mechanical analogue of the virial theorem it is essential that the left-hand side vanish. Under what condition would this happen?

Solution. Summing over repeated indices, we have

$$\begin{aligned} [\mathbf{x} \cdot \mathbf{p}, H] &= [x_i p_i, H] = -[H, x_i p_i] \\ &= -([H, x_i] p_i + x_i [H, p_i]) \\ &= [x_i, H] p_i + x_i [p_i, H] \\ &= \left[x_i, \frac{\mathbf{p}^2}{2m} \right] p_i + x_i [p_i, V(\mathbf{x})] \\ &= i\hbar \frac{p_i p_i}{m} - i\hbar x_i \frac{\partial V}{\partial x_i} \\ &= i\hbar \left(\frac{\mathbf{p}^2}{m} - \mathbf{x} \cdot \nabla V \right) \end{aligned}$$

and then in the Heisenberg picture,

$$\frac{d}{dt}(\mathbf{x} \cdot \mathbf{p}) = \frac{\mathbf{p}^2}{m} - \mathbf{x} \cdot \nabla V.$$

The desired identity follows because the state ket is constant in the Heisenberg picture, so we can sandwich this between a $\langle \psi |$ and a $|\psi \rangle$, then bring the time derivative outside to act on the entire expression. Then

$$\frac{d}{dt}\langle \mathbf{x} \cdot \mathbf{p} \rangle = \left\langle \frac{\mathbf{p}^2}{m} \right\rangle - \langle \mathbf{x} \cdot \nabla V \rangle.$$

The final answer is independent of the choice of picture, since expectation values in the two pictures are the same. The left-hand side vanishes when $\langle \mathbf{x} \cdot \mathbf{p} \rangle$ is a constant; in classical physics, this happens for circular motion, where the velocity is always tangent to the position vector. $\square$

4-22 (Cohen-Tannoudji, Problem LIII-4)

Spreading of a free wave packet.

Consider a free particle.

- Show, applying Ehrenfest's theorem, that $\langle X \rangle$ is a linear function of time, the mean value $\langle P \rangle$ remaining constant.
- Write the equations of motion for the mean values $\langle X^2 \rangle$ and $\langle XP + PX \rangle$. Integrate these equations.
- Show that, with a suitable choice of the time origin, the root-mean-square deviation ΔX is given by:

$$(\Delta X)^2 = \frac{1}{m^2}(\Delta P)_0^2 t^2 + (\Delta X)_0^2$$

where $(\Delta X)_0$ and $(\Delta P)_0$ are the root-mean-square deviations at the initial time. How does the width of the wave packet vary as a function of time? Give a physical interpretation.

Solution. (a) By Ehrenfest's theorem,

$$\begin{aligned} \frac{d\langle P \rangle}{dt} &= \left\langle \frac{1}{i\hbar} [P, H] \right\rangle \\ &= \left\langle \frac{1}{i\hbar} \left[P, \frac{P^2}{2m} \right] \right\rangle \\ &= 0, \end{aligned}$$

so that $\langle P \rangle = \langle P \rangle_0$, the expectation value of P at $t = t_0$. Then

$$\begin{aligned} \frac{d\langle X \rangle}{dt} &= \left\langle \frac{1}{i\hbar} [X, H] \right\rangle \\ &= \left\langle \frac{1}{i\hbar} \left[X, \frac{P^2}{2m} \right] \right\rangle \\ &= \left\langle \frac{1}{2mi\hbar} (2i\hbar P) \right\rangle \\ &= \frac{\langle P \rangle_0}{m} \end{aligned}$$

which implies that $\langle X \rangle$ is a linear function of time,

$$\langle X \rangle = \langle X \rangle_0 + \frac{\langle P \rangle_0 (t - t_0)}{m}.$$

(b) Again by Ehrenfest's theorem,

$$\begin{aligned}
\frac{d\langle X^2 \rangle}{dt} &= \left\langle \frac{1}{i\hbar} [X^2, H] \right\rangle \\
&= \left\langle \frac{1}{i\hbar} \left[X^2, \frac{P^2}{2m} \right] \right\rangle \\
&= \left\langle \frac{1}{2mi\hbar} \left([X^2, P]P + P[X^2, P] \right) \right\rangle \\
&= \left\langle \frac{1}{2mi\hbar} \left((2i\hbar X)P + P(2i\hbar X) \right) \right\rangle \\
&= \frac{\langle XP + PX \rangle}{m},
\end{aligned}$$

where we've used the fact that $[A, BC] = [A, B]C + B[A, C]$. Next, we have

$$\frac{d\langle XP + PX \rangle}{dt} = \left\langle \frac{1}{i\hbar} [XP + PX, H] \right\rangle.$$

We evaluate the two commutators inside the angle brackets:

$$\begin{aligned}
\frac{1}{i\hbar} [XP, H] &= \frac{1}{i\hbar} \left[XP, \frac{P^2}{2m} \right] \\
&= -\frac{1}{2mi\hbar} [P^2, XP] \\
&= -\frac{1}{2mi\hbar} \left([P^2, X]P + X[P^2, P] \right) \\
&= -\frac{1}{2mi\hbar} \left((-2i\hbar P)P \right) \\
&= \frac{P^2}{m}
\end{aligned}$$

$$\begin{aligned}
\frac{1}{i\hbar} [PX, H] &= -\frac{1}{2mi\hbar} [P^2, PX] \\
&= -\frac{1}{2mi\hbar} \left([P^2, P]X + P[P^2, X] \right) \\
&= -\frac{1}{2mi\hbar} \left(P(-2i\hbar P) \right) \\
&= \frac{P^2}{m}
\end{aligned}$$

Thus

$$\frac{d\langle XP + PX \rangle}{dt} = \frac{2\langle P^2 \rangle}{m}.$$

Notice that $\langle P^2 \rangle$ is also conserved because $[P^2, H] = 0$, so that $\langle P^2 \rangle = \langle P^2 \rangle_0$. This allows us to

integrate our equations of motion. First we have

$$\langle XP + PX \rangle = \langle XP + PX \rangle_0 + \frac{2\langle P^2 \rangle_0 (t - t_0)}{m},$$

which implies

$$\langle X^2 \rangle = \langle X^2 \rangle_0 + \frac{1}{m} \left(\langle XP + PX \rangle_0 (t - t_0) + \frac{\langle P^2 \rangle_0 (t - t_0)^2}{m} \right).$$

(c) Using the definitions of ΔX and ΔP , the previous equation becomes

$$(\Delta X)^2 + \langle X \rangle^2 = (\Delta X)_0^2 + \langle X \rangle_0^2 + \frac{1}{m} \langle XP + PX \rangle_0 (t - t_0) + \frac{1}{m^2} \left((\Delta P)_0^2 + \langle P \rangle_0^2 \right) (t - t_0)^2.$$

By our result from part (a), we have

$$\langle X \rangle^2 = \langle X \rangle_0^2 + 2 \frac{\langle X \rangle_0 \langle P \rangle_0}{m} (t - t_0) + \frac{\langle P \rangle_0^2 (t - t_0)^2}{m^2},$$

which we then substitute to obtain

$$(\Delta X)^2 = (\Delta X)_0^2 + \frac{\langle XP + PX \rangle_0 - 2\langle X \rangle_0 \langle P \rangle_0}{m} (t - t_0) + \frac{(\Delta P)_0^2}{m^2} (t - t_0)^2.$$

Then, setting $t_0 = 0$, we get

$$(\Delta X)^2 = (\Delta X)_0^2 + \frac{\langle XP + PX \rangle_0 - 2\langle X \rangle_0 \langle P \rangle_0}{m} t + \frac{(\Delta P)_0^2}{m^2} t^2.$$

If you assume the initial wave packet takes the form $\psi(x, 0) = \phi(x)e^{ikx}$, with a real envelope function $\phi(x)$, then it can be shown that the coefficient of t is zero, so that the desired result is obtained. For example, the Gaussian wave packet is of this form. (See the discussion in Section 11.2 of Schumacher and Westmoreland.)

For our purposes, we note that the root-mean-square deviation ΔX can decrease for some interval of time, depending on the initial conditions, but that $(\Delta X)^2 \sim t^2$ if you wait long enough. [The spreading of free wave packets is not an intrinsically quantum effect; you can make a similar statement in classical mechanics by noting that any initial uncertainty in the velocity will result in a growing position uncertainty. But you can make a statement that holds only in QM if you use the Heisenberg uncertainty principle to write](#)

$$(\Delta X)^2 \geq (\Delta X)_0^2 + \frac{\hbar^2}{4m^2(\Delta X)_0^2} t^2.$$

[Thus, the smaller you make the initial position uncertainty, the faster this uncertainty grows with time. This is a quantum effect, since in classical mechanics, we treat \$X\$ and \$P\$ as uncorrelated variables, and a smaller position uncertainty has no effect on the momentum uncertainty.](#) $\square$

5 Time evolution; Heisenberg and Schrödinger pictures

5-1 (Sakurai, Problem 2.1)

Consider the spin-precession problem discussed in the text. It can also be solved in the Heisenberg picture. Using the Hamiltonian

$$H = -\left(\frac{eB}{mc}\right)S_z = \omega S_z,$$

write the Heisenberg equations of motion for the time-dependent operators $S_x(t)$, $S_y(t)$, and $S_z(t)$. Solve them to obtain $S_{x,y,z}$ as functions of time.

Solution. First we note that the commutators are preserved in the Heisenberg picture:

$$\begin{aligned} [S_i^{(H)}, S_j^{(H)}] &= (U^\dagger S_i U)(U^\dagger S_j U) - (U^\dagger S_j U)(U^\dagger S_i U) \\ &= U^\dagger S_i S_j U - U^\dagger S_j S_i U \\ &= U^\dagger [S_i, S_j] U \\ &= U^\dagger (i\hbar \epsilon_{ijk} S_k) U \\ &= i\hbar \epsilon_{ijk} S_k^{(H)}. \end{aligned}$$

Thus, the Heisenberg equation of motion for the spin operator $S_i^{(H)}$ is

$$\begin{aligned} \frac{dS_i^{(H)}}{dt} &= \frac{1}{i\hbar} [S_i^{(H)}, H] = \frac{1}{i\hbar} [S_i^{(H)}, \omega S_3^{(H)}] \\ &= \omega \epsilon_{i3k} S_k^{(H)}, \end{aligned}$$

which implies

$$\frac{dS_x^{(H)}}{dt} = -\omega S_y^{(H)}, \quad \frac{dS_y^{(H)}}{dt} = \omega S_x^{(H)}, \quad \frac{dS_z^{(H)}}{dt} = 0.$$

Since the spin operators in the Schrödinger and Heisenberg pictures must agree at $t = 0$, the third equation implies that $S_z^{(H)}(t) = S_z$ for all $t \geq 0$. Differentiating the first two equations with respect to time, we get

$$\frac{d^2 S_x^{(H)}}{dt^2} = -\omega^2 S_x^{(H)}, \quad \frac{d^2 S_y^{(H)}}{dt^2} = -\omega^2 S_y^{(H)}$$

which can be solved immediately to give

$$\begin{aligned} S_x^{(H)}(t) &= Ae^{i\omega t} + Be^{-i\omega t}, & \frac{dS_x^{(H)}}{dt} &= i\omega Ae^{i\omega t} - i\omega Be^{-i\omega t}, \\ S_y^{(H)}(t) &= Ce^{i\omega t} + De^{-i\omega t}, & \frac{dS_y^{(H)}}{dt} &= i\omega Ce^{i\omega t} - i\omega De^{-i\omega t}, \end{aligned}$$

Since $S_i^{(H)}(0) = S_i$, it follows that (using the Heisenberg equations of motion)

$$\begin{aligned} S_x &= A + B, & A - B &= iS_y, \\ S_y &= C + D, & C - D &= -iS_x. \end{aligned}$$

These equations can then be solved to give

$$\begin{aligned} A &= \frac{S_x + iS_y}{2}, & C &= \frac{S_y - iS_x}{2}, \\ B &= \frac{S_x - iS_y}{2}, & D &= \frac{S_y + iS_x}{2}. \end{aligned}$$

Then we get

$$\begin{aligned} S_x^{(H)}(t) &= \frac{S_x + iS_y}{2} e^{i\omega t} + \frac{S_x - iS_y}{2} e^{-i\omega t} \\ &= S_x \frac{e^{i\omega t} + e^{-i\omega t}}{2} + iS_y \frac{e^{i\omega t} - e^{-i\omega t}}{2} \\ &= S_x \cos(\omega t) - S_y \sin(\omega t), \\ S_y^{(H)}(t) &= \frac{S_y - iS_x}{2} e^{i\omega t} + \frac{S_y + iS_x}{2} e^{-i\omega t} \\ &= -iS_x \frac{e^{i\omega t} - e^{-i\omega t}}{2} + S_y \frac{e^{i\omega t} + e^{-i\omega t}}{2} \\ &= S_x \sin(\omega t) + S_y \cos(\omega t). \end{aligned}$$

Collecting all our results, we have

$$\begin{aligned} S_x^{(H)}(t) &= S_x \cos(\omega t) - S_y \sin(\omega t), \\ S_y^{(H)}(t) &= S_x \sin(\omega t) + S_y \cos(\omega t), \\ S_z^{(H)}(t) &= S_z. \end{aligned}$$

□

If you insist on using the definition, e.g. $S_x^{(H)} = U^\dagger S_x U$, then you must either work in a basis or apply the Baker-Hausdorff lemma. The former is easy enough; we did it in the lectures, and then you can decompose the final operator into a linear combination of the operators in the Schrödinger picture to get the same result as above. (For this the result of Problem 1.2 is useful.)

5-2 (Sakurai, Problem 2.4)

Let $x(t)$ be the coordinate operator for a free particle in one dimension in the Heisenberg picture. Evaluate

$$[x(t), x(0)].$$

Solution. We first note that

$$\begin{aligned}\frac{dp}{dt} &= \frac{1}{i\hbar}[p, H] \\ &= \frac{1}{2mi\hbar}[p, p^2] \\ &= 0.\end{aligned}$$

In particular, we have $p(t) = p(0)$, where $p(0)$ is the momentum operator in the Schrödinger picture. Then

$$\begin{aligned}\frac{dx}{dt} &= \frac{1}{i\hbar}[x, H] \\ &= \frac{1}{2mi\hbar}[x, p^2] \\ &= \frac{1}{2mi\hbar}([x, p]p + p[x, p]) \\ &= \frac{1}{2mi\hbar}(2i\hbar p) \\ &= \frac{p(0)}{m},\end{aligned}$$

which implies

$$x(t) = x(0) + \frac{p(0)t}{m}.$$

Then

$$[x(t), x(0)] = \frac{t}{m}[p(0), x(0)] = -\frac{i\hbar t}{m}.$$

□

5-3 (Sakurai, Problem 2.8)

Let $|a'\rangle$ and $|a''\rangle$ be eigenstates of a Hermitian operator A with eigenvalues a' and a'' , respectively ($a' \neq a''$). The Hamiltonian operator is given by

$$H = |a'\rangle\delta\langle a''| + |a''\rangle\delta\langle a'|,$$

where δ is just a real number.

- Clearly, $|a'\rangle$ and $|a''\rangle$ are not eigenstates of the Hamiltonian. Write down the eigenstates of the Hamiltonian. What are their energy eigenvalues?
- Suppose the system is known to be in state $|a'\rangle$ at $t = 0$. Write down the state vector in the Schrödinger picture for $t > 0$.
- What is the probability for finding the system in $|a''\rangle$ for $t > 0$ if the system is known to be in state $|a'\rangle$ at $t = 0$?
- Can you think of a physical situation corresponding to this problem?

Solution. Assigned for Problem Set 2.

- (a) The eigenstates of the Hamiltonian are

$$\frac{1}{\sqrt{2}}|a'\rangle + \frac{1}{\sqrt{2}}|a''\rangle \quad \text{and} \quad \frac{1}{\sqrt{2}}|a'\rangle - \frac{1}{\sqrt{2}}|a''\rangle,$$

corresponding to the eigenvalues δ and $-\delta$, respectively.

- (b) The state vector in the Schrödinger picture for $t > 0$ is

$$|\psi(t)\rangle = \cos\left(\frac{\delta t}{\hbar}\right)|a'\rangle - i \sin\left(\frac{\delta t}{\hbar}\right)|a''\rangle.$$

- (c) The probability for finding the system in $|a''\rangle$ for $t > 0$, assuming the system is in state $|a'\rangle$ at $t = 0$, is given by

$$|\langle a''|\psi(t)\rangle|^2 = \sin^2\left(\frac{\delta t}{\hbar}\right).$$

- (d) This identical in form to the Hamiltonian

$$H = \omega S_x = \frac{\hbar\omega}{2}(|+\rangle\langle -| + |- \rangle\langle +|).$$

□

5-4 (Sakurai, Problem 2.10)

Using the one-dimensional simple harmonic oscillator as an example, illustrate the difference between the Heisenberg picture and the Schrödinger picture. Discuss in particular how (a) the dynamic variables x and p and (b) the most general state vector evolve with time in each of the two pictures.

Solution. The Hamiltonian for the one-dimensional simple harmonic oscillator is given by

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2.$$

In the Heisenberg picture, the state vector is fixed in time: $|\psi_H\rangle = |\psi(0)\rangle$. It is the operators x and p which evolve in time according to the Heisenberg equation of motion:

$$\begin{aligned}\frac{dx}{dt} &= \frac{1}{i\hbar}[x, H] = \frac{1}{i\hbar}\left[x, \frac{p^2}{2m}\right] = \frac{1}{2mi\hbar}(2i\hbar p) = \frac{p}{m}, \\ \frac{dp}{dt} &= \frac{1}{i\hbar}[p, H] = \frac{m\omega^2}{2i\hbar}[p, x^2] = \frac{m\omega^2}{2i\hbar}(-2i\hbar x) = -m\omega^2 x.\end{aligned}$$

Then

$$\begin{aligned}\frac{d^2x}{dt^2} &= -\omega^2 x, \\ \frac{d^2p}{dt^2} &= -\omega^2 p,\end{aligned}$$

which is straightforward to solve, and using the initial conditions, one finds

$$\begin{aligned}x(t) &= x(0) \cos(\omega t) + \frac{p(0)}{m\omega} \sin(\omega t), \\ p(t) &= p(0) \cos(\omega t) - m\omega x(0) \sin(\omega t).\end{aligned}$$

If you take expectation values on both sides, then you recover the result for the classical harmonic oscillator. The expectation values are identical to the classical motion because the Hamiltonian for the quantum harmonic oscillator is at most quadratic in X and P ; see the Littlejohn notes for further discussion.

In the Schrödinger picture, the operators x and p are fixed, and the state vectors evolves in time. This is the viewpoint adopted in many textbooks. $\square$

5-5 (Sakurai, Problem 2.11)

Consider a particle subject to a one-dimensional simple harmonic oscillator potential. Suppose at $t = 0$ the state vector is given by

$$\exp\left(\frac{-ipa}{\hbar}\right)|0\rangle,$$

where p is the momentum operator and a is some number with dimension of length. Using the Heisenberg picture, evaluate the expectation value $\langle x \rangle$ for $t \geq 0$.

Solution. In Problem 2.10 we showed that

$$x_H(t) = x \cos(\omega t) - \frac{p}{m\omega} \sin(\omega t),$$

where the operators on the right are the ones in the Schrödinger picture. Then

$$\begin{aligned} \langle x \rangle &= \langle \psi_H | x_H(t) | \psi_H \rangle \\ &= \langle 0 | \exp\left(\frac{ipa}{\hbar}\right) \left(x \cos(\omega t) - \frac{p}{m\omega} \sin(\omega t) \right) \exp\left(\frac{-ipa}{\hbar}\right) | 0 \rangle \\ &= \langle 0 | \exp\left(\frac{ipa}{\hbar}\right) x \exp\left(\frac{-ipa}{\hbar}\right) | 0 \rangle \cos(\omega t) - \frac{1}{m\omega} \langle 0 | p | 0 \rangle \sin(\omega t) \\ &= \langle 0 | \exp\left(\frac{ipa}{\hbar}\right) x \exp\left(\frac{-ipa}{\hbar}\right) | 0 \rangle \cos(\omega t), \end{aligned}$$

where we have used the fact that $\exp(ipa/\hbar)$ and p commute. Now from the relation

$$\left[x, \exp\left(\frac{ipa}{\hbar}\right) \right] = -a \exp\left(\frac{ipa}{\hbar}\right)$$

it follows that

$$(x + a) \exp\left(\frac{ipa}{\hbar}\right) = \exp\left(\frac{ipa}{\hbar}\right) x.$$

Acting $\exp(-ipa/\hbar)$ on the right side of both expressions, we get

$$x + a = \exp\left(\frac{ipa}{\hbar}\right) x \exp\left(\frac{-ipa}{\hbar}\right),$$

which shows that

$$\begin{aligned} \langle x \rangle &= \langle 0 | (x + a) | 0 \rangle \cos(\omega t) \\ &= \cancel{\langle 0 | x | 0 \rangle} \cos \omega t + a \cos(\omega t) \\ &= \boxed{a \cos(\omega t)}. \end{aligned}$$

This makes sense; one can show that the expectation value follows the classical trajectory for the quantum harmonic oscillator (see Littlejohn notes). $\square$

6 Harmonic oscillator

6-1 (Shankar, Exercise 7.3.1)

Consider the question why we tried a power-series solution for Eq. (7.3.11) but not Eq. (7.3.8). By feeding in a series into the latter, verify that a three-term recursion relation between C_{n+2} , C_n , and C_{n-2} obtains, from which the solution does not follow so readily. The problem is that ψ'' has two powers of y less than $2\varepsilon\psi$, while the $-y^2$ piece has two more powers of y . In Eq. (7.3.11) on the other hand, of the three pieces u'' , $-2yu'$, and $(2\varepsilon - 1)u$, the last two have the same powers of y .

Solution. We will try a power-series solution for (7.3.8),

$$\psi'' + (2\varepsilon - y^2)\psi = 0,$$

writing

$$\psi(y) = \sum_{n=0}^{\infty} C_n y^n.$$

The series begins with $n = 0$ because solutions to (7.3.8) go as $A + cy$ (plus higher powers) in the $y \rightarrow 0$ limit. We have

$$\begin{aligned} \psi''(y) &= \sum_{n=0}^{\infty} n(n-1)C_n y^{n-2} \\ &= \sum_{n=2}^{\infty} n(n-1)C_n y^{n-2} \\ &= \sum_{m=0}^{\infty} (m+2)(m+1)C_{m+2} y^m, \\ y^2\psi(y) &= \sum_{n=0}^{\infty} C_n y^{n+2} \\ &= \sum_{m=2}^{\infty} C_{m-2} y^m \\ &= \sum_{m=0}^{\infty} C_{m-2} y^m, \end{aligned}$$

where we can get away with the last reindexing by requiring that $C_{-2} = C_{-1} = 0$. Thus (7.3.8) becomes

$$\sum_{n=0}^{\infty} \left[(n+2)(n+1)C_{n+2} + 2\varepsilon C_n - C_{n-2} \right] y^n = 0,$$

which yields the three-term recursion relation

$$(n+2)(n+1)C_{n+2} + 2\varepsilon C_n - C_{n-2} = 0,$$

as we wanted to obtain. The lesson here is that equations like (7.3.11), where you have more than two differing powers of y in your terms, are not amenable to power-series solutions. $\square$

6-2 (Shankar, Exercise 7.3.2)

Verify that $H_3(y)$ and $H_4(y)$ obey the recursion relation, Eq. (7.3.15).

Solution. We have the recursion relation (7.3.15),

$$C_{m+2} = C_m \frac{2m+1-2\varepsilon}{(m+2)(m+1)},$$

where the allowable energies (recall that $\varepsilon \equiv E/\hbar\omega$) are

$$\varepsilon = \frac{2n+1}{2}, \quad n = 0, 1, 2, 3, \dots$$

Finally, note that

$$H_3(y) = -12(y - \frac{2}{3}y^3), \quad H_4(y) = 12(1 - 4y^2 + \frac{4}{3}y^4).$$

For the Hermite polynomial $H_3(y)$, for which we have $\varepsilon = (2(3) + 1)/2 = 7/2$, we have

$$\begin{aligned} C_3 &= C_1 \frac{2(1) + 1 - 2(7/2)}{(1+2)(1+1)} \\ &= -\frac{2}{3}C_1, \end{aligned}$$

as we expect. Then for $H_4(y)$, which corresponds to $\varepsilon = (2(4) + 1)/2 = 9/2$, we have

$$\begin{aligned} C_2 &= C_0 \frac{2(0) + 1 - 2(9/2)}{(0+2)(0+1)} \\ &= -4C_0, \\ C_4 &= C_2 \frac{2(2) + 1 - 2(9/2)}{(2+2)(2+1)} \\ &= -\frac{4}{12}C_2 \\ &= \frac{4}{3}C_0, \end{aligned}$$

also as expected. $\square$

6-3 (Shankar, Exercise 7.3.3)

If $\psi(x)$ is even and $\phi(x)$ is odd under $x \rightarrow -x$, show that

$$\int_{-\infty}^{\infty} \psi(x)\phi(x) dx = 0$$

Use this to show that $\psi_2(x)$ and $\psi_1(x)$ are orthogonal. Using the values of the Gaussian integrals in Appendix A.2 verify that $\psi_2(x)$ and $\psi_0(x)$ are orthogonal.

Solution. We have

$$\begin{aligned} \int_{-\infty}^{\infty} \psi(x)\phi(x) dx &= - \int_{\infty}^{-\infty} \psi(-u)\phi(-u) du \\ &= \int_{-\infty}^{\infty} \psi(-u)\phi(-u) du \\ &= \int_{-\infty}^{\infty} \psi(u)[- \phi(u)] du \\ &= - \int_{-\infty}^{\infty} \psi(u)\phi(u) du, \end{aligned}$$

where in the first line we make the substitution $u = -x$, in the second line we pick up a minus sign when reversing the order of the integration bounds, and the third line uses the fact that $\psi(x)$ and $\phi(x)$ are even and odd, respectively. This shows that the integral is zero. Then since $\psi_2(x)$ is even and $\psi_1(x)$ is odd, which can be seen by inspecting (7.3.21) and (7.3.22), it follows that these eigenfunctions are orthogonal.

To verify that $\psi_2(x)$ and $\psi_0(x)$ are orthogonal, we write (eliding out overall mult. factors)

$$\begin{aligned} \int_{-\infty}^{\infty} \psi_0(x)\psi_2(x) dx &= (\dots) \int_{-\infty}^{\infty} H_0(y)H_2(y)e^{-y^2} dy \\ &= (\dots) \int_{-\infty}^{\infty} (1 - 2y^2)e^{-y^2} dy \\ &= (\dots) \left(\int_{-\infty}^{\infty} e^{-y^2} dy - 2 \int_{-\infty}^{\infty} y^2 e^{-y^2} dy \right) \\ &= (\dots) (I_0(1) - 2I_2(1)) \\ &= (\dots) \left(\pi^{1/2} - 2 \frac{\pi^{1/2}}{2} \right) \\ &= 0, \end{aligned}$$

where we've used (A.2.1)–(A.2.3). □

6-4 (Shankar, Exercise 7.3.4)

Using Eqs. (7.3.23)–(7.3.25), show that

$$\begin{aligned}\langle n'|X|n\rangle &= \left(\frac{\hbar}{2m\omega}\right)^{1/2} \left[\delta_{n',n+1}(n+1)^{1/2} + \delta_{n',n-1}n^{1/2}\right] \\ \langle n'|P|n\rangle &= \left(\frac{m\omega\hbar}{2}\right)^{1/2} i \left[\delta_{n',n+1}(n+1)^{1/2} - \delta_{n',n-1}n^{1/2}\right]\end{aligned}$$

Solution. There are many factors floating around here which are easy to lose track of. One advantage of the algebraic approach is that it's much cleaner than this.

$$\begin{aligned}\langle n'|X|n\rangle &= \int_{-\infty}^{\infty} x \langle n'|x\rangle \langle x|n\rangle dx \\ &= A_n A_{n'} \frac{\hbar}{m\omega} \int_{-\infty}^{\infty} y H_{n'}(y) H_n(y) e^{-y^2} dy \\ &= A_n A_{n'} \frac{\hbar}{2m\omega} \int_{-\infty}^{\infty} H_{n'}(y) \left(H_{n+1}(y) + 2n H_{n-1}(y)\right) e^{-y^2} dy \\ &= A_n A_{n'} \frac{\hbar}{2m\omega} \left[\delta_{n',n+1} \pi^{1/2} 2^{n'} (n')! + 2n \delta_{n',n-1} \pi^{1/2} 2^{n'} (n')!\right],\end{aligned}$$

where we used (7.3.25) in the third line and (7.3.26) in the last line. If we substitute $n' \rightarrow n+1$ in the first term and $n' \rightarrow n-1$ in the second term, we get

$$\begin{aligned}A_n A_{n+1} \pi^{1/2} 2^{n+1} (n+1)! &= \left[\frac{m\omega}{\pi\hbar 2^{2n} (n!)^2}\right]^{1/4} \left[\frac{m\omega}{\pi\hbar 2^{2n+2} [(n+1)!]^2}\right]^{1/4} \pi^{1/2} 2^{n+1} (n+1)! \\ &= \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} \frac{1}{2^{n+\frac{1}{2}} n! (n+1)^{1/2}} \pi^{1/2} 2^{n+1} (n+1)! \\ &= \left(\frac{2m\omega}{\hbar}\right)^{1/2} (n+1)^{1/2},\end{aligned}$$

$$\begin{aligned}2n A_n A_{n-1} \pi^{1/2} 2^{n-1} (n-1)! &= A_n A_{n-1} \pi^{1/2} 2^n n! \\ &= \left[\frac{m\omega}{\pi\hbar 2^{2n} (n!)^2}\right]^{1/4} \left[\frac{m\omega}{\pi\hbar 2^{2n-2} [(n-1)!]^2}\right]^{1/4} \pi^{1/2} 2^n n! \\ &= \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} \frac{1}{2^{n-\frac{1}{2}} (n-1)! n^{1/2}} \pi^{1/2} 2^n n! \\ &= \left(\frac{2m\omega}{\hbar}\right)^{1/2} n^{1/2},\end{aligned}$$

respectively, where we've used (7.3.23).

This shows that

$$\langle n'|X|n\rangle = \left(\frac{\hbar}{2m\omega}\right)^{1/2} \left[\delta_{n',n+1}(n+1)^{1/2} + \delta_{n',n-1}n^{1/2} \right],$$

as we wanted. Next, we have

$$\begin{aligned} \langle n'|P|n\rangle &= \int_{-\infty}^{\infty} \langle n'|x\rangle \langle x|P|n\rangle dx \\ &= -i\hbar \int_{-\infty}^{\infty} \langle n'|x\rangle \frac{d\langle x|n\rangle}{dx} dx \\ &= -i\hbar \int_{-\infty}^{\infty} \left(A_{n'} e^{-y^2/2} H_{n'}(y) \right) A_n \left(-y e^{-y^2/2} H_n(y) + \frac{dH_n}{dy} e^{-y^2/2} \right) dy \\ &= i\hbar A_n A_{n'} \left[\int_{-\infty}^{\infty} y H_{n'}(y) H_n(y) e^{-y^2} dy - 2n \int_{-\infty}^{\infty} H_{n'}(y) H_{n-1}(y) e^{-y^2} dy \right], \end{aligned}$$

where we used (7.3.24) in the last line. We dealt with the first integral when evaluating $\langle n'|X|n\rangle$:

$$\begin{aligned} i\hbar A_n A_{n'} \int_{-\infty}^{\infty} y H_{n'}(y) H_n(y) e^{-y^2} dy &= i\hbar \left(\frac{m\omega}{2\hbar} \right)^{1/2} \left[\delta_{n',n+1}(n+1)^{1/2} + \delta_{n',n-1}n^{1/2} \right] \\ &= \left(\frac{m\omega\hbar}{2} \right)^{1/2} i \left[\delta_{n',n+1}(n+1)^{1/2} + \delta_{n',n-1}n^{1/2} \right]. \end{aligned}$$

This leaves the second integral:

$$\begin{aligned} -2ni\hbar A_n A_{n'} \int_{-\infty}^{\infty} H_{n'}(y) H_{n-1}(y) e^{-y^2} dy &= -2ni\hbar A_n A_{n'} \delta_{n',n-1} \pi^{1/2} 2^{n'} (n')! \\ &= -i\hbar A_n A_{n-1} \delta_{n',n-1} \pi^{1/2} 2^n n! \\ &= -i\hbar \left(\frac{2m\omega}{\hbar} \right)^{1/2} \delta_{n',n-1} n^{1/2} \\ &= -\left(\frac{m\omega\hbar}{2} \right)^{1/2} 2i \delta_{n',n-1} n^{1/2}, \end{aligned}$$

where in the second line we used (7.3.26), and the third line follows because this expression was dealt with earlier. Therefore

$$\langle n'|P|n\rangle = \left(\frac{m\omega\hbar}{2} \right)^{1/2} i \left[\delta_{n',n+1}(n+1)^{1/2} - \delta_{n',n-1}n^{1/2} \right],$$

which is the result we wanted. □

6-5 (Shankar, Exercise 7.3.5)

Using the symmetry arguments from Exercise 7.3.3 show that $\langle n|X|n\rangle = \langle n|P|n\rangle = 0$ and thus that $\langle X^2\rangle = (\Delta X)^2$ and $\langle P^2\rangle = (\Delta P)^2$ in these states. Show that $\langle 1|X^2|1\rangle = 3\hbar/2m\omega$ and $\langle 1|P^2|1\rangle = \frac{3}{2}m\omega\hbar$. Show that $\psi_0(x)$ saturates the uncertainty bound $\Delta X \cdot \Delta P \geq \hbar/2$.

Solution. We have

$$\langle n|X|n\rangle = \int_{-\infty}^{\infty} x \langle n|x\rangle \langle x|n\rangle dx = \int_{-\infty}^{\infty} x |\langle x|n\rangle|^2 dx = 0$$

by Exercise 7.3.3, since x is odd and $|\langle x|n\rangle|^2$ is even. The same argument shows that $\langle n|P|n\rangle = 0$. Now we compute

$$\begin{aligned} \langle 1|X^2|1\rangle &= \int_{-\infty}^{\infty} x^2 |\langle x|1\rangle|^2 dx \\ &= \int_{-\infty}^{\infty} \left(\frac{m\omega}{\hbar\pi 2^2}\right)^{1/2} x^2 \exp\left(-\frac{m\omega x^2}{\hbar}\right) H_1\left[\left(\frac{m\omega}{\hbar}\right)^{1/2} x\right]^2 dx \\ &= \frac{1}{2\pi^{1/2}} \left(\frac{m\omega}{\hbar}\right)^{1/2} \int_{-\infty}^{\infty} x^2 \exp\left(-\frac{m\omega x^2}{\hbar}\right) H_1\left[\left(\frac{m\omega}{\hbar}\right)^{1/2} x\right]^2 dx \\ &= \frac{1}{2\pi^{1/2}} \frac{\hbar}{m\omega} \int_{-\infty}^{\infty} y^2 e^{-y^2} H_1(y)^2 dy, \end{aligned}$$

where in the last line we change variables to $y = (m\omega/\hbar)^{1/2}x$. Then since $H_1(y) = 2y$, we get

$$\begin{aligned} \langle 1|X^2|1\rangle &= \frac{2}{\pi^{1/2}} \frac{\hbar}{m\omega} \int_{-\infty}^{\infty} y^4 e^{-y^2} dy \\ &= \frac{2}{\pi^{1/2}} \frac{\hbar}{m\omega} \left(\frac{3}{4}\pi^{1/2}\right) \\ &= \frac{3\hbar}{2m\omega}, \end{aligned}$$

where we evaluated the integral via the method of (A.2.3). We also compute

$$\begin{aligned} \langle 1|P^2|1\rangle &= \int_{-\infty}^{\infty} p^2 |\langle p|1\rangle|^2 dp \\ &= \int_{-\infty}^{\infty} \left(\frac{1}{m\omega\hbar\pi 2^2}\right)^{1/2} p^2 \exp\left(-\frac{p^2}{m\omega\hbar}\right) H_1\left[\left(\frac{1}{m\omega\hbar}\right)^{1/2} p\right]^2 dp \\ &= \frac{1}{2\pi^{1/2}} \left(\frac{1}{m\omega\hbar}\right)^{1/2} \int_{-\infty}^{\infty} p^2 \exp\left(-\frac{p^2}{m\omega\hbar}\right) H_1\left[\left(\frac{1}{m\omega\hbar}\right)^{1/2} p\right]^2 dp \\ &= \frac{1}{2\pi^{1/2}} m\omega\hbar \int_{-\infty}^{\infty} y^2 e^{-y^2} H_1(y)^2 dy \\ &= \frac{3}{2}m\omega\hbar, \end{aligned}$$

where in the second line we used the result of Exercise 7.3.7, in the fourth line we changed variables to $y = (m\omega\hbar)^{-1/2}p$, and in the final line we recognize that this is the same integral that was evaluated earlier. [Alternatively, you could use the fact that \$P \rightarrow -i\hbar\(d/dx\)\$ in coordinate space, but then you'd need to differentiate \$\langle x|1\rangle\$ twice with respect to \$x\$, which is a bit more annoying.](#)

A quick way to see that $\psi_0(x)$ saturates the uncertainty bound is to note that $H_0(x) = 1$, so that $\psi_0(x)$ is a Gaussian. Alternatively, note that

$$\begin{aligned}
\langle 0|X^2|0\rangle &= \int_{-\infty}^{\infty} x^2 |\langle x|0\rangle|^2 dx \\
&= \int_{-\infty}^{\infty} x^2 \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} \exp\left(-\frac{m\omega x^2}{\hbar}\right) H_0\left[\left(\frac{m\omega}{\hbar}\right)^{1/2} x\right]^2 dx \\
&= \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} \int_{-\infty}^{\infty} x^2 e^{-m\omega x^2/\hbar} dx \\
&= \frac{1}{\pi^{1/2}} \frac{\hbar}{m\omega} \int_{-\infty}^{\infty} y^2 e^{-y^2} dy \\
&= \frac{\hbar}{2m\omega}, \\
\langle 0|P^2|0\rangle &= \int_{-\infty}^{\infty} p^2 |\langle p|0\rangle|^2 dp \\
&= \int_{-\infty}^{\infty} p^2 \left(\frac{1}{\pi m\omega\hbar}\right)^{1/2} \exp\left(-\frac{p^2}{m\omega\hbar}\right) H_0\left[\left(\frac{1}{m\omega\hbar}\right)^{1/2} p\right]^2 dp \\
&= \frac{1}{\pi^{1/2}} m\omega\hbar \int_{-\infty}^{\infty} y^2 e^{-y^2} dy \\
&= \frac{m\omega\hbar}{2},
\end{aligned}$$

where we've performed the same substitutions as we did in the previous part. We showed earlier that $(\Delta X)^2 = \langle X^2 \rangle$ and $(\Delta P)^2 = \langle P^2 \rangle$ for energy eigenstates, so that for the ground state

$$(\Delta X)^2(\Delta P)^2 = \frac{\hbar}{2m\omega} \frac{m\omega\hbar}{2} = \frac{\hbar^2}{4},$$

which shows that $\psi_0(x)$ saturates the uncertainty bound. □

6-6 (Shankar, Exercise 7.3.6)

Consider a particle in a potential

$$V(x) = \frac{1}{2}m\omega^2 x^2, \quad x > 0$$

$$= \infty, \quad x \leq 0$$

What are the boundary conditions on the wave functions now? Find the eigenvalues and eigenfunctions.

Solution. We require that $\psi(x) = 0$ for $x \leq 0$, in addition to the previous boundary conditions. The solution to the Schrödinger equation for $x > 0$ is the same as before, i.e., it is given by the energy eigenfunctions $\psi_n(x)$ in (7.3.22), but in order for the new boundary condition for $x \leq 0$ to be satisfied, we can choose only those wave functions $\psi_n(x)$ which are odd, since these are automatically zero at $x = 0$. Thus the energy eigenvalues are

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right), \quad n = 1, 3, 5, \dots,$$

corresponding to the eigenfunctions

$$\psi_n(x) = \begin{cases} \left(\frac{m\omega}{\pi\hbar 2^{2n}(n!)^2} \right)^{1/4} \exp\left(-\frac{m\omega x^2}{2\hbar}\right) H_n \left[\left(\frac{m\omega}{\hbar} \right)^{1/2} x \right], & x > 0 \\ 0, & x \leq 0 \end{cases}$$

This numbering is a bit unusual because the eigenfunctions go like ψ_1, ψ_3, ψ_5 , etc. □

6-7 (Shankar, Exercise 7.3.7)

The Oscillator in Momentum Space. By setting up an eigenvalue equation for the oscillator in the P basis and comparing it to Eq. (7.3.2), show that the momentum space eigenfunctions may be obtained from the ones in coordinate space through the substitution $x \rightarrow p$, $m\omega \rightarrow 1/m\omega$. Thus, for example,

$$\psi_0(p) = \left(\frac{1}{m\pi\hbar\omega} \right)^{1/4} e^{-p^2/2m\hbar\omega}$$

There are several other pairs, such as ΔX and ΔP in the state $|n\rangle$, which are related by the substitution $m\omega \rightarrow 1/m\omega$. You may wish to watch out for them. (Refer back to Exercise 7.3.5.)

Solution. The eigenvalue equation in the P basis is

$$\frac{d^2\phi}{dp^2} + \left(\frac{2E}{m\omega^2\hbar^2} - \frac{1}{m^2\omega^2\hbar^2} p^2 \right) \phi = 0,$$

which can be nondimensionalized by setting

$$p = (m\omega\hbar)^{1/2} z,$$

so that the above simplifies to

$$\phi'' + (2\varepsilon - z^2)\phi = 0,$$

where $\varepsilon = E/\hbar\omega$, as before. This equation is identical in form to the one obtained in the X basis, the only difference being that the dimensionless variable was defined via $x = (\hbar/m\omega)^{1/2}y$. Thus, given an eigenfunction $\psi(x)$ in coordinate space, we can get the corresponding momentum space eigenfunction $\phi(p)$ by the replacement $x \rightarrow p$ and $m\omega \rightarrow 1/m\omega$, as claimed. $\square$

6-8 (Shankar, Exercise 7.4.1)

Compute the matrix elements of X and P in the $|n\rangle$ basis and compare with the result from Exercise 7.3.4.

Solution. Using (7.4.28) and (7.4.29),

$$X = \left(\frac{\hbar}{2m\omega}\right)^{1/2} (a^\dagger + a), \quad P = i\left(\frac{m\omega\hbar}{2}\right)^{1/2} (a^\dagger - a),$$

it follows that

$$\begin{aligned} \langle n'|X|n\rangle &= \left(\frac{\hbar}{2m\omega}\right)^{1/2} \langle n'|a + a^\dagger|n\rangle \\ &= \left(\frac{\hbar}{2m\omega}\right)^{1/2} [\langle n'|a|n\rangle + \langle n'|a^\dagger|n\rangle] \\ &= \left(\frac{\hbar}{2m\omega}\right)^{1/2} [\sqrt{n} \langle n'|n-1\rangle + \sqrt{n+1} \langle n'|n+1\rangle] \\ &= \left(\frac{\hbar}{2m\omega}\right)^{1/2} [\sqrt{n} \delta_{n',n-1} + \sqrt{n+1} \delta_{n',n+1}] \end{aligned}$$

and

$$\begin{aligned} \langle n'|P|n\rangle &= i\left(\frac{m\omega\hbar}{2}\right)^{1/2} \langle n'|a^\dagger - a|n\rangle \\ &= i\left(\frac{m\omega\hbar}{2}\right)^{1/2} [\langle n'|a^\dagger|n\rangle - \langle n'|a|n\rangle] \\ &= i\left(\frac{m\omega\hbar}{2}\right)^{1/2} [\sqrt{n+1} \delta_{n',n+1} - \sqrt{n} \delta_{n',n-1}], \end{aligned}$$

which are precisely what we obtained in Exercise 7.3.4. [Justification for these steps is provided in the next exercise.](#) $\square$

6-9 (Shankar, Exercise 7.4.2)

Find $\langle X \rangle$, $\langle P \rangle$, $\langle X^2 \rangle$, $\langle P^2 \rangle$, $\Delta X \cdot \Delta P$ in the state $|n\rangle$.

Solution. Assigned for Problem Set 3. Using (7.4.28) and (7.4.29),

$$X = \left(\frac{\hbar}{2m\omega} \right)^{1/2} (a^\dagger + a), \quad P = i \left(\frac{m\omega\hbar}{2} \right)^{1/2} (a^\dagger - a),$$

it follows that

$$\begin{aligned} \langle X \rangle &= \langle n | X | n \rangle = \left(\frac{\hbar}{2m\omega} \right)^{1/2} \langle n | a + a^\dagger | n \rangle \\ &= \left(\frac{\hbar}{2m\omega} \right)^{1/2} \left[\langle n | a | n \rangle + \langle n | a^\dagger | n \rangle \right] \\ &= \left(\frac{\hbar}{2m\omega} \right)^{1/2} \left[\sqrt{n} \langle n | n-1 \rangle + \sqrt{n+1} \langle n | n+1 \rangle \right] \\ &= 0, \end{aligned}$$

where the third line follows from the fact that $a^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle$ and $a |n\rangle = \sqrt{n} |n-1\rangle$, and the last line follows because the energy eigenbasis is orthonormal, i.e. $\langle n' | n \rangle = \delta_{n'n}$.

Next we have

$$\begin{aligned} \langle X^2 \rangle &= \langle n | X^2 | n \rangle = \frac{\hbar}{2m\omega} \langle n | (a + a^\dagger)^2 | n \rangle \\ &= \frac{\hbar}{2m\omega} \langle n | a^2 + aa^\dagger + a^\dagger a + (a^\dagger)^2 | n \rangle \\ &= \frac{\hbar}{2m\omega} \left[\cancel{\langle n | a^2 | n \rangle} + \langle n | aa^\dagger | n \rangle + \langle n | a^\dagger a | n \rangle + \cancel{\langle n | (a^\dagger)^2 | n \rangle} \right] \\ &= \frac{\hbar}{2m\omega} \left[(n+1) + n \right] \\ &= \frac{\hbar}{2m\omega} (2n+1), \end{aligned}$$

where again we have used the relations $a^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle$, $a |n\rangle = \sqrt{n} |n-1\rangle$, and $\langle n' | n \rangle = \delta_{n'n}$, together with the fact that the annihilation and creation operators are Hermitian conjugates of each other. It follows that

$$\boxed{(\Delta X)^2 = \frac{\hbar}{2m\omega} (2n+1)}.$$

Next, we note that

$$\begin{aligned}
\langle P \rangle &= \langle n|P|n \rangle = i \left(\frac{m\omega\hbar}{2} \right)^{1/2} \langle n|a^\dagger - a|n \rangle \\
&= i \left(\frac{m\omega\hbar}{2} \right)^{1/2} \left[\langle n|a^\dagger|n \rangle - \langle n|a|n \rangle \right] \\
&= 0,
\end{aligned}$$

which vanishes for the same reasons as before. Then

$$\begin{aligned}
\langle P^2 \rangle &= \langle n|P^2|n \rangle = - \left(\frac{m\omega\hbar}{2} \right) \langle n|(a^\dagger - a)^2|n \rangle \\
&= - \left(\frac{m\omega\hbar}{2} \right) \langle n|(a^\dagger)^2 - a^\dagger a - a a^\dagger + a^2|n \rangle \\
&= - \left(\frac{m\omega\hbar}{2} \right) \left[\overline{\langle n|(a^\dagger)^2|n \rangle} - \langle n|a^\dagger a|n \rangle - \langle n|a a^\dagger|n \rangle + \overline{\langle n|a^2|n \rangle} \right] \\
&= - \left(\frac{m\omega\hbar}{2} \right) \left[-n - (n+1) \right] \\
&= \left(\frac{m\omega\hbar}{2} \right) (2n+1).
\end{aligned}$$

We therefore have

$$(\Delta P)^2 = \frac{m\omega\hbar}{2} (2n+1).$$

Finally, we note that

$$\Delta X \cdot \Delta P = \frac{\hbar}{2} (2n+1) \geq \frac{\hbar}{2},$$

and that the bound is saturated for $n = 0$, so that the ground state of the quantum harmonic oscillator is a minimum-uncertainty state. (This was shown explicitly in Exercise 7.3.5.)

All of these findings are consistent with what we obtained in Exercises 7.3.4 and 7.3.5, where we worked in the coordinate basis. You can develop expressions for the matrix elements of higher powers of the position operator, e.g. $\langle n'|X^3|n \rangle$, in the same way as Exercise 7.4.1. This involves a bit more algebra, but is probably less work than the coordinate-space method. The main difficulty I see with extending this to matrix elements of, say, X^n , has to do with the ordering of the operators. You can see a bit of this on page 209. $\square$

6-10 (Shankar, Exercise 7.4.3)

Virial Theorem. The virial theorem in classical mechanics states that for a particle bound by a potential $V(r) = ar^k$, the average (over the orbit) kinetic and potential energies are related by

$$\bar{T} = c(k)\bar{V}$$

where $c(k)$ depends only on k . Show that $c(k) = k/2$ by considering a circular orbit. Using the results from the previous exercise show that for the oscillator ($k = 2$)

$$\langle T \rangle = \langle V \rangle$$

in the quantum state $|n\rangle$.

Solution. If we have a potential $V(r) = ar^k$, then $F = -dV/dr = -kar^{k-1}$. For circular orbits,

$$-kar^{k-1} = -\frac{mv^2}{r}$$

$$kar^k = mv^2$$

$$v^2 = \frac{ka}{m}r^k$$

which implies that

$$\tilde{T} = \frac{1}{2}mv^2 = \frac{1}{2}kar^k$$

and since $\tilde{V} = ar^k$, we have

$$\tilde{T} = \frac{k\tilde{V}}{2},$$

as we wanted to show. In particular, for the simple harmonic oscillator we get $\tilde{T} = \tilde{V}$. To establish the quantum-mechanical analogue of this statement, we use the result of Exercise 7.4.2

to compute

$$\begin{aligned}
\langle T \rangle &= \left\langle \frac{P^2}{2m} \right\rangle \\
&= \frac{1}{2m} \langle P^2 \rangle \\
&= \frac{1}{2m} \left(\frac{m\omega\hbar}{2} \right) (2n+1) \\
&= \frac{2n+1}{4} \hbar\omega, \\
\langle V \rangle &= \left\langle \frac{1}{2} m\omega^2 X^2 \right\rangle \\
&= \frac{1}{2} m\omega^2 \langle X^2 \rangle \\
&= \frac{1}{2} m\omega^2 \left(\frac{\hbar}{2m\omega} \right) (2n+1) \\
&= \frac{2n+1}{4} \hbar\omega,
\end{aligned}$$

which shows that $\langle T \rangle = \langle V \rangle$ for the quantum state $|n\rangle$ of the harmonic oscillator. [For an alternative treatment, see Problem 2.6 in Sakurai.](#) $\square$

6-11 (Shankar, Exercise 7.4.4)

Show that $\langle n|X^4|n\rangle = (\hbar/2m\omega)^2[3 + 6n(n+1)]$.

Solution. This one is a bit involved because $a^\dagger$ and a don't commute, so we need to multiply things out manually, and there are 16 terms to keep track of:

$$\begin{aligned}
\langle n|X^4|n\rangle &= \left(\frac{\hbar}{2m\omega} \right)^2 \langle n|(a^\dagger + a)^4|n\rangle \\
&= \left(\frac{\hbar}{2m\omega} \right)^2 \left\langle n \left| \left[(a^\dagger)^4 + (a^\dagger)^3 a + (a^\dagger)^2 a a^\dagger + (a^\dagger)^2 a^2 + a^\dagger a (a^\dagger)^2 + a^\dagger a a^\dagger a \right. \right. \right. \\
&\quad \left. \left. \left. + a^\dagger a^2 a^\dagger + a^\dagger a^3 + a (a^\dagger)^3 + a (a^\dagger)^2 a + a a^\dagger a a^\dagger \right. \right. \right. \\
&\quad \left. \left. \left. + a a^\dagger a^2 + a^2 (a^\dagger)^2 + a^2 a^\dagger a + a^3 a^\dagger + a^4 \right] \right| n \right\rangle.
\end{aligned}$$

However, the only nonzero terms are those with an equal number of $a^\dagger$'s and a 's, since otherwise we end up with the inner product of two orthogonal kets:

$$\langle n|X^4|n\rangle = \left(\frac{\hbar}{2m\omega} \right)^2 \langle n|(a^\dagger)^2 a^2 + a^\dagger a a^\dagger a + a^\dagger a^2 a^\dagger + a (a^\dagger)^2 a + a a^\dagger a a^\dagger + a^2 (a^\dagger)^2|n\rangle.$$

What remains is an exercise in manipulating commutators. We first recall that the number

operator $N = a^\dagger a$ has the property $N|n\rangle = n|n\rangle$, and that

$$[a, a^\dagger] = 1, \quad [N, a] = -a, \quad [N, a^\dagger] = a^\dagger,$$

which imply

$$aa^\dagger = N + 1, \quad a^\dagger N = (N - 1)a^\dagger, \quad Na^\dagger = a^\dagger(N + 1).$$

Then we examine all six terms appearing in our matrix element:

$$\begin{aligned} (a^\dagger)^2 a^2 &= a^\dagger N a \\ &= (N - 1) a^\dagger a \\ &= (N - 1) N \\ &= N^2 - N, \\ a^\dagger a a^\dagger a &= N^2, \\ a^\dagger a^2 a^\dagger &= N(N + 1) \\ &= N^2 + N, \\ a a^\dagger a a^\dagger &= (N + 1) N \\ &= N^2 + N, \\ a a^\dagger a a^\dagger &= (N + 1)(N + 1) \\ &= N^2 + 2N + 1, \\ a^2 (a^\dagger)^2 &= a(N + 1) a^\dagger \\ &= a N a^\dagger + N + 1 \\ &= a a^\dagger (N + 1) + N + 1 \\ &= (N + 1)(N + 1) + N + 1 \\ &= N^2 + 2N + 1 + N + 1 \\ &= N^2 + 3N + 2. \end{aligned}$$

Thus

$$\begin{aligned} (a^\dagger)^2 a^2 + a^\dagger a a^\dagger a + a^\dagger a^2 a^\dagger + a (a^\dagger)^2 a + a a^\dagger a a^\dagger + a^2 (a^\dagger)^2 \\ = (N^2 - N) + N^2 + (N^2 + N) + (N^2 + N) + (N^2 + 2N + 1) + (N^2 + 3N + 2) \\ = 6N^2 + 6N + 3, \end{aligned}$$

which gives

$$\boxed{\langle n | X^4 | n \rangle = \left(\frac{\hbar}{2m\omega} \right)^2 (6n^2 + 6n + 3)},$$

as we wanted to show. □

6-12 (Shankar, Exercise 7.4.5)

At $t = 0$ a particle starts out in $|\psi(0)\rangle = 1/2^{1/2}(|0\rangle + |1\rangle)$. (1) Find $|\psi(t)\rangle$; (2) find $\langle X(0)\rangle = \langle\psi(0)|X|\psi(0)\rangle$, $\langle P(0)\rangle$, $\langle X(t)\rangle$, $\langle P(t)\rangle$; (3) find $\langle \dot{X}(t)\rangle$ and $\langle \dot{P}(t)\rangle$ using Ehrenfest's theorem and solve for $\langle X(t)\rangle$ and $\langle P(t)\rangle$ and compare with part (2).

Solution. (1) We have

$$\begin{aligned} |\psi(t)\rangle &= e^{-iHt/\hbar}|\psi(0)\rangle \\ &= e^{-i[\hbar\omega(N+1/2)]t/\hbar}|\psi(0)\rangle \\ &= e^{-i\omega t/2}e^{-Ni\omega t}|\psi(0)\rangle \\ &= \frac{1}{\sqrt{2}}e^{-i\omega t/2}e^{-Ni\omega t}(|0\rangle + |1\rangle) \\ &= \frac{1}{\sqrt{2}}\left(e^{-i\omega t/2}|0\rangle + e^{-3i\omega t/2}|1\rangle\right). \end{aligned}$$

(2) Since

$$\begin{aligned} X|\psi(0)\rangle &= \left(\frac{\hbar}{2m\omega}\right)^{1/2}(a + a^\dagger)\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \\ &= \left(\frac{\hbar}{2m\omega}\right)^{1/2}\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle + \sqrt{2}|2\rangle), \end{aligned}$$

we get

$$\begin{aligned} \langle X(0)\rangle &= \langle\psi(0)|X|\psi(0)\rangle = \frac{1}{2}\left(\frac{\hbar}{2m\omega}\right)^{1/2}(1 + 1) \\ &= \left(\frac{\hbar}{2m\omega}\right)^{1/2}. \end{aligned}$$

We also compute

$$\begin{aligned} P|\psi(0)\rangle &= i\left(\frac{m\omega\hbar}{2}\right)^{1/2}(a^\dagger - a)\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \\ &= i\left(\frac{m\omega\hbar}{2}\right)^{1/2}\frac{1}{\sqrt{2}}(|1\rangle + \sqrt{2}|2\rangle - |0\rangle), \end{aligned}$$

which gives

$$\begin{aligned} \langle P(0)\rangle &= \langle\psi(0)|P|\psi(0)\rangle = \frac{1}{2}i\left(\frac{m\omega\hbar}{2}\right)^{1/2}(1 - 1) \\ &= 0. \end{aligned}$$

This is to be expected, since the wave function at $t = 0$ is real; see Exercise 4.2.2. Next, we have

$$\begin{aligned} X|\psi(t)\rangle &= \left(\frac{\hbar}{2m\omega}\right)^{1/2} (a + a^\dagger) \frac{1}{\sqrt{2}} \left(e^{-i\omega t/2} |0\rangle + e^{-3i\omega t/2} |1\rangle \right) \\ &= \left(\frac{\hbar}{2m\omega}\right)^{1/2} \frac{1}{\sqrt{2}} \left(e^{-3i\omega t/2} |0\rangle + e^{-i\omega t/2} |1\rangle + \sqrt{2} e^{-3i\omega t/2} |2\rangle \right), \end{aligned}$$

which implies

$$\begin{aligned} \langle X(t) \rangle &= \langle \psi(t) | X | \psi(t) \rangle = \frac{1}{2} \left(\frac{\hbar}{2m\omega}\right)^{1/2} \left(e^{i\omega t/2} \langle 0| + e^{3i\omega t/2} \langle 1| \right) \\ &\quad \times \left(e^{-3i\omega t/2} |0\rangle + e^{-i\omega t/2} |1\rangle + \sqrt{2} e^{-3i\omega t/2} |2\rangle \right), \\ &= \frac{1}{2} \left(\frac{\hbar}{2m\omega}\right)^{1/2} (e^{-i\omega t} + e^{i\omega t}) \\ &= \left(\frac{\hbar}{2m\omega}\right)^{1/2} \cos(\omega t). \end{aligned}$$

We also have

$$\begin{aligned} P|\psi(t)\rangle &= i \left(\frac{m\omega\hbar}{2}\right)^{1/2} (a^\dagger - a) \frac{1}{\sqrt{2}} \left(e^{-i\omega t/2} |0\rangle + e^{-3i\omega t/2} |1\rangle \right) \\ &= i \left(\frac{m\omega\hbar}{2}\right)^{1/2} \frac{1}{\sqrt{2}} \left(e^{-i\omega t/2} |1\rangle + e^{-3i\omega t/2} \sqrt{2} |2\rangle - e^{-3i\omega t/2} |0\rangle \right), \end{aligned}$$

which gives

$$\begin{aligned} \langle P(t) \rangle &= \langle \psi(t) | P | \psi(t) \rangle = \frac{1}{2} i \left(\frac{m\omega\hbar}{2}\right)^{1/2} \left(e^{i\omega t/2} \langle 0| + e^{3i\omega t/2} \langle 1| \right) \\ &\quad \times \left(e^{-i\omega t/2} |1\rangle + e^{-3i\omega t/2} \sqrt{2} |2\rangle - e^{-3i\omega t/2} |0\rangle \right), \\ &= \frac{1}{2} i \left(\frac{m\omega\hbar}{2}\right)^{1/2} (-e^{-i\omega t} + e^{i\omega t}) \\ &= -\left(\frac{m\omega\hbar}{2}\right)^{1/2} \sin(\omega t). \end{aligned}$$

(3) See Problem 2.10 in Sakurai, which works in the Heisenberg picture to show that

$$\begin{aligned} X(t) &= X(0) \cos(\omega t) + \frac{P(0)}{m\omega} \sin(\omega t), \\ P(t) &= P(0) \cos(\omega t) - m\omega X(0) \sin(\omega t) \end{aligned}$$

for the simple harmonic oscillator. Taking expectation values on both sides and recalling that $\langle P(0) \rangle = 0$ by our earlier analysis, we see that our expressions are consistent with this result.

I lost a lot of time by mistakenly writing $U(t) = e^{iHt/\hbar}$. □

6-13 (Shankar, Exercise 7.4.6)

Show that $\langle a(t) \rangle = e^{-i\omega t} \langle a(0) \rangle$ and that $\langle a^\dagger(t) \rangle = e^{i\omega t} \langle a^\dagger(0) \rangle$.

Solution. Assigned as a class exercise.

We have

$$\begin{aligned} [a, H] &= [a, \hbar\omega(N + 1/2)] \\ &= \hbar\omega[a, N] \\ &= \hbar\omega a. \end{aligned}$$

Then by Ehrenfest's theorem,

$$\begin{aligned} \frac{d\langle a \rangle}{dt} &= \frac{1}{i\hbar} \langle [a, H] \rangle \\ &= \frac{1}{i\hbar} \langle \hbar\omega a \rangle \\ &= -i\omega \langle a \rangle, \end{aligned}$$

and the solution to this differential equation is

$$\boxed{\langle a(t) \rangle = \langle a(0) \rangle e^{-i\omega t}},$$

as we wanted to show. The analysis for the creation operator is the more or less the same, except $[a^\dagger, N] = -a^\dagger$, so the sign in the exponential is different. $\square$

6-14 (Shankar, Exercise 7.4.7)

Verify Eq. (7.4.40) for the case

$$(1) \quad \Omega = X, \quad \Lambda = X^2 + P^2$$

$$(2) \quad \Omega = X^2, \quad \Lambda = P^2$$

The second case illustrates the ordering ambiguity.

Solution. (1) If we take $\omega = x$ and $\lambda = x^2 + p^2$, then their Poisson bracket is

$$\begin{aligned} \{\omega, \lambda\} &= \frac{\partial \omega}{\partial x} \frac{\partial \lambda}{\partial p} - \frac{\partial \omega}{\partial p} \frac{\partial \lambda}{\partial x} \\ &= (1)(2p) - (0)(2x) \\ &= 2p, \end{aligned}$$

so by Eq. (7.4.40) we expect $[\Omega, \Lambda] = i\hbar(2P)$. Now

$$\begin{aligned} [\Omega, \Lambda] &= [X, X^2 + P^2] \\ &= [X, P^2] \\ &= i\hbar(2P), \end{aligned}$$

which checks out.

(2) Here we take $\omega = x^2$ and $\Lambda = p^2$, which have Poisson bracket

$$\begin{aligned} \{\omega, \lambda\} &= \frac{\partial \omega}{\partial x} \frac{\partial \lambda}{\partial p} - \frac{\partial \omega}{\partial p} \frac{\partial \lambda}{\partial x} \\ &= (2x)(2p) - (0)(0) \\ &= 4xp. \end{aligned}$$

There is an ordering ambiguity when going from the classical to the quantum-mechanical picture, since X and P do not commute. However, we can compute

$$\begin{aligned} [\Omega, \Lambda] &= [X^2, P^2] \\ &= [X^2, P]P + P[X^2, P] \\ &= 2i\hbar XP + P(2i\hbar X) \\ &= i\hbar \left(4 \frac{XP + PX}{2} \right), \end{aligned}$$

and observe that the prescription (7.4.40) does carry over, as long as $xp \rightarrow (XP + PX)/2$. $\square$

6-15 (Shankar, Exercise 7.4.8)

Consider the three angular momentum variables in classical mechanics:

$$l_x = yp_z - zp_y$$

$$l_y = zp_x - xp_z$$

$$l_z = xp_y - yp_x$$

- (1) Construct L_x , L_y , and L_z , the quantum counterparts, and note that there are no ordering ambiguities.
- (2) Verify that $\{l_x, l_y\} = l_z$.
- (3) Verify that $[L_x, L_y] = i\hbar L_z$.

Solution. (1) Following Postulate II, the corresponding operators are

$$L_x = YP_z - ZP_y,$$

$$L_y = ZP_x - XP_z,$$

$$L_z = XP_y - YP_x,$$

where there are no ordering ambiguities since $[X_i, P_j] = i\hbar\delta_{ij}$.

(2)

$$\begin{aligned} \{l_x, l_y\} &= \sum_i \left(\frac{\partial l_x}{\partial q_i} \frac{\partial l_y}{\partial p_i} - \frac{\partial l_x}{\partial p_i} \frac{\partial l_y}{\partial q_i} \right) \\ &= \left(\frac{\partial l_x}{\partial x} \frac{\partial l_y}{\partial p_x} + \frac{\partial l_x}{\partial y} \frac{\partial l_y}{\partial p_y} + \frac{\partial l_x}{\partial z} \frac{\partial l_y}{\partial p_z} \right) - \left(\frac{\partial l_x}{\partial p_x} \frac{\partial l_y}{\partial x} + \frac{\partial l_x}{\partial p_y} \frac{\partial l_y}{\partial y} + \frac{\partial l_x}{\partial p_z} \frac{\partial l_y}{\partial z} \right) \\ &= [0 + 0 + (-p_y)(-x)] - [0 + 0 + (y)(p_x)] \\ &= xp_y - yp_x \\ &= l_z. \end{aligned}$$

(3)

$$\begin{aligned} [L_1, L_2] &= [X_2P_3 - X_3P_2, X_3P_1 - X_1P_3] \\ &= [X_2P_3, X_3P_1] - [X_2P_3, X_1P_3] - [X_3P_2, X_3P_1] + [X_3P_2, X_1P_3] \\ &= (X_2P_3X_3P_1 - X_3P_1X_2P_3) + (X_3P_2X_1P_3 - X_1P_3X_3P_2) \\ &= X_2P_1(P_3X_3 - X_3P_3) + X_1P_2(X_3P_3 - P_3X_3) \\ &= i\hbar(X_1P_2 - X_2P_1) \\ &= i\hbar L_3. \end{aligned}$$

□

6-16 (Shankar, Exercise 7.4.9)

Consider the unconventional (but fully acceptable) operator choice

$$X \rightarrow x$$

$$P \rightarrow -i\hbar \frac{d}{dx} + f(x)$$

in the X basis.

- (1) Verify that the canonical commutation relation is satisfied.
- (2) It is possible to interpret the change in the operator assignment as a result of a unitary change of the X basis:

$$|x\rangle \rightarrow |\tilde{x}\rangle = e^{ig(X)/\hbar}|x\rangle = e^{ig(x)/\hbar}|x\rangle$$

where

$$g(x) = \int^x f(x') dx'.$$

First verify that

$$\langle \tilde{x}|X|\tilde{x}'\rangle = x\delta(x - x')$$

i.e.,

$$X \xrightarrow{\text{new } X \text{ basis}} x$$

Next verify that

$$\langle \tilde{x}|P|\tilde{x}'\rangle = \left[-i\hbar \frac{d}{dx} + f(x) \right] \delta(x - x')$$

i.e.,

$$P \xrightarrow{\text{new } X \text{ basis}} -i\hbar \frac{d}{dx} + f(x)$$

This exercise teaches us that the “ X basis” is not unique; given a basis $|x\rangle$, we can get another $|\tilde{x}\rangle$, by multiplying by a phase factor which changes neither the norm nor the orthogonality. The matrix elements of P change with f , the standard choice corresponding to $f = 0$. Since the presence of f is related to a change of basis, the invariance of the physics under a change in f (from zero to nonzero) follows. What is novel here is that we are changing from one X basis to another X basis rather than to some other Ω basis. Another lesson to remember is that two different differential operators $\omega(x, -i\hbar d/dx)$ and $\omega(x, -i\hbar d/dx + f)$ can have the same eigenvalues and a one-to-one correspondence between their eigenfunctions, since they both represent the same abstract operator $\Omega(X, P)$.

Solution. (1) Act $[X, P]$ on an arbitrary ket $|\psi\rangle$ whose representation in the X basis is $\psi(x)$:

$$\begin{aligned}
[X, P]|\psi\rangle &= (XP - PX)|\psi\rangle \\
&\rightarrow \left[x \left(-i\hbar \frac{d}{dx} + f(x) \right) - \left(-i\hbar \frac{d}{dx} + f(x) \right) x \right] \psi(x) \\
&= -i\hbar x \frac{d\psi}{dx} + \cancel{x f(x) \psi(x)} + i\hbar \frac{d(x\psi(x))}{dx} - \cancel{f(x) x \psi(x)} \\
&= i\hbar \psi(x) \\
&\rightarrow i\hbar |\psi\rangle.
\end{aligned}$$

Since $|\psi\rangle$ was arbitrary, we conclude that $[X, P] = i\hbar$; that is, the canonical commutation relation is satisfied for this operator choice.

(2) The matrix elements of X in this new basis are

$$\begin{aligned}
\langle \tilde{x} | X | \tilde{x}' \rangle &= e^{-ig(x)/\hbar} \langle x | \left(X e^{ig(x')/\hbar} | x' \rangle \right) \\
&= e^{-ig(x)/\hbar} e^{ig(x')/\hbar} \langle x | X | x' \rangle \\
&= e^{-ig(x)/\hbar} e^{ig(x')/\hbar} x \delta(x - x') \\
&= x \delta(x - x'),
\end{aligned}$$

as we wanted to show. The matrix elements of P in the same basis are

$$\begin{aligned}
\langle \tilde{x} | P | \tilde{x}' \rangle &= e^{-ig(x)/\hbar} \langle x | \left(P e^{ig(x')/\hbar} | x' \rangle \right) \\
&= -i\hbar e^{-ig(x)/\hbar} \frac{d}{dx} \left(e^{ig(x')/\hbar} \langle x | x' \rangle \right) \\
&= -i\hbar e^{-ig(x)/\hbar} \frac{d}{dx} \left(e^{ig(x)/\hbar} \delta(x - x') \right) \\
&= -i\hbar e^{-ig(x)/\hbar} \left(\frac{i}{\hbar} \frac{dg}{dx} e^{ig(x)/\hbar} \delta(x - x') + e^{ig(x)/\hbar} \delta'(x - x') \right) \\
&= f(x) \delta(x - x') - i\hbar \delta(x - x') \frac{d}{dx'},
\end{aligned}$$

where we've used the definition of $g(x)$, and the derivative of the delta function is taken as in (1.10.28). A point that caused some confusion for me is that the two parts of this problem take different viewpoints, so that the representation of P in these parts differ. To be more precise, in (1) we change the representation of the *operator*, while in (2), we change the *basis* itself. The equivalence of these two viewpoints is analogous to how we can view the same transformation as either passive or active. $\square$

6-17 (Shankar, Exercise 7.4.10)

Solution. (1) This part is all exposition, but for later reference we note that

$$P_\rho \rightarrow -i\hbar \frac{\partial}{\partial \rho}.$$

(2) We have, by the previous choice,

$$\begin{aligned} \langle \psi_1 | P_\rho | \psi_2 \rangle &= \int_0^\infty \int_0^{2\pi} \psi_1^* \left(-i\hbar \frac{\partial \psi_2}{\partial \rho} \right) \rho \, d\rho \, d\phi \\ &= \int_0^{2\pi} \left[\cancel{-i\hbar \psi_1^* \psi_2 \rho} \Big|_0^\infty + i\hbar \int_0^\infty \psi_2 \left(\rho \frac{\partial \psi_1^*}{\partial \rho} + \psi_1^* \right) d\rho \right] d\phi \\ &= \int_0^\infty \int_0^{2\pi} \left(-i\hbar \frac{\partial \psi_1}{\partial \rho} \right)^* \psi_2 \rho \, d\rho \, d\phi + \int_0^\infty \int_0^{2\pi} i\hbar \psi_1^* \psi_2 \, d\rho \, d\phi \\ &\neq \langle \mathbf{P}_\rho \psi_1 | \psi_2 \rangle, \end{aligned}$$

where we integrated by parts in the second line, choosing $u = \psi_1^* \rho$ and $dv = -i\hbar(\partial \psi_2 / \partial \rho) d\rho$. The equivalence in the last line is spoiled by the appearance of the integral involving $\psi_1^* \psi_2$ in the third line. However, we can choose P_ρ in a clever way to make it Hermitian:

$$P_\rho \rightarrow -i\hbar \left(\frac{\partial}{\partial \rho} + \frac{1}{2\rho} \right)$$

This still satisfies the canonical commutation rule, since the additional term is just a function of ρ (compare Exercise 7.4.9), and we can check that this choice yields a Hermitian operator:

$$\begin{aligned} \langle \psi_1 | P_\rho | \psi_2 \rangle &= \int_0^\infty \int_0^{2\pi} \psi_1^* \left(-i\hbar \frac{\partial \psi_2}{\partial \rho} - i\hbar \frac{\psi_2}{2\rho} \right) \rho \, d\rho \, d\phi \\ &= \int_0^\infty \int_0^{2\pi} \left(-i\hbar \frac{\partial \psi_1}{\partial \rho} \right)^* \psi_2 \rho \, d\rho \, d\phi + \frac{1}{2} \int_0^\infty \int_0^{2\pi} i\hbar \psi_1^* \psi_2 \, d\rho \, d\phi \\ &= \int_0^\infty \int_0^{2\pi} \left[-i\hbar \left(\frac{\partial \psi_1}{\partial \rho} + \frac{\psi_1}{2\rho} \right) \right]^* \psi_2 \rho \, d\rho \, d\phi \\ &= \langle \mathbf{P}_\rho \psi_1 | \psi_2 \rangle, \end{aligned}$$

where we used our result from earlier.

(3) In Exercise 7.4.9 we implicitly assumed that $f(x)$ was real, which is what led to the same matrix elements and hence the same physical effects. In general, the matrix elements pick up an overall phase $e^{i[g(x)-g(x)^*]/\hbar}$, so it is plausible that the new imaginary term added to $-i\hbar \partial / \partial \rho$ is what causes the elements $\langle \psi_1 | P_\rho | \psi_2 \rangle$ to change. *Although it's not clear to me which matrix elements change, since I don't know if there are kets like $|\rho\rangle$ or $|\phi\rangle$.*

(4) This follows from the fact that

$$\begin{aligned}\left(\frac{\partial}{\partial \rho} + \frac{1}{2\rho}\right)^2 f &= \frac{\partial^2 f}{\partial \rho^2} + \frac{1}{2\rho} \frac{\partial f}{\partial \rho} + f \left(-\frac{1}{2\rho^2}\right) + \frac{1}{2\rho} \frac{\partial f}{\partial \rho} + \frac{1}{4\rho^2} f \\ &= \frac{\partial^2 f}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial f}{\partial \rho} - \frac{1}{4\rho^2} f.\end{aligned}$$

Initially I got one of the terms wrong because I forgot to put in a placeholder function. □

6-18 (Shankar, Exercise 7.5.1)

Project Eq. (7.5.1) on the P basis and obtain $\psi_0(p)$.

Solution. We have

$$\begin{aligned}0 &= \langle p|a|0\rangle \\ &= \left(\frac{m\omega}{2\hbar}\right)^{1/2} \langle p|X|0\rangle + i\left(\frac{1}{2m\omega\hbar}\right)^{1/2} \langle p|P|0\rangle \\ &= i\hbar\left(\frac{m\omega}{2\hbar}\right)^{1/2} \frac{d\psi_0}{dp} + i\left(\frac{1}{2m\omega\hbar}\right)^{1/2} p\psi_0(p) \\ &= \frac{i}{2^{1/2}} \left(\frac{d\psi_0}{dz} + z\psi_0(z)\right),\end{aligned}$$

where we set $p = (m\omega\hbar)^{1/2}z$ in the last line. But this is the same as (7.5.5), so we get

$$\psi_0(z) = A_0 e^{-z^2/2},$$

and by restoring dimensions and normalizing, we get

$$\boxed{\psi_0(p) = \left(\frac{1}{\pi m\omega\hbar}\right)^{1/4} \exp\left(-\frac{p^2}{2m\omega\hbar}\right)}.$$

Note that this is consistent with the result of Exercise 7.3.7; the main idea is that the equations look the same in the X and P bases after nondimensionalizing, so the solutions only differ by the scale factor that was used. □

6-19 (Shankar, Exercise 7.5.2)

Project the relation

$$a|n\rangle = n^{1/2}|n-1\rangle$$

on the X basis and derive the recursion relation

$$H'_n(y) = 2nH_{n-1}(y)$$

using Eq. (7.3.22).

Solution. First project onto the X basis:

$$\begin{aligned}\langle x|a|n\rangle &= n^{1/2}\langle x|n-1\rangle \\ \left(\frac{m\omega}{2\hbar}\right)^{1/2}\langle x|X|n\rangle + i\left(\frac{1}{2m\omega\hbar}\right)^{1/2}\langle x|P|n\rangle &= n^{1/2}\langle x|n-1\rangle \\ \left(\frac{m\omega}{2\hbar}\right)^{1/2}x\psi_n(x) + i\left(\frac{1}{2m\omega\hbar}\right)^{1/2}\left(-i\hbar\frac{d\psi_n}{dx}\right) &= n^{1/2}\psi_{n-1}(x)\end{aligned}$$

Then nondimensionalize by setting $x = (h/m\omega)^{1/2}y$ to get

$$\frac{1}{2^{1/2}}\left(y\psi_n(y) + \frac{d\psi_n}{dy}\right) = n^{1/2}\psi_{n-1}(y).$$

Using (7.3.22), we have

$$\begin{aligned}\frac{d\psi_n}{dy} &= \frac{d}{dy}\left(A_n e^{-y^2/2} H_n(y)\right) \\ &= A_n\left(-ye^{-y^2/2} H_n(y) + e^{-y^2/2} H'_n(y)\right) \\ &= -yA_n e^{-y^2/2} H_n(y) + A_n e^{-y^2/2} H'_n(y) \\ &= -y\psi_n(y) + A_n e^{-y^2/2} H'_n(y),\end{aligned}$$

which implies that

$$y\psi_n(y) + \frac{d\psi_n}{dy} = A_n e^{-y^2/2} H'_n(y),$$

which we substitute back into our original equation to get

$$\begin{aligned}A_n e^{-y^2/2} H'_n(y) &= (2n)^{1/2} A_{n-1} e^{-y^2/2} H_{n-1}(y) \\ H'_n(y) &= (2n)^{1/2} \frac{A_{n-1}}{A_n} H_{n-1}(y).\end{aligned}$$

Finally, by (7.3.23) we have

$$\begin{aligned}\frac{A_{n-1}}{A_n} &= \left(\frac{2^{2n}(n!)^2}{2^{2(n-1)}[(n-1)!]^2}\right)^{1/4} \\ &= (2^2 n^2)^{1/4} \\ &= (2n)^{1/2},\end{aligned}$$

so that

$$\boxed{H'_n(y) = 2n H_{n-1}(y)},$$

as we wanted to show. □

6-20 (Shankar, Exercise 7.5.3)

Starting with

$$a + a^\dagger = 2^{1/2}y$$

and

$$(a + a^\dagger)|n\rangle = n^{1/2}|n-1\rangle + (n+1)^{1/2}|n+1\rangle$$

and Eq. (7.3.22), derive the relation

$$H_{n+1}(y) = 2yH_n(y) - 2nH_{n-1}(y)$$

Solution. Project the second equation onto the X basis and use (7.3.22):

$$\begin{aligned}\langle x|(a + a^\dagger)|n\rangle &= n^{1/2}\langle x|n-1\rangle + (n+1)^{1/2}\langle x|n+1\rangle \\ &= n^{1/2}A_{n-1}e^{-y^2/2}H_{n-1}(y) + (n+1)^{1/2}A_{n+1}e^{-y^2/2}H_{n+1}(y).\end{aligned}$$

But by the first equation,

$$\begin{aligned}\langle x|(a + a^\dagger)|n\rangle &= 2^{1/2}y\langle x|n\rangle \\ &= 2^{1/2}yA_n e^{-y^2/2}H_n(y),\end{aligned}$$

so that

$$2^{1/2}yA_n H_n(y) = n^{1/2}A_{n-1}H_{n-1}(y) + (n+1)^{1/2}A_{n+1}H_{n+1}(y).$$

By (7.3.23), this is equivalent to

$$\begin{aligned}2^{1/2}y\left(\frac{1}{n^2}\right)^{1/4}H_n(y) \\ = n^{1/2}\left(\frac{1}{2^{-2}}\right)^{1/4}H_{n-1}(y) + (n+1)^{1/2}\left(\frac{1}{2^2n^2(n+1)^2}\right)^{1/4}H_{n+1}(y),\end{aligned}$$

and after multiplying through by $[2n(n+1)]^{1/2}$ we get

$$2y(n+1)^{1/2}H_n(y) = 2n(n+1)^{1/2}H_{n-1}(y) + (n+1)^{1/2}H_{n+1}(y).$$

Cancelling the common factor $(n+1)^{1/2}$ and rearranging, we obtain the relation

$$\boxed{H_{n+1}(y) = 2yH_n(y) - 2nH_{n-1}(y)},$$

as we wanted. □

6-21 (Shankar, Exercise 7.5.4)

Thermodynamics of Oscillators.

Solution. (1) This follows because

$$\frac{\partial}{\partial \beta} \ln Z = \frac{1}{Z} \frac{\partial Z}{\partial \beta} = -\frac{1}{Z} \sum_i E(i) e^{-\beta E(i)} = -\sum_i E(i) P(i).$$

(2) We have

$$\begin{aligned} Z_{\text{cl}} &= \iint dx dp e^{-\beta E(x,p)} \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dx dp e^{-\frac{\beta p^2}{2m}} e^{-\frac{1}{2}\beta m\omega^2 x^2} \\ &= \left(\int_{-\infty}^{\infty} dx e^{-\frac{1}{2}\beta m\omega^2 x^2} \right) \left(\int_{-\infty}^{\infty} dp e^{-\frac{\beta p^2}{2m}} \right) \\ &= \frac{\pi}{\left(\frac{1}{2}\beta m\omega^2 \times \frac{\beta}{2m} \right)^{1/2}} \\ &= \frac{2\pi}{\omega\beta}, \end{aligned}$$

where we used the results of Appendix A.2. Then

$$\bar{E}_{\text{cl}} = -\frac{\partial}{\partial \beta} \ln Z = -\frac{1}{Z} \frac{\partial Z}{\partial \beta} = -\frac{\omega\beta}{2\pi} \left(-\frac{2\pi}{\omega\beta^2} \right) = \frac{1}{\beta} \equiv kT,$$

as we wanted.

(3) For the quantum oscillator, which has energy levels given by

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right), \quad n = 0, 1, 2, 3, \dots$$

we have

$$\begin{aligned} Z_{\text{qu}} &= \sum_{n=0}^{\infty} e^{-\beta E_n} \\ &= e^{-\beta\hbar\omega/2} \sum_{n=0}^{\infty} \left(e^{-\beta\hbar\omega} \right)^n \\ &= e^{-\beta\hbar\omega/2} (1 - e^{-\beta\hbar\omega})^{-1}, \end{aligned}$$

because $e^{-\beta\hbar\omega} < 1$ for all $\beta > 0$. Then

$$\begin{aligned}\bar{E}_{\text{qu}} &= -\frac{\partial}{\partial\beta} \ln Z \\ &= -\frac{1}{Z} \frac{\partial Z}{\partial\beta} \\ &= -e^{\beta\hbar\omega/2}(1 - e^{-\beta\hbar\omega}) \frac{\partial}{\partial\beta} \left[e^{-\beta\hbar\omega/2}(1 - e^{-\beta\hbar\omega})^{-1} \right],\end{aligned}$$

and since

$$\begin{aligned}\frac{\partial}{\partial\beta} \left[e^{-\beta\hbar\omega/2}(1 - e^{-\beta\hbar\omega})^{-1} \right] &= -\frac{\hbar\omega}{2} e^{-\beta\hbar\omega/2}(1 - e^{-\beta\hbar\omega})^{-1} + e^{-\beta\hbar\omega/2}(-1)(1 - e^{-\beta\hbar\omega})^{-2}(-1)(-\hbar\omega)e^{-\beta\hbar\omega} \\ &= -\frac{\hbar\omega}{2} e^{-\beta\hbar\omega/2}(1 - e^{-\beta\hbar\omega})^{-1} - \hbar\omega e^{-3\beta\hbar\omega/2}(1 - e^{-\beta\hbar\omega})^{-2} \\ &= -e^{-\beta\hbar\omega/2}(1 - e^{-\beta\hbar\omega})^{-1} \hbar\omega \left[\frac{1}{2} + e^{-\beta\hbar\omega}(1 - e^{-\beta\hbar\omega})^{-1} \right],\end{aligned}$$

it follows that

$$\bar{E}_{\text{qu}} = \hbar\omega \left(\frac{1}{2} + \frac{e^{-\beta\hbar\omega}}{1 - e^{-\beta\hbar\omega}} \right) = \hbar\omega \left(\frac{1}{2} + \frac{1}{e^{\beta\hbar\omega} - 1} \right),$$

as we wanted.

(4) If the temperature is large enough, so that $\beta\hbar\omega \ll 1$ (equivalently $T \gg \hbar\omega/k$), then

$$\bar{E}_{\text{qu}} = \hbar\omega \left(\frac{1}{2} + \frac{1}{[1 + \beta\hbar\omega + (\beta\hbar\omega)^2 + \dots] - 1} \right) \xrightarrow{T \rightarrow \infty} \hbar\omega \left(\frac{1}{\beta\hbar\omega} \right) = kT = \bar{E}_{\text{cl}},$$

as we expect.

(5) If we treat all N_0 atoms in the crystal as decoupled classical oscillators with 3 degrees of freedom, then the mean thermal energy of the crystal is

$$\bar{E}_{\text{crystal}} = 3N_0 \bar{E}_{\text{cl}} \implies \boxed{C_{\text{cl}}(T) = \frac{1}{N_0} \frac{\partial \bar{E}_{\text{crystal}}}{\partial T} = 3k}.$$

If we treat the atoms quantum mechanically, assuming that all the normal modes have the same frequency ω , then we get

$$\bar{E}_{\text{crystal}} = 3N_0 \bar{E}_{\text{qu}} = 3N_0 \hbar\omega \left(\frac{1}{2} + \frac{1}{e^{\theta_E/T} - 1} \right),$$

where we've defined the Einstein temperature $\theta_E = \hbar\omega/k$. Now

$$\frac{\partial \bar{E}_{\text{crystal}}}{\partial T} = 3N_0 \hbar\omega (-1) \frac{1}{(e^{\theta_E/T} - 1)^2} \left(-\frac{\theta_E}{T^2} e^{\theta_E/T} \right),$$

which shows that

$$C_{\text{qu}}(T) = 3k \left(\frac{\theta_E}{T} \right)^2 \frac{e^{\theta/T}}{(e^{\theta_E/T} - 1)^2},$$

as we wanted. The low-temperature behavior of this model is certainly more accurate than that predicted by the classical picture, but it still disagrees with experiment. A better model was proposed by Debye. $\square$

7 Path integrals

7-1 (Shankar, Exercise 8.6.1)

Verify that

$$U(x, t; x', 0) = A(t) \exp(iS_{\text{cl}}/\hbar), \quad A(t) = \left(\frac{m}{2\pi\hbar it} \right)^{1/2}$$

agrees with the exact result, Eq. (5.4.31), for $V(x) = -fx$.

Solution. This is the kind of exercise that “builds character” when done by hand instead of being delegated to a computer. (It isn’t very exciting, though.) We want to evaluate

$$S_{\text{cl}} = \int_0^t dt'' \mathcal{L}(x_{\text{cl}}, \dot{x}_{\text{cl}}),$$

given that

$$x_{\text{cl}}(t'') = x_0 + v_0 t'' + \frac{f(t'')^2}{2m},$$

$$\mathcal{L}(x, \dot{x}) = \frac{1}{2}m\dot{x}^2 + fx.$$

Substituting the classical path $x_{\text{cl}}(t'')$ into the Lagrangian and grouping like powers of t'' gives

$$\mathcal{L}(x_{\text{cl}}, \dot{x}_{\text{cl}}) = \frac{1}{2}mv_0^2 + fx_0 + 2v_0ft'' + \frac{f^2t''^2}{m}.$$

This is just a polynomial in t'' , so the integration for S_{cl} is trivial, and yields

$$S_{\text{cl}} = \left(\frac{1}{2}mv_0^2 + fx_0 \right) t + v_0ft^2 + \frac{f^2t^3}{3m}.$$

To simplify further, note that the constants x_0 and v_0 are fixed by the boundary conditions $x_{\text{cl}}(0) = x'$ and $x_{\text{cl}}(t) = x$. In particular,

$$x_0 = x',$$

$$v_0 = \frac{x - x'}{t} - \frac{ft}{2m}.$$

Then we look at each of the terms in our expression for S_{cl} :

$$\begin{aligned}\frac{1}{2}mv_0^2 + fx_0 &= \frac{1}{2}m \left[\frac{(x-x')^2}{t^2} - \frac{f(x-x')}{m} + \frac{f^2 t^2}{4m^2} \right] + fx' \\ \Rightarrow \left(\frac{1}{2}mv_0^2 + fx_0 \right) t &= \frac{m(x-x')^2}{2t} - \frac{1}{2}ftx + \frac{3}{2}ftx' + \frac{1}{8} \frac{f^2 t^3}{m}, \\ v_0 f t^2 &= \left[\frac{x-x'}{t} - \frac{ft}{2m} \right] f t^2 \\ &= ftx - ftx' - \frac{f^2 t^3}{2m}.\end{aligned}$$

Upon combining everything, we get

$$S_{\text{cl}} = \frac{m(x-x')^2}{2t} + \frac{1}{2}ft(x+x') - \frac{1}{24} \frac{f^2 t^3}{m},$$

as we wanted to show. □

7-2 (Shankar, Exercise 8.6.2)

Show that for the harmonic oscillator with

$$\begin{aligned}\mathcal{L} &= \frac{1}{2}m\dot{x}^2 - \frac{1}{2}m\omega^2 x^2 \\ U(x, t; x') &= A(t) \exp \left\{ \frac{im\omega}{2\hbar \sin \omega t} [(x^2 + x'^2) \cos \omega t - 2xx'] \right\}\end{aligned}$$

where $A(t)$ is an unknown function. (Recall Exercise 2.8.7.)

Solution. The result follows from (8.6.12),

$$U(x, t; x') = e^{iS_{\text{cl}}/\hbar} A(t),$$

and the classical action S_{cl} for the harmonic oscillator, which was computed in Exercise 2.8.7, where we identify $x_1 \rightarrow x'$, $x_2 \rightarrow x$, and $T \rightarrow t$. *The unknown function $A(t)$ can be determined by working with its Fourier expansion; see the P241 lecture notes.* □

8 Uncertainty relations

8-1 (Sakurai, Problem 1.22)

Estimate the rough order of magnitude of the length of time that an ice pick can be balanced on its point if the only limitation is that set by the Heisenberg uncertainty principle. Assume that the point is sharp and that the point and the surface on which it rests is hard. You may make approximations which do not alter the general order of magnitude of the result. Assume reasonable values for the dimensions and weight of the ice pick. Obtain an approximate numerical result and express it *in seconds*.

Solution. Also Problem 3.13 in Morin. I had my own solution, and then peeked at Morin's and thought it was a lot cleaner, so this is adapted from there. It's quite remarkable how much harder dimensional analysis is when you give your expressions nondescript labels – in my case, α for a quantity with units of inverse time. An ice pick may be modeled as a point mass m at the end of a rigid, massless rod of length ℓ . We will assume that $m = 0.1$ kg and $\ell = 0.1$ m. From Newton's second law, the equation of motion for the angle θ the rod makes with the vertical is

$$mg \sin \theta = m\ell \ddot{\theta}.$$

We assume that θ is small, so that $\sin \theta \approx \theta$. Then we have

$$\ddot{\theta} = \frac{1}{\tau^2} \theta,$$

where $\tau^2 \equiv \ell/g$. The solution to this equation is

$$\theta(t) = \frac{1}{2}(\theta_0 + \tau\omega_0)e^{t/\tau} + \frac{1}{2}(\theta_0 - \tau\omega_0)e^{-t/\tau},$$

where θ_0 and ω_0 are the initial angle and angular velocity, respectively. Then since $\tau = \sqrt{\ell/g} \approx 0.10$ s, the second term is rapidly decaying and can be neglected:

$$\theta(t) = \frac{1}{2}(\theta_0 + \tau\omega_0)e^{t/\tau}.$$

Now, regardless of how carefully we try to balance the ice pick on its point, there will be a bound on how certain we can be about both the position and momentum of the pick. More quantitatively, since $\Delta x \Delta p \geq \hbar/2$, we have that

$$(\ell\theta_0)(m\ell\omega_0) \geq \hbar.$$

We've dropped the factor of 1/2, which doesn't change the order of magnitude of the result. Thus

$$\theta(t) \geq \frac{1}{2} \left(\theta_0 + \frac{\hbar\tau}{m\ell^2\theta_0} \right) e^{t/\tau}.$$

Notice that we cannot make the coefficient of $e^{t/\tau}$ arbitrarily small by setting $\theta_0 \rightarrow 0$; a small

uncertainty in θ_0 implies a large uncertainty in ω_0 , and vice versa. However, we can minimize this with respect to θ_0 , and a short calculation shows that this minimum value is attained when

$$\theta_0^2 = \frac{\hbar}{m\ell^2} \sqrt{\frac{\ell}{g}}.$$

This yields

$$\theta(t) \geq \sqrt{\frac{\hbar}{m\ell^2}} \left(\frac{\ell}{g}\right)^{1/4} e^{t/\tau}.$$

Setting $\theta(t)$ to unity and solving, we obtain

$$t \leq \frac{1}{4} \sqrt{\frac{\ell}{g}} \ln\left(\frac{m^2 \ell^3 g}{\hbar^2}\right).$$

For the values $m = 0.1 \text{ kg}$, $\ell = 0.1 \text{ m}$, $g = 9.81 \text{ m s}^{-2}$, and $\hbar = 1.06 \times 10^{-34} \text{ J s}$, this gives an upper bound of 3.72 seconds for the time it takes $\theta(t)$ to become of order unity. In other words, you cannot balance an ice pick on its tip for longer than around 3.7 seconds. □

8-2 (Shankar, Exercise 9.4.1)

Consider the oscillator in the state $|n = 1\rangle$ and verify that

$$\left\langle \frac{1}{X^2} \right\rangle \approx \frac{1}{\langle X^2 \rangle} \approx \frac{m\omega}{\hbar}$$

Solution. In the state $|n = 1\rangle$, we have

$$\begin{aligned} \left\langle \frac{1}{X^2} \right\rangle &= \int_{-\infty}^{\infty} \frac{1}{x^2} |\langle x|1\rangle|^2 dx \\ &= \int_{-\infty}^{\infty} \frac{1}{x^2} \left(\frac{m\omega}{\pi\hbar^2} \right)^{1/2} \exp\left(-\frac{m\omega x^2}{\hbar}\right) \left[2\left(\frac{m\omega}{\hbar}\right)^{1/2} x \right]^2 dx \\ &= 2\left(\frac{m\omega}{\hbar}\right) \frac{1}{\pi^{1/2}} \int_{-\infty}^{\infty} \frac{1}{x^2} \exp\left(-\frac{m\omega x^2}{\hbar}\right) x^2 dx \\ &= 2\left(\frac{m\omega}{\hbar}\right)^{3/2} \frac{1}{\pi^{1/2}} \int_{-\infty}^{\infty} \left(\frac{\hbar}{m\omega}\right)^{1/2} \exp(-y^2) dy \\ &= \frac{2m\omega}{\hbar} \frac{1}{\sqrt{\pi}} \sqrt{\pi} \\ &= \frac{2m\omega}{\hbar}, \end{aligned}$$

where we used (7.3.22) for the oscillator eigenstates in the second line, and made the substitution $y = (m\omega/\hbar)^{1/2}x$ in the fourth line. Then since (see Exercise 7.4.2)

$$\langle X^2 \rangle = \frac{\hbar}{2m\omega}(2n+1),$$

it follows that for $|n = 1\rangle$, we have

$$\frac{1}{\langle X^2 \rangle} = \frac{2m\omega}{3\hbar} \approx \frac{2m\omega}{\hbar} = \left\langle \frac{1}{X^2} \right\rangle;$$

the two expectation values differ by a factor of order unity. □

8-3 (Shankar, Exercise 9.4.2)

(1) By referring to the table of integrals in Appendix A.2, verify that

$$\psi = \frac{1}{(\pi a_0^3)^{1/2}} e^{-r/a_0}, \quad r = (x^2 + y^2 + z^2)^{1/2}$$

is a normalized wave function (of the ground state of hydrogen). Note that in three dimensions, the normalization condition is

$$\begin{aligned}\langle\psi|\psi\rangle &= \int \psi^*(r, \theta, \phi) \psi(r, \theta, \phi) r^2 dr d(\cos \theta) d\phi \\ &= 4\pi \int \psi^*(r) \psi(r) r^2 dr = 1\end{aligned}$$

for a function of just r .

- (2) Calculate $(\Delta X)^2$ in this state [argue that $(\Delta X)^2 = \frac{1}{3}\langle r^2 \rangle$] and regain the result quoted in Eq. (9.4.9).
 (3) Show that $\langle 1/r \rangle \approx 1/\langle r \rangle \approx me^2/\hbar^2$ in this state.

Solution. (1) We compute

$$\begin{aligned}\langle\psi|\psi\rangle &= 4\pi \int \psi^*(r) \psi(r) r^2 dr \\ &= 4\pi \int_0^\infty \frac{1}{\pi a_0^3} e^{-2r/a_0} r^2 dr \\ &= \frac{4}{a_0^3} \int_0^\infty r^2 e^{-2r/a_0} dr.\end{aligned}$$

From Appendix A.2, we have

$$\int_0^\infty r^n e^{-\alpha r} dr = \frac{n!}{\alpha^{n+1}},$$

and applying this result with $n = 2$ and $\alpha = 2/a_0$ gives

$$\langle\psi|\psi\rangle = \frac{4}{a_0^3} \frac{2!}{(2/a_0)^3} = \frac{4}{a_0^3} \times \frac{2a_0^3}{8} = 1,$$

which shows that ψ is normalized, as we wanted.

- (2) We have

$$\langle r^2 \rangle = \langle X^2 + Y^2 + Z^2 \rangle = \langle X^2 \rangle + \langle Y^2 \rangle + \langle Z^2 \rangle,$$

and in the ground state $\langle X^2 \rangle = (\Delta X)^2 + \langle X \rangle^2$ (with similar expressions for $\langle Y^2 \rangle$ and $\langle Z^2 \rangle$) because $\langle X \rangle = \langle Y \rangle = \langle Z \rangle = 0$ due to spherical symmetry. Finally, we have $(\Delta X)^2 = (\Delta Y)^2 = (\Delta Z)^2$, also due to the symmetry. [Can this argument be extended to all excited states?](#) Thus

$$\langle r^2 \rangle = 3(\Delta X)^2,$$

which is what wanted to show first. Then

$$\begin{aligned}
(\Delta X)^2 &= \frac{1}{3} \langle r^2 \rangle \\
&= \frac{4\pi}{3} \int_0^\infty \psi^*(r) \psi(r) r^4 dr \\
&= \frac{4\pi}{3} \int_0^\infty \frac{1}{\pi a_0^3} e^{-2r/a_0} r^4 dr \\
&= \frac{4}{3a_0^3} \int_0^\infty r^4 e^{-2r/a_0} dr \\
&= \frac{4}{3a_0^3} \times \frac{4!}{(2/a_0)^5} \\
&= a_0^2,
\end{aligned}$$

where we used the result from Appendix A.2. Thus

$$\Delta X = a_0 = \frac{\hbar^2}{me^2},$$

which reproduces Eq. (9.4.9).

(3) We calculate and then compare both of these expectation values:

$$\begin{aligned}
\left\langle \frac{1}{r} \right\rangle &= 4\pi \int_0^\infty \frac{1}{r} \frac{1}{\pi a_0^3} e^{-2r/a_0} r^2 dr \\
&= \frac{4}{a_0^3} \int_0^\infty r e^{-2r/a_0} dr \\
&= \frac{4}{a_0^3} \times \frac{1!}{(2/a_0)^2} \\
&= \frac{1}{a_0},
\end{aligned}$$

$$\begin{aligned}
\langle r \rangle &= 4\pi \int_0^\infty \frac{1}{\pi a_0^3} e^{-2r/a_0} r^3 dr \\
&= \frac{4}{a_0^3} \times \frac{3!}{(2/a_0)^4} \\
&= \frac{4 \times 3 \times 2}{16} a_0 \\
&= \frac{3}{2} a_0.
\end{aligned}$$

Thus

$$\left\langle \frac{1}{r} \right\rangle = \frac{1}{a_0} \approx \frac{2}{3a_0} = \frac{1}{\langle r \rangle},$$

as we wanted to show. The point is that these rough approximations get within an order of magnitude of the exact result. $\square$

8-4 (Shankar, Exercise 9.4.3)

Ignore the fact that the hydrogen atom is a three-dimensional system and pretend that

$$H = \frac{P^2}{2m} - \frac{e^2}{(R^2)^{1/2}} \quad (P^2 = P_x^2 + P_y^2 + P_z^2, R^2 = X^2 + Y^2 + Z^2)$$

corresponds to a one-dimensional problem. Assuming

$$\Delta P \cdot \Delta R \geq \hbar/2$$

estimate the ground-state energy.

Solution. We begin from the uncertainty relation:

$$\Delta P \cdot \Delta R \geq \frac{\hbar}{2} \quad \implies \quad (\Delta P)^2 \geq \frac{\hbar^2}{4(\Delta R)^2},$$

so that we have

$$\begin{aligned} \langle H \rangle &= \frac{\langle P^2 \rangle}{2m} - e^2 \left\langle \frac{1}{(R^2)^{1/2}} \right\rangle \\ &= \frac{(\Delta P)^2}{2m} - e^2 \left\langle \frac{1}{(R^2)^{1/2}} \right\rangle \\ &\approx \frac{(\Delta P)^2}{2m} - \frac{e^2}{\langle (R^2)^{1/2} \rangle} \\ &= \frac{(\Delta P)^2}{2m} - \frac{e^2}{[(\Delta R)^2]^{1/2}} \\ &\geq \frac{\hbar^2}{8m(\Delta R)^2} - \frac{e^2}{[(\Delta R)^2]^{1/2}}, \end{aligned}$$

where we assume that the ground state has $\langle P \rangle = 0$ and $\langle R \rangle = 0$ in lines 2 and 4, we approximate the expectation value $\langle 1/(R^2)^{1/2} \rangle$ in the third line, and we apply our bound from the uncertainty relation in the last line.

Differentiating $\langle H \rangle$ with respect to ΔR to find the minimum yields

$$\begin{aligned} -\frac{2\hbar^2}{8m(\Delta R)^3} + \frac{e^2}{(\Delta R)^2} &= 0, \\ -\frac{\hbar^2}{4m} + e^2(\Delta R) &= 0, \\ \Delta R &= \frac{\hbar^2}{4me^2}. \end{aligned}$$

Thus, we have

$$\begin{aligned}\langle H \rangle &\gtrsim \frac{\hbar^2}{8m \left(\frac{\hbar^4}{16m^2 e^4} \right)} - \frac{e^2}{\hbar^2} 4me^2 \\ &= \frac{2me^4}{\hbar^2} - \frac{4me^4}{\hbar^2} \\ &= -\frac{2me^4}{\hbar^2}.\end{aligned}$$

This gives us our estimate for the ground-state energy:

$$\boxed{\langle H \rangle \approx -\frac{2me^4}{\hbar^2}}.$$

□

8-5 (Shankar, Exercise 9.4.4)

Compute $\Delta T \cdot \Delta X$, where $T = P^2/2m$. Why is this relation not so famous?

Solution. We start from the bound (9.2.12),

$$(\Delta\Omega)^2(\Delta\Lambda)^2 \geq \frac{1}{4}\langle\psi|\{\Omega, \Lambda\}|\psi\rangle^2 + \frac{1}{4}\langle\psi|\Gamma|\psi\rangle^2,$$

where Ω and Λ are Hermitian operators and $[\Omega, \Lambda] = i\Gamma$. Since

$$\begin{aligned}[T, X] &= \left[\frac{P^2}{2m}, X \right] \\ &= \frac{1}{2m} [P^2, X] \\ &= \frac{1}{2m} (P[P, X] + [P, X]P) \\ &= i \left(-\frac{\hbar}{m} P \right),\end{aligned}$$

we can make the identification $\Omega = T$, $\Lambda = X$, and $\Gamma = -\hbar P/m$ to get

$$(\Delta T)^2(\Delta X)^2 \geq \frac{1}{4}\langle\psi|\{T, X\}|\psi\rangle^2 + \frac{\hbar^2}{4m^2}\langle P \rangle^2.$$

The first term on the right is positive definite, so it follows that for any state ψ , we have

$$\Delta T \Delta X \geq \frac{\hbar}{2m} \langle P \rangle.$$

But this relation isn't very interesting, because – unlike the Heisenberg uncertainty relation – it doesn't provide a strong lower bound on the product $\Delta T \Delta X$. For states with $\langle P \rangle = 0$, this relation says that $\Delta T \Delta X \geq 0$, which provides no information. [The upshot is that these uncertainty relations are most interesting for canonically conjugate operators.](#) □

9 Systems with N degrees of freedom

9-1 (Shankar, Exercise 10.1.1)

Solution. (1) Consider the action of this commutator on an arbitrary direct product ket:

$$\begin{aligned} & \left[\Omega_1^{(1)} \otimes I^{(2)}, I^{(1)} \otimes \Lambda_2^{(2)} \right] (|\alpha_1\rangle \otimes |\alpha_2\rangle) \\ &= \left(\left(\Omega_1^{(1)} \otimes I^{(2)} \right) \left(I^{(1)} \otimes \Lambda_2^{(2)} \right) - \left(I^{(1)} \otimes \Lambda_2^{(2)} \right) \left(\Omega_1^{(1)} \otimes I^{(2)} \right) \right) (|\alpha_1\rangle \otimes |\alpha_2\rangle) \end{aligned}$$

Now consider both terms. The first term is

$$\begin{aligned} & \left(\Omega_1^{(1)} \otimes I^{(2)} \right) \left(I^{(1)} \otimes \Lambda_2^{(2)} \right) (|\alpha_1\rangle \otimes |\alpha_2\rangle) = \left(\Omega_1^{(1)} \otimes I^{(2)} \right) (|I^{(1)}\alpha_1\rangle \otimes |\Lambda_2^{(2)}\alpha_2\rangle) \\ &= |\Omega_1^{(1)}I^{(1)}\alpha_1\rangle \otimes |I^{(2)}\Lambda_2^{(2)}\alpha_2\rangle \\ &= |\Omega_1^{(1)}\alpha_1\rangle \otimes |\Lambda_2^{(2)}\alpha_2\rangle, \end{aligned}$$

where we've applied (10.1.14) twice, and the last line follows because I is the identity operator. By the same reasoning, the second term is

$$\begin{aligned} & \left(I^{(1)} \otimes \Lambda_2^{(2)} \right) \left(\Omega_1^{(1)} \otimes I^{(2)} \right) (|\alpha_1\rangle \otimes |\alpha_2\rangle) = \left(I^{(1)} \otimes \Lambda_2^{(2)} \right) (|\Omega_1^{(1)}\alpha_1\rangle \otimes |I^{(2)}\alpha_2\rangle) \\ &= |I^{(1)}\Omega_1^{(1)}\alpha_1\rangle \otimes |\Lambda_2^{(2)}I^{(2)}\alpha_2\rangle \\ &= |\Omega_1^{(1)}\alpha_1\rangle \otimes |\Lambda_2^{(2)}\alpha_2\rangle, \end{aligned}$$

which shows that the commutator sends $|\alpha_1\rangle \otimes |\alpha_2\rangle$ to zero. Since you can construct a basis for $V_{1\otimes 2}$ out of direct product kets [this is the content of (10.1.22b)], it follows that the commutator sends all kets $|\psi\rangle \in V_{1\otimes 2}$ to zero; that is, the commutator is zero.

(2) Consider an arbitrary direct product ket and apply (10.1.14) in succession:

$$\begin{aligned} & \left(\Omega_1^{(1)} \otimes \Gamma_2^{(2)} \right) \left(\theta_1^{(1)} \otimes \Lambda_2^{(2)} \right) (|\alpha_1\rangle \otimes |\alpha_2\rangle) = \left(\Omega_1^{(1)} \otimes \Gamma_2^{(2)} \right) (|\theta_1^{(1)}\alpha_1\rangle \otimes |\Lambda_2^{(2)}\alpha_2\rangle) \\ &= |\Omega_1^{(1)}\theta_1^{(1)}\alpha_1\rangle \otimes |\Gamma_2^{(2)}\Lambda_2^{(2)}\alpha_2\rangle \\ &= (\Omega\theta)_1^{(1)} \otimes (\Gamma\Lambda)_2^{(2)} (|\alpha_1\rangle \otimes |\alpha_2\rangle). \end{aligned}$$

This proves the operator equality.

(3) We have

$$\begin{aligned}
\left[\Omega_1^{(1) \otimes (2)}, \Lambda_1^{(1) \otimes (2)} \right] &= \Omega_1^{(1) \otimes (2)} \Lambda_1^{(1) \otimes (2)} - \Lambda_1^{(1) \otimes (2)} \Omega_1^{(1) \otimes (2)} \\
&= (\Omega_1^{(1)} \otimes I^{(2)}) (\Lambda_1^{(1)} \otimes I^{(2)}) - (\Lambda_1^{(1)} \otimes I^{(2)}) (\Omega_1^{(1)} \otimes I^{(2)}) \\
&= (\Omega \Lambda)_1^{(1)} \otimes I^{(2)} - (\Lambda \Omega)_1^{(1)} \otimes I^{(2)} \\
&= [\Omega_1^{(1)}, \Lambda_1^{(1)}] \otimes I^{(2)} \\
&= \Gamma_1^{(1)} \otimes I^{(2)},
\end{aligned}$$

where in the third line we used the result of (2), and in the last line we use the given commutator. The same reasoning applies with $1 \rightarrow 2$ in the subscripts.

(4) Another direct calculation:

$$\begin{aligned}
\left(\Omega_1^{(1) \otimes (2)} + \Omega_2^{(1) \otimes (2)} \right)^2 &= \Omega_1^{(1) \otimes (2)} \Omega_1^{(1) \otimes (2)} + \Omega_1^{(1) \otimes (2)} \Omega_2^{(1) \otimes (2)} + \Omega_2^{(1) \otimes (2)} \Omega_1^{(1) \otimes (2)} + \Omega_2^{(1) \otimes (2)} \Omega_2^{(1) \otimes (2)} \\
&= (\Omega_1^2)^{(1)} \otimes I^{(2)} + I^{(1)} \otimes (\Omega_2^2)^{(2)} + \Omega_1^{(1) \otimes (2)} \Omega_2^{(1) \otimes (2)} + \Omega_2^{(1) \otimes (2)} \Omega_1^{(1) \otimes (2)} \\
&= (\Omega_1^2)^{(1)} \otimes I^{(2)} + I^{(1)} \otimes (\Omega_2^2)^{(2)} + 2\Omega_1^{(1)} \otimes \Omega_2^{(2)},
\end{aligned}$$

where we used (2) in the second line and (1) in the last line. □

9-2 (Shankar, Exercise 10.1.2)

Imagine a fictitious world in which the single-particle Hilbert space is two-dimensional. Let us denote the basis vectors by $|+\rangle$ and $|-\rangle$. Let

$$\sigma_1^{(1)} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad \text{and} \quad \sigma_2^{(2)} = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$$

be operators in $\mathbb{V}_1$ and $\mathbb{V}_2$, respectively, with the basis ordering $\{|+\rangle, |-\rangle\}$, so that $b = \langle + | \sigma_1^{(1)} | - \rangle$, etc. The space $\mathbb{V}_1 \otimes \mathbb{V}_2$ is spanned by four vectors $|+\rangle \otimes |+\rangle$, $|+\rangle \otimes |-\rangle$, $|-\rangle \otimes |+\rangle$, and $|-\rangle \otimes |-\rangle$.

Solution. Handwritten solution attached below. □

Exercise 10.1.2

$$\sigma_i^{(1)} \otimes I^{(2)} = \sigma_i^{(1)} \otimes I^{(2)}$$

By orthonormality, it is clear that the off-diagonal entries are zero:

$$\sigma_i^{(1)} \otimes I^{(2)} = \begin{matrix} & \begin{matrix} ++ & +- & -+ & -- \end{matrix} \\ \begin{matrix} ++ \\ +- \\ -+ \\ -- \end{matrix} & \begin{bmatrix} & 0 & & 0 \\ 0 & & 0 & \\ & 0 & & 0 \\ 0 & & 0 & \end{bmatrix} \end{matrix}$$

this means only 8 (at most) nonzero matrix elements. But by symm. some of these are the same.

then automatically we can read — since

$$\sigma_1^{(1)} = \begin{matrix} + & - \\ \left[\begin{array}{cc} a & b \\ c & d \end{array} \right] \end{matrix}$$

that we have

$$\sigma_1^{(1) \otimes (2)} = \begin{matrix} & ++ & +- & -+ & -- \\ \begin{matrix} ++ \\ +- \\ -+ \\ -- \end{matrix} & \left[\begin{array}{cccc} a & 0 & b & 0 \\ b & a & 0 & b \\ c & 0 & d & 0 \\ 0 & c & 0 & d \end{array} \right] \end{matrix}$$

we also have, first identifying zeros,

$$\sigma_2^{(1) \otimes (2)} = \begin{matrix} & ++ & +- & -+ & -- \\ \begin{matrix} ++ \\ +- \\ -+ \\ -- \end{matrix} & \left[\begin{array}{cccc} & & 0 & 0 \\ & & 0 & 0 \\ 0 & 0 & & \\ 0 & 0 & & \end{array} \right] \end{matrix}$$

and since

$$\sigma_2^{(2)} = \begin{matrix} + & - \\ \left[\begin{array}{cc} e & f \\ g & h \end{array} \right] \end{matrix}$$

we get

$$\sigma_2^{(1) \otimes (2)} = \left[\begin{array}{cccc} e & f & 0 & 0 \\ g & h & 0 & 0 \\ 0 & 0 & e & f \\ 0 & 0 & g & h \end{array} \right]$$

Now we do part (3) in two ways. The first
direct matrix mult:

$$(\sigma_1 \sigma_2)^{(1) \otimes (2)} = \sigma_1^{(1)} \otimes \sigma_2^{(2)}$$

$$= \begin{bmatrix} a & 0 & b & 0 \\ 0 & a & 0 & b \\ c & 0 & d & 0 \\ 0 & c & 0 & d \end{bmatrix} \begin{bmatrix} e & f & 0 & 0 \\ g & h & 0 & 0 \\ 0 & 0 & e & f \\ 0 & 0 & g & h \end{bmatrix}$$

$$= \begin{bmatrix} ae & af & be & bf \\ ag & ah & bg & bh \\ ce & cf & de & df \\ cg & ch & dg & dh \end{bmatrix}$$

we can also compute matrix el's directly.

$$\langle ++ | (\sigma_1 \sigma_2)^{(1) \otimes (2)} | ++ \rangle = \langle + | \sigma_1^{(1)} | + \rangle \langle + | \sigma_2^{(2)} | + \rangle$$

$$= ae$$

$$\langle +- | \dots | ++ \rangle = \cancel{ag} \quad ag$$

$$\langle ++ | \dots | +- \rangle = \cancel{af} \quad af$$

$$\langle +- | \dots | +- \rangle = ah$$

and so on. For this example, either
is easy enough, but the direct method
~~seems less error-prone~~ easier to do.

9-3 (Shankar, Exercise 10.1.3)

Consider the Hamiltonian of the coupled mass system:

$$\mathcal{H} = \frac{p_1^2}{2m} + \frac{p_2^2}{2m} + \frac{1}{2} m\omega^2 [x_1^2 + x_2^2 + (x_1 - x_2)^2]$$

We know from Example 1.8.6 that $\mathcal{H}$ can be decoupled if we use normal coordinates

$$x_{\text{I,II}} = \frac{x_1 \pm x_2}{2^{1/2}}$$

and the corresponding momenta

$$p_{\text{I,II}} = \frac{p_1 \pm p_2}{2^{1/2}}$$

(1) Rewrite $\mathcal{H}$ in terms of normal coordinates. Verify that the normal coordinates are also canonical, i.e., that

$$\{x_i, p_j\} = \delta_{ij} \text{ etc.}; \quad i, j = \text{I, II}$$

Now quantize the system, promoting these variables to operators obeying

$$[X_i, P_j] = i\hbar\delta_{ij} \text{ etc.}; \quad i, j = \text{I, II}$$

Write the eigenvalue equation for H in the simultaneous eigenbasis of X_{I} and X_{II} .

(2) Quantize the system directly, by promoting x_1 , x_2 , p_1 , and p_2 to quantum operators. Write the eigenvalue equation for H in the simultaneous eigenbasis of X_1 and X_2 . Now change from x_1 , x_2 (and of course $\partial/\partial x_1$, $\partial/\partial x_2$) to x_{I} , x_{II} , (and $\partial/\partial x_{\text{I}}$, $\partial/\partial x_{\text{II}}$) *in the differential equation*. You should end up with the result from part (1).

Solution. (1) From the definitions, we compute

$$x_1 = \frac{1}{\sqrt{2}}(x_{\text{I}} + x_{\text{II}}),$$

$$x_2 = \frac{1}{\sqrt{2}}(x_{\text{I}} - x_{\text{II}}),$$

and similar expressions hold for p_1 and p_2 . Direct substitution then gives

$$\mathcal{H} = \frac{p_{\text{I}}^2}{2m} + \frac{p_{\text{II}}^2}{2m} + \frac{1}{2} m\omega^2 [x_{\text{I}}^2 + 3x_{\text{II}}^2].$$

It is straightforward to check that these variables are canonical. The eigenvalue equation for H in the simultaneous eigenbasis of X_{I} and X_{II} is then

$$\left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x_{\text{I}}^2} - \frac{\hbar^2}{2m} \frac{\partial^2}{\partial x_{\text{II}}^2} + \frac{1}{2} m\omega^2 [x_{\text{I}}^2 + 3x_{\text{II}}^2] \right) \psi_E(x_{\text{I}}, x_{\text{II}}) = E\psi_E(x_{\text{I}}, x_{\text{II}})$$

where $\psi_E(x_{\text{I}}, x_{\text{II}}) = \langle x_{\text{I}} x_{\text{II}} | E \rangle$.

(2) We can also quantize directly:

$$\begin{aligned} H &= \frac{P_1^2}{2m} + \frac{P_2^2}{2m} + \frac{1}{2}m\omega^2[X_1^2 + X_2^2 + (X_1^2 - X_1X_2 - X_2X_1 + X_2^2)] \\ &= \frac{P_1^2}{2m} + \frac{P_2^2}{2m} + \frac{1}{2}m\omega^2[2X_1^2 + 2X_2^2 - X_1X_2 - X_2X_1], \end{aligned}$$

where we've been careful to promote $2x_1x_2 \rightarrow X_1X_2 + X_2X_1$. Using the chain rule,

$$\begin{aligned} \frac{\partial}{\partial x_1} &= \frac{\partial x_I}{\partial x_1} + \frac{\partial x_{II}}{\partial x_1} \frac{\partial}{\partial x_{II}} \\ &= \frac{1}{\sqrt{2}} \left(\frac{\partial}{\partial x_I} + \frac{\partial}{\partial x_{II}} \right), \\ \frac{\partial^2}{\partial x_1^2} &= \frac{1}{2} \left(\frac{\partial^2}{\partial x_I^2} + 2 \frac{\partial^2}{\partial x_I \partial x_{II}} + \frac{\partial^2}{\partial x_{II}^2} \right), \\ \frac{\partial}{\partial x_2} &= \frac{1}{\sqrt{2}} \left(\frac{\partial}{\partial x_I} - \frac{\partial}{\partial x_{II}} \right), \\ \frac{\partial^2}{\partial x_2^2} &= \frac{1}{2} \left(\frac{\partial^2}{\partial x_I^2} - 2 \frac{\partial^2}{\partial x_I \partial x_{II}} + \frac{\partial^2}{\partial x_{II}^2} \right), \end{aligned}$$

which shows that when P_1^2 and P_2^2 are expressed in the simultaneous eigenbasis of X_I and X_{II} , the mixed partial derivatives cancel out, and we recover the same differential operators as (1). For the part in square brackets, we have

$$\begin{aligned} 2X_1^2 + 2X_2^2 - X_1X_2 - X_2X_1 &\rightarrow 2x_1^2 + 2x_2^2 - 2x_1x_2 \\ &= x_I^2 + 3x_{II}^2 \end{aligned}$$

by direct substitution, which shows that we recover the same eigenvalue equation as (1). The reason we can either perform the canonical transformation first, and then quantize, or quantize first and then perform the corresponding *unitary* transformation, is discussed in Chapter 11. $\square$

9-4 (Shankar, Exercise 10.2.1)

Particle in a Three-Dimensional Box. Consider a particle confined in a three-dimensional cubic box of volume L^3 . Choosing as the origin one of the corners, and the x , y , and z axes along the three edges meeting there, show that the normalized energy eigenfunctions are

$$\psi_E(x, y, z) = \left(\frac{2}{L}\right)^{1/2} \sin\left(\frac{n_x\pi x}{L}\right) \left(\frac{2}{L}\right)^{1/2} \sin\left(\frac{n_y\pi y}{L}\right) \left(\frac{2}{L}\right)^{1/2} \sin\left(\frac{n_z\pi z}{L}\right)$$

where

$$E = \frac{\hbar^2\pi^2}{2mL^2}(n_x^2 + n_y^2 + n_z^2)$$

and n_i are positive integers.

Solution. This is a separable Hamiltonian:

$$H = H(X, P_x) + H(Y, P_y) + H(Z, P_z).$$

It follows [see discussion preceding (10.1.28c)] that the eigenstates take the form

$$|E\rangle = |E_x\rangle \otimes |E_y\rangle \otimes |E_z\rangle,$$

with $E = E_x + E_y + E_z$, and the result we want follows because all sides have the same length. (The most general result has L_x, L_y, L_z not necessarily equal.) $\square$

9-5 (Shankar, Exercise 10.2.2)

Quantize the two-dimensional oscillator for which

$$\mathcal{H} = \frac{p_x^2 + p_y^2}{2m} + \frac{1}{2}m\omega_x^2 x^2 + \frac{1}{2}m\omega_y^2 y^2$$

(1) Show that the allowed energies are

$$E = (n_x + 1/2)\hbar\omega_x + (n_y + 1/2)\hbar\omega_y, \quad n_x, n_y = 0, 1, 2, \dots$$

(2) Write down the corresponding wave functions in terms of single oscillator wave functions. Verify that they have definite parity (even/odd) number $x \rightarrow -x, y \rightarrow -y$ and that the parity depends only on $n = n_x + n_y$.

(3) Consider next the *isotropic* oscillator ($\omega_x = \omega_y$). Write *explicit*, normalized eigenfunctions of the first three *states* (that is, for the cases $n = 0$ and 1). Reexpress your results in terms of polar coordinates ρ and ϕ (for later use). Show that the degeneracy of a level with $E = (n + 1)\hbar\omega$ is $(n + 1)$.

Solution. (1) The quantized Hamiltonian is

$$H = \frac{P_x^2 + P_y^2}{2m} + \frac{1}{2}m\omega_x^2 X^2 + \frac{1}{2}m\omega_y^2 Y^2,$$

which is separable, so the eigenstates take the form

$$|E\rangle = |E_x\rangle \otimes |E_y\rangle,$$

with energies

$$\begin{aligned} E &= E_x + E_y \\ &= (n_x + \tfrac{1}{2})\hbar\omega_x + (n_y + \tfrac{1}{2})\hbar\omega_y, \quad n_x, n_y = 0, 1, 2, \dots \end{aligned}$$

as we wanted. Note that in general there is no degeneracy unless ω_x/ω_y is rational. We'll see this for the isotropic case $\omega_x = \omega_y$.

(2) Using the result of (1), we have

$$\begin{aligned}\psi_E(x, y) &= \langle xy|E\rangle \\ &= \langle x|E_x\rangle\langle y|E_y\rangle.\end{aligned}$$

The wave function $\psi_E(x, y)$ has definite parity under $x \rightarrow -x$, $y \rightarrow -y$ because the single-particle wave functions do. Note further that if n_x and n_y have the same parity, then the sign of the product $\langle x|E_x\rangle\langle y|E_y\rangle$ is unchanged by the transformation, i.e., $\psi_E(x, y)$ has even parity. Similarly, if n_x and n_y have different parity, then $\psi_E(x, y)$ has odd parity.

But if n_x, n_y have the same parity, then $n = n_x + n_y$ is even, and if n_x, n_y have differing parity, then $n = n_x + n_y$ is odd. Therefore,

ψ_E has definite parity, and is even (odd) if and only if $n = n_x + n_y$ is even (odd).

(3) For the isotropic oscillator $\omega_x = \omega_y \equiv \omega$, we use the result of (2) and Eq. (7.3.22) to write

$$\begin{aligned}\psi_E(x, y) &= \left(\frac{m\omega}{\pi\hbar 2^{n_x}(n_x!)^2}\right)^{1/4} \left(\frac{m\omega}{\pi\hbar 2^{n_y}(n_y!)^2}\right)^{1/4} \exp\left(-\frac{m\omega}{2\hbar}(x^2 + y^2)\right) \\ &\quad \times H_{n_x}\left[\left(\frac{m\omega}{\hbar}\right)^{1/2} x\right] H_{n_y}\left[\left(\frac{m\omega}{\hbar}\right)^{1/2} y\right].\end{aligned}$$

The energies then take the form

$$E = (n_x + n_y + 1)\hbar\omega \equiv (n + 1)\hbar\omega, \quad n_x, n_y = 0, 1, 2, \dots$$

The first three states, which correspond to the cases $n = 0$ and $n = 1$, are

$$\begin{aligned}\langle xy|n_x = 0, n_y = 0\rangle &= \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} \exp\left(-\frac{m\omega}{2\hbar}(x^2 + y^2)\right), \\ \langle xy|n_x = 1, n_y = 0\rangle &= \left(\frac{m\omega}{\pi\hbar 2^2}\right)^{1/4} \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \exp\left(-\frac{m\omega}{2\hbar}(x^2 + y^2)\right) \\ &\quad \times 2\left(\frac{m\omega}{\hbar}\right)^{1/2} x \\ &= \sqrt{\frac{2}{\pi}} \frac{m\omega}{\hbar} x \exp\left(-\frac{m\omega}{2\hbar}(x^2 + y^2)\right), \\ \langle xy|n_x = 0, n_y = 1\rangle &= \sqrt{\frac{2}{\pi}} \frac{m\omega}{\hbar} y \exp\left(-\frac{m\omega}{2\hbar}(x^2 + y^2)\right),\end{aligned}$$

where the ground state $n = 0$ has energy $\hbar\omega$ and the excited state $n = 1$ is twofold degenerate

with energy $2\hbar\omega$. In polar coordinates, these can be written

$$\begin{aligned}\langle \rho\phi | n_x = 0, n_y = 0 \rangle &= \frac{1}{\sqrt{\pi\rho_0^2}} \exp\left[-\frac{1}{2}\left(\frac{\rho}{\rho_0}\right)^2\right], \\ \langle \rho\phi | n_x = 1, n_y = 0 \rangle &= \sqrt{\frac{2}{\pi\rho_0^2}} \exp\left[-\frac{1}{2}\left(\frac{\rho}{\rho_0}\right)^2\right] \frac{\rho}{\rho_0} \cos\phi, \\ \langle \rho\phi | n_x = 0, n_y = 1 \rangle &= \sqrt{\frac{2}{\pi\rho_0^2}} \exp\left[-\frac{1}{2}\left(\frac{\rho}{\rho_0}\right)^2\right] \frac{\rho}{\rho_0} \sin\phi,\end{aligned}$$

where $\rho_0 \equiv \sqrt{\hbar/m\omega}$, and it is straightforward to check normalization using either form. [There is probably a nicer way of writing these down.](#)

It follows from our equation for E in terms of n_x and n_y that the degeneracy of a level with $E = (n+1)\hbar\omega$ is $(n+1)$. (There are $n+1$ ways to pick two numbers $n_x, n_y = 0, 1, 2, \dots, n$ that sum to n ; there are $n+1$ choices for the first number, and then the second one is fixed by this choice.) $\square$

9-6 (Shankar, Exercise 10.2.3)

Quantize the three-dimensional *isotropic oscillator* for which

$$\mathcal{H} = \frac{p_x^2 + p_y^2 + p_z^2}{2m} + \frac{1}{2}m\omega^2(x^2 + y^2 + z^2)$$

- (1) Show that $E = (n + 3/2)\hbar\omega$; $n = n_x + n_y + n_z$; $n_x, n_y, n_z = 0, 1, 2, \dots$
- (2) Write the corresponding eigenfunctions in terms of single-oscillator wave functions and verify that the parity of the level with a given n is $(-1)^n$. Reexpress the first four states in terms of spherical coordinates. Show that the degeneracy of a level with energy $E = (n + 3/2)\hbar\omega$ is $(n+1)(n+2)/2$.

Solution. (1) This follows from the same argument as the previous exercise.

(2) Same steps as part (3) of the previous exercise:

$$\begin{aligned}\psi_E(x, y, z) &= \left(\frac{m\omega}{\pi\hbar}\right)^{3/4} \left(\frac{1}{2^{2(n_x+n_y+n_z)}(n_x!)^2(n_y!)^2(n_z!)^2}\right)^{1/4} \exp\left(-\frac{m\omega}{2\hbar}(x^2 + y^2 + z^2)\right) \\ &\quad \times H_{n_x}\left[\left(\frac{m\omega}{\hbar}\right)^{1/2}x\right] H_{n_y}\left[\left(\frac{m\omega}{\hbar}\right)^{1/2}y\right] H_{n_z}\left[\left(\frac{m\omega}{\hbar}\right)^{1/2}z\right].\end{aligned}$$

The same argument for parity applies; you can draw a table to verify that the parity of a level with given n is equal to $(-1)^n$ for all possible combinations, i.e., when n is the sum of 3 odd, 2

odd, etc. The first four states in terms of spherical coordinates are

$$\begin{aligned}\psi_0(r, \theta, \phi) &= \left(\frac{m\omega}{\pi\hbar}\right)^{3/4} \exp\left(-\frac{m\omega}{2\hbar} r^2\right), \\ \psi_1(r, \theta, \phi) &= \begin{cases} \frac{1}{2^{1/2}} \left(\frac{m\omega}{\pi\hbar}\right)^{3/4} \exp\left(-\frac{m\omega}{2\hbar} r^2\right) 2\left(\frac{m\omega}{\hbar}\right)^{1/2} r \sin\theta \cos\phi, \\ \frac{1}{2^{1/2}} \left(\frac{m\omega}{\pi\hbar}\right)^{3/4} \exp\left(-\frac{m\omega}{2\hbar} r^2\right) 2\left(\frac{m\omega}{\hbar}\right)^{1/2} r \sin\theta \sin\phi, \\ \frac{1}{2^{1/2}} \left(\frac{m\omega}{\pi\hbar}\right)^{3/4} \exp\left(-\frac{m\omega}{2\hbar} r^2\right) 2\left(\frac{m\omega}{\hbar}\right)^{1/2} r \cos\theta, \end{cases}\end{aligned}$$

There is a bit of abuse of notation here, but I can't seem to settle on a nice way of writing. One can see that the first excited state $n = 1$ is threefold degenerate.

To determine the degeneracy of a level with energy $E = (n + 3/2)\hbar\omega$, we ask how many ways there are to choose $n_x, n_y, n_z = 0, 1, 2, \dots, n$ that sum to n . The answer is

$$\binom{n+3-1}{3-1} = \binom{n+2}{2} = \frac{(n+1)(n+2)}{2},$$

which is a standard combinatorial result. □

9-7 (Shankar, Exercise 10.3.1)

Two identical bosons are found to be in states $|\phi\rangle$ and $|\psi\rangle$. Write down the normalized state vector describing the system when $\langle\phi|\psi\rangle \neq 0$.

Solution. This is

$$\boxed{\frac{1}{\sqrt{2}}(|\phi\rangle \otimes |\psi\rangle + |\psi\rangle \otimes |\phi\rangle)},$$

since the result must be symmetric. □

9-8 (Shankar, Exercise 10.3.2)

When an energy measurement is made on a system of three bosons in a box, the n values obtained were 3, 3, and 4. Write down a symmetrized, normalized state vector.

Solution. We have

$$\boxed{|\psi\rangle = \frac{1}{\sqrt{3}}(|334\rangle + |343\rangle + |433\rangle)},$$

which is manifestly symmetric under the exchange of any two particles, and is easily seen to be normalized. □

9-9 (Shankar, Exercise 10.3.3)

Imagine a situation in which there are three particles and only three states a , b , and c available to them. Show that the total number of allowed, distinct configurations for this system is

- (1) 27 if they are labeled
- (2) 10 if they are bosons
- (3) 1 if they are fermions

Solution. We count the ways that states can be assigned to the three particles.

- (1) In this case, we have

$$3 \times 3 \times 3 = 27$$

possible configurations, since each particle can be one of a , b , or c .

- (2) Bosons mean that only symmetric states are allowed. We enumerate:

$$\begin{aligned}
 & |aaa\rangle, \quad |bbb\rangle, \quad |ccc\rangle \\
 & \frac{1}{\sqrt{3}}(|aab\rangle + |aba\rangle + |baa\rangle), \quad \frac{1}{\sqrt{3}}(|aac\rangle + |aca\rangle + |caa\rangle), \quad \frac{1}{\sqrt{3}}(|bbc\rangle + |bcb\rangle + |cbb\rangle), \\
 & \frac{1}{\sqrt{3}}(|ccb\rangle + |cbc\rangle + |bcc\rangle), \quad \frac{1}{\sqrt{3}}(|cca\rangle + |cac\rangle + |acc\rangle), \quad \frac{1}{\sqrt{3}}(|bba\rangle + |bab\rangle + |abb\rangle), \\
 & \frac{1}{\sqrt{3}}(|abc\rangle + |cab\rangle + |bca\rangle + |bac\rangle + |acb\rangle + |cba\rangle).
 \end{aligned}$$

This is a total of 10 distinct configurations.

- (3) If we have fermions, then only the totally antisymmetric configuration is allowed:

$$\frac{1}{\sqrt{3}}(|abc\rangle + |cab\rangle + |bca\rangle - |bac\rangle - |acb\rangle - |cba\rangle).$$

□

9-10 (Shankar, Exercise 10.3.4)

Two identical particles of mass m are in a one-dimensional box of length l . Energy measurement of the system yields the value $E_{\text{sys}} = \hbar^2\pi^2/mL^2$. Write down the state vector of the system. Repeat for $E_{\text{sys}} = 5\hbar^2\pi^2/2mL^2$. (There are two possible vectors in this case.) You are not told if they are bosons or fermions. You may assume that the only degrees of freedom are orbital.

Solution. By the same reasoning as Exercise 10.2.1, the energy of the system is given by

$$E = \frac{\hbar^2\pi^2}{2mL^2}(n_1^2 + n_2^2)$$

where $n_1, n_2 = 1, 2, 3, \dots$ are the energy quantum numbers for particles 1 and 2, respectively. In the first case we have

$$E_{\text{sys}} = \frac{\hbar^2\pi^2}{mL^2} = \frac{\hbar^2\pi^2}{2mL^2}(n_1^2 + n_2^2),$$

hence $n_1^2 + n_2^2 = 2$, which has the unique solution $n_1 = n_2 = 1$. Thus the state vector is

$$\boxed{|11\rangle},$$

and the particles must be bosons. Similarly, if

$$E_{\text{sys}} = \frac{\hbar^2\pi^2}{mL^2} = \frac{5\hbar^2\pi^2}{2mL^2}(n_1^2 + n_2^2),$$

then $n_1^2 + n_2^2 = 5$, which has the two solutions $n_1 = 1$ and $n_2 = 2$ or $n_1 = 2$ and $n_2 = 1$. The state vector is either

$$\boxed{\frac{1}{\sqrt{2}}(|12\rangle + |21\rangle) \quad \text{or} \quad \frac{1}{\sqrt{2}}(|12\rangle - |21\rangle)},$$

corresponding to the particles being either bosons or fermions, respectively. □

9-11 (Shankar, Exercise 10.3.5)

Consider the *exchange operator* P_{12} whose action on the X basis is

$$P_{12}|x_1, x_2\rangle = |x_2, x_1\rangle$$

- (1) Show that P_{12} has eigenvalues ± 1 . (It is Hermitian and unitary.)
- (2) Show that its action on the basis ket $|\omega_1, \omega_2\rangle$ is also to exchange the labels 1 and 2, and hence that $\mathbb{V}_{S/A}$ are its eigenspaces with eigenvalues ± 1 .
- (3) Show that $P_{12}X_1P_{12} = X_2$, $P_{12}X_2P_{12} = X_1$ and similarly for P_1 and P_2 . Then show that $P_{12}\Omega(X_1, P_1; X_2, P_2)P_{12} = \Omega(X_2, P_2; X_1, P_1)$.
- (4) Show that the Hamiltonian and propagator for two *identical* particles are left unaffected under $H \rightarrow P_{12}HP_{12}$ and $U \rightarrow P_{12}UP_{12}$. Given this, show that any eigenstate of P_{12} continues to remain an eigenstate with the same eigenvalue as time passes, i.e., elements of $\mathbb{V}_{S/A}$ never leave the symmetric or antisymmetric subspaces they start in.

Solution. (1) It is straightforward to see that P_{12}^2 is the identity operator. Now suppose that $|\omega_1\omega_2\rangle$ is an eigenket of P_{12} , so that $P_{12}|\omega_1\omega_2\rangle = \Omega|\omega_1\omega_2\rangle$. Then

$$\begin{aligned} |\omega_1\omega_2\rangle &= P_{12}^2|\omega_1\omega_2\rangle \\ &= P_{12}(P_{12}|\omega_1\omega_2\rangle) \\ &= P_{12}(\Omega|\omega_1\omega_2\rangle) \\ &= \Omega^2|\omega_1\omega_2\rangle. \end{aligned}$$

It follows that $\Omega^2 = 1$, thus $\Omega = \pm 1$, as we wanted.

- (2) Expand $|\omega_1\omega_2\rangle$ in the position basis, and then act the exchange operator:

$$\begin{aligned} P_{12}|\omega_1\omega_2\rangle &= \int dx_1 dx_2 (P_{12}|x_1x_2\rangle) \langle x_1x_2|\omega_1\omega_2\rangle \\ &= \int dx_1 dx_2 |x_2x_1\rangle \langle x_1x_2|\omega_1\omega_2\rangle \\ &= \left(\int dx_2 |x_2\rangle \langle x_2|\omega_2\rangle \right) \otimes \left(\int dx_1 |x_1\rangle \langle x_1|\omega_1\rangle \right) \\ &= |\omega_2\rangle \otimes |\omega_1\rangle \\ &= |\omega_2\omega_1\rangle. \end{aligned}$$

Since $|\omega_1\omega_2\rangle$ was arbitrary, this must also hold for the eigenkets of P_{12} , so for any given eigenket

$$P_{12}|\omega_1\omega_2\rangle = |\omega_2\omega_1\rangle = \pm|\omega_1\omega_2\rangle,$$

where the second equality follows from part (1). Thus the elements of the eigenspace of P_{12} belong to either $\mathbb{V}_S$ or $\mathbb{V}_A$, corresponding to the plus and minus sign above, respectively.

(3) We have

$$\begin{aligned} P_{12}X_1P_{12}|x_1, x_2\rangle &= P_{12}X_1|x_2, x_1\rangle \\ &= P_{12}(x_2|x_2, x_1\rangle) \\ &= x_2|x_1, x_2\rangle \\ &= X_2|x_1, x_2\rangle, \end{aligned}$$

hence $P_{12}X_1P_{12} = X_2$, and similar arguments apply for the other operators.

(3) Suppose that $\Omega(X_1, P_1; X_2, P_2)$ can be expanded in a power series; it then suffices to prove the equality for an arbitrary term in the expansion. Note that for any n

$$X_1^n = \overbrace{P_{12}X_2P_{12} \cdots P_{12}X_2P_{12}}^{n \text{ times}} = P_{12}X_2^nP_{12},$$

using part (3) and the fact that $P_{12}^2 = 1$. Using this identity, we have, e.g.,

$$\begin{aligned} P_{12}(X_1^i X_2^j P_1^k P_2^\ell)P_{12} &= P_{12}X_1^i(P_{12}X_1^jP_{12})(P_{12}P_2^kP_{12})P_2^\ell P_{12} \\ &= (P_{12}X_1^iP_{12})X_1^j(P_{12}P_{12})P_2^k(P_{12})P_2^\ell P_{12} \\ &= X_2^i X_1^j P_2^k P_1^\ell, \end{aligned}$$

which looks like the first term in parentheses, but with the 1 and 2 indices swapped. A similar argument can be applied to all power-series terms, which proves the equality.

(4) If the particles are identical, then the Hamiltonian must be symmetric in indices 1 and 2. It follows that H is invariant under $H \rightarrow P_{12}HP_{12}$, for this transformation merely exchanges the 1 and 2 indices, as we proved in part (3). The same goes for U , since it is a function of H .

Now suppose $|\omega_1, \omega_2\rangle$ is an eigenstate of P_{12} , so that $P_{12}|\omega_1, \omega_2\rangle = \pm|\omega_1, \omega_2\rangle$, then note that

$$\begin{aligned} P_{12}(U|\omega_1, \omega_2\rangle) &= P_{12}P_{12}UP_{12}|\omega_1, \omega_2\rangle \\ &= U(P_{12}|\omega_1, \omega_2\rangle) \\ &= \pm U|\omega_1, \omega_2\rangle, \end{aligned}$$

which shows that $|\omega_1, \omega_2\rangle$ remains an eigenstate of P_{12} under time evolution.

Alternatively, since the Hamiltonian is symmetric in indices 1 and 2, we have $[P_{12}, H] = 0$ and therefore $d\langle P_{12} \rangle/dt = 0$. In particular, states that start out symmetric/antisymmetric remain so under time evolution. $\square$

9-12 (Shankar, Exercise 10.3.6)

Consider a composite object such as the hydrogen atom. Will it behave as a boson or fermion? Argue in general that objects containing an even/odd number of fermions will behave as bosons/fermions.

Solution. The hydrogen atom is composed of a proton and an electron, both of which are fermions. Suppose we have two identical hydrogen atoms, so that the wave function for the system is (in the X basis)

$$\psi(x_{p1}, x_{e1}; x_{p2}, x_{e2}),$$

where the 1 and 2 indices label atoms 1 and 2. Acting the exchange operator serves to swap the indices, as we showed in the previous exercise, and we pick up two overall minus signs, which cancel out. Thus

$$\psi(x_{p1}, x_{e1}; x_{p2}, x_{e2}) = \psi(x_{p2}, x_{e2}; x_{p1}, x_{e1}),$$

that is, the hydrogen atom behaves as a boson. The same argument applies to show that objects containing an even/odd number of fermions behave as bosons/fermions. (Note that the number of bosons in the object is immaterial, since these don't contribute a sign change.) $\square$

10 Symmetries

10-1 (Shankar, Exercise 11.2.1)

Verify Eq. (11.2.11b).

Solution. We have

$$\begin{aligned}\langle \psi_\epsilon | P | \psi_\epsilon \rangle &= \int dx \int dx' \langle \psi_\epsilon | x \rangle \langle x | P | x' \rangle \langle x' | \psi_\epsilon \rangle \\ &= \int dx \int dx' e^{-i\epsilon g(x)/\hbar} \psi^*(x - \epsilon) \left(-i\hbar \delta(x - x') \frac{d}{dx'} \right) \left(e^{i\epsilon g(x')/\hbar} \psi(x' - \epsilon) \right).\end{aligned}$$

The derivative acting on the last set of parentheses becomes

$$\frac{i\epsilon}{\hbar} \frac{dg}{dx'} e^{i\epsilon g(x')/\hbar} \psi(x' - \epsilon) + e^{i\epsilon g(x')/\hbar} \frac{d}{dx'} \psi(x' - \epsilon),$$

and upon substituting into the integral and integrating over x' , we have

$$\begin{aligned}\langle \psi_\epsilon | P | \psi_\epsilon \rangle &= -i\hbar \int dx \cancel{e^{-i\epsilon g(x)/\hbar}} \psi^*(x - \epsilon) \left(\frac{i\epsilon}{\hbar} \frac{dg}{dx} \cancel{e^{i\epsilon g(x)/\hbar}} \psi(x - \epsilon) + \cancel{e^{i\epsilon g(x)/\hbar}} \frac{d}{dx} \psi(x - \epsilon) \right) \\ &= \int dx \psi^*(x - \epsilon) \left(-i\hbar \frac{d}{dx} \psi(x - \epsilon) \right) + \epsilon \int dx \psi^*(x - \epsilon) \frac{dg}{dx} \psi(x - \epsilon) \\ &= \langle P \rangle + \epsilon \langle f(x) \rangle,\end{aligned}$$

where we've used the fact that $f = dg/dx$. □

10-2 (Shankar, Exercise 11.2.2)

Using $T^\dagger(\epsilon)T(\epsilon) = I$ to order ϵ , deduce that $G^\dagger = G$.

Solution. We have

$$\begin{aligned}T^\dagger(\epsilon)T(\epsilon) &= \left(I + \frac{i\epsilon}{\hbar} G^\dagger \right) \left(I - \frac{i\epsilon}{\hbar} G \right) \\ &= I + \frac{i\epsilon}{\hbar} (G^\dagger - G) + O(\epsilon^2),\end{aligned}$$

and since we demand that the left-hand side is the identity to order ϵ , it follows that $G^\dagger = G$. □

10-3 (Shankar, Exercise 11.2.3)

Solution. The effect of a finite translation a due to the generator p is given by the following sum of nested Poisson brackets:

$$\omega \rightarrow \omega + a\{\omega, p\} + \frac{a^2}{2!}\{\{\omega, p\}, p\} + \dots$$

Now we consider a finite rotation through the angle θ generated by $g = xp_y - yp_x$. First we work out the Poisson brackets

$$\begin{aligned}\{x, g\} &= \{x, xp_y - yp_x\} \\ &= \{x, xp_y\} - \{x, yp_x\} \\ &= (\cancel{\{x, x\}}p_y + x\cancel{\{x, p_y\}}) - (\cancel{\{x, y\}}p_x + y\{x, p_x\}) \\ &= -y,\end{aligned}$$

where we've used the properties of Poisson brackets and the fact that our variables are canonical, and

$$\begin{aligned}\{y, g\} &= \{y, xp_y - yp_x\} \\ &= \{y, xp_y\} - \{y, yp_x\} \\ &= (\cancel{\{y, x\}}p_y + \{y, p_y\}x) - (\cancel{\{y, y\}}p_x + \cancel{\{y, p_x\}}y) \\ &= x.\end{aligned}$$

From here we can work out all of the nested Poisson brackets:

$$\begin{aligned}\{x, g\} &= -y, \\ \{\{x, g\}, g\} &= \{-y, g\} \\ &= -x, \\ \{\{\{x, g\}, g\}, g\} &= \{-x, g\} \\ &= y, \\ \{\{\{\{x, g\}, g\}, g\}, g\} &= \{y, g\} \\ &= x,\end{aligned}$$

and so on. Thus, we have

$$\begin{aligned}
x &\rightarrow x - \theta y + \frac{\theta^2}{2!}(-x) + \frac{\theta^3}{3!}y + \frac{\theta^4}{4!}x + \cdots \\
&= \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!}\right)x - \left(\theta - \frac{\theta^3}{3!} + \cdots\right)y \\
&= x \cos \theta - y \sin \theta,
\end{aligned}$$

as we wanted to show. The transformation rule for y is obtained in a similar way. $\square$

10-4 (Shankar, Exercise 11.4.1)

Solution. Suppose $|\psi\rangle$ is an eigenstate of parity, i.e.,

$$\Pi|\psi\rangle = \pm|\psi\rangle.$$

Then if Π and H commute for all times t , we have

$$[\Pi, H] = 0 \implies [\Pi, U(t)] = 0.$$

Thus

$$\begin{aligned}
\Pi(U(t)|\psi\rangle) &= U(t)\Pi|\psi\rangle \\
&= U(t)(\pm|\psi\rangle) \\
&= \pm U(t)|\psi\rangle.
\end{aligned}$$

Thus eigenstates of parity remain eigenstates under time evolution, as we wanted. $\square$

10-5 (Shankar, Exercise 11.4.2)

Solution. No, momentum is not conserved. The argument for conservation makes use of infinitesimal translation invariance, which we don't have here. Alternatively, note that obviously $[H, P] \neq 0$, since P involves a spatial derivative in the X basis, and $V(x)(d/dx) \neq dV/dx$. Thus $\langle \dot{P} \rangle \neq 0$ by Ehrenfest's theorem. $\square$

10-6 (Shankar, Exercise 11.4.4)

Solution. A reflection through a mirror in the x - y plane sends $(x, y, z) \rightarrow (x, y, -z)$, and a subsequent rotation by 180° about the z axis serves to flip the signs of x and y . Thus, the net effect is to send $(x, y, z) \rightarrow (-x, -y, -z)$, which is equivalent to parity. $\square$

11 Rotations and spin

11-1 (Littlejohn, Problem 1)

Prove the commutation relations (11.34), using Eq. (11.23) and the properties of the Levi-Civita symbol ϵ_{ijk} .

Solution. We have

$$\begin{aligned}[J_i, J_j]_{k\ell} &= (J_i J_j)_{k\ell} - (J_j J_i)_{k\ell} \\&= \delta_{i\ell} \delta_{kj} - \cancel{\delta_{ij} \delta_{k\ell}} - \delta_{j\ell} \delta_{ki} + \cancel{\delta_{ji} \delta_{k\ell}} \\&= \delta_{i\ell} \delta_{kj} - \delta_{j\ell} \delta_{ki} \\&= \epsilon_{mij} \epsilon_{m\ell k} \\&= -\epsilon_{mij} \epsilon_{mk\ell} \\&= \epsilon_{mij} (J_m)_{k\ell} \\&= \epsilon_{ijm} (J_m)_{k\ell},\end{aligned}$$

where we used (11.33) in the second line. This shows that

$$[J_i, J_j] = \epsilon_{ijk} J_k,$$

as we wanted. □

11-2 (Littlejohn, Problem 2)

Some useful identities.

(a) Show that if $\mathbf{R} \in O(3)$, then

$$(\mathbf{R}\mathbf{a}) \cdot (\mathbf{R}\mathbf{b}) = \mathbf{a} \cdot \mathbf{b}.$$

Also show that

$$\mathbf{a} \cdot (\mathbf{R}\mathbf{b}) = (\mathbf{R}^{-1}\mathbf{a}) \cdot \mathbf{b}.$$

These formulas hold for all vectors $\mathbf{a}$ and $\mathbf{b}$.

(b) Show that if $\mathbf{R} \in SO(3)$, then

$$\mathbf{R}(\mathbf{a} \times \mathbf{b}) = (\mathbf{R}\mathbf{a}) \times (\mathbf{R}\mathbf{b}).$$

Hint: Use the fact that if $\mathbf{M}$ is any 3×3 matrix, then

$$\epsilon_{ijk} \det \mathbf{M} = \epsilon_{lmn} M_{il} M_{jm} M_{kn}.$$

Solution. (a) Let $\mathbf{a}, \mathbf{b}$ be arbitrary vectors, and let $\mathbf{R} \in O(3)$, so that $\mathbf{R}^t = \mathbf{R}^{-1}$. Then

$$\begin{aligned} (\mathbf{R}\mathbf{a}) \cdot (\mathbf{R}\mathbf{b}) &= R_{ij} a_j R_{ik} b_k \\ &= R_{ij} R_{ik} a_j b_k \\ &= (\mathbf{R}^t)_{ji} R_{ik} a_j b_k \\ &= \delta_{jk} a_j b_k \\ &= a_j b_j \\ &= \mathbf{a} \cdot \mathbf{b}, \end{aligned}$$

and also

$$\begin{aligned} \mathbf{a} \cdot (\mathbf{R}\mathbf{b}) &= a_j R_{jk} b_k \\ &= a_j R_{kj}^{-1} b_k \\ &= R_{kj}^{-1} a_j b_k \\ &= (\mathbf{R}^{-1}\mathbf{a}) \cdot \mathbf{b}, \end{aligned}$$

as we wanted to show. Geometrically, the first result says that the dot product of two vectors is preserved under a rotation.

Note that these results still hold if we replace the Euclidean dot product $\mathbf{a} \cdot \mathbf{b}$ by the inner product $\mathbf{a}^\dagger \mathbf{b}$; we'll use this fact in Problem 4.

(b) First note that

$$\begin{aligned} [\mathbf{R}(\mathbf{a} \times \mathbf{b})]_i &= R_{ij}(\mathbf{a} \times \mathbf{b})_j \\ &= R_{ij}\epsilon_{jkl}a_kb_l. \end{aligned}$$

Now since $\mathbf{R} \in SO(3)$, its inverse is also in $SO(3)$, and by the hint it follows that

$$\epsilon_{ijk} = \epsilon_{ijk} \det(\mathbf{R}^{-1}) = \epsilon_{lmn} R_{il}^{-1} R_{jm}^{-1} R_{kn}^{-1}.$$

Then we get

$$\begin{aligned} [\mathbf{R}(\mathbf{a} \times \mathbf{b})]_i &= R_{ij}(\epsilon_{pqr} R_{jp}^{-1} R_{kq}^{-1} R_{lr}^{-1}) a_k b_l \\ &= R_{ij} R_{jp}^{-1} \epsilon_{pqr} (R_{kq}^{-1} a_k) (R_{lr}^{-1} b_l) \\ &= \delta_{ip} \epsilon_{pqr} (R_{qk} a_k) (R_{rl} b_l) \\ &= \epsilon_{ipr} (\mathbf{R}\mathbf{a})_q (\mathbf{R}\mathbf{b})_r \\ &= [(\mathbf{R}\mathbf{a}) \times (\mathbf{R}\mathbf{b})]_i, \end{aligned}$$

as we wanted to show. □

11-3 (Littlejohn, Problem 3)

The geometrical meaning of Eq. (11.44) is illustrated in Fig. 4. The rotation leaves the component of $\mathbf{u}$ along the axis $\hat{\mathbf{n}}$ invariant, while rotating the orthogonal component in the plane perpendicular to $\hat{\mathbf{n}}$. By expressing powers of the $\mathbf{J}$ matrices in terms of lower powers, sum the exponential series (11.40) and obtain another proof of Eq. (11.44).

Solution. The series we wish to sum is

$$\begin{aligned} \mathbf{R}(\hat{\mathbf{n}}, \theta) &= \exp(\theta \hat{\mathbf{n}} \cdot \mathbf{J}) \\ &= \sum_{k=0}^{\infty} \frac{(\theta \hat{\mathbf{n}} \cdot \mathbf{J})^k}{k!} \\ &= 1 + \theta \hat{\mathbf{n}} \cdot \mathbf{J} + \frac{\theta^2}{2!} (\hat{\mathbf{n}} \cdot \mathbf{J})^2 + \frac{\theta^3}{3!} (\hat{\mathbf{n}} \cdot \mathbf{J})^3 + \dots \end{aligned}$$

We have the result

$$(\hat{\mathbf{n}} \cdot \mathbf{J})\mathbf{u} = \hat{\mathbf{n}} \times \mathbf{u},$$

and by the definitions and (11.33), we have

$$\begin{aligned}
[(\hat{\mathbf{n}} \cdot \mathbf{J})^2 \mathbf{u}]_k &= (n_i J_i n_j J_j)_{k\ell} u_\ell \\
&= n_i n_j (J_i J_j)_{k\ell} u_\ell \\
&= n_i n_j (\delta_{i\ell} \delta_{kj} - \delta_{ij} \delta_{k\ell}) u_\ell \\
&= n_i u_i n_k - n_i n_i u_k \\
&= [\hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \mathbf{u}) - \mathbf{u}]_k.
\end{aligned}$$

From these results we have

$$\begin{aligned}
(\hat{\mathbf{n}} \cdot \mathbf{J})^3 \mathbf{u} &= (\hat{\mathbf{n}} \cdot \mathbf{J})[\hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \mathbf{u}) - \mathbf{u}] \\
&= (\hat{\mathbf{n}} \times \hat{\mathbf{n}})(\hat{\mathbf{n}} \cdot \mathbf{u}) - \hat{\mathbf{n}} \times \mathbf{u} \\
&= -\hat{\mathbf{n}} \times \mathbf{u}, \\
(\hat{\mathbf{n}} \cdot \mathbf{J})^4 \mathbf{u} &= (\hat{\mathbf{n}} \cdot \mathbf{J})^2 [\hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \mathbf{u}) - \mathbf{u}] \\
&= (\hat{\mathbf{n}} \cdot \mathbf{u})(\hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \hat{\mathbf{n}}) - \hat{\mathbf{n}}) - (\hat{\mathbf{n}} \cdot \mathbf{J})^2 \mathbf{u} \\
&= -(\hat{\mathbf{n}} \cdot \mathbf{J})^2 \mathbf{u}.
\end{aligned}$$

We then infer that for $j = 1, 2, 3, \dots$

$$\begin{aligned}
(\hat{\mathbf{n}} \cdot \mathbf{J})^{2j+1} \mathbf{u} &= (-1)^j (\hat{\mathbf{n}} \times \mathbf{u}), \\
(\hat{\mathbf{n}} \cdot \mathbf{J})^{2j} \mathbf{u} &= (-1)^{j+1} [\hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \mathbf{u}) - \mathbf{u}].
\end{aligned}$$

This allows us to sum the series:

$$\begin{aligned}
R(\hat{\mathbf{n}}, \theta) \mathbf{u} &= \mathbf{u} + \left(\sum_{j=0}^{\infty} \frac{\theta^{2j+1}}{(2j+1)!} (\hat{\mathbf{n}} \cdot \mathbf{J})^{2j+1} \right) \mathbf{u} + \left(\sum_{j=1}^{\infty} \frac{\theta^{2j}}{(2j)!} (\hat{\mathbf{n}} \cdot \mathbf{J})^{2j} \right) \mathbf{u} \\
&= \mathbf{u} + \sum_{j=0}^{\infty} \frac{\theta^{2j+1}}{(2j+1)!} (-1)^j (\hat{\mathbf{n}} \times \mathbf{u}) + \sum_{j=1}^{\infty} \frac{\theta^{2j}}{(2j)!} (-1)^{j+1} [\hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \mathbf{u}) - \mathbf{u}] \\
&= \mathbf{u} + \underbrace{\left(\sum_{j=0}^{\infty} \frac{\theta^{2j+1}}{(2j+1)!} (-1)^j \right)}_{\sin \theta} (\hat{\mathbf{n}} \times \mathbf{u}) - \underbrace{\left(-1 + \sum_{j=0}^{\infty} \frac{\theta^{2j}}{(2j)!} (-1)^j \right)}_{\cos \theta} [\hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \mathbf{u}) - \mathbf{u}] \\
&= \mathbf{u} + \sin \theta (\hat{\mathbf{n}} \times \mathbf{u}) + (1 - \cos \theta) [\hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \mathbf{u}) - \mathbf{u}] \\
&= \mathbf{u} + \sin \theta (\hat{\mathbf{n}} \times \mathbf{u}) + \hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \mathbf{u})(1 - \cos \theta) - \mathbf{u} \cos \theta \\
&= \hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \mathbf{u})(1 - \cos \theta) + \mathbf{u} \cos \theta + (\hat{\mathbf{n}} \times \mathbf{u}) \sin \theta.
\end{aligned}$$

This proves (11.44). □

11-4 (Littlejohn, Problem 4)

The axis $\hat{\mathbf{n}}$ of a proper rotation $\mathbf{R}$ is invariant under the action of $\mathbf{R}$, that is, $\mathbf{R}\hat{\mathbf{n}} = \hat{\mathbf{n}}$. Therefore $\hat{\mathbf{n}}$ is a real, normalized eigenvector of $\mathbf{R}$ with eigenvalue $+1$. Prove that every proper rotation has an axis. Show that the axis is unique (apart from the change in sign $\hat{\mathbf{n}} \rightarrow -\hat{\mathbf{n}}$) as long as the angle of rotation is nonzero. This proof is the essential step in showing that every rotation can be expressed in axis-angle form. Do proper rotations in 4 dimensions have an axis?

Solution. Suppose $\mathbf{u}$ is an eigenvector of $\mathbf{R}$ with eigenvalue λ . Then

$$\begin{aligned}\mathbf{R}\mathbf{u} &= \lambda\mathbf{u} \\ \implies (\mathbf{R}\mathbf{u})^\dagger(\mathbf{R}\mathbf{u}) &= |\lambda|^2\mathbf{u}^\dagger\mathbf{u} \\ \implies |\lambda|^2 &= 1,\end{aligned}$$

where we used the result of Problem 2(a) in the last line. Thus all eigenvalues of a proper rotation have magnitude 1. Furthermore, since $\det(\mathbf{R} - \lambda\mathbf{I})$ has at least one real root (it is a degree 3 polynomial), it follows that one eigenvalue is either $+1$ or -1 . This proves the claim that every proper rotation has an axis.

Suppose there is another axis $\hat{\mathbf{n}}'$, so that $\mathbf{R}\hat{\mathbf{n}}' = \hat{\mathbf{n}}'$ and $\hat{\mathbf{n}} \times \hat{\mathbf{n}}' \neq 0$. Then by Problem 2(b)

$$\begin{aligned}\mathbf{R}(\hat{\mathbf{n}} \times \hat{\mathbf{n}}') &= (\mathbf{R}\hat{\mathbf{n}}) \times (\mathbf{R}\hat{\mathbf{n}}') \\ &= \hat{\mathbf{n}} \times \hat{\mathbf{n}}',\end{aligned}$$

so that $\hat{\mathbf{n}} \times \hat{\mathbf{n}}'$ is a third axis of rotation. But then $\mathbf{R}$ is just the identity, contradicting our assumption that $\mathbf{R}$ rotates through a nonzero angle. Thus the axis $\hat{\mathbf{n}}$ is unique.

Proper rotations in 4 dimensions do not always have an axis, because then $\det(\mathbf{R} - \lambda\mathbf{I})$ is a degree 4 polynomial, and we are not guaranteed to have real roots. $\square$

11-5 (Littlejohn, Problem 5)

It is claimed that every proper rotation can be written in Euler angle form. Find the Euler angles (α, β, γ) for the rotation $\mathbf{R}(\hat{\mathbf{x}}, \pi/2)$.

Solution. See attached scan. $\square$

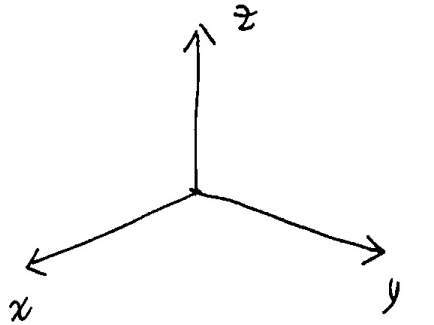
11.5

$$R(\alpha, \beta, \gamma) = R(\hat{z}, \alpha) R(\hat{y}, \beta) R(\hat{z}, \gamma)$$

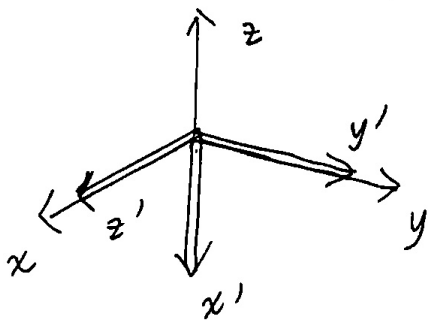
$$R_1 = R(\hat{z}, \alpha) R(\hat{y}, \beta)$$

$$R = R(\hat{z}', \gamma) R_1$$

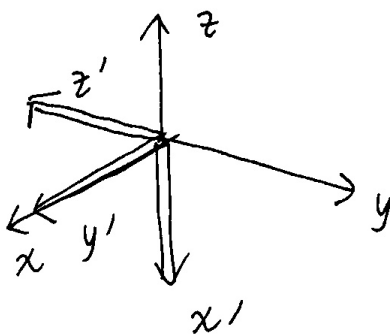
via Euler param:



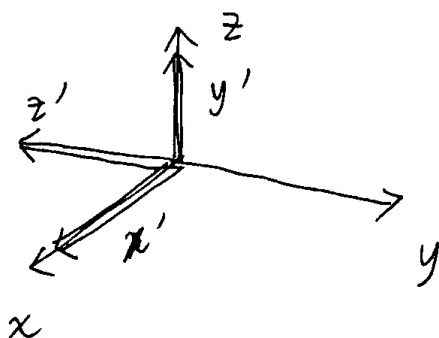
$\Downarrow R(\hat{y}, \frac{\pi}{2})$



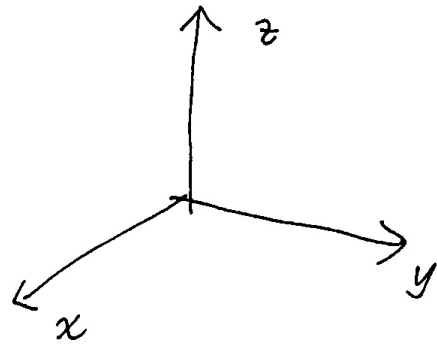
$\Downarrow R(\hat{z}, -\frac{\pi}{2})$



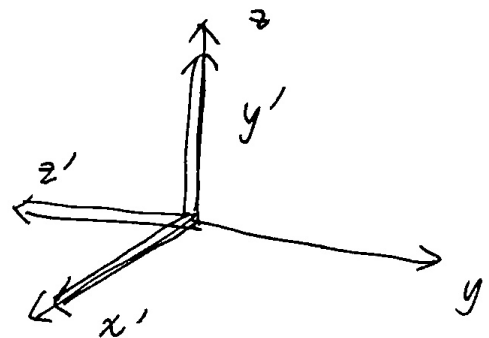
$\Downarrow R(\hat{z}', \frac{\pi}{2})$



Axis-angle:



$\Downarrow R(\hat{x}, \frac{\pi}{2})$



thus we have

$$\alpha = -\frac{\pi}{2}$$

$$\beta = \frac{\pi}{2}$$

$$\gamma = \frac{\pi}{2},$$

or

$$R(\hat{x}, \frac{\pi}{2}) = R\left(-\frac{\pi}{2}, \frac{\pi}{2}, \frac{\pi}{2}\right).$$

11-6 (Griffiths 1st ed., Problem 9.19)

A electron is at rest at the origin, in the presence of a magnetic field whose *magnitude* (B_0) is constant but whose *direction* rides around at constant angular velocity ω on the lip of a cone of opening angle α :

$$\mathbf{B}(t) = B_0 \left[\sin \alpha \cos(\omega t) \hat{\mathbf{i}} + \sin \alpha \sin(\omega t) \hat{\mathbf{j}} + \cos \alpha \hat{\mathbf{k}} \right]$$

- Construct the 2×2 Hamiltonian matrix for this system.
- Find the *exact* solution to the (time-dependent) Schrödinger equation, assuming the particle starts out with spin up.
- Now treat the same problem by (first-order) time-dependent perturbation theory: use Equation 9.17 to calculate the (approximate) probability of a transition from spin up (the initial state) to spin down, as a function of time, and compare the exact answer (from part b). State the criterion on the strength of the field that determines whether perturbation theory is applicable in this case.

This problem is actually quite involved. Later editions of Griffiths give you the general solution and ask you to verify it instead of deriving it from scratch.

Solution. (a) Since

$$\begin{aligned} \hat{H} &= -\hat{\boldsymbol{\mu}} \cdot \mathbf{B} = -\gamma \mathbf{B} \cdot \mathbf{S} \\ &= -\gamma B_0 \left[\sin \alpha \cos(\omega t) \hat{S}_x + \sin \alpha \sin(\omega t) \hat{S}_y + \cos \alpha \hat{S}_z \right] \end{aligned}$$

where $\gamma \equiv -e/m$, we obtain

$$\begin{aligned} \hat{H} &\xrightarrow{S_z \text{ basis}} \frac{e\hbar B_0}{2m} \begin{pmatrix} \cos \alpha & \sin \alpha [\cos(\omega t) - i \sin(\omega t)] \\ \sin \alpha [\cos(\omega t) + i \sin(\omega t)] & -\cos \alpha \end{pmatrix} \\ &= \boxed{\frac{e\hbar B_0}{2m} \begin{pmatrix} \cos \alpha & e^{-i\omega t} \sin \alpha \\ e^{i\omega t} \sin \alpha & -\cos \alpha \end{pmatrix}}. \end{aligned}$$

- Working in the S_z basis, we have

$$|\psi(t)\rangle \rightarrow \begin{pmatrix} a(t) \\ b(t) \end{pmatrix} \quad \text{and} \quad |\psi(0)\rangle \rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Then by substituting into the Schrödinger equation $\hat{H}|\psi(t)\rangle = i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle$, we get

$$\frac{eB_0}{2m} \begin{pmatrix} \cos \alpha & e^{-i\omega t} \sin \alpha \\ e^{i\omega t} \sin \alpha & -\cos \alpha \end{pmatrix} \begin{pmatrix} a(t) \\ b(t) \end{pmatrix} = i \begin{pmatrix} \dot{a}(t) \\ \dot{b}(t) \end{pmatrix},$$

and in what follows, we'll write $\Omega \equiv eB_0/m$. At this point, Griffiths makes a comment about how the 'rotating wave approximation' is exact in this case, and says that we can just lift a result from a previous problem. But I'm fairly certain this isn't true! The problem he is referencing makes the additional assumption that the diagonal elements of the Hamiltonian matrix are zero, but this doesn't hold in our case.

First we'll look for expressions for b and $\dot{b}$ in terms of a and $\dot{a}$. We can rewrite the matrix equation above as

$$\dot{a} = -\frac{i\Omega}{2}(a \cos \alpha + b e^{-i\omega t} \sin \alpha) \quad (1)$$

$$\dot{b} = -\frac{i\Omega}{2}(a e^{i\omega t} \sin \alpha - b \cos \alpha) \quad (2)$$

Multiplying (1) by $e^{i\omega t}/\tan \alpha$ and adding to (2), then rearranging gives

$$\dot{b} = -\frac{\dot{a} e^{i\omega t}}{\tan \alpha} - i \frac{\Omega a e^{i\omega t}}{2 \sin \alpha}.$$

We use this in (2) to eliminate $\dot{b}$, and obtain

$$b = -\frac{a e^{i\omega t}}{\tan \alpha} + i \frac{2 \dot{a} e^{i\omega t}}{\Omega \sin \alpha}.$$

Then we differentiate (1) and then insert our expressions for b and $\dot{b}$ to get

$$\begin{aligned} \ddot{a} &= -\frac{i\Omega}{2} \left(\dot{a} \cos \alpha - i\omega b e^{-i\omega t} \sin \alpha + \dot{b} e^{-i\omega t} \sin \alpha \right) \\ &= -\frac{i\Omega}{2} \left[\dot{a} \cos \alpha - i\omega \left(-\frac{a e^{i\omega t}}{\tan \alpha} + i \frac{2 \dot{a} e^{i\omega t}}{\Omega \sin \alpha} \right) e^{-i\omega t} \sin \alpha + \left(-\frac{\dot{a} e^{i\omega t}}{\tan \alpha} - i \frac{\Omega a e^{i\omega t}}{2 \sin \alpha} \right) e^{-i\omega t} \sin \alpha \right] \\ &= -\frac{i\Omega}{2} \left[\cancel{\dot{a} \cos \alpha} + i\omega a \cos \alpha + \frac{2\omega \dot{a}}{\Omega} - \cancel{\dot{a} \cos \alpha} - i \frac{\Omega a}{2} \right] \\ &= -i\omega \dot{a} + \left(\frac{\omega \Omega \cos \alpha}{2} + \frac{\Omega^2}{4} \right) a, \end{aligned}$$

which leaves us with

$$\ddot{a} + i\omega \dot{a} + \frac{\Omega}{2} \left(\frac{\Omega}{2} - \omega \cos \alpha \right) a = 0.$$

For convenience, we define new constants γ and β so that the equation above becomes

$$\ddot{a} + 2\gamma \dot{a} + \beta a = 0.$$

Then the general solution is

$$a(t) = e^{-\gamma t} \left(A e^{t\sqrt{\gamma^2 - \beta}} + B e^{-t\sqrt{\gamma^2 - \beta}} \right),$$

and we also observe that

$$\begin{aligned} a(0) &= A + B, \\ \dot{a}(0) &= -\gamma(A + B) + \sqrt{\gamma^2 - \beta}(A - B). \end{aligned}$$

Now, the electron starts with spin up, so our initial conditions are $a(0) = 1$ and [using Eqs. (1) and (2)] $\dot{a}(0) = -i\Omega \cos \alpha/2$. That is, we have

$$\begin{cases} A + B = 1 \\ -\gamma(A + B) + \sqrt{\gamma^2 - \beta}(A - B) = -i \frac{\Omega \cos \alpha}{2}. \end{cases}$$

A bit of algebra shows that this has the solution

$$\begin{aligned} A &= \frac{\gamma}{2\sqrt{\gamma^2 - \beta}} + \frac{1}{2} - i \frac{\Omega \cos \alpha}{4\sqrt{\gamma^2 - \beta}}, \\ B &= -\frac{\gamma}{2\sqrt{\gamma^2 - \beta}} + \frac{1}{2} + i \frac{\Omega \cos \alpha}{4\sqrt{\gamma^2 - \beta}}. \end{aligned}$$

Substituting this back in to our expression for $a(t)$, we get

$$\begin{aligned} a(t) &= e^{-\gamma t} \left[\left(\frac{\gamma}{2\sqrt{\gamma^2 - \beta}} + \frac{1}{2} - i \frac{\Omega \cos \alpha}{4\sqrt{\gamma^2 - \beta}} \right) e^{t\sqrt{\gamma^2 - \beta}} \right. \\ &\quad \left. + \left(-\frac{\gamma}{2\sqrt{\gamma^2 - \beta}} + \frac{1}{2} + i \frac{\Omega \cos \alpha}{4\sqrt{\gamma^2 - \beta}} \right) e^{-t\sqrt{\gamma^2 - \beta}} \right] \\ &= e^{-\gamma t} \left[\left(\frac{\gamma}{\sqrt{\gamma^2 - \beta}} - i \frac{\Omega \cos \alpha}{2\sqrt{\gamma^2 - \beta}} \right) \left(\frac{e^{t\sqrt{\gamma^2 - \beta}} - e^{-t\sqrt{\gamma^2 - \beta}}}{2} \right) \right. \\ &\quad \left. + \frac{e^{t\sqrt{\gamma^2 - \beta}} + e^{-t\sqrt{\gamma^2 - \beta}}}{2} \right]. \end{aligned}$$

Next we note that since $2\gamma \equiv i\omega$ and $\beta \equiv \Omega(\Omega - 2\omega \cos \alpha)/4$, we have

$$\begin{aligned} \sqrt{\gamma^2 - \beta} &= \sqrt{-\omega^2/4 - \Omega(\Omega - 2\omega \cos \alpha)/4} \\ &= \frac{i\sqrt{\omega^2 + \Omega^2 - 2\omega\Omega \cos \alpha}}{2} \\ &\equiv \frac{i\lambda}{2}. \end{aligned}$$

Therefore,

$$\begin{aligned} a(t) &= e^{-\gamma t} \left[\left(\frac{\gamma}{\sqrt{\gamma^2 - \beta}} - i \frac{\Omega \cos \alpha}{2\sqrt{\gamma^2 - \beta}} \right) (i \sin(\lambda t/2)) + \cos(\lambda t/2) \right] \\ &= e^{-i\omega t/2} \left[i \left(\frac{\omega - \Omega \cos \alpha}{\lambda} \right) \sin(\lambda t/2) + \cos(\lambda t/2) \right]. \end{aligned}$$

To find $b(t)$ we use our expression in terms of a and $\dot{a}$, using Mathematica to help with the

algebra:

$$\begin{aligned} \text{In[33]} &:= \mathbf{a}[\mathbf{t_}] := \mathbf{e}^{\frac{-\mathbf{i} \omega \mathbf{t}}{2}} \left(\mathbf{i} \left(\frac{\omega - \Omega \cos[\alpha]}{\lambda} \right) \sin\left[\frac{\lambda \mathbf{t}}{2}\right] + \cos\left[\frac{\lambda \mathbf{t}}{2}\right] \right) \\ \text{In[34]} &:= -\frac{\mathbf{a}[\mathbf{t}] \mathbf{e}^{\mathbf{i} \omega \mathbf{t}}}{\sin[\alpha]} \cos[\alpha] + \mathbf{i} \frac{2 \mathbf{a}'[\mathbf{t}] \mathbf{e}^{\mathbf{i} \omega \mathbf{t}}}{\Omega \sin[\alpha]} // \text{Simplify} \\ \text{Out[34]} &:= \frac{\mathbf{i} \mathbf{e}^{\frac{\mathbf{i} \mathbf{t} \omega}{2}} \left(\Omega (-2 \omega + \Omega \cos[\alpha]) \cot[\alpha] + (-\lambda^2 + \omega^2) \csc[\alpha] \right) \sin\left[\frac{\mathbf{t} \lambda}{2}\right]}{\lambda \Omega} \end{aligned}$$

Replace λ^2 with $\omega^2 + \Omega^2 - 2\omega\Omega \cos \alpha$ to make things look nicer:

$$\begin{aligned} \text{In[35]} &:= \frac{\mathbf{i} \mathbf{e}^{\frac{\mathbf{i} \mathbf{t} \omega}{2}} \left(\Omega (-2 \omega + \Omega \cos[\alpha]) \cot[\alpha] + (-\left(\omega^2 + \Omega^2 - 2 \omega \Omega \cos[\alpha]\right) + \omega^2) \csc[\alpha] \right) \sin\left[\frac{\mathbf{t} \lambda}{2}\right]}{\lambda \Omega} // \\ &\text{Simplify} \\ \text{Out[35]} &:= -\frac{\mathbf{i} \mathbf{e}^{\frac{\mathbf{i} \mathbf{t} \omega}{2}} \Omega \sin[\alpha] \sin\left[\frac{\mathbf{t} \lambda}{2}\right]}{\lambda} \end{aligned}$$

We then conclude that in the S_z basis,

$$|\psi(t)\rangle \rightarrow \begin{pmatrix} e^{-i\omega t/2} \left[i \left(\frac{\omega - \Omega \cos \alpha}{\lambda} \right) \sin(\lambda t/2) + \cos(\lambda t/2) \right] \\ -i e^{i\omega t/2} \frac{\Omega \sin \alpha \sin(\lambda t/2)}{\lambda} \end{pmatrix},$$

where $\Omega \equiv eB_0/m$ and $\lambda \equiv \sqrt{\omega^2 + \Omega^2 - 2\omega\Omega \cos \alpha}$. It's clear that this satisfies the initial conditions, and we can also verify that the solution is normalized:

$$\begin{aligned} |a(t)|^2 + |b(t)|^2 &= \cos^2\left(\frac{\lambda t}{2}\right) + \underbrace{\left[\frac{(\omega - \Omega \cos \alpha)^2}{\lambda^2} + \frac{\Omega^2 \sin^2 \alpha}{\lambda^2} \right]}_{= (\omega^2 - 2\omega\Omega \cos \alpha + \Omega^2)/\lambda^2 = 1} \sin^2\left(\frac{\lambda t}{2}\right) \\ &= \cos^2\left(\frac{\lambda t}{2}\right) + \sin^2\left(\frac{\lambda t}{2}\right) \\ &= 1. \end{aligned}$$

Finally, we observe that the transition rate to the spin down state is

$$P_{|\uparrow\rangle \rightarrow |\downarrow\rangle} = |b(t)|^2 = \frac{\Omega^2 \sin^2 \alpha}{\lambda^2} \sin^2\left(\frac{\lambda t}{2}\right).$$

(c) We worked hard for an exact solution, but now we'll use perturbation theory. In this case,

the entire Hamiltonian is to be thought of the perturbation. Now

$$\begin{aligned}
 H'_{ab} &= \langle \uparrow | \hat{H} | \downarrow \rangle \\
 &= \begin{pmatrix} 1 & 0 \end{pmatrix} \frac{\Omega \hbar}{2} \begin{pmatrix} \cos \alpha & e^{-i\omega t} \sin \alpha \\ e^{i\omega t} \sin \alpha & -\cos \alpha \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\
 &= \frac{\Omega \hbar}{2} e^{-i\omega t} \sin \alpha,
 \end{aligned}$$

so to first order,

$$\begin{aligned}
 b(t) &= -\frac{i}{\hbar} \int_0^t \frac{\Omega \hbar}{2} e^{-i\omega t'} \sin \alpha dt' \\
 &= \frac{\Omega \sin \alpha}{2\omega} (e^{-i\omega t} - 1).
 \end{aligned}$$

It follows that the approximate transition rate from spin up to spin down is

$$\begin{aligned}
 |b(t)|^2 &= \frac{\Omega^2 \sin^2 \alpha}{4\omega^2} (1 - 2\cos(\omega t) + 1) \\
 &= \frac{\Omega^2 \sin^2 \alpha}{2\omega^2} (1 - \cos(\omega t)) \\
 &= \boxed{\frac{\Omega^2 \sin^2 \alpha}{\omega^2} \sin^2 \left(\frac{\omega t}{2} \right)}.
 \end{aligned}$$

We can see that the approximation uses ω , whereas the exact answer uses λ . When is perturbation theory applicable? We'll assume that α and ω are fixed, so that this becomes a question about the field strength Ω . Now

$$\begin{aligned}
 \lambda^2 &= \omega^2 + \Omega^2 - 2\omega\Omega \cos \alpha \\
 &= \omega^2 \left(1 - \frac{2\Omega}{\omega} \cos \alpha + \left(\frac{\Omega}{\omega} \right)^2 \right),
 \end{aligned}$$

so to first order in Ω/ω , the exact answer from part (b) becomes

$$P_{|\uparrow\rangle \rightarrow |\downarrow\rangle} = \frac{\Omega^2 \sin^2 \alpha}{\omega^2 (1 - \frac{2\Omega}{\omega} \cos \alpha)} \sin^2 \left(\frac{\omega \sqrt{1 - \frac{2\Omega}{\omega} \cos \alpha}}{2} t \right).$$

Recalling that $\Omega \equiv eB_0/m$, we see that

perturbation theory is applicable when $B_0 \ll \frac{m\omega}{2e \cos \alpha}$.

It's neat to see that this approximation works reasonably well, because it requires only a tiny fraction of the effort needed to get the exact answer. □

11-7 (Griffiths, Problem 4.8)

Starting from the Rodrigues formula, derive the orthonormality condition for Legendre polynomials:

$$\int_{-1}^1 P_\ell(x) P_{\ell'}(x) dx = \left(\frac{2}{2\ell + 1} \right) \delta_{\ell\ell'}.$$

Solution. A slightly cleaner solution uses the following lemma: if $\ell \geq r \geq 0$, then

$$\frac{d^r}{dx^r} (x^2 - 1)^\ell = (x^2 - 1)^{\ell-r} Q_{\ell,r}(x)$$

where $Q_{\ell,r}(x)$ is some polynomial of degree r . In particular, whenever $0 \leq r < \ell$, one has

$$\left. \frac{d^r}{dx^r} (x^2 - 1)^\ell \right|_{x=\pm 1} = 0.$$

This is the approach that was used in David Skinner's lecture notes on mathematical methods.

First we show that the integral is zero when $\ell \neq \ell'$. Suppose, without loss of generality, that $\ell > \ell'$. Now

$$\int_{-1}^1 P_\ell(x) P_{\ell'}(x) dx = \frac{1}{2^\ell 2^{\ell'} \ell! (\ell')!} \int_{-1}^1 \left[\left(\frac{d}{dx} \right)^\ell (x^2 - 1)^\ell \right] \left[\left(\frac{d}{dx} \right)^{\ell'} (x^2 - 1)^{\ell'} \right] dx.$$

We want to integrate by parts in order to transfer derivatives from $(d/dx)^\ell$ onto $(d/dx)^{\ell'}$ until we can differentiate the second term down to zero. This works because each of the boundary terms that arise in this process is zero, as we'll demonstrate. First note that by the Leibniz rule,

$$(fg)^{(n)} = \sum_{k=0}^n \binom{n}{k} f^{(n-k)} g^{(k)}.$$

Thus, the n th derivative of $(x^2 - 1)^\ell$ is given by

$$\begin{aligned} \left(\frac{d}{dx} \right)^n (x^2 - 1)^\ell &= \left(\frac{d}{dx} \right)^n \left[(x+1)^\ell (x-1)^\ell \right] \\ &= \sum_{k=0}^n \binom{n}{k} \left(\frac{d}{dx} \right)^{n-k} (x+1)^\ell \left(\frac{d}{dx} \right)^k (x-1)^\ell. \end{aligned}$$

Now if $n < \ell$, then $n - k$ and k are strictly less than ℓ , so that at least one factor of $x + 1$ and $x - 1$ appear in each term of the sum, and we can factor out an overall $x^2 - 1$ from the sum. Therefore, if $n < \ell$, then the n th derivative of $(x^2 - 1)^\ell$ vanishes at $x = \pm 1$. This result allows us to discard our boundary terms.

Returning to our integral (and omitting the overall multiplicative factor), we integrate by

parts to transfer a derivative from $(d/dx)^\ell$ onto $(d/dx)^{\ell'}$:

$$\begin{aligned}
& \int_{-1}^1 \left[\left(\frac{d}{dx} \right)^\ell (x^2 - 1)^\ell \right] \left[\left(\frac{d}{dx} \right)^{\ell'} (x^2 - 1)^{\ell'} \right] dx \\
&= \cancel{\left(\frac{d}{dx} \right)^{\ell'} (x^2 - 1)^{\ell'} \left(\frac{d}{dx} \right)^{\ell-1} (x^2 - 1)^\ell \Big|_{-1}^1} \\
&\quad - \int_{-1}^1 \left(\frac{d}{dx} \right)^{\ell-1} (x^2 - 1)^\ell \left(\frac{d}{dx} \right)^{\ell'+1} (x^2 - 1)^{\ell'} dx \\
&= - \int_{-1}^1 \left(\frac{d}{dx} \right)^{\ell-1} (x^2 - 1)^\ell \left(\frac{d}{dx} \right)^{\ell'+1} (x^2 - 1)^{\ell'} dx.
\end{aligned}$$

The boundary term vanishes because $\ell - 1 < \ell$, and we repeat this procedure ℓ' more times:

$$\begin{aligned}
& \int_{-1}^1 \left[\left(\frac{d}{dx} \right)^\ell (x^2 - 1)^\ell \right] \left[\left(\frac{d}{dx} \right)^{\ell'} (x^2 - 1)^{\ell'} \right] dx \\
&= (-1)^{\ell'+1} \int_{-1}^1 \left[\left(\frac{d}{dx} \right)^{\ell-\ell'-1} (x^2 - 1)^\ell \right] \left[\left(\frac{d}{dx} \right)^{2\ell'+1} (x^2 - 1)^{\ell'} \right] dx \\
&= 0.
\end{aligned}$$

Note that the first bracketed term is well-defined because $\ell \geq \ell' + 1$ by assumption, and the second bracketed term vanishes because $(x^2 - 1)^{\ell'}$ is a polynomial of degree $2\ell'$. Thus, we get

$$\int_{-1}^1 P_\ell(x) P_{\ell'}(x) dx = 0 \quad \text{for } \ell \neq \ell',$$

which was what we wanted. It remains to show that when $\ell = \ell'$ we get

$$\int_{-1}^1 P_\ell(x) P_\ell(x) dx = \frac{2}{2\ell + 1}.$$

In this case, we perform the integration by parts ℓ' times to get

$$\begin{aligned}
\int_{-1}^1 P_\ell(x) P_\ell(x) dx &= \frac{1}{(2^\ell \ell!)^2} (-1)^\ell \int_{-1}^1 (x^2 - 1)^\ell \left(\frac{d}{dx} \right)^{2\ell} (x^2 - 1)^\ell dx \\
&= \frac{(2\ell)!}{(2^\ell \ell!)^2} (-1)^\ell \int_{-1}^1 (x^2 - 1)^\ell dx \\
&= \frac{(2\ell)!}{(2^\ell \ell!)^2} \underbrace{\int_0^\pi (\sin \theta)^{2\ell+1} d\theta}_{\equiv I_\ell},
\end{aligned}$$

where in the second line, the 2ℓ -times differentiation brings down a factor of $(2\ell)!$, and in the third line we make the substitution $x = \cos \theta$. To evaluate I_ℓ , we integrate by parts, taking

$u = (\sin \theta)^{2\ell}$ and $dv = \sin \theta d\theta$ to get

$$\begin{aligned} I_\ell &= \left[\cancel{-(\sin \theta)^{2\ell} \cos \theta} \Big|_0^\pi + 2\ell \int_0^\pi (\sin \theta)^{2\ell-1} (1 - \sin^2 \theta) d\theta \right] \\ &= 2\ell \int_0^\pi (\sin \theta)^{2\ell-1} d\theta - 2\ell \int_0^\pi (\sin \theta)^{2\ell+1} d\theta \\ &= 2\ell I_{\ell-1} - 2\ell I_\ell, \end{aligned}$$

which yields the recursion relation

$$I_\ell = \frac{2\ell}{2\ell+1} I_{\ell-1}.$$

It is straightforward to see from the definition that $I_0 = 2$, so that we get

$$\begin{aligned} I_\ell &= \frac{(2\ell)!!}{(2\ell+1)!!} \times 2 = \frac{2^\ell \ell!}{(2\ell+1)!/(2^\ell \ell!)} \times 2 \\ &= (2^\ell \ell!)^2 \times \frac{2}{(2\ell+1)!}, \end{aligned}$$

and then upon substitution back into our integral we have

$$\begin{aligned} \int_{-1}^1 P_\ell(x) P_\ell(x) dx &= \frac{(2\ell)!}{(2^\ell \ell!)^2} \times (2^\ell \ell!)^2 \times \frac{2}{(2\ell+1)!} \\ &= \frac{2}{2\ell+1}, \end{aligned}$$

as we wanted. Thus the Legendre polynomials are orthonormal:

$$\boxed{\int_{-1}^1 P_\ell(x) P_{\ell'}(x) dx = \left(\frac{2}{2\ell+1} \right) \delta_{\ell\ell'}}.$$

□

11-8 (Sakurai, Problem 1.8)

Using the orthonormality of $|+\rangle$ and $|-\rangle$, prove

$$[S_i, S_j] = i\epsilon_{ijk} \hbar S_k, \quad \{S_i, S_j\} = \left(\frac{\hbar^2}{2} \right) \delta_{ij},$$

where

$$\begin{aligned} S_x &= \frac{\hbar}{2} (|+\rangle\langle-| + |-\rangle\langle+|), & S_y &= \frac{i\hbar}{2} (-|+\rangle\langle-| + |-\rangle\langle+|), \\ S_z &= \frac{\hbar}{2} (|+\rangle\langle+| - |-\rangle\langle-|). \end{aligned}$$

Solution. Handwritten solution attached below.

□

$$[S_x, S_y] = S_x S_y - S_y S_x$$

$$= \frac{i\hbar^2}{4} \left[(|+\rangle\langle-| + |-\rangle\langle+|) (-|+\rangle\langle-| + |-\rangle\langle+|) \right. \\ \left. - (-|+\rangle\langle-| + |-\rangle\langle+|) (|+\rangle\langle-| + |-\rangle\langle+|) \right]$$

$$= \frac{i\hbar^2}{4} \left[|+\rangle\langle+| - |-\rangle\langle-| \right. \\ \left. + |+\rangle\langle+| - |-\rangle\langle-| \right]$$

$$= i\hbar \frac{\hbar}{2} (|+\rangle\langle+| - |-\rangle\langle-|)$$

$$= i\hbar S_z.$$

$$[S_x, S_z] = S_x S_z - S_z S_x$$

$$= \frac{\hbar^2}{4} \left[(|+\rangle\langle-| + |-\rangle\langle+|) (|+\rangle\langle+| - |-\rangle\langle-|) \right. \\ \left. - (|+\rangle\langle+| - |-\rangle\langle-|) (|+\rangle\langle-| + |-\rangle\langle+|) \right]$$

$$= \frac{\hbar^2}{4} \left[-|+\rangle\langle-| + |-\rangle\langle+| \right. \\ \left. - |+\rangle\langle-| + |-\rangle\langle+| \right]$$

$$= -\frac{i\hbar \hbar}{2} (-|+\rangle\langle-| + |-\rangle\langle+|)$$

$$= -i\hbar S_y.$$

This is consistent because $\epsilon_{xyz} = -1$.

$$\begin{aligned}
[S_y, S_z] &= \frac{i\hbar^2}{4} \left[(-|+\rangle\langle-| + |-\rangle\langle+|) (|+\rangle\langle+| - |-\rangle\langle-|) \right. \\
&\quad \left. - (|+\rangle\langle+| - |-\rangle\langle-|) (-|+\rangle\langle-| + |-\rangle\langle+|) \right] \\
&= \frac{i\hbar^2}{4} \left[|+\rangle\langle-| + |-\rangle\langle+| \right. \\
&\quad \left. + |+\rangle\langle-| + |-\rangle\langle+| \right] \\
&= i\hbar \frac{\hbar}{2} (|+\rangle\langle-| + |-\rangle\langle+|) \\
&= i\hbar S_x.
\end{aligned}$$

thus

$$[S_i, S_j] = i\hbar \epsilon_{ijk} S_k,$$

as we wanted to show.

Most of the work has been done; for the anticommutators we just need to reverse some signs.

For example,

$$\{S_x, S_y\} = S_x S_y + S_y S_x$$

$$= \frac{\hbar^2}{4} \left[\cancel{1+} \cancel{<+1} - \cancel{1-} \cancel{<-1} - \cancel{1+} \cancel{<-1} + \cancel{1-} \cancel{<+1} \right]$$

$$= 0$$

and you can see that all the other $\{S_i, S_j\}$ with $i \neq j$ also vanish. Now

$$\{S_x, S_x\} = 2S_x^2$$

$$= \frac{\hbar^2}{2} (1+<-1 + 1-<+1)(1+<-1 + 1-<+1)$$

$$= \frac{\hbar^2}{2} \left[\underbrace{1+<+1 + 1-<-1}_I \right]$$

$$= \frac{\hbar^2}{2} \cdot I$$

$$\{S_y, S_y\} = -\frac{\hbar^2}{2} (-1+<-1 + 1-<+1)(-1+<-1 + 1-<+1)$$

$$= -\frac{\hbar^2}{2} (-1+<+1 - 1-<-1)$$

$$= \frac{\hbar^2}{2}$$

$$\{S_z, S_z\} = \frac{\hbar^2}{2} (1+<+1 - 1-<-1)(1+<+1 - 1-<-1)$$

$$= \frac{\hbar^2}{2} (1+<+1 + 1-<-1)$$

$$= \frac{\hbar^2}{2} \cdot I$$

Thus $\{S_i, S_j\} = \frac{\hbar^2}{2} \delta_{ij}$.

□

11-9 (Sakurai, Problem 1.9)

Construct $|\mathbf{S} \cdot \hat{\mathbf{n}}\rangle$ such that

$$\mathbf{S} \cdot \hat{\mathbf{n}}|\mathbf{S} \cdot \hat{\mathbf{n}}; +\rangle = \frac{\hbar}{2}|\mathbf{S} \cdot \hat{\mathbf{n}}; +\rangle$$

where $\hat{\mathbf{n}}$ is characterized by the angles shown in the figure. Express your answer as a linear combination of $|+\rangle$ and $|-\rangle$. [Note: Treat the problem as a straightforward eigenvalue problem. Also do not use rotation operators, which we will introduce later in this book.]

Solution. Assigned for Problem Set 1, but with $\alpha \rightarrow \theta$ and $\beta \rightarrow \phi$.

Working in the $|S_z; \pm\rangle$ basis, we have

$$\begin{aligned}\mathbf{S} \cdot \hat{\mathbf{n}} &= S_x \cos \alpha \sin \beta + S_y \sin \alpha \sin \beta + S_z \cos \beta \\ &\stackrel{*}{=} \frac{\hbar}{2}(\cos \alpha \sin \beta \sigma_x + \sin \alpha \sin \beta \sigma_y + \cos \beta \sigma_z) \\ &= \frac{\hbar}{2} \left[\cos \alpha \sin \beta \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \sin \alpha \sin \beta \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + \cos \beta \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right] \\ &= \frac{\hbar}{2} \begin{pmatrix} \cos \beta & e^{-i\alpha} \sin \beta \\ e^{i\alpha} \sin \beta & -\cos \beta \end{pmatrix},\end{aligned}$$

where we used the Euler identity in the last line. To compute the eigenvalues λ of this operator, we solve

$$\begin{aligned}0 = \det(\mathbf{S} \cdot \hat{\mathbf{n}} - \lambda I) &= \left(\lambda - \frac{\hbar}{2} \cos \beta \right) \left(\lambda + \frac{\hbar}{2} \cos \beta \right) - \frac{\hbar^2}{4} \sin^2 \beta \\ &= \lambda^2 - \frac{\hbar^2}{4}.\end{aligned}$$

This shows that the eigenvalues of $\mathbf{S} \cdot \hat{\mathbf{n}}$ are $\lambda = \pm \hbar/2$. To find an eigenvector corresponding to $\lambda = \hbar/2$, we note that the equation

$$\begin{pmatrix} 1 - \cos \beta & -e^{-i\alpha} \sin \beta \\ -e^{i\alpha} \sin \beta & 1 + \cos \beta \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

has the solution

$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 + \cos \beta \\ e^{i\alpha} \sin \beta \end{pmatrix}.$$

To find an eigenvector corresponding to $\lambda = -\hbar/2$, we note that

$$\begin{pmatrix} 1 + \cos \beta & e^{-i\alpha} \sin \beta \\ e^{i\alpha} \sin \beta & 1 - \cos \beta \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

has solution

$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 - \cos \beta \\ -e^{i\alpha} \sin \beta \end{pmatrix}.$$

A direct calculation shows that the first eigenvector has a norm squared of $2(1 + \cos \beta) = 4 \cos^2(\beta/2)$, and the second has a norm squared of $2(1 - \cos \beta) = 4 \sin^2(\beta/2)$. Then we can rewrite both eigenvectors using standard trigonometric identities to get

$$\begin{pmatrix} 2 \cos^2(\beta/2) \\ 2e^{i\alpha} \sin(\beta/2) \cos(\beta/2) \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 2 \sin^2(\beta/2) \\ -2e^{i\alpha} \sin(\beta/2) \cos(\beta/2) \end{pmatrix}$$

corresponding to $\pm \hbar/2$, respectively. After dividing through by their norms, we find that

Working in terms of the $|S_z; \pm\rangle$ basis, the eigenvectors of $\mathbf{S} \cdot \hat{\mathbf{n}}$ are

$$\begin{pmatrix} \cos(\beta/2) \\ e^{i\alpha} \sin(\beta/2) \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} \sin(\beta/2) \\ -e^{i\alpha} \cos(\beta/2) \end{pmatrix},$$

corresponding to $\hbar/2$ and $-\hbar/2$, respectively. In the abstract ket notation,

$$|\mathbf{S} \cdot \hat{\mathbf{n}}; +\rangle = \cos\left(\frac{\beta}{2}\right)|+\rangle + e^{i\alpha} \sin\left(\frac{\beta}{2}\right)|-\rangle,$$

$$|\mathbf{S} \cdot \hat{\mathbf{n}}; -\rangle = \sin\left(\frac{\beta}{2}\right)|+\rangle - e^{i\alpha} \cos\left(\frac{\beta}{2}\right)|-\rangle.$$

Strictly speaking, we should have been more careful about the cases where $\beta = 0$ or π , because then one of our two (non-normalized) eigenvectors degenerates to the zero vector. But it appears that the process of normalization leads to a pair of eigenvectors that are orthonormal for all $0 \leq \beta \leq \pi$, so there seems to be no issue.

One can check that these reduce to the eigenvectors of S_x , S_y , and S_z in the appropriate cases. For instance, when $\alpha = 0$ and $\beta = \pi/2$, so that $\mathbf{S} \cdot \hat{\mathbf{n}} = \mathbf{S} \cdot \hat{\mathbf{i}} = S_x$, these become

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \end{pmatrix}^T \quad \text{and} \quad \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \end{pmatrix}^T,$$

as we expect. □

11-10 (Sakurai, Problem 1.10)

The Hamiltonian operator for a two-state system is given by

$$H = a(|1\rangle\langle 1| - |2\rangle\langle 2| + |1\rangle\langle 2| + |2\rangle\langle 1|),$$

where a is a number with the dimensions of energy. Find the energy eigenvalues and the corresponding energy eigenkets (as linear combinations of $|1\rangle$ and $|2\rangle$.)

Solution. Using the orthonormality of the base kets, it's straightforward to see that the matrix representation of H in this basis is

$$\begin{pmatrix} a & a \\ a & -a \end{pmatrix}.$$

Then we have $\det(H - \lambda I) = (\lambda - a)(\lambda + a) - a^2 = \lambda^2 - 2a^2$, so the eigenvalues are $\boxed{\lambda = \pm\sqrt{2}a}$. It is straightforward to verify that

$$\begin{pmatrix} (1 - \sqrt{2})a & a \\ a & -(1 + \sqrt{2})a \end{pmatrix} \begin{pmatrix} 1 + \sqrt{2} \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$
$$\begin{pmatrix} (1 + \sqrt{2})a & a \\ a & -(1 - \sqrt{2})a \end{pmatrix} \begin{pmatrix} 1 - \sqrt{2} \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

which shows that the energy eigenkets are

$$(1 + \sqrt{2})|1\rangle + |2\rangle \quad \text{and} \quad (1 - \sqrt{2})|1\rangle + |2\rangle,$$

corresponding to the eigenvalues $\sqrt{2}a$ and $-\sqrt{2}a$, respectively. □

11-11 (Sakurai, Problem 1.12)

A spin $\frac{1}{2}$ system is known to be in an eigenstate of $\mathbf{S} \cdot \hat{\mathbf{n}}$ with eigenvalue $\hbar/2$, where $\hat{\mathbf{n}}$ is a unit vector lying in the xz -plane that makes an angle γ with the positive z -axis.

- Suppose S_x is measured. What is the probability of getting $+\hbar/2$?
- Evaluate the dispersion in S_x , that is

$$\langle (S_x - \langle S_x \rangle)^2 \rangle.$$

(For your own peace of mind check your answers for the special cases $\gamma = 0$, $\pi/2$, and π .)

Solution. (a) We will use the result of Problem 1.9, noting that since $\hat{\mathbf{n}}$ lies in the xz -plane we have $\alpha = 0$, and since $\beta = \gamma$ we have

$$|\mathbf{S} \cdot \hat{\mathbf{n}}; +\rangle = \cos\left(\frac{\gamma}{2}\right)|+\rangle + \sin\left(\frac{\gamma}{2}\right)|-\rangle.$$

The amplitude we want to compute is

$$\begin{aligned} \langle S_x; + | \mathbf{S} \cdot \hat{\mathbf{n}}; + \rangle &= \frac{1}{\sqrt{2}} \langle + | \mathbf{S} \cdot \hat{\mathbf{n}}; + \rangle + \frac{1}{\sqrt{2}} \langle - | \mathbf{S} \cdot \hat{\mathbf{n}}; + \rangle \\ &= \frac{1}{\sqrt{2}} \cos\left(\frac{\gamma}{2}\right) + \frac{1}{\sqrt{2}} \sin\left(\frac{\gamma}{2}\right), \end{aligned}$$

and the corresponding probability is

$$\begin{aligned} |\langle S_x; + | \mathbf{S} \cdot \hat{\mathbf{n}}; + \rangle|^2 &= \frac{1}{2} \left(\sin^2(\gamma/2) + \cos^2(\gamma/2) + 2 \sin(\gamma/2) \cos(\gamma/2) \right) \\ &= \boxed{\frac{1 + \sin \gamma}{2}}. \end{aligned}$$

This makes sense; when $\gamma = \pi/2$ we have $\mathbf{S} \cdot \hat{\mathbf{n}} = S_x$, so that the probability is 1. For $\gamma = 0$ we have $\mathbf{S} \cdot \hat{\mathbf{n}} = S_z$, so that both eigenvalues for S_x are equally likely to be measured.

- We want to compute

$$\begin{aligned} \langle (S_x - \langle S_x \rangle)^2 \rangle &= \langle S_x^2 - 2S_x \langle S_x \rangle + \langle S_x \rangle^2 \rangle \\ &= \langle S_x^2 \rangle - 2\langle S_x \rangle^2 + \langle S_x \rangle^2 \\ &= \langle S_x^2 \rangle - \langle S_x \rangle^2. \end{aligned}$$

First, we have

$$\langle S_x^2 \rangle = \langle \mathbf{S} \cdot \hat{\mathbf{n}}; + | S_x^2 | \mathbf{S} \cdot \hat{\mathbf{n}}; + \rangle, \quad (3)$$

and since (doing the calculation in the S_z basis)

$$\begin{aligned}
S_x|\mathbf{S} \cdot \hat{\mathbf{n}}; +\rangle &\stackrel{*}{=} \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \cos(\gamma/2) \\ \sin(\gamma/2) \end{pmatrix} \\
&= \frac{\hbar}{2} \begin{pmatrix} \sin(\gamma/2) \\ \cos(\gamma/2) \end{pmatrix} \\
&\stackrel{*}{=} \frac{\hbar}{2} \left[\sin\left(\frac{\gamma}{2}\right)|+\rangle + \cos\left(\frac{\gamma}{2}\right)|-\rangle \right]
\end{aligned}$$

and $S_x = S_x^\dagger$, it follows from the orthonormality of $|\pm\rangle$ that

$$\langle S_x^2 \rangle = \frac{\hbar^2}{4} \left(\sin^2(\gamma/2) + \cos^2(\gamma/2) \right) = \frac{\hbar^2}{4}.$$

It follows immediately from our expression for $S_x|\mathbf{S} \cdot \hat{\mathbf{n}}; +\rangle$ that

$$\begin{aligned}
\langle S_x \rangle &= \langle \mathbf{S} \cdot \hat{\mathbf{n}}; + | S_x | \mathbf{S} \cdot \hat{\mathbf{n}}; + \rangle \\
&= \frac{\hbar}{2} \left[\sin\left(\frac{\gamma}{2}\right) \langle + | + \cos\left(\frac{\gamma}{2}\right) \langle - | \right] \left[\cos\left(\frac{\gamma}{2}\right) | + \rangle + \sin\left(\frac{\gamma}{2}\right) | - \rangle \right] \\
&= \frac{\hbar}{2} \sin \gamma,
\end{aligned}$$

which implies $\langle S_x \rangle^2 = (\hbar^2/4) \sin^2 \gamma$. Thus, the dispersion for this state is

$$\boxed{\langle (S_x - \langle S_x \rangle)^2 \rangle = \frac{\hbar^2}{4} \cos^2 \gamma}.$$

This result makes sense: when $\gamma = \pi/2$, so that the system is in a definite S_x state (in particular, the one corresponding to $+\hbar/2$), the dispersion vanishes. The dispersion attains a maximum value of $\hbar^2/4$ at $\gamma = 0$ and π , which correspond to states of definite S_z . This result agrees with Eq. (1.4.52) in Sakurai. $\square$

11-12 (Sakurai, Problem 1.13)

A beam of spin $\frac{1}{2}$ atoms goes through a series of Stern-Gerlach-type measurements as follows:

- The first measurement accepts $s_z = \hbar/2$ atoms and rejects $s_z = -\hbar/2$ atoms.
- The second measurement accepts $s_n = \hbar/2$ atoms and rejects $s_n = -\hbar/2$ atoms, where s_n is the eigenvalue of the operator $\mathbf{S} \cdot \hat{\mathbf{n}}$, with $\hat{\mathbf{n}}$ making an angle β in the xz -plane with respect to the z -axis.
- The third measurement accepts $s_z = -\hbar/2$ atoms and rejects $s_z = \hbar/2$ atoms.

What is the intensity of the final $s_z = -\hbar/2$ beam when the $s_z = \hbar/2$ beam surviving the first measurement is normalized to unity? How must we orient the second measuring apparatus if we are to maximize the intensity of the final $s_z = -\hbar/2$ beam?

Solution. For the orientation $\hat{\mathbf{n}}$ specified in the problem, we have

$$\begin{aligned} |\mathbf{S} \cdot \hat{\mathbf{n}}; +\rangle &= \cos\left(\frac{\beta}{2}\right)|+\rangle + \sin\left(\frac{\beta}{2}\right)|-\rangle, \\ |\mathbf{S} \cdot \hat{\mathbf{n}}; -\rangle &= \sin\left(\frac{\beta}{2}\right)|+\rangle - \cos\left(\frac{\beta}{2}\right)|-\rangle. \end{aligned}$$

This implies that

$$\cos\left(\frac{\beta}{2}\right)|\mathbf{S} \cdot \hat{\mathbf{n}}; +\rangle + \sin\left(\frac{\beta}{2}\right)|\mathbf{S} \cdot \hat{\mathbf{n}}; -\rangle = |+\rangle.$$

Thus, after the second measurement filters out the $s_n = -\hbar/2$ atoms, we are left with the state

$$\cos\left(\frac{\beta}{2}\right)|\mathbf{S} \cdot \hat{\mathbf{n}}; +\rangle.$$

Decomposing this ket into its $|\pm\rangle$ components then filtering out the $s_z = \hbar/2$ part gives

$$\sin\left(\frac{\beta}{2}\right)\cos\left(\frac{\beta}{2}\right)|-\rangle = \frac{1}{2}\sin\beta|-\rangle.$$

Thus, the intensity of the final $s_z = -\hbar/2$ beam is $\boxed{\frac{1}{4}\sin^2\beta}$. To maximize this intensity, choose $\boxed{\beta = \pi/2}$, i.e., orient the second measuring apparatus perpendicular to the others. $\square$

11-13 (Calculating D-matrix elements)

Solution. Following Sankara's convention for the Euler angles, a general rotation matrix element in the $|jm\rangle$ basis is given by

$$D_{m'm}^j(\alpha, \beta, \gamma) = e^{-im'\alpha} e^{-im\gamma} d_{m'm}^j(\beta),$$

where the matrix element $d_{m'm}^j(\beta)$ is given by the sum

$$d_{m'm}^j(\beta) = \sum_{k=k_{\min}}^{k_{\max}} \frac{(-1)^{k-m+m'} \sqrt{(j+m)!(j-m)!(j+m')!(j-m')!}}{(j+m-k)!k!(j-k-m')!(k-m+m')!} \\ \times \left(\cos \frac{\beta}{2}\right)^{2j-2k+m-m'} \left(\sin \frac{\beta}{2}\right)^{2k-m+m'},$$

and where the bounds

$$k_{\min} = \max(0, m - m'), \quad k_{\max} = \min(j + m, j - m')$$

are chosen to ensure that none of the arguments of factorials in the denominator are negative. These sums are tedious to evaluate by hand, but relatively straightforward to program into Mathematica. The matrix for $j = 3/2$ is given on the following page.² $\square$

²At first, I spent a long time wondering why my matrices didn't match the expressions given in Sakurai. Then I remembered that the conventional ordering is to put the highest value of m first, as one does when writing the spin-1/2 basis as $\{|+\rangle, |-\rangle\}$. It's easy to get confused if you're used to having your matrix indices increment upward as you go down or to the right!

$$\begin{aligned}
In[6]:= & d[j_][mp_ , m_][\beta_] := Sum[\\
& (((-1)^{k-m+mp} \sqrt{((j+m)!(j-m)!(j+mp)!(j-mp)!)) / ((j+m-k)!(k)!(j-k-mp)!(k-m+mp)!)} \\
& \left(\cos\left[\frac{\beta}{2}\right]\right)^{2j-2k+m-mp} \left(\sin\left[\frac{\beta}{2}\right]\right)^{2k-m+mp}, \\
& \{k, \text{Max}[0, m-mp], \text{Min}[j+m, j-mp]\} \\
&] \\
& d[j_][mp_ , m_][\alpha_ , \beta_ , \gamma_] := e^{-i mp \alpha} e^{-i m \gamma} d[j][mp, m][\beta]
\end{aligned}$$

$$In[6]:= j = 3/2;$$

$$Table[d[j][mp, m][\alpha, \beta, \gamma], \{mp, j, -j, -1\}, \{m, j, -j, -1\}] // FullSimplify // MatrixForm$$

Out[6]/MatrixForm=

$$\begin{pmatrix}
e^{\frac{3}{2}i(\alpha+\gamma)} \cos\left[\frac{\beta}{2}\right]^3 & -\frac{1}{4}\sqrt{3}e^{\frac{1}{2}i(3\alpha+\gamma)} \csc\left[\frac{\beta}{2}\right] \sin[\beta]^2 & \frac{1}{2}\sqrt{3}e^{\frac{1}{2}i(3\alpha-\gamma)} \sin\left[\frac{\beta}{2}\right] \sin[\beta] & -e^{\frac{3}{2}i(\alpha-\gamma)} \sin\left[\frac{\beta}{2}\right]^3 \\
\frac{1}{4}\sqrt{3}e^{\frac{1}{2}i(\alpha+3\gamma)} \csc\left[\frac{\beta}{2}\right] \sin[\beta]^2 & \frac{1}{4}e^{\frac{1}{2}i(\alpha+\gamma)} \left(\cos\left[\frac{\beta}{2}\right] + 3\cos\left[\frac{3\beta}{2}\right]\right) & \frac{1}{4}e^{\frac{1}{2}i(\alpha-\gamma)} \left(\sin\left[\frac{\beta}{2}\right] - 3\sin\left[\frac{3\beta}{2}\right]\right) & \frac{1}{2}\sqrt{3}e^{\frac{1}{2}i(\alpha-3\gamma)} \sin\left[\frac{\beta}{2}\right] \sin[\beta] \\
\frac{1}{2}\sqrt{3}e^{\frac{1}{2}i(\alpha-3\gamma)} \sin\left[\frac{\beta}{2}\right] \sin[\beta] & \frac{1}{2}e^{\frac{1}{2}i(\alpha-\gamma)} (1+3\cos[\beta]) \sin\left[\frac{\beta}{2}\right] & \frac{1}{2}e^{\frac{1}{2}i(\alpha+\gamma)} \cos\left[\frac{\beta}{2}\right] (-1+3\cos[\beta]) & -\frac{1}{4}\sqrt{3}e^{\frac{1}{2}i(\alpha+3\gamma)} \csc\left[\frac{\beta}{2}\right] \sin[\beta]^2 \\
e^{\frac{3}{2}i(\alpha-\gamma)} \sin\left[\frac{\beta}{2}\right]^3 & \frac{1}{2}\sqrt{3}e^{\frac{1}{2}i(3\alpha-\gamma)} \sin[\beta] & \frac{1}{4}\sqrt{3}e^{\frac{1}{2}i(3\alpha+\gamma)} \csc\left[\frac{\beta}{2}\right] \sin[\beta]^2 & e^{\frac{3}{2}i(\alpha+\gamma)} \cos\left[\frac{\beta}{2}\right]^3
\end{pmatrix}$$

11-14 (Constructing orbital angular momentum eigenstates)

Solution. To prove this equality, we will evaluate the matrix element $\langle \theta, \phi | L_-^{\ell-m} | \ell, \ell \rangle$. We first note that by the definition of L_- and the properties of the orbital angular momentum operators L_x and L_y , we have

$$\begin{aligned} \langle \mathbf{x}' | L_- | \alpha \rangle &= e^{-i\phi} \left(-\frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \phi} \right) \langle \mathbf{x}' | \alpha \rangle \\ &\equiv D_- \langle \mathbf{x}' | \alpha \rangle. \end{aligned}$$

It follows that $\langle \mathbf{x}' | L_-^n | \alpha \rangle = D_-^n \langle \mathbf{x}' | \alpha \rangle$, and in particular

$$\langle \theta, \phi | L_-^{\ell-m} | \ell, \ell \rangle = D_-^{\ell-m} \langle \theta, \phi | \ell, \ell \rangle. \quad (1)$$

The differential operator on the right acts on the spherical harmonic $Y_\ell^\ell(\theta, \phi) \equiv \langle \theta, \phi | \ell, \ell \rangle$, which we showed in class to be

$$\langle \theta, \phi | \ell, \ell \rangle = c_\ell e^{i\ell\phi} \sin^\ell \theta, \quad (2)$$

where c_ℓ is a normalization constant given by

$$c_\ell = \frac{(-1)^\ell}{2^\ell \ell!} \sqrt{\frac{(2\ell+1)(2\ell)!}{4\pi}}. \quad (3)$$

Now, we showed in the Feb 13 class exercise (see attached scans) that

$$D_-^n [f_\ell(\theta) e^{im\phi}] = \sin^{n-m} \theta \frac{d^n}{d(\cos \theta)^n} (\sin^m \theta f_\ell(\theta)) e^{i(m-n)\phi}.$$

Applying this identity to (2) with $n \rightarrow \ell - m$, $f_\ell(\theta) \rightarrow c_\ell \sin^\ell \theta$, and $m \rightarrow \ell$, we get

$$\begin{aligned} D_-^{\ell-m} [c_\ell \sin^\ell \theta e^{i\ell\phi}] &= c_\ell \sin^{(\ell-m)-\ell} \theta \frac{d^{\ell-m}}{d(\cos \theta)^{\ell-m}} (\sin^{2\ell} \theta) e^{i(\ell-(\ell-m))\phi} \\ &= \frac{(-1)^\ell}{2^\ell \ell!} \sqrt{\frac{(2\ell+1)(2\ell)!}{4\pi}} \frac{e^{im\phi}}{\sin^m \theta} \frac{d^{\ell-m}}{d(\cos \theta)^{\ell-m}} (\sin^{2\ell} \theta), \end{aligned}$$

where we used (3) in the second line. It follows from (1) that

$$\langle \theta, \phi | L_-^{\ell-m} | \ell, \ell \rangle = \frac{(-1)^\ell}{2^\ell \ell!} \sqrt{\frac{(2\ell+1)(2\ell)!}{4\pi}} \frac{e^{im\phi}}{\sin^m \theta} \frac{d^{\ell-m}}{d(\cos \theta)^{\ell-m}} (\sin^{2\ell} \theta),$$

which in turn implies

$$\boxed{\sqrt{\frac{(\ell+m)!}{(2\ell)!(\ell-m)!}} \langle \theta, \phi | L_-^{\ell-m} | \ell, \ell \rangle = \frac{(-1)^\ell}{2^\ell \ell!} \sqrt{\frac{2\ell+1}{4\pi}} \frac{(\ell+m)!}{(\ell-m)!} \frac{e^{im\phi}}{\sin^m \theta} \frac{d^{\ell-m}}{d(\cos \theta)^{\ell-m}} (\sin^{2\ell} \theta)}$$

as we wanted to show. □

EXERCISE

$$D^{-2} [f_\rho(\theta) e^{im\phi}] = ?$$

$$D^{-3} [f_\rho(\theta) e^{im\phi}] = ?$$

$$D^{-n} [f_\rho(\theta) e^{im\phi}] = ?$$

$$(a) D_- D_- [f_\rho(\theta) e^{im\phi}]$$

$$= D_- \left[\sin^{1-m} \theta \frac{d}{d(\cos \theta)} (\sin^m \theta f_\rho) e^{i(m-1)\phi} \right]$$

$$= \sin^{1-m'} \theta \frac{d}{d(\cos \theta)} (\sin^{m'} \theta g_\rho) e^{i(m'-1)\phi}$$

$$= \sin^{1-(m-1)} \theta \frac{d}{d(\cos \theta)} (\sin^{m-1} \theta g_\rho) e^{i(m-1-1)\phi}$$

$$= \sin^{2-m} \theta \frac{d}{d(\cos \theta)} (\sin^{m-1} \theta g_\rho) e^{i(m-2)\phi}$$

$$= \sin^{2-m} \theta \frac{d}{d(\cos \theta)} \left(\cancel{\sin^{m-1} \theta} \cancel{\sin^{1-m'}} \theta \frac{d}{d(\cos \theta)} (\sin^m \theta f_\rho) \right)$$

$$= \sin^{2-m} \theta \frac{d^2}{d(\cos \theta)^2} (\sin^m \theta f_\rho) e^{i(m-2)\phi}$$

Repeat for D_-^3

(b) $D_- D_-^2 [f_\phi e^{im\phi}]$

55

$$= D_- \left[\sin^{2-m} \theta \frac{d^2}{d(\cos\theta)^2} (\sin^m f_1) e^{i(m-2)\phi} \right]$$

$m' = m - 2$

$$= \sin^{1-m'} \theta \frac{d}{d(\cos\theta)} \left(\sin^{m'} \theta g_1 \right) e^{i(m'-1)\phi}$$

$$= \sin^{3-m} \theta \frac{d}{d(\cos\theta)} \left(\sin^{m-2} \theta g_1 \right) e^{i(m-3)\phi}$$

$$= \sin^{3-m} \theta \frac{d^3}{d(\cos\theta)^3} (\sin^m f_1) e^{i(m-3)\phi}$$

(c) $D_-^n [f_\phi e^{im\phi}] = \sin^{n-m} \theta \frac{d^n}{d(\cos\theta)^n} (\sin^m f_1) e^{i(m-n)\phi}$

11-15 (Orbital angular momentum eigenstates have integer eigenvalues)

Solution. (a) Since $|\frac{1}{2}\frac{1}{2}\rangle$ has the maximum allowed value of ℓ , it follows that

$$\begin{aligned}
 & \langle \hat{n} | L_+ | \frac{1}{2}\frac{1}{2} \rangle = 0 \\
 \Longleftrightarrow & -i\hbar e^{i\phi} \left(i \frac{\partial}{\partial \theta} - \cot \theta \frac{\partial}{\partial \phi} \right) \langle \hat{n} | \frac{1}{2}\frac{1}{2} \rangle = 0 \\
 \Longleftrightarrow & i \frac{\partial Y_{1/2}^{1/2}}{\partial \theta} - \cot \theta \frac{\partial Y_{1/2}^{1/2}}{\partial \phi} = 0 \\
 \Longleftrightarrow & i \frac{\partial Y_{1/2}^{1/2}}{\partial \theta} - \cot \theta \left(-\frac{i}{2} \right) Y_{1/2}^{1/2} = 0 \\
 \Longleftrightarrow & \frac{\partial Y_{1/2}^{1/2}}{\partial \theta} - \frac{1}{2} \cot \theta Y_{1/2}^{1/2} = 0.
 \end{aligned}$$

In the second-to-last line, we use the fact that $Y_\ell^m(\theta, \phi) \sim e^{im\phi}$, which was derived in class by applying L_z to the ket $|\ell, m\rangle$ and solving the resulting differential equation. Now, the differential equation in the last line has solution

$$Y_{1/2}^{1/2}(\theta, \phi) = c_{1/2} e^{i\phi/2} \sqrt{\sin \theta},$$

where $c_{1/2}$ is a normalization constant; compare (3.6.34)–(3.6.35) of Sakurai.

(b) We then construct $Y_{1/2}^{-1/2}(\theta, \phi)$ by applying the lowering operator L_- to the ket $|\frac{1}{2}\frac{1}{2}\rangle$:

$$\begin{aligned}
 \langle \hat{n} | \frac{1}{2}, -\frac{1}{2} \rangle &= \frac{\langle \hat{n} | L_- | \frac{1}{2}\frac{1}{2} \rangle}{\sqrt{(\frac{1}{2} + \frac{1}{2})(\frac{1}{2} - \frac{1}{2} + 1)\hbar}} \\
 &= e^{-i\phi} \left(-\frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \phi} \right) \langle \hat{n} | \frac{1}{2}\frac{1}{2} \rangle \\
 &= e^{-i\phi} \left(-\frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \phi} \right) (c_{1/2} e^{i\phi/2} \sqrt{\sin \theta}) \\
 &= -c_{1/2} e^{-i\phi/2} \cot \theta \sqrt{\sin \theta}.
 \end{aligned}$$

That is,

$$Y_{1/2}^{-1/2}(\theta, \phi) = -c_{1/2} e^{-i\phi/2} \cot \theta \sqrt{\sin \theta}.$$

(c) Another way to construct $Y_{1/2}^{-1/2}(\theta, \phi)$ is by applying L_- to $|\frac{1}{2}, -\frac{1}{2}\rangle$, which gives

$$\begin{aligned}
0 &= \langle \hat{\mathbf{n}} | L_- | \frac{1}{2}, -\frac{1}{2} \rangle \\
&= \left(-\frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \phi} \right) Y_{1/2}^{-1/2} \\
&= -\frac{\partial Y_{1/2}^{-1/2}}{\partial \theta} + i \cot \theta \left(-\frac{i}{2} \right) Y_{1/2}^{-1/2} \\
&= -\frac{\partial Y_{1/2}^{-1/2}}{\partial \theta} + \frac{1}{2} \cot \theta Y_{1/2}^{-1/2}.
\end{aligned}$$

But this differential equation is identical to the one we solved in part (a), and has solution

$$Y_{1/2}^{-1/2}(\theta, \phi) = c_{-1/2} e^{-i\phi/2} \sqrt{\sin \theta}.$$

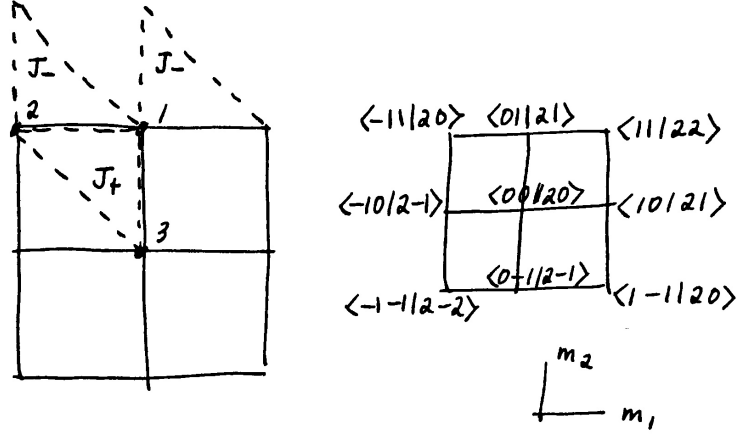
This is inconsistent with the result of part (b), since the θ dependence is different. □

11-16 (Calculating Clebsch-Gordan coefficients)

Solution. First we collect our recursion relations, corresponding to J_+ and J_- respectively:

$$\begin{aligned} \sqrt{(j \mp m)(j \pm m + 1)} \langle m_1 m_2 | j, m \pm 1 \rangle \\ = \sqrt{(j_1 \mp m_1 + 1)(j_1 \pm m_1)} \langle m_1 \mp 1, m_2 | jm \rangle \\ + \sqrt{(j_2 \mp m_2 + 1)(j_2 \pm m_2)} \langle m_1, m_2 \mp 1 | jm \rangle. \end{aligned}$$

We will calculate the Clebsch-Gordan coefficients, in terms of the upper-right coefficient $\langle 1, 1 | 2, 2 \rangle$, in the following order:



There are only three triangles in this picture, due to the identities

$$\langle j_1 j_2 m_1 m_2 | jm \rangle = (-1)^{j_1 + j_2 - j} \langle j_1 j_2 m_2 m_1 | jm \rangle, \quad (1)$$

$$\langle j_1 j_2 m_1 m_2 | jm \rangle = (-1)^{j_1 + j_2 - j} \langle j_1 j_2, -m_1, -m_2 | j, -m \rangle. \quad (2)$$

In words, identity (1) specifies how the Clebsch-Gordan coefficients change under a reflection about $m_1 = m_2$, while identity (2) specifies how the coefficients change under an inversion, $(m_1, m_2) \rightarrow (-m_1, -m_2)$. It should be clear from inspection that the three coefficients in the picture, together with the coefficient $\langle 1, 1 | 2, 2 \rangle$ and identities (1)–(2), are enough to specify the other four.³

Triangle 1:

$$m = 2, \quad m_1 = 0, \quad m_2 = 1$$

$$\begin{aligned} \sqrt{(2+2)(2-2+1)} \langle 0, 1 | 2, 1 \rangle \\ = \sqrt{(1+0+1)(1-0)} \langle 1, 1 | 2, 2 \rangle \\ + \sqrt{1+1+1} (1-1) \langle 0, 2 | 2, 2 \rangle \end{aligned}$$

³I've also attached my first, handwritten solution, where I do not make use of identities (1) or (2), to the end of this problem set. For some reason, I use nine triangles when eight would have been sufficient.

$$\begin{aligned}\Rightarrow \quad 2\langle 0, 1|2, 1\rangle &= \sqrt{2}\langle 1, 1|2, 2\rangle \\ \Rightarrow \quad \langle 0, 1|2, 1\rangle &= \frac{1}{\sqrt{2}}\langle 1, 1|2, 2\rangle.\end{aligned}$$

Triangle 2:

$$m = 1, \quad m_1 = -1, \quad m_2 = 1$$

$$\begin{aligned}\sqrt{(2+1)(2-1+1)}\langle -1, 1|2, 0\rangle \\ = \sqrt{(\cancel{1}-\cancel{1}+1)(1+1)}\langle 0, 1|2, 1\rangle \\ + \sqrt{(\cancel{1}+\cancel{1}+1)(1-1)}\langle -1, 2|2, 1\rangle\end{aligned}$$

$$\begin{aligned}\Rightarrow \quad \sqrt{6}\langle -1, 1|2, 0\rangle &= \sqrt{2}\langle 0, 1|2, 1\rangle \\ \Rightarrow \quad \langle -1, 1|2, 0\rangle &= \frac{1}{\sqrt{3}}\langle 0, 1|2, 1\rangle \\ &= \sqrt{\frac{2}{3}}\langle 1, 1|2, 2\rangle.\end{aligned}$$

Triangle 3:

$$m = 0, \quad m_1 = 0, \quad m_2 = 1$$

$$\begin{aligned}\sqrt{(2-0)(2+0+1)}\langle 0, 1|2, 1\rangle \\ = \sqrt{(1-0+1)(1+0)}\langle -1, 1|2, 0\rangle \\ + \sqrt{(\cancel{1}-\cancel{1}+1)(1+1)}\langle 0, -1|2, 0\rangle\end{aligned}$$

$$\begin{aligned}\Rightarrow \quad \sqrt{6}\langle 0, 1|2, 1\rangle &= \sqrt{2}\langle -1, 1|2, 0\rangle + \sqrt{2}\langle 0, 0|2, 0\rangle \\ \Rightarrow \quad \langle 0, 0|2, 0\rangle &= \sqrt{3}\langle 0, 1|2, 1\rangle - \langle -1, 1|2, 0\rangle \\ &= \left(\sqrt{\frac{3}{2}} - \sqrt{\frac{2}{3}}\right)\langle 1, 1|2, 2\rangle \\ &= \sqrt{\frac{1}{6}}\langle 1, 1|2, 2\rangle.\end{aligned}$$

These three coefficients, together with identities (1) and (2), imply that the Clebsch-Gordan

coefficients for $j_1 = j_2 = 1$ and $j = 2$ are given by

$$\begin{aligned}
 \langle 0, 1|2, 1\rangle &= \frac{1}{\sqrt{2}}\langle 1, 1|2, 2\rangle, \\
 \langle 1, 0|2, 1\rangle &= \frac{1}{\sqrt{2}}\langle 1, 1|2, 2\rangle, & \langle 0, -1|2, -1\rangle &= \frac{1}{\sqrt{2}}\langle 1, 1|2, 2\rangle, \\
 \langle 0, 0|2, 0\rangle &= \sqrt{\frac{2}{3}}\langle 1, 1|2, 2\rangle, & \langle -1, 0|2, -1\rangle &= \frac{1}{\sqrt{2}}\langle 1, 1|2, 2\rangle, \\
 \langle -1, 1|2, 0\rangle &= \frac{1}{\sqrt{6}}\langle 1, 1|2, 2\rangle, & \langle -1, -1|2, -2\rangle &= \langle 1, 1|2, 2\rangle. \\
 \langle 1, -1|2, 0\rangle &= \frac{1}{\sqrt{6}}\langle 1, 1|2, 2\rangle,
 \end{aligned}$$

□

12 Scattering theory

12-1 (Taylor, Problem 1.2)

Solution. For the second statement, we observe that

$$\begin{aligned}
 \psi_{\boldsymbol{\rho}}(\mathbf{p}) &= \langle \mathbf{p} | \psi_{\boldsymbol{\rho}} \rangle \\
 &= \int d^3x \langle \mathbf{p} | \mathbf{x} \rangle \langle \mathbf{x} | \psi_{\boldsymbol{\rho}} \rangle \\
 &= \int d^3x \frac{1}{(2\pi)^{3/2}} \exp(-i\mathbf{p} \cdot \mathbf{x}) \psi(\mathbf{x} - \boldsymbol{\rho}) \\
 &= \exp(-i\mathbf{p} \cdot \boldsymbol{\rho}) \int d^3x' \frac{1}{(2\pi)^{3/2}} \exp(-i\mathbf{p} \cdot \mathbf{x}') \psi(\mathbf{x}') \\
 &= \exp(-i\mathbf{p} \cdot \boldsymbol{\rho}) \psi(\mathbf{p}),
 \end{aligned}$$

where in the third line we changed $\mathbf{x} \rightarrow \mathbf{x}' + \boldsymbol{\rho}$. With the momentum-space representation, it is easy to see that $|\psi_{\boldsymbol{\rho}}\rangle$ does not go to zero in the $\boldsymbol{\rho} \rightarrow \infty$ limit; in fact, the norm is preserved under translations:

$$\|\psi_{\boldsymbol{\rho}}\|^2 = \int d^3p \langle \psi_{\boldsymbol{\rho}} | \mathbf{p} \rangle \langle \mathbf{p} | \psi_{\boldsymbol{\rho}} \rangle = \int d^3p |\psi(\mathbf{p})|^2 = \|\psi\|^2.$$

To examine the overlap $\langle \phi | \psi_{\boldsymbol{\rho}} \rangle$, we work in the position representation:

$$\begin{aligned}
 \langle \phi | \psi_{\boldsymbol{\rho}} \rangle &= \int d^3x \langle \phi | \mathbf{x} \rangle \langle \mathbf{x} | \psi_{\boldsymbol{\rho}} \rangle \\
 &= \int d^3x \phi(\mathbf{x})^* \psi(\mathbf{x} - \boldsymbol{\rho}).
 \end{aligned}$$

Now, assuming $\psi(\mathbf{x})$ is localized in space, which should be true if it is to be normalizable,⁴ it follows that for any fixed $\mathbf{x}$, the wave function $\psi(\mathbf{x} - \boldsymbol{\rho})$ is vanishingly small in the $\boldsymbol{\rho} \rightarrow \infty$ limit. Of course, there will still be some points where $\psi(\mathbf{x} - \boldsymbol{\rho})$ is non-negligible, but the idea is that in sending $\boldsymbol{\rho}$ to infinity, those points move away from where $\phi(\mathbf{x})$ is non-negligible (for the same reasons, $\phi(\mathbf{x})$ must be localized in space).

Thus $\phi(\mathbf{x})^* \psi(\mathbf{x} - \boldsymbol{\rho}) \xrightarrow{\boldsymbol{\rho} \rightarrow \infty} 0$ for all points $\mathbf{x}$, so the whole integral vanishes, and

$$\langle \phi | \psi_{\boldsymbol{\rho}} \rangle \xrightarrow{\boldsymbol{\rho} \rightarrow \infty} 0.$$

□

⁴To be more precise, we can use the result that any $|\psi\rangle \in \mathcal{L}^2(\mathbb{R}^3)$ can be arbitrarily well-approximated by a finite sum of Gaussians. Then the wave function $\psi(\mathbf{x})$ surely goes to zero if we send $\mathbf{x} \rightarrow \infty$, since it is a sum of exponentially decaying functions.

12-2 (Taylor, Problem 1.3)

Solution.

(1) Suppose $|\psi_t\rangle \rightarrow |\psi\rangle$ as $t \rightarrow \infty$, which means that

$$\|\psi_t - \psi\| \xrightarrow{t \rightarrow \infty} 0.$$

Applying the reverse triangle inequality, we have

$$0 \leq \left| \|\psi_t\| - \|\psi\| \right| \leq \|\psi_t - \psi\| \xrightarrow{t \rightarrow \infty} 0,$$

which shows that $\|\psi_t\| \rightarrow \|\psi\|$ as $t \rightarrow \infty$. □

(2) Suppose U_t is a unitary operator that tends to the limit Ω , so that for all $|\psi\rangle$ in $\mathcal{H}$,

$$U_t|\psi\rangle \xrightarrow{t \rightarrow \infty} \Omega|\psi\rangle.$$

Then Ω is norm-preserving,

$$\|\Omega\psi\| = \left\| \lim_{t \rightarrow \infty} U_t\psi \right\| = \lim_{t \rightarrow \infty} \underbrace{\|U_t\psi\|}_{\|\psi\|, \text{ by unitarity}} = \|\psi\|.$$

By construction, Ω is defined for all vectors in $\mathcal{H}$, so we conclude that Ω is isometric. □

12-3 (Free-particle evolution of a Gaussian wavepacket)

Solution. In the following calculation we'll set $\hbar = 1$. We have

$$\begin{aligned}
 \langle \mathbf{x} | U^0(\tau) | \psi_{\text{in}} \rangle &= \int d^3p \langle \mathbf{x} | U^0(\tau) | \mathbf{p} \rangle \langle \mathbf{p} | \psi_{\text{in}} \rangle \\
 &= \int d^3p \exp\left(-\frac{ip^2\tau}{2m}\right) \langle \mathbf{x} | \mathbf{p} \rangle \langle \mathbf{p} | \psi_{\text{in}} \rangle \\
 &= \frac{1}{(2\pi)^{3/2}} \int d^3p \exp\left(-\frac{ip^2\tau}{2m}\right) \exp(i\mathbf{p} \cdot \mathbf{x}) \langle \mathbf{p} | \psi_{\text{in}} \rangle,
 \end{aligned} \tag{4}$$

where the second line uses the fact that $U^0(\tau) = \exp(-iP^2\tau/2m)$,⁵ and the third line writes out $\langle \mathbf{x} | \mathbf{p} \rangle$ explicitly. Next we find the momentum-space representation of $|\psi_{\text{in}}\rangle$:

$$\begin{aligned}
 \langle \mathbf{p} | \psi_{\text{in}} \rangle &= \int d^3x \langle \mathbf{p} | \mathbf{x} \rangle \langle \mathbf{x} | \psi_{\text{in}} \rangle \\
 &= \frac{1}{(2\pi)^{3/2}} \int d^3x \exp(-i\mathbf{p} \cdot \mathbf{x}) \exp\left(-\frac{(\mathbf{x} - \mathbf{a})^2}{2\xi^2}\right).
 \end{aligned}$$

We can regard this as the product of three integrals in x, y, z , respectively. First we have

$$\begin{aligned}
 \int_{-\infty}^{\infty} dx \exp(-ip_x x) \exp\left(-\frac{(x - a_x)^2}{2\xi^2}\right) \\
 &= \int_{-\infty}^{\infty} dx \exp\left(-\frac{x^2}{2\xi^2} + \frac{a_x x}{\xi^2} - ip_x x - \frac{a_x^2}{2\xi^2}\right) \\
 &= \exp\left(-\frac{a_x^2}{2\xi^2}\right) \int_{-\infty}^{\infty} dx \exp\left[-\frac{x^2}{2\xi^2} + \left(\frac{a_x - i\xi^2 p_x}{\xi^2}\right)x\right] \\
 &\equiv \exp\left(-\frac{a_x^2}{2\xi^2}\right) \underbrace{\int_{-\infty}^{\infty} dx \exp(-\alpha x^2 + \beta x)}_{\equiv I(\alpha, \beta)},
 \end{aligned}$$

where in the last line we define $\alpha \equiv 1/2\xi^2$ and $\beta \equiv (a_x - i\xi^2 p_x)/\xi^2$. Since α and β are complex numbers, this integral can be treated via the method of residues, but here we'll just quote Appendix A.2 of Shankar:

$$I(\alpha, \beta) = e^{\beta^2/4\alpha} \left(\frac{\pi}{\alpha}\right)^{1/2}, \quad \text{provided } \text{Re } \alpha > 0. \tag{5}$$

From this result (and a bit of algebra), we obtain

$$\int_{-\infty}^{\infty} dx \exp(-ip_x x) \exp\left(-\frac{(x - a_x)^2}{2\xi^2}\right) = (2\pi)^{1/2} \xi \exp\left(-\frac{p_x^2 \xi^2}{2} - ia_x p_x\right),$$

⁵Here we denote the square of the momentum operator by a capital letter, P^2 , to avoid confusion with the eigenvalue p^2 .

with similar results for the y and z integrals. Putting everything together, we find that the momentum-space representation of $|\psi_{\text{in}}\rangle$ is given by

$$\langle \mathbf{p} | \psi_{\text{in}} \rangle = \xi^3 \exp(-i\mathbf{p} \cdot \mathbf{a}) \exp\left(-\frac{p^2 \xi^2}{2}\right).$$

Inserting this back into Eq. (4), we get

$$\langle \mathbf{x} | \mathbf{U}^0(\tau) | \psi_{\text{in}} \rangle = \frac{\xi^3}{(2\pi)^{3/2}} \int d^3p \exp(i\mathbf{p} \cdot (\mathbf{x} - \mathbf{a})) \exp\left(-\frac{p^2 \xi^2}{2} - \frac{ip^2 \tau}{2m}\right), \quad (6)$$

and again we can regard this as the product of three integrals. The integral over p_x is

$$\begin{aligned} \int_{-\infty}^{\infty} dp_x \exp(ip_x(x - a_x)) \exp\left(-\frac{p_x^2 \xi^2}{2} - \frac{ip_x^2 \tau}{2m}\right) \\ = \int_{-\infty}^{\infty} dp_x \exp\left[-\left(\frac{\xi^2}{2} + \frac{i\tau}{2m}\right)p_x^2 + i(x - a_x)p_x\right] \\ \equiv \underbrace{\int_{-\infty}^{\infty} dp_x \exp(-\alpha p_x^2 + \beta p_x)}_{\equiv I(\alpha, \beta)} \end{aligned}$$

where we've defined $\alpha \equiv \xi^2/2 + i\tau/2m$ and $\beta \equiv i(x - a_x)$. Applying Eq. (5) again and using Mathematica to help with the algebra, we get

$$\begin{aligned} \int_{-\infty}^{\infty} dp_x \exp(i(p_x - a_x)x) \exp\left(-\frac{p_x^2 \xi^2}{2} - \frac{ip_x^2 \tau}{2m}\right) \\ = (2\pi)^{1/2} \left(\xi^2 + \frac{i\tau}{m}\right)^{-1/2} \exp\left[-\frac{m^2 \xi^2 (x - a_x)^2}{2(m^2 \xi^4 + \tau^2)} + i \frac{m\tau(x - a_x)^2}{2(m^2 \xi^4 + \tau^2)}\right] \\ = (2\pi)^{1/2} \left(\xi^4 + \frac{\tau^2}{m^2}\right)^{-1/4} \exp\left[-\frac{(x - a_x)^2}{2(\xi^2 + \tau^2/m^2 \xi^2)} + i \left(\frac{m\tau(x - a_x)^2}{2(m^2 \xi^4 + \tau^2)} - \frac{\theta}{2}\right)\right], \end{aligned}$$

where $\theta \equiv \tan^{-1}(\tau/m\xi^2)$. The integrals over p_y and p_z contribute similar terms, and so from Eq. (6) we obtain

$$\langle \mathbf{x} | \mathbf{U}^0(\tau) | \psi_{\text{in}} \rangle = \xi^3 \left(\xi^4 + \frac{\tau^2}{m^2}\right)^{-3/4} \exp\left[-\frac{(\mathbf{x} - \mathbf{a})^2}{2(\xi^2 + \tau^2/m^2 \xi^2)} + i \left(\frac{m\tau(\mathbf{x} - \mathbf{a})^2}{2(m^2 \xi^4 + \tau^2)} - \frac{3\theta}{2}\right)\right]$$

Finally, moving ξ^3 inside the parentheses and multiplying by the complex conjugate gives

$$\boxed{|\langle \mathbf{x} | \mathbf{U}^0(\tau) | \psi_{\text{in}} \rangle|^2 = \left(1 + \frac{\tau^2}{m^2 \xi^4}\right)^{-3/2} \exp\left[-\frac{(\mathbf{x} - \mathbf{a})^2}{\xi^2 + \tau^2/m^2 \xi^2}\right]},$$

which was what we wanted. □

12-4 (Taylor, Problem 2.1)

Solution. (a) Consider a finite linear combination of Gaussian vectors $|\phi^\alpha\rangle$, and call it $|\phi\rangle$:

$$|\phi\rangle = \sum_{\alpha} c_{\alpha} |\phi^{\alpha}\rangle.$$

Section 2-c showed that the asymptotic condition is satisfied for each Gaussian vector individually. That is, to each $|\phi^\alpha\rangle$ there is a corresponding $|\psi^\alpha\rangle$ such that

$$U(t)|\psi^\alpha\rangle - U^0(t)|\phi^\alpha\rangle \xrightarrow[t \rightarrow -\infty]{} 0. \quad (7)$$

To prove the asymptotic condition for $|\phi\rangle$, we need to find some vector which is asymptotic to the free orbit $U^0(t)|\phi\rangle$ as $t \rightarrow -\infty$. We claim that this is

$$|\chi\rangle = \sum_{\alpha} c_{\alpha} |\psi^{\alpha}\rangle.$$

This is straightforward to show by applying linearity of both evolution operators:

$$\begin{aligned} U(t)|\chi\rangle - U^0(t)|\phi\rangle &= U(t) \left(\sum_{\alpha} c_{\alpha} |\psi^{\alpha}\rangle \right) - U^0(t) \left(\sum_{\alpha} c_{\alpha} |\phi^{\alpha}\rangle \right) \\ &= \left(\sum_{\alpha} c_{\alpha} U(t)|\psi^{\alpha}\rangle \right) - \left(\sum_{\alpha} c_{\alpha} U^0(t)|\phi^{\alpha}\rangle \right) \\ &= \sum_{\alpha} c_{\alpha} [U(t)|\psi^{\alpha}\rangle - U^0(t)|\phi^{\alpha}\rangle]. \end{aligned}$$

Since each expression in square brackets tends to zero as $t \rightarrow -\infty$ [cf. Eq. (7)], the entire sum tends to zero in the same limit, which was what we wanted.

(b) Let $|\psi_{\text{in}}\rangle$ be an arbitrary vector in $\mathcal{L}^2(\mathbb{R}^3)$. Then there exists a finite sum of Gaussian vectors $|\phi\rangle$ such that, for any $\epsilon > 0$, we have $\|\psi_{\text{in}} - \phi\| \leq \epsilon$. Now by the result of part (a), the vector $|\phi\rangle$ satisfies the asymptotic condition: there is some $|\chi\rangle$ such that

$$U(t)|\chi\rangle - U^0(t)|\phi\rangle \xrightarrow[t \rightarrow -\infty]{} 0.$$

Now we observe that

$$\begin{aligned} \|U(t)|\chi\rangle - U^0(t)|\psi_{\text{in}}\rangle\| &= \|U(t)|\chi\rangle - U^0(t)|\phi\rangle + U^0(t)(|\phi\rangle - |\psi_{\text{in}}\rangle)\| \\ &\leq \|U(t)|\chi\rangle - U^0(t)|\phi\rangle\| + \|\phi - \psi_{\text{in}}\| \\ &\leq \|U(t)|\chi\rangle - U^0(t)|\phi\rangle\| + \epsilon. \end{aligned}$$

In the first line, we add and subtract the vector $U^0(t)|\phi\rangle$. In the second line, we apply the triangle inequality and the unitarity of $U^0(t)$. Finally, in the last line, we use the fact that $|\psi_{\text{in}}\rangle$ is approximated arbitrarily well by $|\phi\rangle$. As we showed earlier, the final quantity on the right

goes to zero as $t \rightarrow -\infty$, and since we can make ϵ arbitrarily small, we conclude

$$U(t)|\chi\rangle - U^0(t)|\psi_{\text{in}}\rangle \xrightarrow[t \rightarrow -\infty]{} 0.$$

Thus $|\psi_{\text{in}}\rangle$, which was arbitrary, satisfies the asymptotic condition. This completes the proof in Section 2-c. $\square$

12-5 (Taylor, Problem 2.2)

Solution. As we discussed in the lectures, the asymptotic condition will be satisfied if

$$\int_{-\infty}^0 d\tau \|VU^0(\tau)\psi_{\text{in}}\| < \infty \quad (8)$$

is satisfied. We then showed that

$$\begin{aligned} \|VU^0(\tau)\psi_{\text{in}}\|^2 &= \langle \psi_{\text{in}} | U^0(\tau)^\dagger V^\dagger V U^0(\tau) | \psi_{\text{in}} \rangle \\ &= \int d^3x' \int d^3x \langle \psi_{\text{in}} | U^0(\tau)^\dagger | x' \rangle \langle x' | V^\dagger V | x \rangle \langle x | U^0(\tau) | \psi_{\text{in}} \rangle. \end{aligned}$$

Since $V = |\zeta\rangle\langle\zeta|$, where $|\zeta\rangle$ is a fixed vector (which we'll assume is normalized), we have

$$\begin{aligned} \langle x' | V^\dagger V | x \rangle &= \langle x' | \zeta \rangle \langle \zeta | \zeta \rangle \langle \zeta | x \rangle \\ &= \langle x' | \zeta \rangle \langle \zeta | x \rangle \\ &= \zeta(\mathbf{x}') \zeta(\mathbf{x})^*. \end{aligned}$$

Thus, if we assume that $|\zeta\rangle$ has a Gaussian wave function, then we can write (up to an overall normalization factor)

$$\langle x' | V^\dagger V | x \rangle = \exp\left[-\frac{(\mathbf{x}' - \mathbf{b})^2}{2\sigma^2}\right] \exp\left[-\frac{(\mathbf{x} - \mathbf{b})^2}{2\sigma^2}\right].$$

Now by the result of Problem 3, we have

$$\begin{aligned} \langle \psi_{\text{in}} | U^0(\tau)^\dagger | \mathbf{x}' \rangle &= \langle \mathbf{x}' | U^0(\tau) | \psi_{\text{in}} \rangle^* \\ &= \xi^3 \left(\xi^4 + \frac{\tau^2}{m^2} \right)^{-3/4} \exp\left[-\frac{(\mathbf{x}' - \mathbf{a})^2}{2(\xi^2 + \tau^2/m^2\xi^2)} - i\left(\frac{m\tau(\mathbf{x}' - \mathbf{a})^2}{2(m^2\xi^4 + \tau^2)} - \frac{3\theta}{2} \right) \right], \end{aligned}$$

so that [notice that the terms involving $\theta \equiv \tan^{-1}(\tau/m\xi^2)$ cancel out]

$$\begin{aligned} \|VU^0(\tau)\psi_{\text{in}}\|^2 &= \left(1 + \frac{\tau^2}{m^2\xi^4} \right)^{-3/2} \\ &\times \int d^3x' \exp\left[-\frac{(\mathbf{x}' - \mathbf{a})^2}{2(\xi^2 + \tau^2/m^2\xi^2)} - \frac{(\mathbf{x}' - \mathbf{b})^2}{2\sigma^2} - i\left(\frac{m\tau(\mathbf{x}' - \mathbf{a})^2}{2(m^2\xi^4 + \tau^2)} \right) \right] \\ &\times \int d^3x \exp\left[-\frac{(\mathbf{x} - \mathbf{a})^2}{2(\xi^2 + \tau^2/m^2\xi^2)} - \frac{(\mathbf{x} - \mathbf{b})^2}{2\sigma^2} + i\left(\frac{m\tau(\mathbf{x} - \mathbf{a})^2}{2(m^2\xi^4 + \tau^2)} \right) \right]. \end{aligned}$$

The resulting integrals are messy, so I'll make extensive use of Mathematica here, together with Eq. (5). (See the notebook attached to the end of this problem set.) We find that

$$\begin{aligned} \int d^3x' (\dots) &= (2\pi)^{3/2} \left(\frac{1}{\sigma^2} + \frac{1}{\xi^2 - \frac{i\tau}{m}} \right)^{-3/2} \\ &\times \exp\left[-\frac{(\mathbf{a} - \mathbf{b})^2 m^2 (\xi^2 + \sigma^2)}{2[m^2(\xi^2 + \sigma^2)^2 + \tau^2]} - \frac{i(\mathbf{a} - \mathbf{b})^2 m\tau}{2[m^2(\xi^2 + \sigma^2)^2 + \tau^2]} \right], \end{aligned}$$

$$\int d^3x (\dots) = (2\pi)^{3/2} \left(\frac{1}{\sigma^2} + \frac{1}{\xi^2 + \frac{i\tau}{m}} \right)^{-3/2} \\ \times \exp \left[-\frac{(\mathbf{a} - \mathbf{b})^2 m^2 (\xi^2 + \sigma^2)}{2[m^2(\xi^2 + \sigma^2)^2 + \tau^2]} + \frac{i(\mathbf{a} - \mathbf{b})^2 m \tau}{2[m^2(\xi^2 + \sigma^2)^2 + \tau^2]} \right].$$

Finally, multiplying these together gives

$$\|V\mathbf{U}^0(\tau)|\psi_{\text{in}}\rangle\|^2 = (2\pi)^3 \left(1 + \frac{\tau^2}{m^2 \xi^4} \right)^{-3/2} \left[\left(\frac{1}{\sigma^2} + \frac{\xi^2}{\xi^4 + \tau^2/m^2} \right)^2 + \left(\frac{\tau/m}{\xi^4 + \tau^2/m^2} \right)^2 \right]^{-3/2} \\ \times \exp \left[-\frac{(\mathbf{a} - \mathbf{b})^2 m^2 (\xi^2 + \sigma^2)}{m^2 (\xi^2 + \sigma^2)^2 + \tau^2} \right] \\ \leq (2\pi)^3 \left(1 + \frac{\tau^2}{m^2 \xi^4} \right)^{-3/2} \left[\left(\frac{1}{\sigma^2} + \frac{\xi^2}{\xi^4 + \tau^2/m^2} \right)^2 + \left(\frac{\tau/m}{\xi^4 + \tau^2/m^2} \right)^2 \right]^{-3/2} \\ = 8\pi^3 \left(\frac{\xi^2 + \sigma^2 + \tau^2/m^2}{\xi^4 \sigma^2} \right)^{-3/2}.$$

This decays fast enough as $t \rightarrow -\infty$ for the integral in (8) to converge, so we conclude that the asymptotic condition is satisfied for the (nonlocal) potential $V = |\zeta\rangle\langle\zeta|$.⁶ $\square$

⁶This shows that there are potentials which don't satisfy conditions I–III, but for which the asymptotic condition still holds.

In[1]:= $\$Assumptions = \{px \in \text{Reals}, ax \in \text{Reals}, \xi > 0, m > 0, \tau \in \text{Reals}\};$

In[2]:= $(\star \text{ xp integral } \star)$

$$-\frac{(xp - ax)^2}{2 \left(\xi^2 + \frac{\tau^2}{m^2 \xi^2} \right)} - \frac{(xp - bx)^2}{2 \sigma^2} - i \left(\frac{m \tau (xp - ax)^2}{2 (m^2 \xi^4 + \tau^2)} \right) // \text{Expand} // \text{Collect}[\#, xp] \&$$

$$\text{Out[2]} = -\frac{bx^2}{2 \sigma^2} - \frac{i ax^2 m \tau}{2 (m^2 \xi^4 + \tau^2)} - \frac{ax^2}{2 \left(\xi^2 + \frac{\tau^2}{m^2 \xi^2} \right)} +$$

$$xp^2 \left(-\frac{1}{2 \sigma^2} - \frac{i m \tau}{2 (m^2 \xi^4 + \tau^2)} - \frac{1}{2 \left(\xi^2 + \frac{\tau^2}{m^2 \xi^2} \right)} \right) + xp \left(\frac{bx}{\sigma^2} + \frac{i ax m \tau}{m^2 \xi^4 + \tau^2} + \frac{ax}{\xi^2 + \frac{\tau^2}{m^2 \xi^2}} \right)$$

In[3]:= $(\star \text{ xp integral } \star)$

$$\frac{\text{Exp} \left[-\frac{bx^2}{2 \sigma^2} - \frac{i ax^2 m \tau}{2 (m^2 \xi^4 + \tau^2)} - \frac{ax^2}{2 \left(\xi^2 + \frac{\tau^2}{m^2 \xi^2} \right)} \right] \text{Exp} \left[\frac{\beta^2}{4 \alpha} \right] \sqrt{\pi}}{\sqrt{\alpha}} /. \left\{ \alpha \rightarrow - \left(-\frac{1}{2 \sigma^2} - \frac{i m \tau}{2 (m^2 \xi^4 + \tau^2)} - \frac{1}{2 \left(\xi^2 + \frac{\tau^2}{m^2 \xi^2} \right)} \right), \right.$$

$$\left. \beta \rightarrow \left(\frac{bx}{\sigma^2} + \frac{i ax m \tau}{m^2 \xi^4 + \tau^2} + \frac{ax}{\xi^2 + \frac{\tau^2}{m^2 \xi^2}} \right) \right\} // \text{FullSimplify}$$

$$\text{Out[3]} = \frac{e^{-\frac{(ax-bx)^2 m}{2 m (\xi^2 + \sigma^2) - 2 i \tau}} \sqrt{2 \pi}}{\sqrt{\frac{1}{\sigma^2} + \frac{m}{m \xi^2 - i \tau}}}$$

In[4]:= $(\star \text{ further simplification } \star)$

$$-\frac{(ax - bx)^2 m}{2 m (\xi^2 + \sigma^2) - 2 i \tau} // \text{ComplexExpand}$$

$$\text{Out[4]} = -\frac{2 ax^2 m^2 \xi^2}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} + \frac{4 ax bx m^2 \xi^2}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} - \frac{2 bx^2 m^2 \xi^2}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} -$$

$$\frac{2 ax^2 m^2 \sigma^2}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} + \frac{4 ax bx m^2 \sigma^2}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} - \frac{2 bx^2 m^2 \sigma^2}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} +$$

$$i \left(-\frac{2 ax^2 m \tau}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} + \frac{4 ax bx m \tau}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} - \frac{2 bx^2 m \tau}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} \right)$$

In[5]:= (* real part *)

$$-\frac{2 a x^2 m^2 \xi^2}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} + \frac{4 a x b x m^2 \xi^2}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} - \frac{2 b x^2 m^2 \xi^2}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} -$$

$$\frac{2 a x^2 m^2 \sigma^2}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} + \frac{4 a x b x m^2 \sigma^2}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} - \frac{2 b x^2 m^2 \sigma^2}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} // \text{Simplify}$$

$$\text{Out[5]} = -\frac{(a x - b x)^2 m^2 (\xi^2 + \sigma^2)}{2 (m^2 (\xi^2 + \sigma^2)^2 + \tau^2)}$$

In[6]:= (* imaginary part *)

$$\mathbf{i} \left(-\frac{2 a x^2 m \tau}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} + \frac{4 a x b x m \tau}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} - \frac{2 b x^2 m \tau}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} \right) // \text{Simplify}$$

$$\text{Out[6]} = -\frac{\mathbf{i} (a x - b x)^2 m \tau}{2 (m^2 (\xi^2 + \sigma^2)^2 + \tau^2)}$$

In[7]:= (* x argument *)

$$-\frac{(x - a x)^2}{2 (\xi^2 + \frac{\tau^2}{m^2 \xi^2})} - \frac{(x - b x)^2}{2 \sigma^2} + \mathbf{i} \left(\frac{m \tau (x - a x)^2}{2 (m^2 \xi^4 + \tau^2)} \right) // \text{Expand} // \text{Collect}[\#, x] \&$$

$$\text{Out[7]} = -\frac{b x^2}{2 \sigma^2} + \frac{\mathbf{i} a x^2 m \tau}{2 (m^2 \xi^4 + \tau^2)} - \frac{a x^2}{2 (\xi^2 + \frac{\tau^2}{m^2 \xi^2})} +$$

$$x^2 \left(-\frac{1}{2 \sigma^2} + \frac{\mathbf{i} m \tau}{2 (m^2 \xi^4 + \tau^2)} - \frac{1}{2 (\xi^2 + \frac{\tau^2}{m^2 \xi^2})} \right) + x \left(\frac{b x}{\sigma^2} - \frac{\mathbf{i} a x m \tau}{m^2 \xi^4 + \tau^2} + \frac{a x}{\xi^2 + \frac{\tau^2}{m^2 \xi^2}} \right)$$

In[8]:= (* x integral *)

$$\frac{\text{Exp} \left[-\frac{b x^2}{2 \sigma^2} + \frac{\mathbf{i} a x^2 m \tau}{2 (m^2 \xi^4 + \tau^2)} - \frac{a x^2}{2 (\xi^2 + \frac{\tau^2}{m^2 \xi^2})} \right] \text{Exp} \left[\frac{\beta^2}{4 \alpha} \right] \sqrt{\pi}}{\sqrt{\alpha}} /.$$

$$\left\{ \alpha \rightarrow -\left(-\frac{1}{2 \sigma^2} + \frac{\mathbf{i} m \tau}{2 (m^2 \xi^4 + \tau^2)} - \frac{1}{2 (\xi^2 + \frac{\tau^2}{m^2 \xi^2})} \right), \right.$$

$$\left. \beta \rightarrow \left(\frac{b x}{\sigma^2} - \frac{\mathbf{i} a x m \tau}{m^2 \xi^4 + \tau^2} + \frac{a x}{\xi^2 + \frac{\tau^2}{m^2 \xi^2}} \right) \right\} // \text{FullSimplify}$$

$$\text{Out[8]} = \frac{e^{-\frac{(a x - b x)^2 m}{2 m (\xi^2 + \sigma^2) + 2 \mathbf{i} \tau}} \sqrt{2 \pi}}{\sqrt{\frac{1}{\sigma^2} + \frac{m}{m \xi^2 + \mathbf{i} \tau}}}$$

In[9]:= (* further simplifying *)

$$\begin{aligned}
 & - \frac{(ax - bx)^2 m}{2 m (\xi^2 + \sigma^2) + 2 i \tau} // \text{ComplexExpand} \\
 \text{Out[9]} = & - \frac{2 ax^2 m^2 \xi^2}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} + \frac{4 ax bx m^2 \xi^2}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} - \frac{2 bx^2 m^2 \xi^2}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} - \\
 & \frac{2 ax^2 m^2 \sigma^2}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} + \frac{4 ax bx m^2 \sigma^2}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} - \frac{2 bx^2 m^2 \sigma^2}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} + \\
 & i \left(\frac{2 ax^2 m \tau}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} - \frac{4 ax bx m \tau}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} + \frac{2 bx^2 m \tau}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} \right)
 \end{aligned}$$

In[10]:= (* real part *)

$$\begin{aligned}
 & - \frac{2 ax^2 m^2 \xi^2}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} + \frac{4 ax bx m^2 \xi^2}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} - \frac{2 bx^2 m^2 \xi^2}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} - \\
 & \frac{2 ax^2 m^2 \sigma^2}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} + \frac{4 ax bx m^2 \sigma^2}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} - \frac{2 bx^2 m^2 \sigma^2}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} // \text{Simplify} \\
 \text{Out[10]} = & - \frac{(ax - bx)^2 m^2 (\xi^2 + \sigma^2)}{2 (m^2 (\xi^2 + \sigma^2)^2 + \tau^2)}
 \end{aligned}$$

In[11]:= (* imaginary part *)

$$\begin{aligned}
 & i \left(\frac{2 ax^2 m \tau}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} - \frac{4 ax bx m \tau}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} + \frac{2 bx^2 m \tau}{4 m^2 (\xi^2 + \sigma^2)^2 + 4 \tau^2} \right) // \text{Simplify} \\
 \text{Out[11]} = & \frac{i (ax - bx)^2 m \tau}{2 (m^2 (\xi^2 + \sigma^2)^2 + \tau^2)}
 \end{aligned}$$

12-6 (Taylor, Problem 6.2)

Solution. First, we note that for any pair of initial and final states $|\phi\rangle, |\chi\rangle$, we have

$$\langle\chi|S|\phi\rangle = \langle\phi_T|S|\chi_T\rangle \quad (\text{T invariance})$$

$$\langle\chi|S|\phi\rangle = \langle\chi_P|S|\phi_P\rangle \quad (\text{P invariance})$$

(See the discussion on pages 91 and 95 of Taylor.)

(a) By the definition of T and P, for any momentum eigenvector $|\mathbf{p}\rangle$ we have $T|\mathbf{p}\rangle = |-\mathbf{p}\rangle$ and $P|\mathbf{p}\rangle = |-\mathbf{p}\rangle$. Thus

$$\langle\mathbf{p}'|S|\mathbf{p}\rangle = \langle-\mathbf{p}'|S|-\mathbf{p}\rangle \quad (\text{T invariance})$$

$$= \langle\mathbf{p}'|S|\mathbf{p}\rangle \quad (\text{P invariance}),$$

that is, the momentum-space S matrix is symmetric.

(b) From the definition (6.6), the orbital angular momentum eigenstates $|E, l, m\rangle$ correspond to the spatial wave functions

$$\langle\mathbf{x}|E, l, m\rangle = i^l \left(\frac{2m}{\pi p}\right)^{1/2} \frac{1}{r} \hat{j}_l(pr) Y_l^m(\hat{\mathbf{x}}),$$

where $\hat{j}_l(z) = z j_l(z)$, with $j_l(z)$ the spherical Bessel function. Now, the effect of T for a spinless particle is to send a position-space wave function to its complex conjugate, so

$$\begin{aligned} \langle\mathbf{x}|T|E, l, m\rangle &= (-i)^l \left(\frac{2m}{\pi p}\right)^{1/2} \frac{1}{r} \hat{j}_l(pr) Y_l^m(\hat{\mathbf{x}})^* \\ &= (-1)^{l-m} i^l \left(\frac{2m}{\pi p}\right)^{1/2} \frac{1}{r} \hat{j}_l(pr) Y_l^{-m}(\hat{\mathbf{x}}) \\ &= (-1)^{l-m} \langle\mathbf{x}|E, l, -m\rangle, \end{aligned}$$

where in the second line we use the property $Y_l^m(\hat{\mathbf{x}})^* = (-1)^m Y_l^{-m}(\hat{\mathbf{x}})$, and in the third line we use the definition (6.6). Since $|\mathbf{x}\rangle$ was arbitrary, we conclude that

$$T|E, l, m\rangle = (-1)^{l-m} |E, l, -m\rangle.$$

Next we consider the parity operator P:

$$\begin{aligned} \langle\mathbf{x}|P|E, l, m\rangle &= i^l \left(\frac{2m}{\pi p}\right)^{1/2} \frac{1}{r} \hat{j}_l(pr) Y_l^m(-\hat{\mathbf{x}}) \\ &= (-1)^l \langle\mathbf{x}|E, l, m\rangle \end{aligned}$$

using the parity of the spherical harmonics, $Y_l^m(-\hat{\mathbf{x}}) = (-1)^l Y_l^m(\hat{\mathbf{x}})$. This shows that

$$P|E, l, m\rangle = (-1)^l |E, l, m\rangle.$$

Thus, assuming parity and time reversal invariance, the S -matrix elements in the angular-momentum representation are related by

$$\begin{aligned}\langle E', l', m' | S | E, l, m \rangle &= (-1)^{l'+l} \langle E', l', m' | S | E, l, m \rangle \\ &= (-1)^{l'+l} (-1)^{l'-m'} (-1)^{l-m} \langle E, l, -m | S | E', l', -m' \rangle \\ &= (-1)^{m-m'} \langle E, l, -m | S | E', l', -m' \rangle,\end{aligned}$$

where the first line uses $\mathbf{P}$ invariance and the second line uses $\mathbf{T}$ invariance.

(c) No, we showed that the $\mathbf{S}$ operator is represented by a symmetric matrix in the momentum representation, but that it is not symmetric in the angular momentum representation. $\square$

It is worth pointing out that the relation [Eq. (6.7) in Taylor]

$$\langle E', l', m' | S | E, l, m \rangle = \delta(E' - E) \delta_{l'l} \delta_{m'm} s_l(E)$$

does not hold in the present case, because that result was obtained using the assumption that $\mathbf{S}$ commutes with $\mathbf{L}$, which in turn followed from rotational invariance.

12-7 (Taylor, Problem 6.3a)

Solution. The rotation operator $R(\alpha)$ can be rewritten in the following way:

$$\begin{aligned}
 R(\alpha) &= \exp(-i\alpha \cdot \mathbf{S}) \\
 &\rightarrow \exp(-i\alpha \hat{\alpha} \cdot \boldsymbol{\sigma}/2) \\
 &= \sum_{n=0}^{\infty} \left(-\frac{i\alpha}{2}\right)^n \frac{1}{n!} (\hat{\alpha} \cdot \boldsymbol{\sigma})^n \\
 &= \left[\sum_{n=0}^{\infty} (-i)^{2n} \left(\frac{\alpha}{2}\right)^{2n} \frac{1}{(2n)!} (\hat{\alpha} \cdot \boldsymbol{\sigma})^{2n} \right] + \left[\sum_{n=0}^{\infty} (-i)^{2n+1} \left(\frac{\alpha}{2}\right)^{2n+1} \frac{1}{(2n+1)!} (\hat{\alpha} \cdot \boldsymbol{\sigma})^{2n+1} \right] \\
 &= \left[\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \left(\frac{\alpha}{2}\right)^{2n} \right] I - i \left[\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \left(\frac{\alpha}{2}\right)^{2n+1} \right] (\hat{\alpha} \cdot \boldsymbol{\sigma}) \\
 &= \cos(\alpha/2) I - i \sin(\alpha/2) (\hat{\alpha} \cdot \boldsymbol{\sigma}),
 \end{aligned}$$

where in the second line we move to the S_z basis, and in the second-to-last line we use the fact that $(\hat{\alpha} \cdot \boldsymbol{\sigma})^2 = I$ for any unit vector $\hat{\alpha}$. (See page 381 of Shankar for a proof.) Using this result, we have

$$R(\pi \hat{\mathbf{y}}) \rightarrow -i\sigma_y = -i \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Thus, the effect of this rotation on the spin-up and spin-down states is given by

$$\begin{aligned}
 R(\pi \hat{\mathbf{y}})|+\rangle &\rightarrow \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \rightarrow |-\rangle, \\
 R(\pi \hat{\mathbf{y}})|-\rangle &\rightarrow \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow (-1)|+\rangle.
 \end{aligned}$$

Since $s = 1/2$ and $m = \pm 1/2$, it is equivalent to write

$$\boxed{R(\pi \hat{\mathbf{y}})|m\rangle = (-1)^{s-m}|-m\rangle,}$$

which was what we wanted to prove. □

12-8 (Taylor, Problem 6.9)

Solution. The scattering amplitude for the process $(\mathbf{p}', \xi' \leftarrow \mathbf{p}, \xi)$, where ξ and ξ' stand for the initial and final spin states of the system, respectively, can be obtained from

$$\langle \mathbf{p}', \xi' | (S - 1) | \mathbf{p}, \xi \rangle = \frac{i}{2\pi m} \delta(E_{p'} - E_p) f(\mathbf{p}', \xi' \leftarrow \mathbf{p}, \xi). \quad (9)$$

Writing the left-hand side as an expansion in angular-momentum states, we get

$$\begin{aligned} \langle \mathbf{p}', \xi' | (S - 1) | \mathbf{p}, \xi \rangle &= \int dE \sum_{l,j} \langle \mathbf{p}', \xi' | (S - 1) | E, l, s, j, m \rangle \langle E, l, s, j, m | \mathbf{p}, \xi \rangle \\ &= \int dE \sum_{l,j} (e^{2i\delta_l^j} - 1) \langle \mathbf{p}', \xi' | E, l, s, j, m \rangle \langle E, l, s, j, m | \mathbf{p}, \xi \rangle, \end{aligned} \quad (10)$$

where in the first line we can omit the sum over s because total spin is conserved, and omit the sum over m because total angular momentum is conserved due to rotational invariance. The second line follows from the fact that the eigenvectors $|E, l, s, j, m\rangle$ of $\mathbf{J}^2$ and J_z are also eigenvectors of $\mathbf{S}$ for rotationally invariant systems [see (6.48) and (6.49)].

Following the discussion starting on page 98 of Taylor, we can make use of R and P invariance to choose $\mathbf{p}$ along the z axis and write the amplitude matrix $F(\mathbf{p}' \leftarrow \mathbf{p})$ as

$$F(E, \theta, \phi) = \begin{pmatrix} a(E, \theta) & b(E, \theta)e^{-i\phi} \\ -b(E, \theta)e^{i\phi} & a(E, \theta) \end{pmatrix},$$

where $a(E, \theta)$ and $b(E, \theta)$ correspond to the spin-nonflip and flip amplitudes, respectively, and the latter is obtained by choosing $\mathbf{p}'$ to lie in the half plane $\phi = 0$. We will compute both of these amplitudes.

To begin, we suppose (without loss of generality) that the initial state is spin-up, and we note that $\hat{\mathbf{p}} = \hat{\mathbf{z}}$ fixes $m_l = 0$. This implies $m = m_l + m_s = 1/2$. Next, we note that for the final state

$$\begin{aligned} \langle \mathbf{p}', \xi' | E, l, s, j, m \rangle &= \sum_{m_l, m_s} \langle \mathbf{p}', \xi' | E, l, s, m_l, m_s \rangle \langle l s m_l m_s | j m \rangle \\ &= \sum_{m_l, m_s} \langle \mathbf{p}' | E, l, m_l \rangle \langle \xi' | s, m_s \rangle \langle l s m_l m_s | j m \rangle \\ &= \sum_{m_l, m_s} (m p')^{-1/2} \delta(E_{p'} - E) Y_l^{m_l}(\hat{\mathbf{p}}') \langle \xi' | s, m_s \rangle \langle l s m_l m_s | j m \rangle \\ &= \sum_{m_s} (m p')^{-1/2} \delta(E_{p'} - E) Y_l^{-m_s + \frac{1}{2}}(\hat{\mathbf{p}}') \langle \xi' | s, m_s \rangle \langle l s, m_l = -m_s + \frac{1}{2}, m_s | j m \rangle, \end{aligned}$$

where the first line uses (6.47), the second line uses the equation preceding (6.47), the third line uses (6.10), and the final line uses the conservation of total angular momentum $m = m_l + m_s =$

1/2. By the same procedure, we find for the initial state

$$\begin{aligned}\langle E, l, s, j, m | \mathbf{p}, + \rangle &= \sum_{m_s} (mp)^{-1/2} \delta(E_p - E) Y_l^0(\hat{\mathbf{p}}) \langle s, m_s | + \rangle \langle ls, m_l = 0, m_s | jm \rangle \\ &= (mp)^{-1/2} \delta(E_p - E) Y_l^0(\hat{\mathbf{p}}) \langle ls, m_l = 0, m_s = \frac{1}{2} | jm \rangle.\end{aligned}$$

Substituting these results into Eq. (10) and performing the integral over E , we obtain

$$\begin{aligned}\langle \mathbf{p}', \xi' | (S - 1) | \mathbf{p}, + \rangle &= \frac{1}{mp} \delta(E_{p'} - E_p) \sum_{l, j, m_s} (e^{2i\delta_l^j} - 1) Y_l^0(\hat{\mathbf{p}}) Y_l^{-m_s + \frac{1}{2}}(\hat{\mathbf{p}}') \langle \xi' | s, m_s \rangle \\ &\quad \times \langle ls, m_l = -m_s + \frac{1}{2}, m_s | jm \rangle \langle ls, m_l = 0, m_s = \frac{1}{2} | jm \rangle \\ &= \frac{2i}{m} \delta(E_{p'} - E_p) \sum_{l, j, m_s} Y_l^0(\hat{\mathbf{p}}) Y_l^{-m_s + \frac{1}{2}}(\hat{\mathbf{p}}') \langle \xi' | s, m_s \rangle \\ &\quad \times f_l^j(E_p) \langle ls, m_l = -m_s + \frac{1}{2}, m_s | jm \rangle \langle ls, m_l = 0, m_s = \frac{1}{2} | jm \rangle \\ &= \frac{2i}{m} \delta(E_{p'} - E_p) \sum_{l, m_s} \sqrt{\frac{2l+1}{4\pi}} Y_l^{-m_s + \frac{1}{2}}(\hat{\mathbf{p}}') \langle \xi' | s, m_s \rangle \\ &\quad \times \left[f_l^+(E_p) \langle ls, m_l = -m_s + \frac{1}{2}, m_s | j = l + \frac{1}{2}, m \rangle \langle ls, m_l = 0, m_s = \frac{1}{2} | j = l + \frac{1}{2}, m \rangle \right. \\ &\quad \left. + f_l^-(E_p) \langle ls, m_l = -m_s + \frac{1}{2}, m_s | j = l - \frac{1}{2}, m \rangle \langle ls, m_l = 0, m_s = \frac{1}{2} | j = l - \frac{1}{2}, m \rangle \right],\end{aligned}$$

where in the third line we perform the summation over j , and we've used the following:

$$f_l^j(E) \equiv \frac{e^{2i\delta_l^j} - 1}{2ip}, \quad Y_l^0(\hat{\mathbf{z}}) = \sqrt{\frac{2l+1}{4\pi}}.$$

Now we can compute the scattering amplitudes. Referring back to Eq. (9) and using the provided Clebsch-Gordan coefficients, we see that for the spin-nonflip case [this uses the notation of (5.7)]

$$\begin{aligned}f_{++}(\mathbf{p}' \leftarrow \mathbf{p}) &= 4\pi \sum_l \sqrt{\frac{2l+1}{4\pi}} Y_l^0(\hat{\mathbf{p}}') \left(f_l^+(E) \frac{l+1}{2l+1} + f_l^-(E) \frac{l}{2l+1} \right) \\ &= 4\pi \sum_l \frac{2l+1}{4\pi} P_l(\cos \theta) \frac{1}{2l+1} [(l+1)f_l^+(E) + lf_l^-(E)] \\ &= \sum_l [(l+1)f_l^+(E) + lf_l^-(E)] P_l(\cos \theta).\end{aligned}$$

Finally, since $f_{++} = a$ [see (6.40)], we conclude that

$$\boxed{a(E, \theta) = \sum_{l=0}^{\infty} [(l+1)f_l^+(E) + lf_l^-(E)] P_l(\cos \theta)},$$

as we wanted to show. Meanwhile, for the spin-flip case we have $\xi' = -1/2$, so that

$$\begin{aligned} f_{-+}(\mathbf{p}' \leftarrow \mathbf{p}) &= 4\pi \sum_l \sqrt{\frac{2l+1}{4\pi}} Y_l^1(\hat{\mathbf{p}}') \left[f_l^+(E) \left(\frac{l}{2l+1} \right)^{1/2} \left(\frac{l+1}{2l+1} \right)^{1/2} \right. \\ &\quad \left. - f_l^-(E) \left(\frac{l+1}{2l+1} \right)^{1/2} \left(\frac{l}{2l+1} \right)^{1/2} \right] \\ &= \sum_l \sqrt{4\pi \frac{l(l+1)}{2l+1}} [f_l^+(E) - f_l^-(E)] Y_l^1(\hat{\mathbf{p}}'). \end{aligned}$$

By the discussion preceding (6.36), the amplitude $b(E, \theta)$ is obtained by choosing $\mathbf{p}'$ to lie in the half-plane $\phi = 0$. [In fact $b(E, \theta)$ was defined to be the part of $f_{-+}(E, \theta, \phi)$ with the ϕ dependence peeled off.] Thus

$$b(E, \theta) = \sum_l \sqrt{4\pi \frac{l(l+1)}{2l+1}} [f_l^+(E) - f_l^-(E)] Y_l^1(\theta, \phi = 0). \quad (11)$$

Then we have [see (12.5.41) of Shankar]

$$\begin{aligned} Y_l^1(\theta, \phi = 0) &= \sqrt{\frac{(2l+1)(l-1)!}{4\pi(l+1)!}} (-1) P_l^1(\cos \theta) \\ &= \sqrt{\frac{1}{4\pi} \frac{2l+1}{l(l+1)}} (1 - \cos^2 \theta)^{1/2} P_l'(\cos \theta), \end{aligned}$$

where the second line uses the definition of the associated Legendre polynomials $P_l^m(\cos \theta)$. Substituting this back into Eq. (11), we obtain

$$b(E, \theta) = \sum_{l=1}^{\infty} [f_l^+(E) - f_l^-(E)] \sin \theta P_l'(\cos \theta),$$

which was what we wanted to show.⁷ □

⁷The indexing starts at $l = 1$ because spin-flip corresponds to $m_s = -1/2$, so that $m_l = m - m_s = 1$. But $|m_l| \leq l$, so only $l \geq 1$ can contribute. Alternatively, the Legendre polynomial P_0 is constant, so its derivative is zero, and the first nontrivial contribution comes at $l = 1$.